Rigidity of homogeneous mechanical families with common invariant force means

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Abstract

We study deformations of natural Hamiltonians for which all invariant probabilities on a fixed energy shell give the same mean parameter force. In two normal dimensions, a connected jointly C^2 family of positive homogeneous wells of degree three, four, or six containing one radial member remains radial. No polynomial assumption, central symmetry, or joint convexity is imposed for this classification. A second classification concerns binary quartics near $x^4 + 12x^2y^2 + 16y^4$: after phase-volume normalization, a connected family entering a suitable coefficient neighborhood consists of rotations of a single well. The radial and nonradial mechanisms are different. The former uses exact periodic actions and, in degree six, a mechanical brake-boundary response that excludes high-frequency escape; the latter uses constrained action curvature through a rank jump. These classifications produce controlled force correctors. Under joint convexity of the original potential and an actual compact selected tail, the correctors imply point selection for damping $(1 + 2/p)/t$. The selected limit maximizes the frozen phase-volume root, while the physical mechanical energy has a positive t^{-2} leading asymptotic. An explicit radial family supplies a checkable nontrivial set of initial data for the compact-tail hypothesis.

Keywords. Homogeneous Hamiltonian; invariant force mean; finite deformation rigidity; periodic action; dissipative selection.

Reconstruction convention. The main text is editable LaTeX. Complete long technical proofs are preserved as vector PDF source folios in the proof archive of this document. Source numbers are explicitly prefixed and are not the new theorem numbers. This preserves formulas from the only available original, a 250-page PDF. The archive has not been independently recertified in this reconstruction. The two new consequences in Sections 7.4 and 8 are proved here.

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1 The mechanical question and the main results

Let $S \subset \mathbb{R}^m$ be open and let $P \in C^2(S \times \mathbb{R}^2)$ be positive away from the normal origin and positively homogeneous of degree $p > 2$:

$$P(s, rZ) = r^p P(s, Z), \qquad H_s(Z, W) = \tfrac{1}{2}|W|^2 + P(s, Z), \qquad r > 0.$$

The parameter s labels a frozen well. The force exerted by the normal motion on this parameter is $-P_s$. We impose

$$\int P_s(s, Z)\, d\mu = b(s) \quad \text{for every invariant probability on } \{H_s = 1\}. \tag{CM}$$

This is equality between statistical states at the same parameter and energy. It is not equality of instantaneous forces, and it makes no ergodicity assumption. By homogeneity the common mean on $H_s = h > 0$ is $hb(s)$. The central question is whether such equality permits a genuine change of the shape of a mechanical well.

The answer below concerns finite families, not merely their derivatives at one well. An infinitesimal condition can fail to exclude a family whose first derivative vanishes at the base; in an infinite-dimensional angular class it can also fail to control oscillation at increasing frequencies. Both distinctions are essential here.

Theorem 1.1 (Radial finite rigidity)**.** *Let $p \in \{3, 4, 6\}$ and let $S \subset \mathbb{R}^m$ be connected and open, with $m < \infty$. Suppose that $P \in C^2(S \times \mathbb{R}^2)$ is positive for $Z \neq 0$, positively homogeneous of degree p in Z, and satisfies (CM) throughout S. If*

$$P(s_*, Z) = \kappa_* |Z|^p \quad \text{for some } s_* \in S,\ \kappa_* > 0,$$

then $P(s, Z) = \kappa(s)|Z|^p$ on $S \times \mathbb{R}^2$, where $\kappa > 0$ is C^2.

The theorem is [source Theorems 1, 5, 17]. Positivity on the unit circle gives local uniform coercivity. No angular C^3 hypothesis is hidden in the statement, including at degree three. Joint convexity belongs to the subsequent dynamical application and is not a frozen rigidity hypothesis.

Let $\mathcal{V}_4$ be the five-dimensional coefficient space of real binary quartic forms and set

$$Q_0(x, y) = x^4 + 12x^2y^2 + 16y^4, \qquad \mathcal{I}(P) = \int_0^{2\pi} P(e_\theta)^{-1/2}\, d\theta.$$

Use $I_0 = \mathcal{I}(Q_0)$ and normalize by

$$V(s, Z) = \left(\frac{\mathcal{I}(P(s, \cdot))}{I_0}\right)^2 P(s, Z). \tag{1}$$

Theorem 1.2 (Quartic coefficient-neighborhood rigidity)**.** *There is a coefficient-open neighborhood $\mathcal{U} \subset \mathcal{V}_4$ of Q_0 with the following property. Let $P \in C^2(S \times \mathbb{R}^2)$ be a positive binary-quartic family satisfying (CM) on connected open S. If $V(s_*, \cdot) \in \mathcal{U}$ at one parameter and $Q = V(s_*, \cdot)$, then*

$$V(s, \cdot) \in \{Q(R_\theta \cdot) : \theta \in \mathbb{R}\} \qquad (s \in S).$$

The stabilizer of Q in $SO(2)$ is $\{I, -I\}$. Locally $P(s, Z) = \alpha(s) Q(R_{\theta(s)} Z)$ with positive C^2 amplitude and C^2 angle. If S is simply connected the angle can be chosen globally.

This is [source Theorem 2]; its proof is retained in Archive A.5. No explicit radius for $\mathcal{U}$ is claimed. Only one normalized member must enter the neighborhood. Nearby members need not remain integrable, parabolically separable, or reflection symmetric. On general S we use intrinsic rotation coefficients instead of assuming a global angle.

1.1 A dynamical application, with its additional hypotheses

The original dissipative equations are

$$\ddot{s} + \frac{1+2/p}{t}\dot{s} + P_s(s,q) = 0, \qquad \ddot{q} + \frac{1+2/p}{t}\dot{q} + P_q(s,q) = 0, \qquad t > 0. \tag{2}$$

For $\nu = 1/2 + 1/p$ define

$$D(s) = \mathrm{vol}_4\{H_s \le 1\}^{1/2}, \qquad I(s,h) = D(s)h^\nu, \qquad U_I(s) = \left(\frac{I}{D(s)}\right)^{1/\nu}. \tag{3}$$

Theorem 1.3 (Selection on an actual compact tail)**.** *Let $U \subset \mathbb{R}^m$ be open and convex, and let $P \in C^2(U \times \mathbb{R}^2)$ be jointly convex in the original variables and p-homogeneous, $p \in \{3,4,6\}$. Assume normal coercivity near an actual selected tail $s(t) \in K \Subset U$. Suppose that there is a connected open $S \subset U$ with $K \Subset S$ on which Theorem 1.1 applies, or on which $p = 4$ and Theorem 1.2 applies. Put*

$$a = 2/p, \quad \sigma = \log t, \quad \varepsilon = t^{-a}, \quad q = \varepsilon Z,$$
$$W = \varepsilon Z_\sigma, \quad k = s_\sigma, \quad H = \tfrac{1}{2}|W|^2 + P(s,Z), \quad E = H + \tfrac{1}{2}|k|^2.$$

If the normal state is not identically zero, then

$$E \longrightarrow E_\infty > 0, \qquad I(s,H) \longrightarrow I_\infty > 0, \qquad I - I_\infty = O(t^{-2/p}),$$

and $s(t) \to s_\infty \in \arg\min_U U_{I_\infty} = \arg\max_U D$. Also $q(t) \to 0$.

The compact-tail condition is an assumption on an actual solution, not a conclusion for all initial data. Section 8 supplies a concrete sufficient condition in one family. The zero-normal branch satisfies $k' = -ak$ and converges to $s(\sigma_0) + k(\sigma_0)/a$; this point need not maximize D.

The coefficient $1 + 2/p$ is the balance value for the exact homogeneous rescaling used here. We do not prove it to be a sharp convergence threshold among all damping coefficients. Since $p > 2$, the normal minimum is degenerate and the whole set $q = 0$ has zero potential. Decay of the original energy alone cannot distinguish points of that set. The additional selection mechanism is the dependence of the frozen normal phase volume on s.

1.2 Position relative to existing methods

The fixed-energy action variation used below is classical; see Brännström and Gelfreich [BG08]. Its use here is to constrain a family, rather than to prescribe a finite slow-time drift. Jarzynski [J13] formulates classical adiabatic transport through a centered generator, and Deffner, Jarzynski, and del Campo [DJdC14] construct counterdiabatic protocols for scale-invariant driving. Here the generator is obtained after a rigidity classification and is used in (2) without adding an auxiliary Hamiltonian. The quartic base has a known separation [RML99]; separation is a tool at the base, not an assumption on all neighboring wells.

The degree-six proof uses the minimal average action of twist maps and the rational differentiability criterion recorded in [source references 18, 19, 22]. What must first be proved mechanically is preservation of the actual minimum actions on one physical annulus. The source references and their original numbering are reproduced at the end of the archive.

2 Action, phase volume, and normalization

For a differentiable periodic branch of energy h, period T, and unnormalized action $J = \oint W \cdot dZ$, variation on a fixed parameter circle gives

$$J_h = T, \qquad J_{s_j} = -\int_0^T P_{s_j}(s, Z(t))\, dt, \qquad J(s,h) = h^\nu J(s,1). \tag{4}$$

Indeed the variation of the loop integral is $\int(\delta W\cdot\dot Z-\dot W\cdot\delta Z)\,dt$. Hamilton's equations identify the integrand with $\delta h-P_s\cdot\delta s$. The scaling $Z=h^{1/p}z$, $W=h^{1/2}w$ gives the last formula.

The Hamiltonian generator $L_s=W\cdot\partial_Z-P_Z\cdot\partial_W$ satisfies

$$L_s(Z\cdot W)=2H_s-(p+2)P.$$

Every invariant probability on the compact shell annihilates this derivative, so

$$\int P\,d\mu=\frac{2h}{p+2}. \tag{5}$$

In normal dimension n, let $\Omega(s,h)=\mathrm{vol}_{2n}\{H_s\le h\}$ and $D=\Omega(s,1)^{1/n}$. Homogeneity and coarea give

$$\Omega(s,h)=h^{n\nu}\Omega(s,1),\qquad b=-\frac{1}{n\nu}\nabla\log\Omega(s,1)=-\frac{1}{\nu}\nabla\log D. \tag{6}$$

The measure used in this argument is the normalized coarea measure $|\nabla H_s|^{-1}d\Sigma$, not normalized Euclidean surface area. Positive shells are regular: $W=0$ and $P_Z=0$ would imply $P=0$ by Euler's identity.

In the planar quartic case, $\Omega(s,h)=(2\pi/3)h^{3/2}\mathcal{I}(P)$. Consequently (1) makes all invariant force means zero. For general planar degree p one may use

$$\mathcal{J}_p(s)=\langle P(s,e_\theta)^{-2/p}\rangle,\qquad V=\mathcal{J}_p^{p/2}P,\qquad \langle V(s,e_\theta)^{-2/p}\rangle=1,$$

where $\langle\cdot\rangle$ denotes normalized angular integration. Scaling transports every invariant probability, so the zero-mean conclusion holds for all such probabilities, not only for the microcanonical one.

Lemma 2.1 (Convex effective energy)**.** *If the original $P\in C^2(U\times\mathbb{R}^n)$ is jointly convex, homogeneous of degree $p>1$, and coercive at one parameter of convex open U, then D is positive, C^2, and concave on U. For every $I>0$, U_I is convex and C^2.*

Proof. A zero normal direction at one parameter propagates to all parameters by joint convexity and homogeneity, contradicting coercivity. Positivity is therefore local-uniform. The bodies $K_s=\{P(s,\cdot)\le 1\}$ satisfy

$$(1-\theta)K_{s_0}+\theta K_{s_1}\subset K_{(1-\theta)s_0+\theta s_1}.$$

The phase sublevel volume is a fixed positive constant times $\mathrm{vol}_n K_s$. Brunn–Minkowski gives concavity of its nth root, hence of D. Composition with the decreasing convex function $x\mapsto I^{1/\nu}x^{-1/\nu}$ proves convexity of U_I. Differentiation of the angular integral gives C^2 regularity. □

The exponent $1/n$ is the normal-dimensional root, not the phase-dimensional root. Joint convexity is imposed on P, never substituted by a claim about the normalized V.

3 From common means to exact periodic constraints

The finite rigidity proof first converts force means into exact actions on compact periodic families. The compact-envelope step is short enough to state independently.

Lemma 3.1 (Compact envelope)**.** *Let N be a compact boundaryless manifold and $a\in C^1((-\epsilon,\epsilon)\times N)$. Suppose that $d_m a(r,m)=0$ implies $\partial_r a(r,m)=0$, and $a(0,m)=a_0$. Then $a(r,m)=a_0$ throughout its domain.*

Proof. The maximum and minimum over N are locally Lipschitz. Compare their values at r and $r+h$ using an extremizer at each parameter. Compactness gives convergent subsequences of extremizers; every limit is an extremizer at r and is a critical point. The bounding difference quotients tend to zero by continuity of $\partial_r a$. Thus both envelopes have derivative zero, and their common initial value forces every intermediate value to be a_0. □

For the free-period mechanical action

$$\mathscr{A}_s(q,T) = \int_0^1 \left(\frac{|\dot{q}|^2}{2T} - TP(s,q) + T \right) du,$$

critical points have physical energy one. On a compact Morse–Bott family with a uniformly invertible normal Hessian, normal reduction produces a finite-dimensional reduced action. Its critical parameter derivative is the negative integrated force, hence zero after normalization. The envelope lemma then fixes the entire reduced action. The construction is on H^1 loops together with the period variable; the latter cannot be suppressed from the Hessian. [source Lemmas 4–6, pp. 12–17] give the functional-analytic construction and kernel computations.

4 Finite radial isolation in degrees three and four

Write a nearby normalized well as $V(\rho e_\theta) = \rho^p(1+f(\theta))$ and split

$$e(\theta) = \tfrac{1}{2}(f(\theta)+f(\theta+\pi)), \qquad o(\theta) = \tfrac{1}{2}(f(\theta)-f(\theta+\pi)), \qquad \delta = \|f\|_{C^2}.$$

The exact radial and circular action conditions and normalized volume are the three finite constraints. The normal reductions have a remainder of the form

$$|R| \le C\delta \|f\|_{H^1}^2. \tag{7}$$

This norm is decisive. A small bound in a stronger norm would not exclude angular frequencies escaping to infinity.

The radial action and its derivative along the finite-dimensional rotation orbit give

$$\|e\|_{H^1} \le C\delta \|f\|_{H^1}, \qquad \|f\|_{H^1} \le 2\|o\|_{H^1} \tag{8}$$

for small δ. This does not differentiate a C^2-valued parameter map. For $p=3$, the pointwise estimate

$$\|D^2|x|^3 - D^2|y|^3\| \le 12|x-y|$$

supplies the needed action Taylor remainder through the physical origin. An angular third derivative is not required.

Let w_n be the squared real Fourier amplitude of o at positive odd n. The two remaining response identities read

$$\sum_{n \ modd} D_{p,n} w_n = O(\delta \|o\|_{H^1}^2), \qquad \sum_{n \ modd} (C_p n^2 - 1/p) w_n = O(\delta \|o\|_{H^1}^2),$$

where

$$D_{p,n} = \frac{(p+3)n^2 - (p+2)}{2p(p+2-n^2)}.$$

For the scalar radial oscillator $q'' + p|q|^{p-2}q = 0$ with $q(0)=1$, $q'(0)=0$, let Q_p be its first zero time. The coefficient C_p is defined by

$$z'' + pq^{p-2}z = q^{p-1}, \quad z'(0) = z'(Q_p) = 0, \qquad C_p = \frac{p+2}{2Q_p} \int_0^{Q_p} q^{p-1} z \, dt.$$

The source estimates imply a fixed $c_p > 0$ such that, with $\lambda_3 = 1$ and $\lambda_4 = 2/3$,

$$C_p n^2 - 1/p + \lambda_p D_{p,n} \geq c_p(1 + n^2) \qquad (n \geq 1 \text{ odd}).$$

Adding the response identities and using Parseval forces $\|o\|_{H^1}^2 \leq C\delta\|o\|_{H^1}^2$. For sufficiently small δ, $o = 0$, and (8) gives $f = 0$. The radial locus is closed by angular C^2 continuity and open by this isolation, so connectedness proves Theorem 1.1 for $p = 3, 4$.

The normal reductions, coefficient bounds, and exact remainders are retained in Archive A.2. This argument uses finite differences of shapes. It is stronger than applying a first-jet theorem at the radial member.

5 Degree six: complete actions and a boundary response

For degree six, the preceding scalar response equations alone admit nontrivial compatible Fourier weights. The missing information comes from the full family of mechanical actions. This section states the additional inputs and carries their final closure in the main text.

5.1 One physical annulus, for all denominators

Normalize with $V = \mathcal{J}_6^3 P$. On the large-radius apoapsis section use the angle and angular momentum (θ, L). The section satisfies $W \cdot dQ = L\, d\theta$. Its inverse return map has a C^2 generating action with positive twist on one fixed collar; the parameter neighborhood is independent of rational denominator.

Proposition 5.1 (Preserved minimum actions)**.** *On one neighborhood of a radial parameter the physical inverse apoapsis generator has a global exact positive-twist extension whose minimal average action satisfies*

$$\beta_s(r) = h_0(2\pi r), \qquad -\tfrac{1}{2} \leq r \leq -\tfrac{2}{5}.$$

Every rational rotation in the interior has a physical invariant circle of periodic minimizing points. Their marked actions and periods equal the radial values.

The full proof is [source Proposition A.1, pp. 67–69]. The boundary graphs first localize every intermediate-rotation minimizer in the physical annulus. On such a minimizer, differentiating the actual generating action gives the integrated normalized force; its sum vanishes. The minimum-envelope argument proves invariance for each denominator in the same parameter neighborhood. Continuity gives equality of β on the full interval; the twist-map rational differentiability criterion then supplies the circles. Arbitrary parameter-dependent additive constants in the generator would invalidate this argument.

Proposition 5.2 (Exact brake-boundary invariant)**.** *The brake graph is $L = 0$. Write the inverse return on it as $G_s(\theta, 0) = (F_s(\theta), 0)$, with increasing lift satisfying $F_s(F_s(\theta)) = \theta - 2\pi$. Every brake leg has the same physical duration T. If $b_s(\theta) = \partial_L \pi_\theta G_s(\theta, 0) > 0$ and b_0 is its radial value, then*

$$\langle \sqrt{F_s'/b_s} \rangle = b_0^{-1/2}. \tag{9}$$

This is [source Proposition A.2, pp. 69–72]. Mechanical reversal identifies the brake graph. Conjugating its order-two lift to a half-turn gives a continuous quadratic coefficient in the discrete action. The asymptotic minimum over n steps is $nc + J^2/(2n) + o(1/n)$. Comparing with the preserved radial action identifies J and yields (9). Only a C^1 conjugacy is used; no second derivative of that conjugacy is assumed.

5.2 The estimates that prevent high-frequency escape

Set $h = f''$, $\beta = f'$, $a_* = \|o'\|_2$, and let A_β be the self-adjoint angular transport generator of [source Appendix C]. Its precise realization and response norm are retained in Archive A.4. Write

$$S_h = \|(1 + iA_\beta)^{-1/2} h\|_2.$$

Proposition 5.3 (Closed mechanical response)**.** *For normalized members of an actual C^2 common-mean family sufficiently close to the radial sextic well,*

$$S_h \le Ca_*, \qquad \|f - \langle f \rangle\|_{H^{3/2}} \le Ca_*. \tag{10}$$

The constants are independent of the angular Fourier support.

The first estimate is [source Theorem 16]; the second is [source Lemma 49]. Their mechanical inputs are the exact invariant (9), an actual-transport comparison, and the centered coherent sign

$$Q_{\mathrm{mod}}(h) \le -\frac{1}{2200} S_h^2 + C(a_* S_h + \delta S_h^2).$$

The invariant reduction bounds the same form by $C(a_*^2 + a_* S_h + \delta S_h^2)$. Absorbing the small terms and using Young's inequality gives (10) without dividing by a_*. Interpolation of the transport graph scale gives the half-derivative control.

These estimates retain the physical origin layer, the actual clock, the fixed-state parameter derivatives, and the signed lower-order matrix terms. Replacing the actual regularizer by a radial singular expression inside the origin layer is not permitted. The long calculations establishing precisely these facts occupy source Appendices B–D, preserved in the archive.

Proposition 5.4 (Finite Fourier constraints)**.** *For the same finite shape, with $w_n = 4|\widehat{f}(n)|^2$ for positive odd n, there are fixed $c, C > 0$ such that*

$$\|f\|_{H^1} \le Ca_*, \qquad |\widehat{f}(n)| \le Ca_*^2 \ (n \ge 5 \ \textit{odd}), \qquad ca_*^2 \le \sum_{n \ge 5,\ n\ \textit{odd}} n^2 w_n.$$

This is [source Lemma 22]. Constant-increment competitors for all rational minimum actions are nonnegative after subtraction of the radial minimum. Their Fourier coefficients are bounded by their mean. Uniform one-cycle remainders and a uniform nonzero orbital multiplier yield the coefficient bound. The radial and circular response equations then force a positive fraction of derivative energy into modes $n \ge 5$.

5.3 The finite isolation argument

By Proposition 5.3, $\sum n^3 w_n \le C_2 a_*^2$. Split the sum of Proposition 5.4 at an integer $M \ge 5$:

$$ca_*^2 \le C_1 a_*^4 M^3 + \frac{C_2}{M} a_*^2.$$

Choose M once so that $C_2/M \le c/2$. If $a_* > 0$, division now gives $a_*^2 \ge c/(2C_1 M^3)$, contradicting $a_* \le C\delta$ for sufficiently small δ. If $a_* = 0$, the H^1 bound directly gives $f = 0$. Local isolation and the open–closed argument prove degree-six rigidity on the connected parameter set. This is a finite-family conclusion; the denominator-uniform action control and the transport response are both needed.

6 The nonradial quartic: why curvature is necessary

Let $J(x,y)=(-y,x)$ and $RP=\nabla P\cdot JZ$. At Q_0,

$$RQ_0 = 20x^3y + 40xy^3.$$

There are exactly two real projective angular critical lines, the coordinate axes. Let C_x, C_y be their continued critical values and $G=(C_x,C_y,\mathcal{I})$. Then

$$\operatorname{rank} DG(Q_0)=3, \qquad \ker DG(Q_0)=\operatorname{span}\{x^3y, xy^3\}.$$

After quotienting out rotation, one nonrotational direction remains. A first-order rank argument alone cannot eliminate it.

The known parabolic coordinates

$$x=uv, \quad y=(u^2-v^2)/2, \quad S=u^2+v^2,$$
$$S(H-1)=\tfrac{1}{2}(p_u^2+p_v^2)+U(u)+U(v), \quad U(u)=u^{10}-u^2,$$

with physical clock $dt=S\,d\tau$, provide a nondegenerate curved brake orbit at $v=v_0=5^{-1/8}$. Its continued action is denoted $A(P)$. For the genuine residual direction $g=x^3y$, a jet $P_r=Q_0+rg+(r^2/2)B+o(r^2)$ satisfying $G''=0$ must have the even compensator prescribed by the axial and volume constraints. The second action variation is

$$A''=-\int_0^{4Q}(SB+F_u\xi_u+F_v\xi_v)\,d\tau, \qquad F=Sg.$$

The factor S and its derivatives represent the physical clock and cannot be dropped. Eliminating the longitudinal variation and retaining the negative transverse spectral contribution gives the strict bound

$$\frac{1}{v_0^2}\int_0^{Q}(SB_{\text{even}}-B_{\text{aux}}+fL^{-1}f)\,d\tau>\frac{1}{2500}, \qquad A''<-v_0^2/625. \tag{11}$$

The rational interval recipe and reported enclosure are in [source Appendix P.1]. They are preserved, not represented here as newly executed certificates.

The four invariants (G,A) constrain a family at both full-rank and deficient-rank points. At deficient rank, the nonzero constrained curvature excludes the residual direction. The local rotation-orbit conclusion then continues globally: the parameter set mapping into the fixed compact rotation orbit is closed, and the local result makes it open. This proves Theorem 1.2. The full rank-jump argument is in Archive A.5.

7 Controlled generators and dissipative selection

The classifications provide a regular solution of the centered-force equation; equality of invariant means by itself would provide only obstruction identities.

7.1 The corrector and its parameter derivative

In the radial case $b_j=2\partial_j\log\kappa/(p+2)$ and set $c_j=0$. In the quartic case the intrinsic identity is

$$P_{s_j}-3b_jP=c_jRP, \qquad c_j=\frac{\langle P_{s_j}-3b_jP, RP\rangle_{\text{coeff}}}{\|RP\|^2_{\text{coeff}}}.$$

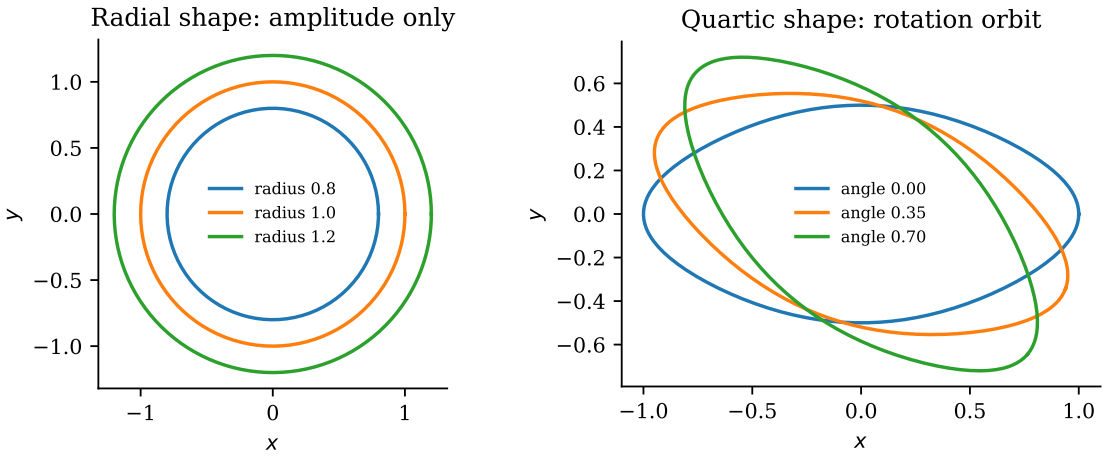


Figure 1: Allowed shape changes in the two classified settings. Left: changing the amplitude of a radial well changes the radius of a fixed equipotential. Right: normalized quartic equipotentials rotate. The figure illustrates the conclusions; it is not a test of the common-mean condition.

The denominator is nonzero on the classified nonradial orbit. Thus c_j is globally defined, without a global angle. Define

$$A_j = -c_j W \cdot JZ - \frac{b_j}{2} Z \cdot W. \tag{12}$$

The identities $L_s(W \cdot JZ) = -RP$ and the virial identity give

$$L_s A = P_s - H_s b, \qquad |A| + |A_s| \le C H_s^{\nu}. \tag{13}$$

Here A_s is differentiated at fixed (Z, W). Coercivity bounds $|Z||W|$ by CH_s^{ν}, and the coefficient derivatives are bounded on a neighborhood of the compact tail. The corrector extends by zero at the normal origin.

7.2 Exact rescaling and positive residual energy

The change of variables in Theorem 1.3 gives, with primes denoting σ derivatives,

$$Z' = \varepsilon^{-1} W, \quad W' = -\varepsilon^{-1} P_Z, \quad s' = k, \quad k' = -ak - P_s, \qquad H' = k \cdot P_s, \quad E' = -a|k|^2. \tag{14}$$

These are exact equations, not an averaged system. Energy and coercivity bound all rescaled states. Hence $k \in L^2$ and $k \to 0$ by uniform continuity. Thus $H \to E_\infty \ge 0$.

For an excited solution H never vanishes at finite time, by uniqueness. Since $|P_s| \le CH$ on the compact tail, $|(\log H)'| \le C|k| \to 0$. For any fixed $0 < d < a$, eventually

$$H(u) \le H(\sigma) e^{d(\sigma - u)} \quad (u \le \sigma).$$

Variation of constants in the equation for k then yields $|k| \le CH$. If $E_\infty = 0$, this also gives $E = H + O(H^2)$ and $|H'| \le CH^2$, so H cannot decay faster than a reciprocal affine function of σ.

Put $B = A/H$ and

$$\mathcal{J} = \log(D(s) H^{\nu}) - \nu \varepsilon k \cdot B.$$

Using (6), (13), and (14), full differentiation gives

$$\mathcal{J}' = \nu\varepsilon\big[(2ak + P_s) \cdot B - k \cdot (B_s k)\big], \qquad |B| + |B_s| \le C H^{\nu - 1}.$$

Since $|k| \le CH$,

$$|\varepsilon k \cdot B| \le C\varepsilon H^\nu, \qquad |\mathcal{J}'| \le C\varepsilon(H^\nu + H^{\nu+1}).$$

The right side is integrable and has an $O(\varepsilon)$ tail. Thus $\log I$ tends to a finite real value, $I_\infty > 0$, and $I - I_\infty = O(\varepsilon)$. Boundedness of D above and below proves $E_\infty > 0$. Positivity has been obtained after controlling division by H, not assumed in advance.

7.3 Effective dynamics and convergence to one point

Let $v = k + \varepsilon A$ and $U_\infty = U_{I_\infty}$. Direct differentiation and the action estimate give

$$s' = v + r_s, \qquad v' = -av - \nabla U_\infty(s) + r_v, \qquad |r_s| + |r_v| \le C\varepsilon, \quad v \to 0, \quad U_\infty(s) \to E_\infty.$$

Every selected cluster point u is stationary: integrate the second equation over a fixed-length interval starting at a subsequence converging to u. The first equation keeps s close to u there, while all remaining integrated terms vanish. Lemma 2.1 makes u a global minimizer.

For any such u, set $y = s + v/a$ and

$$Q_u = \tfrac{1}{2}|y - u|^2 + U_\infty(s)/a^2.$$

An exact cancellation gives

$$Q_u' = -\frac{(s-u)\cdot \nabla U_\infty(s)}{a} + (y-u)\cdot(r_s + r_v/a) + \frac{\nabla U_\infty(s)\cdot r_s}{a^2}.$$

The first term is nonpositive and the other terms are integrable. Hence Q_u has a limit, and so does $|s-u|$. Choosing u to be a cluster point forces this limit to be zero. This proves Theorem 1.3. It gives neither finite length nor a rate for $s - s_\infty$.

7.4 Physical energy and normal amplitudes

Corollary 7.1 (Physical mechanical asymptotics)**.** *Under the excited-tail assumptions of Theorem 1.3, define*

$$\mathscr{E}_{\mathrm{phys}}(t) = \tfrac{1}{2}|\dot s(t)|^2 + \tfrac{1}{2}|\dot q(t)|^2 + P(s(t), q(t)).$$

Then

$$t^2\mathscr{E}_{\mathrm{phys}}(t) \to E_\infty > 0, \qquad q(t) = O(t^{-2/p}), \quad \dot q(t) = O(t^{-1}), \quad \dot s(t) = o(t^{-1}).$$

Proof. Since $q = \varepsilon Z$, $\dot q = t^{-1}(W - a\varepsilon Z)$, $\dot s = t^{-1}k$, and $\varepsilon^p = t^{-2}$, expansion gives the exact identity

$$t^2\mathscr{E}_{\mathrm{phys}} = E - a\varepsilon Z\cdot W + \tfrac{1}{2}a^2\varepsilon^2|Z|^2.$$

Boundedness of Z, W and $\varepsilon \to 0$ prove the limit. The three amplitude estimates follow from the same formulas and $k \to 0$. □

For every fixed $I > 0$, the strict decrease of $x \mapsto (I/x)^{1/\nu}$ gives $\arg\min U_I = \arg\max D$. The candidate set is independent of the positive limiting action. This does not imply independence of the selected point from the initial state, nor a thermodynamic entropy law.

8 A nontrivial example with a checkable compact tail

Take one selected coordinate and

$$P(s,q) = e^{\eta s^2}|q|^p, \qquad \eta > 0, \quad p \in \{3,4,6\}. \tag{15}$$

Choose $\delta > 0$ with $\delta^2 < (p-1)/(2\eta)$ and set $U = (-\delta, \delta)$. The potential is C^2 including $q = 0$.

Proposition 8.1 (Joint convexity and explicit trapping)**.** *The potential* (15) *is jointly convex on* $U \times \mathbb{R}^2$ *and satisfies* (CM)*. Start at* $t_0 = 1$ *and define*

$$s_0 = s(1), \quad k_0 = \dot{s}(1), \quad Z_0 = q(1), \quad W_0 = \dot{q}(1) + aq(1), \quad a = 2/p,$$

$$E_0 = \tfrac{1}{2}|W_0|^2 + e^{\eta s_0^2}|Z_0|^p + \tfrac{1}{2}k_0^2, \qquad B_0 = \tfrac{1}{2}s_0^2 + \frac{(s_0k_0)_+}{a} + \frac{E_0}{a^2}.$$

If $B_0 < \delta^2/2$*, then the solution exists for all* $t \geq 1$*, its selected coordinate remains in* $[-\sqrt{2B_0}, \sqrt{2B_0}] \Subset U$*, and every excited solution satisfies* $s(t) \to 0$ *and Corollary 7.1.*

Proof. Write $\alpha = e^{\eta s^2}$ and $r = |q|$. At $q \neq 0$,

$$P_{qq} = \alpha p r^{p-2}(I + (p-2)\widehat{q}\widehat{q}^T), \quad P_{sq} = \alpha' p r^{p-2} q, \quad P_{ss} = \alpha'' r^p.$$

The normal block is positive definite. Its Schur complement is

$$r^p\left(\alpha'' - \frac{p}{p-1}\frac{(\alpha')^2}{\alpha}\right) = P\left(2\eta - \frac{4\eta^2 s^2}{p-1}\right) > 0.$$

At $q = 0$ all Hessian blocks extend continuously to zero. Thus joint convexity holds throughout the strip. The virial identity gives $b(s) = 4\eta s/(p+2)$, independently of the invariant probability.

In logarithmic time let $h = s^2/2$. The exact equations give

$$h'' + ah' = k^2 - 2\eta s^2 P(s,Z) \leq k^2, \qquad E' = -ak^2.$$

Integrating the first differential inequality twice yields

$$h(\sigma) \leq h(0) + \frac{1 - e^{-a\sigma}}{a}h'(0) + \frac{1}{a}\int_0^\sigma (1 - e^{-a(\sigma-u)})k(u)^2 du \leq B_0.$$

The energy identity gives $\int_0^\sigma k^2 \leq E_0/a$. Strictness of the bound excludes first exit from the strip. Energy and local uniform coercivity bound Z, W, k while s stays in this fixed interior compact interval. On every finite logarithmic-time interval the vector field is locally Lipschitz on the resulting bounded set, giving continuation for all $\sigma \geq 0$.

The phase-volume root is $D(s) = D(0)e^{-\eta s^2/p}$ and has its unique maximum at zero. Apply Theorem 1.3 on a connected open interval containing the trapped set. The excitation condition excludes only the invariant zero-normal branch. □

For example, take $p = 4$, $\eta = 1$, $\delta = 0.75$, $s_0 = 0.08$, $k_0 = 0.02$, $Z_0 = (0.45, 0.12)$, and $W_0 = (0.12, -0.06)$. Then $\dot{q}(1) = (-0.105, -0.12)$, $E_0 < 0.056577$, and $B_0 < 0.232708 < \delta^2/2 = 0.28125$. Both the selected motion and normal excitation are nonzero. These conservative decimal upper bounds follow from $e^{0.0064} < 1.007$. The analytic bound, rather than numerical simulation, proves trapping.

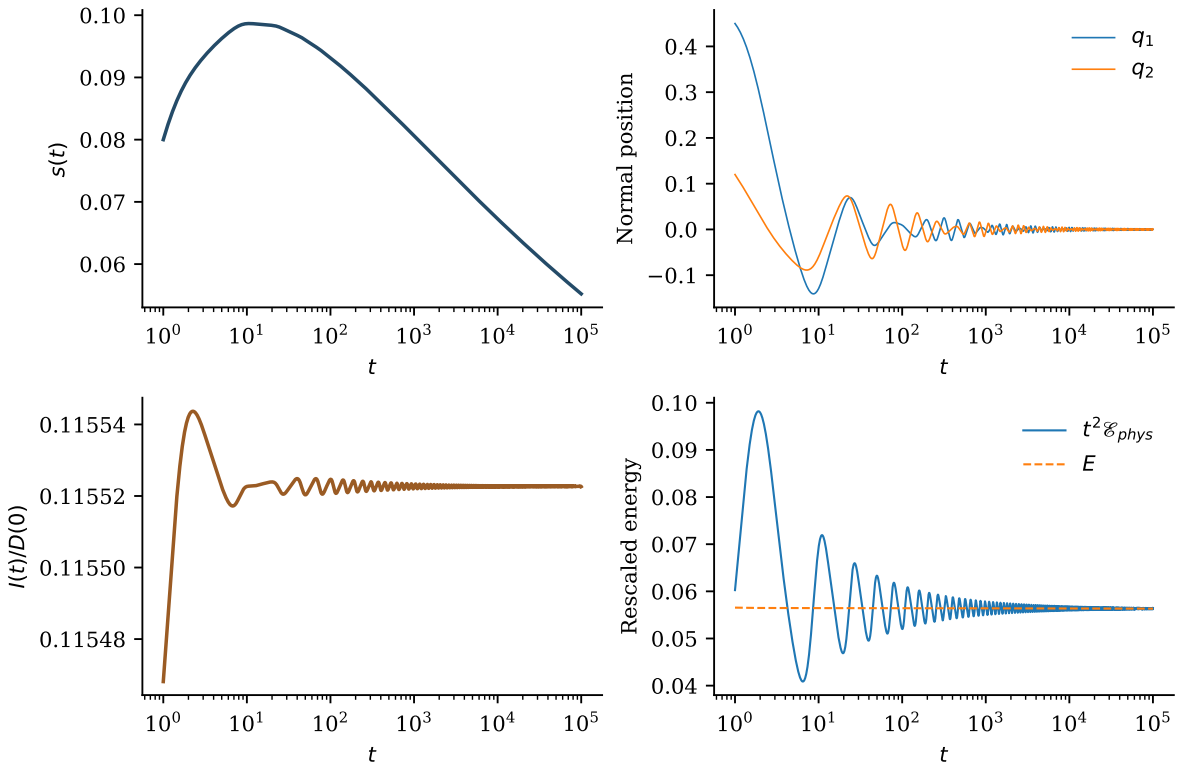


Figure 2: Finite-time illustration of Proposition 8.1 for the stated initial data, integrated to $t = 10^5$. The action is divided by the constant $D(0)$. A second, tighter-tolerance integration checks numerical stability; neither integration proves an infinite-time limit. The asymptotic statements follow from the preceding analysis.

9 Scope, proof dependencies, and remaining questions

The finite classifications concern connected families satisfying common means at every parameter. The radial theorem requires that the family meet the radial locus. The nonradial theorem has a polynomial neighborhood hypothesis and does not classify unrestricted angular perturbations. Four conserved test values alone do not imply (CM): the source constructs a nonpolynomial compensated family preserving those values. That construction is a limitation of four tests, not a counterexample to the classification.

The proof dependencies are now separated by purpose:

$$\text{common means} \implies \text{normalized exact actions} \implies \text{finite radial or quartic rigidity},$$
$$\text{rigidity} \implies \text{controlled corrector} \implies \text{positive action limit and point selection}.$$

For degree six the middle implication includes the fixed annulus, exact boundary invariant, and uniform response estimates in Section 5. None of these implications requires identification of an unrestricted kernel at Q_0. That fixed-base problem is assigned to the companion reconstruction.

The dynamical theorem keeps the actual compact-tail and original joint-convexity assumptions. The example verifies those assumptions for a concrete set of data; it does not remove them in general. Approximate-data stability, other degrees and dimensions, arbitrary nonpolynomial neighborhoods, and other damping regimes are separate problems. We do not extend the present conclusions to them by analogy.

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A Proof archive and source-number concordance

The following folios preserve the original equations and full long proofs. They are part of this research draft, not omitted supplementary evidence. Original page numbers and source theorem numbers remain visible. The editable main text supplies the new organization and the new consequences; the folios are retained because the original LaTeX was unavailable. Their old section headings describe provenance, not the organization of this main text. Boundary pages can contain adjacent source context; that context is not an additional claim of this paper.

New result or proof task	Source location	Role
Theorem 1.1, $p = 3, 4$	Section 3, pp. 12–29	Full normal reduction, response gap, finite isolation
Section 5	Section 9, pp. 51–58; A–D, pp. 62–132	Resonant admission, exact actions, actual transport, finite closure
Theorem 1.2	Section 4, pp. 29–35; P, pp. 247–248	Constrained curvature and connected continuation
Theorem 1.3	Section 8, pp. 46–51	Controlled-corrector argument; separate supporting implication
Corollary 7.1	New proof, from source (94)–(101)	Original mechanical variables
Proposition 8.1	New proof using source dynamics	Explicit initial-data region
Limits of finite tests	Section 10, pp. 58–62	Scope; not full common-mean counterexamples

domains with C^9 boundaries sufficiently close to a circle, using their convention $r = 8$. The invariant-distribution theorem of Avila and Kocsard [7] concerns orientation-preserving C^∞ circle diffeomorphisms with irrational rotation number; the invariant probability spans all invariant distributions, while smooth cohomological stability requires a Diophantine rotation number. Neither result is a theorem for the natural Hamiltonians considered here. They provide context for why periodic tests and regular solvability must be distinguished. The mechanical content proved below is the classification of allowed potential deformations, followed by a controlled corrector and an actual critical-flow argument.

The sextic action argument uses the classical minimal-average action of twist maps. We use Mather's rational differentiability criterion in the C^1 twist-map formulation recorded by Siburg [19, 22]; the C^1 map and C^2 generating-function framework is also explicit in [18]. The parameter invariance of the physical minimum actions is proved here from the actual force condition. It is distinct from the classical existence and differentiability results used after that invariance has been established.

For vanishing damping, energy convergence, objective-value decay and point convergence have different hypotheses. Cabot, Engler and Gadat [13] distinguish energy convergence from convergence to a minimizer and identify a necessary damping condition for the latter. Attouch, Chbani and Riahi [4] prove value decay $O(t^{-2\alpha/3})$ for $0 < \alpha \leq 3$ in the general convex setting; that estimate alone does not select one point of a minimizing valley. Aujol, Dossal and Rondepierre [5] obtain finite-length convergence under coercivity, matched flatness and growth of order $\gamma > 2$, and $\alpha \geq (\gamma + 2)/(\gamma - 2)$. The continuous-time result of Aujol, Dossal, Labarrière and Rondepierre [6] permits nonunique minimizers under growth and second-order regularity of the minimizing set, with $\alpha > 9/2 + 6/(\gamma - 2)$. The latter two finite-length convergence regimes do not contain the damping $\alpha = 1 + 2/p < 2$ treated here. Moreover, homogeneity only in the normal variables is not the full-state flatness hypothesis of [5]. Our conclusion uses the mechanical common-mean structure and an actual compact tail; it asserts neither a general finite-length theorem nor a selection-coordinate rate.

2 Frozen action identities and volume normalization

Lemma 1 (Fixed-energy action variation)**.** *Let a differentiable family of periodic orbits of $H_s = |W|^2/2 + P(s, Z)$ have energy h, period T, and unnormalized action $J = \oint W \cdot \mathrm{d}Z$. Then*

$$J_h = T, \qquad J_{s_j} = -\int_0^T P_{s_j}(s, Z(t))\,\mathrm{d}t. \tag{6}$$

For a homogeneous potential, $J(s, h) = h^\nu J(s, 1)$, where $\nu = 1/2 + 1/p$. In particular, the mean force along the branch is

$$-\frac{J_s}{J_h} = -\frac{h}{\nu}\nabla_s \log J(s, 1).$$

Proof. Parametrize the loop on a fixed circle. Variation and integration by parts give $\delta J = \int (\delta W \cdot \dot{Z} - \dot{W} \cdot \delta Z)\,\mathrm{d}t$. Hamilton's equations and variation of $H_s = h$ identify the integrand with $\delta h - P_s \cdot \delta s$. Periodicity removes the boundary term. Finally, $Z = h^{1/p} z$, $W = h^{1/2} w$, and $t = h^{1/p-1/2}\tau$ give the claimed scaling. These are the standard action identities; see [11]. ∎

For any invariant probability measure on $H_s = h$, the virial identity follows by integrating $\mathcal{L}_s(Z \cdot W) = 2H_s - (p+2)P$:

$$\int P \,\mathrm{d}\mu = \frac{2h}{p+2}. \tag{7}$$

This applies to all invariant measures, including measures supported on periodic orbits.

Let n be the normal dimension and define

$$\Omega(s,h) = \mathrm{vol}_{2n}\{H_s \le h\}, \qquad \Omega_1(s) = \Omega(s,1), \quad D(s) = \Omega_1(s)^{1/n}.$$

The power $1/n$ is the normal-dimensional root, not the phase-dimensional root $1/(2n)$.

Lemma 2 (Microcanonical identification and normalization)**.** *The positive-energy shells are regular, and*

$$\Omega(s,h) = h^{n\nu}\Omega_1(s), \qquad b(s) = -\frac{1}{n\nu}\nabla \log \Omega_1(s) = -\frac{1}{\nu}\nabla \log D(s) \tag{8}$$

whenever (CM) *holds. The functions Ω_1, D, b are respectively C^2, C^2, C^1. For $n = 2, p = 4$,*

$$\Omega(s,h) = \frac{2\pi}{3} h^{3/2} \mathcal{I}(P(s,\cdot)), \qquad b = -\frac{2}{3}\nabla \log \mathcal{I}(P). \tag{9}$$

The normalized family (1) *has $\mathcal{I}(V) = \mathcal{I}_0$ and every invariant force mean equal to zero. Its differentiable periodic branches have constant fixed-energy action.*

Proof. If the normal gradient of H_s vanished on $H_s = h > 0$, then $W = 0$, $P_Z = 0$, and Euler's identity $Z \cdot P_Z = pP$ would give $h = 0$. Coarea and variation of a compact regular sublevel therefore give

$$\Omega_h = \int_{H_s = h} \frac{\mathrm{d}\Sigma}{|\nabla H_s|}, \qquad \Omega_{s_j} = -\int_{H_s = h} \frac{P_{s_j}\,\mathrm{d}\Sigma}{|\nabla H_s|}.$$

The probability $\Omega_h^{-1}|\nabla H_s|^{-1}\,\mathrm{d}\Sigma$ is invariant, since the Hamiltonian flow preserves both Lebesgue volume and energy. It is the microcanonical measure, rather than normalized Euclidean surface area. The homogeneous coordinate change has Jacobian $h^{n\nu}$, so its mean force is $-h\nabla \log \Omega_1/(n\nu)$, proving (8).

If $Q(s) = \mathrm{vol}_n\{P(s,Z) \le 1\}$, polar integration gives

$$Q(s) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} P(s,\omega)^{-n/p}\,\mathrm{d}\omega, \qquad \Omega_1(s) = \frac{(2\pi)^{n/2}\Gamma(1+n/p)}{\Gamma(1+n\nu)} Q(s). \tag{10}$$

Local uniform coercivity permits two parameter derivatives under the integral. For $n = 2, p = 4$, $Q = \mathcal{I}/2$ and the constant in (10) is $4\pi/3$, yielding (9).

Write $V = c(s)P$. The common mean of V_{s_j} on $H_V = 1$ is $b_j + (1/3)\partial_j \log c$. Indeed the term cP_{s_j} has mean b_j after transport from the P shell of energy $1/c$, while $c_{s_j}P$ has mean $\partial_j \log c/3$ by (7). For $c = (\mathcal{I}(P)/\mathcal{I}_0)^2$, this sum vanishes by (9). The action conclusion follows from Lemma 1. ∎

An equality of invariant means alone does not reverse the implication obtained by integrating $\mathcal{L}_s A = P_s - H_s b$: it gives necessary obstruction identities for A, not its existence or a bound for A_s. The classifications below resolve this issue in two particular mechanical classes.

2.1 Convex geometry of the effective energy

Lemma 3. *Let $U \subset \mathbb{R}^m$ be open and convex and $P \in C^2(U \times \mathbb{R}^n)$ jointly convex and p-homogeneous, $p > 1$. If P is coercive at one parameter, it is positive away from zero at every parameter, locally uniformly coercive, and $D = \Omega_1^{1/n}$ is positive, C^2 and concave on U. Hence*

$$U_I(s) = (I/D(s))^{1/\nu} \quad (I > 0)$$

is convex and C^2 on U. If D is affine on any nontrivial selected segment, then $P(s, \cdot)$ is constant on that segment. In particular all frozen potentials agree on $M = \operatorname{argmax}_U D$ when this set is nonempty.

Proof. Convexity and homogeneity give $P(s, 0) = 0$, $P_Z(s, 0) = 0$, and $P \geq 0$. If $P(s_0, z) = 0$ for $z \neq 0$, write an arbitrary $s \in U$ as $s = \theta s_0 + (1 - \theta)s_1$ with $s_1 \in U$ and $0 < \theta < 1$. Joint convexity at $(s_0, z/\theta)$ and $(s_1, 0)$ gives $P(s, z) \leq 0$. Thus that direction would vanish at every parameter, contradicting coercivity at one. Compactness of the sphere then gives local uniform coercivity.

The configuration bodies $K_s = \{P(s, \cdot) \leq 1\}$ obey $(1 - \theta)K_{s_0} + \theta K_{s_1} \subset K_{s_\theta}$. The Brunn–Minkowski inequality [10] and (10) make D concave. Composition with the decreasing convex function $x \mapsto I^{1/\nu} x^{-1/\nu}$ gives convexity of U_I.

For completeness, the strict equality consequence uses additional homogeneous structure. For $w \neq 0$ let $h_s(w)$ be the support function of K_s. The partial Legendre transform in Z is concave in s by joint convexity, and radial maximization gives

$$\sup_Z (w \cdot Z - P(s, Z)) = (p - 1)p^{-p/(p-1)} h_s(w)^{p/(p-1)}.$$

If $D(s_\theta) = (1 - \theta)D(s_0) + \theta D(s_1)$, equality of volumes in the inclusion and Brunn–Minkowski imply $K_{s_\theta} = (1 - \theta)K_{s_0} + \theta K_{s_1}$. Nested full-dimensional compact convex bodies of equal volume coincide. Thus the support at s_θ is the affine average of its endpoint supports. Concavity of its $p/(p - 1)$ power and strict convexity of that scalar power force $h_{s_0}(w) = h_{s_1}(w)$ for every w. The endpoint bodies and potentials coincide. A

concave function meeting its chord at an interior point agrees with the chord throughout the segment; repeating the support argument proves constancy throughout. Applying this to segments in the convex maximizing set M proves the last assertion. ∎

At every $u \in M$, if $T = \operatorname{span}(M - M)$, joint Hessian positivity also gives

$$\Pi_T P_s(u, Z) = 0, \qquad P_{ss}(u, Z)v = P_{Zs}(u, Z)v = 0 \quad (v \in T). \tag{11}$$

Indeed restriction to each segment in M is constant, its second derivative vanishes, and a vector with zero quadratic form for a positive-semidefinite matrix belongs to its kernel. Continuity includes segment endpoints. This geometric fact does not by itself supply a force corrector off M.

3 Exact action reduction and finite radial rigidity

The normal configuration space in this section is $\mathbb{R}^2$. The parameter space may have any finite dimension. Homogeneity always means positive homogeneity in the normal variable; neither antipodal symmetry nor polynomial dependence is assumed. Convexity is not needed for the frozen rigidity statements below. In particular, amplitude normalization will not be assumed to preserve joint convexity.

3.1 Compact reduction for the mechanical action

We first give the reduction argument that supplies the periodic continuations used in the proof. This also specifies the function spaces and the role of the period variable.

Lemma 4 (Compact envelope)**.** *Let N be a compact boundaryless C^1 manifold and let $a \in C^1((-\varepsilon, \varepsilon) \times N)$. Suppose*

$$d_m a(r, m) = 0 \quad \Longrightarrow \quad \partial_r a(r, m) = 0, \qquad a(0, m) = a_0.$$

Then $a(r, m) = a_0$ everywhere. The same conclusion holds in a sufficiently small convex finite-dimensional parameter neighborhood if the implication holds for every parameter derivative.

Proof. Put $M(r) = \max_N a(r, \cdot)$ and $m_*(r) = \min_N a(r, \cdot)$. Both are locally Lipschitz, since $\partial_r a$ is bounded on compact parameter intervals times N. Every extremizer is a critical point, because N has no boundary. If x_0 maximizes at r and x_h maximizes at $r + h$, then, for $h > 0$,

$$a(r + h, x_0) - a(r, x_0) \le M(r + h) - M(r) \le a(r + h, x_h) - a(r, x_h).$$

Every accumulation point of x_h as $h \to 0$ maximizes at r. Divide by h, write the outer differences as integrals of $\partial_r a$, and use compactness and continuity. Both bounds tend to

concave function meeting its chord at an interior point agrees with the chord throughout the segment; repeating the support argument proves constancy throughout. Applying this to segments in the convex maximizing set M proves the last assertion. ∎

At every $u \in M$, if $T = \operatorname{span}(M - M)$, joint Hessian positivity also gives

$$\Pi_T P_s(u, Z) = 0, \qquad P_{ss}(u, Z)v = P_{Zs}(u, Z)v = 0 \quad (v \in T). \tag{11}$$

Indeed restriction to each segment in M is constant, its second derivative vanishes, and a vector with zero quadratic form for a positive-semidefinite matrix belongs to its kernel. Continuity includes segment endpoints. This geometric fact does not by itself supply a force corrector off M.

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Then $a(r, m) = a_0$ everywhere. The same conclusion holds in a sufficiently small convex finite-dimensional parameter neighborhood if the implication holds for every parameter derivative.

Proof. Put $M(r) = \max_N a(r, \cdot)$ and $m_*(r) = \min_N a(r, \cdot)$. Both are locally Lipschitz, since $\partial_r a$ is bounded on compact parameter intervals times N. Every extremizer is a critical point, because N has no boundary. If x_0 maximizes at r and x_h maximizes at $r + h$, then, for $h > 0$,

$$a(r + h, x_0) - a(r, x_0) \le M(r + h) - M(r) \le a(r + h, x_h) - a(r, x_h).$$

Every accumulation point of x_h as $h \to 0$ maximizes at r. Divide by h, write the outer differences as integrals of $\partial_r a$, and use compactness and continuity. Both bounds tend to

zero. The argument for $h < 0$, and then for the minimum, is identical. Thus $M' = m'_* = 0$ everywhere. Their initial values coincide, so $m_* = a = M = a_0$. For several parameters apply the interval assertion on each segment from the base parameter in a small convex neighborhood. ∎

Lemma 5 (Normal reduction with zero critical parameter derivative)**.** *Let E be a real Hilbert space, $\mathcal{O} \subset E$ open, and $F \in C^2(U \times \mathcal{O})$, where U is a neighborhood of 0 in a finite-dimensional parameter space. Let $N \subset \mathcal{O}$ be a compact, connected, boundaryless, finite-dimensional C^2 embedded manifold of critical points of $F(0, \cdot)$. Suppose its Riesz Hessian L_m is self-adjoint Fredholm and*

$$\ker L_m = T_m N \qquad (m \in N).$$

Suppose also that every nearby critical point of $F(s, \cdot)$ satisfies $\partial_{s_j} F(s, y) = 0$ for every parameter component. After restricting U, there is a unique small C^1 normal graph

$$Y(s, m) = m + \psi(s, m), \qquad \psi(s, m) \in E_m := (T_m N)^{\perp}, \qquad \psi(0, m) = 0,$$

whose points are all critical and satisfy $F(s, Y(s, m)) = F(0, N)$. Every sufficiently close critical point belongs to this graph.

Proof. Self-adjointness and the closed range property imply $\operatorname{ran} L_m = (\ker L_m)^{\perp} = E_m$. The restriction of L_m to E_m is therefore boundedly invertible. In local normal frames the Hessians and tangent projections vary continuously. Compactness gives a uniform inverse bound. The C^1 implicit function theorem applied to

$$\Pi_{E_m} \nabla_y F(s, m + \psi) = 0$$

produces the normal graph. Uniqueness glues the local constructions and compactness gives one parameter neighborhood and one tube. The reduced function $a(s, m) = F(s, Y(s, m))$ is C^1. Its tangent graph, together with E_m, spans E, since it is C^1 close to the zero section. Consequently

$$d_m a(s, m) = 0 \quad \Longleftrightarrow \quad d_y F(s, Y(s, m)) = 0.$$

Moreover, $\partial_{s_j} \psi \in E_m$, so the normal equation gives $\partial_{s_j} a = \partial_{s_j} F(s, Y)$. The base critical value is constant on connected N. Lemma 4 now makes a identically that value, and the equivalence just proved makes all graph points full critical points. Normal uniqueness includes every critical point in a sufficiently small common tube. ∎

For a jointly C^2 natural potential $V(s, Z)$ define, on $E = H^1(\mathbb{R}/\mathbb{Z}; \mathbb{R}^2) \times \mathbb{R}$ with $T > 0$,

$$J_s(q, T) = \int_0^1 \left(\frac{|\dot{q}|^2}{2T} - TV(s, q) + T \right) du. \tag{12}$$

A dot in this formula denotes differentiation in u. Physical time is $t = Tu$. On a tube about a compact family of loops and positive periods, $H^1 \hookrightarrow C^0$ confines all loop ranges to one compact ball. The ordinary first and second derivative formulas for the potential, together with uniform continuity of its second derivatives on that ball, show that J is twice continuously Fréchet differentiable. The kinetic term is smooth for $T > 0$. Only quadratic velocity dependence is used. The first variations are

$$d_q J[\eta] = T^{-1} \int_0^1 \dot{q} \cdot \dot{\eta}\, du - T \int_0^1 V_Z(s,q) \cdot \eta\, du,$$
$$\partial_T J = \int_0^1 \left(1 - \frac{|\dot{q}|^2}{2T^2} - V(s,q) \right) du.$$

The first equation gives $\ddot{q} + T^2 V_Z(s,q) = 0$. A weak solution is classical by this equation. Its energy is constant, and the second equation sets it equal to one. Thus critical points are exactly parametrized unit-energy periodic orbits, with the specified period (which need not be primitive). At a critical point,

$$\partial_{s_j} J_s = -\int_0^T V_{s_j}(s, Q(t))\, dt. \tag{13}$$

The full Hessian, evaluated at a critical point, is

$$D^2 J[(\eta,\tau),(\xi,\sigma)] = T^{-1} \int_0^1 \dot{\eta} \cdot \dot{\xi}\, du - T \int_0^1 \eta \cdot V_{ZZ} \xi\, du$$
$$- 2\tau \int_0^1 V_Z \cdot \xi\, du - 2\sigma \int_0^1 V_Z \cdot \eta\, du + \frac{\tau\sigma}{T^3} \int_0^1 |\dot{q}|^2\, du. \tag{14}$$

The leading loop form becomes coercive on adding $T^{-1} \int \eta \cdot \xi$ and a positive period form. All corrections are compact: loop terms factor through the compact embedding $H^1 \hookrightarrow L^2$, and period terms are finite rank. The Riesz Hessian is therefore self-adjoint Fredholm. For a p-homogeneous potential, the virial identity and unit energy give, at every critical point,

$$J_s = \int_0^T |\dot{Q}|^2\, dt = \frac{T}{\nu}, \qquad \nu = \frac{1}{2} + \frac{1}{p}. \tag{15}$$

In particular, the conclusion of Lemma 5 gives constant periods as well as constant actions in the homogeneous case.

3.2 Centering and the two radial-base critical manifolds

For a positive p-homogeneous planar potential set

$$\mathscr{I}(s) = \int_0^{2\pi} P(s, e_\theta)^{-2/p}\, d\theta, \qquad e_\theta = (\cos\theta, \sin\theta).$$

Direct integration first in momentum and then in the radial position variable gives the phase volume

$$\Omega(s,E) = \frac{\pi p}{p+2} E^{1+2/p} \mathscr{I}(s). \tag{16}$$

Differentiating this finite angular integral, or the sublevel volume by coarea, gives the microcanonical mean of an actual parameter force:

$$b_j(s) = -\frac{p}{p+2}\,\partial_{s_j} \log \mathscr{I}(s). \tag{17}$$

The common-mean assumption identifies this with the mean for every invariant probability. Positivity on the angular circle and joint C^2 regularity justify both parameter derivatives of $\mathscr{I}$.

For clarity, the normalization used below can be checked on all invariant probabilities, not only on the microcanonical one. At a fixed parameter, the map $(Z, W) \mapsto (c^{-1/p}Z, W)$ takes the unit shell for P to that for cP and conjugates the flows with a constant time factor. It therefore transports invariant probabilities bijectively. Euler's identity and invariance give

$$\int P\, d\mu = \frac{2}{p+2}$$

on the unit shell: integrate the derivative of $Z \cdot W$ and combine with energy. Homogeneity yields

$$(cP)_{s_j}(c^{-1/p}Z) = P_{s_j}(Z) + (\partial_{s_j} \log c)P(Z).$$

Thus the mean for cP is $b_j + 2\partial_{s_j} \log c/(p+2)$. Choosing

$$V(s, Z) = \left(\frac{\mathscr{I}(s)}{\mathscr{I}(s_0)}\right)^{p/2} P(s, Z) \tag{18}$$

makes every normalized actual force have zero mean on every invariant probability, and makes the angular integral independent of s. A constant amplitude rescaling may first arrange a given radial member to be $|Z|^p$. For this normalization the constant angular integral is 2π. The potential V remains jointly C^2; no assertion of its joint convexity is involved.

Lemma 6 (Radial-base Morse–Bott manifolds)**.** *Let $p > 2$ and $V_0(Z) = |Z|^p$. The complete family N_R of parametrized unit-energy radial-line orbits, with their primitive period, is a compact, connected, boundaryless C^2 submanifold of the space in* (12)*. Its full free-period Hessian kernel consists precisely of time translation and rotation. If $\sqrt{p+2}$ is not an integer, the family N_C of parametrizations of the positively oriented unit-energy circle has Hessian kernel precisely its time-translation direction. Consequently both are Morse–Bott critical manifolds for* (12)*.*

Proof. Let q solve

$$q'' = -p|q|^{p-2}q, \qquad q(0) = 1, \quad q'(0) = 0.$$

Its first zero occurs at Q_p, its period is $T_R = 4Q_p$, and

$$q'^2 = 2(1 - q^p), \quad q'(Q_p) = -\sqrt{2}, \quad Q_p = \int_0^1 \frac{dx}{\sqrt{2(1-x^p)}} \tag{19}$$

on the positive quarter orbit. The loops $q(T_R u + \phi)e_\theta$ are parametrized by a two-torus with the free identification $(\theta, \phi) \sim (\theta + \pi, \phi + T_R/2)$. After this identification the map is injective; its two derivatives $q'e_\theta$ and $qe_\theta^\perp$ are independent. Compactness makes it an embedded manifold. The scalar force is C^1 for $p > 2$, so q is C^3 even at its zero crossings. The phase map into H^1 is C^2: its second derivative is a translate of q'', whose weak derivative q''' is continuous. This verifies the stated regularity without a third spatial derivative of the potential.

The transverse Jacobi equation is

$$y'' + p|q|^{p-2}y = 0.$$

Besides q, let j be its solution with $j(0) = 0$, $j'(0) = 1$. On $[0, Q_p]$ put $w = -q'/p$. Then $w(0) = 0$, $w'(0) = 1$, $w > 0$ on $(0, Q_p]$, $w'(Q_p) = 0$, and w solves the longitudinal Jacobi equation with coefficient $p(p-1)q^{p-2}$. While $j > 0$,

$$(j'w - jw')' = p(p-2)q^{p-2}jw > 0, \qquad (j/w)' = (j'w - jw')/w^2 > 0.$$

The ratio tends to one at zero, so j has no first zero; integration at Q_p also gives $d := j'(Q_p) > 0$. The q, j Wronskian gives $j(Q_p) = 1/\sqrt{2}$. Reflection symmetry about Q_p then yields the fundamental matrices

$$M_{Q_p} = \begin{pmatrix} 0 & 1/\sqrt{2} \\ -\sqrt{2} & d \end{pmatrix}, \qquad M_{2Q_p} = \begin{pmatrix} -1 & \sqrt{2}d \\ 0 & -1 \end{pmatrix}, \qquad M_{T_R} = \begin{pmatrix} 1 & -2\sqrt{2}d \\ 0 & 1 \end{pmatrix}.$$

Thus the periodic transverse kernel is exactly the rotational direction q. The longitudinal equation has independent solutions

$$q', \qquad q + \left(\frac{p}{2} - 1\right) tq',$$

the second obtained by differentiating the amplitude-scaled orbit $\lambda q(\lambda^{p/2-1}t)$. Its energy derivative at the turning point is p, whereas the first has zero energy derivative.

To include the period coordinate explicitly, a free-period kernel vector (η, τ) corresponds in physical time to the ordinary Jacobi field

$$\xi(t) = \eta(t/T_R) - \frac{\tau t}{T_R}\dot{Q}(t).$$

Linearizing the period equation gives zero first energy variation. Periodicity of η gives

$$\xi(T_R) - \xi(0) = -\tau\dot{Q}(0), \qquad \dot{\xi}(T_R) - \dot{\xi}(0) = -\tau\ddot{Q}(0). \tag{20}$$

These conditions are also sufficient, by reversing the substitution and the linearized Euler–Lagrange equations. At the turning point, zero energy variation removes the longitudinal amplitude solution. The surviving q' is periodic, and $\ddot{Q}(0) = -pe_\theta$ forces $\tau = 0$. The nonzero transverse shear removes j. Exactly time shift and rotation remain.

For the circle write

$$Q(t) = Re_{\omega t}, \qquad R^p = \frac{2}{p+2}, \qquad \omega^2 = pR^{p-2}, \qquad T_C = \frac{2\pi}{\omega}.$$

The unforced Jacobi equations in the rotating frame $\xi = ue_{\omega t} + ve^{\perp}_{\omega t}$ are

$$u'' - 2\omega v' + (p-2)\omega^2 u = 0, \qquad v'' + 2\omega u' = 0.$$

Zero first energy variation is $v' + 2\omega u = 0$, whence $u'' + (p+2)\omega^2 u = 0$. The endpoint conditions (20) give $u(T_C) = u(0)$ and $u'(T_C) = u'(0)$. Since $\sqrt{p+2} \notin \mathbb{Z}$, $u = 0$. Then $v' = 0$, and the position endpoint condition forces $\tau = 0$. The remaining constant v is time translation. The circular family is an embedded circle of parametrizations. Together with the Fredholm calculation in (14), these kernel computations prove both assertions. ∎

In a normalized common-mean family, (13) vanishes at every nearby critical point. Lemmas 5 and 6 therefore continue each of these entire compact manifolds as a C^1 graph of periodic orbits with unchanged action and period. This conclusion is obtained from reduced extrema, rather than from an a priori choice of radial periodic branches.

3.3 The diagonal response, including radial displacement

Restrict the normalized parameter family to a line through a radial member, and write

$$V(r, Z) = |Z|^p + rg(Z) + \tfrac{1}{2}r^2 B(Z) + o(r^2), \qquad g(\rho e_\theta) = \rho^p \gamma(\theta), \quad B(\rho e_\theta) = \rho^p A(\theta). \tag{21}$$

Joint C^2 regularity gives $\gamma \in C^1(S^1)$ and $A \in C^0(S^1)$. It also gives the continuous Cartesian gradient of g, with size $O(\rho^{p-1})$ at zero. No derivative of A is used. Write

$$\langle h \rangle = \frac{1}{2\pi} \int_0^{2\pi} h(\theta)\, d\theta.$$

The mean of g on every base radial line is zero. Its two half-periods have equal length and $\langle |q|^p \rangle_{\text{time}} = 2/(p+2)$, so

$$\gamma(\theta + \pi) = -\gamma(\theta). \tag{22}$$

Consequently

$$\gamma(\theta) = \sum_{\substack{n \geq 1 \\ n \text{ odd}}} (a_n \cos n\theta + b_n \sin n\theta), \qquad w_n = a_n^2 + b_n^2,$$

with

$$\langle \gamma^2 \rangle = \tfrac{1}{2} \sum w_n, \qquad \langle (\gamma')^2 \rangle = \tfrac{1}{2} \sum n^2 w_n < \infty. \tag{23}$$

Proposition 3.1 (Three diagonal identities)**.** *If $\sqrt{p+2} \notin \mathbb{Z}$, the jets in* (21) *satisfy*

$$\langle A \rangle = \left(1 + \frac{2}{p}\right) \langle \gamma^2 \rangle, \tag{24}$$

$$\langle A\rangle = \sum_{n \text{ odd}} K_{p,n} w_n, \qquad K_{p,n} = \frac{(p+1)(p+2)+n^2}{2p(p+2-n^2)}, \tag{25}$$

$$A_{\text{even}}(\theta) = \left(1+\frac{1}{p}\right)\gamma(\theta)^2 + C_p\gamma'(\theta)^2, \tag{26}$$

where $A_{\text{even}}(\theta) = (A(\theta)+A(\theta+\pi))/2$. *Here* C_p *is defined by the uniquely solvable quarter-orbit problem*

$$z'' + pq^{p-2}z = q^{p-1}, \qquad z'(0) = z'(Q_p) = 0, \qquad C_p = \frac{p+2}{2Q_p}\int_0^{Q_p} q^{p-1} z\, dt. \tag{27}$$

In particular,

$$\sum_{n \text{ odd}} D_{p,n} w_n = 0, \qquad \sum_{n \text{ odd}} \left(C_p n^2 - \frac{1}{p}\right) w_n = 0, \qquad D_{p,n} = \frac{(p+3)n^2-(p+2)}{2p(p+2-n^2)}. \tag{28}$$

All series in these formulas converge absolutely.

Proof. The normalized angular volume is constant. Twice differentiating $\langle V(r,e_\theta)^{-2/p}\rangle = 1$ gives (24) by ordinary differentiation on the compact angular circle.

For the circle, a unit cosine mode $\gamma = \cos n\theta$ forces

$$u'' - 2\omega v' + (p-2)\omega^2 u = -pR^{p-1}\cos(n\omega t), \qquad v'' + 2\omega u' = nR^{p-1}\sin(n\omega t).$$

A periodic fixed-energy solution is

$$u = U_n\cos(n\omega t), \quad v = V_n\sin(n\omega t), \qquad U_n = -\frac{R(p+2)}{p(p+2-n^2)}, \quad V_n = \frac{R(p+2+n^2)}{np(p+2-n^2)}. \tag{29}$$

Substitution checks both differential equations. The energy variation, after division by $R^p\cos(n\omega t)$, is $pnV_n/R + 2pU_n/R + 1 = 0$. The phase ambiguity has zero force pairing. Averaging $\nabla g\cdot\xi$ gives

$$\frac{R^{p-1}}{2}(pU_n - nV_n) = -R^p K_{p,n}.$$

Sine modes are phase shifts and distinct Fourier modes are orthogonal. The actual periodic branch has constant period, by (15) and the compact reduction. Differentiating its exact zero force mean therefore gives $0 = R^p\langle A\rangle - R^p\sum K_{p,n}w_n$, proving (25).

This Fourier calculation is valid for C^1 angular data. Fourier partial sums converge in H^1, so their circle gradient forcings converge in L^2. The fixed-gauge periodic Jacobi inverse is bounded on the complement of its phase kernel, as follows either from its finite-dimensional nonresonant monodromy or from (29). Indeed $U_n = O(n^{-2})$ and $V_n = O(n^{-1})$ per unit coefficient, so the solutions converge in H^2 for H^1 angular data. The bilinear force pairing then converges. Since $K_{p,n}$ is bounded and (23) holds, the displayed series is absolutely convergent.

For the radial response fix θ. Half-turn oddness gives, also by continuous extension at $q = 0$,

$$g(qe_\theta) = \operatorname{sgn}(q)|q|^p\gamma(\theta), \qquad \nabla g(qe_\theta) = |q|^{p-1}\big(p\gamma(\theta)e_\theta + \gamma'(\theta)e_\theta^\perp\big).$$

The longitudinal and transverse forced equations decouple. To evaluate the longitudinal pairing, use the scalar potential $|x|^p + r\gamma x|x|^{p-1}$. Its positive and negative well coefficients are $1 + r\gamma$ and $1 - r\gamma$. Scaling in each half-well gives its exact mean of the parameter force:

$$\frac{2\gamma}{p+2}\frac{(1+r\gamma)^{-1-1/p} - (1-r\gamma)^{-1-1/p}}{(1+r\gamma)^{-1/p} + (1-r\gamma)^{-1/p}}.$$

Its derivative at zero is $-2(p+1)\gamma^2/[p(p+2)]$. This scalar family has zero second potential derivative and zero first period variation, so the derivative is exactly the longitudinal orbit-displacement pairing at fixed energy and period.

For the transverse component the left-Neumann homogeneous solution is a multiple of q, and $q'(Q_p) = -\sqrt{2}$. Thus adjusting the initial value achieves the terminal Neumann condition uniquely in (27). Reflection extends z to a $2Q_p$-periodic solution of $z'' + p|q|^{p-2}z = |q|^{p-1}$. The transverse displacement is $-\gamma'(\theta)z$, and its full-period averaged force pairing equals

$$-\frac{1}{Q_p}\left(\int_0^{Q_p} q^{p-1}z\,dt\right)(\gamma')^2 = -\frac{2C_p}{p+2}(\gamma')^2.$$

The time-translation kernel contributes the integral of a full-time derivative. The rotational kernel contributes the angular derivative of the identically zero base line mean. Hence neither changes the pairing. The actual continued radial orbit has constant period, and the raw curvature mean on the base line is $2A_{\text{even}}/(p+2)$. Differentiating its zero force mean now proves (26). Only B, ∇g, and the first derivative of the C^1 normal graph have entered. Finally average (26), subtract (24), and use Parseval; subtracting the volume identity from the circle identity gives the other equation in (28). ∎

3.4 Spectral certificates for the cubic and quartic coefficients

We record an integral identity for C_p which will also be useful for the supporting first-jet results. Put

$$I_k = \int_0^{Q_p} q^k\,dt = \frac{1}{p\sqrt{2}}\,\mathrm{B}\left(\frac{k+1}{p}, \frac{1}{2}\right), \qquad c_p = \frac{\sqrt{2}Q_p}{p+2}.$$

The beta integral follows by $x = q(t)$ in (19). Its recurrences and the gamma reflection formula give

$$I_p = \frac{2Q_p}{p+2}, \qquad I_{p-1} = \frac{\sqrt{2}}{p}, \qquad Q_pI_{p-2} = \frac{\pi\cot(\pi/p)}{p(p-2)}. \tag{30}$$

Let

$$\mathcal{L}_p = -\frac{d^2}{dt^2} - pq^{p-2}$$

on $[0, Q_p]$, with Neumann condition at zero and Dirichlet condition at Q_p. The positive interior function q is its zero ground state. For a smooth mixed-boundary test function the substitution $v = qh$ and integration by parts give the nonnegative form $\int q^2|h'|^2$; density extends it to the form domain. The kernel is exactly the span of q. Comparison by the min–max principle with the second Neumann–Dirichlet Laplace eigenvalue gives

$$\lambda_1(\mathcal{L}_p) \geq \left(\frac{3\pi}{2Q_p}\right)^2 - p. \tag{31}$$

When the right side is positive, it bounds the inverse on $q^\perp$.

The Wronskian identity $(qz' - q'z)' = q^p$ gives $z(Q_p) = c_p$. Set

$$\rho_p = q^{p-1} - pc_p q^{p-2}, \qquad d = z - c_p.$$

Then $\rho_p \perp q$ by (30), d satisfies the mixed boundary conditions, and $\mathcal{L}_p d = -\rho_p$. Thus $d_\perp = -\mathcal{L}_{p,\perp}^{-1}\rho_p$. Integrating the Neumann problem for z also gives $\int q^{p-2}z = I_{p-1}/p$. Consequently

$$\begin{aligned}\int q^{p-1}z &= \int \rho_p z + pc_p \int q^{p-2}z \\ &= 2c_p I_{p-1} - pc_p^2 I_{p-2} - \langle \rho_p, \mathcal{L}_{p,\perp}^{-1}\rho_p\rangle_{L^2}.\end{aligned}$$

The possible q component of d has disappeared because $\rho_p \perp q$. Multiplying by $(p+2)/(2Q_p)$ gives the exact identity

$$C_p = \frac{2}{p} - \frac{\pi\cot(\pi/p)}{(p-2)(p+2)} - \frac{p+2}{2Q_p}\langle \rho_p, \mathcal{L}_{p,\perp}^{-1}\rho_p\rangle_{L^2}. \tag{32}$$

The last quadratic form is nonnegative. This statement concerns the specified positive mixed-boundary inverse only; it imposes no sign on a general periodic mechanical Hessian.

Lemma 7 (Strict quartic bound)**.** *The coefficient $C = C_4$ in (27) satisfies*

$$C \geq \frac{912399186767}{3884615949375} > \frac{7}{30}, \qquad C - \frac{7}{30} \geq \frac{11977597159}{7769231898750} > 0. \tag{33}$$

Proof. Write $Q = Q_4$ and $c = Q/(3\sqrt{2})$. The beta identities specialize to

$$I_3 = \frac{1}{2\sqrt{2}}, \quad I_4 = \frac{Q}{3}, \quad I_5 = \frac{\pi}{8\sqrt{2}}, \quad I_6 = \frac{3}{5}I_2, \quad QI_2 = \frac{\pi}{8}.$$

Expanding $\rho = q^3 - 4cq^2$ gives

$$\|\rho\|_{L^2}^2 = I_6 - 8cI_5 + 16c^2 I_4 = \frac{3\pi}{40Q} - \frac{\pi Q}{6} + \frac{8Q^3}{27}.$$

Equations (31) and (32) therefore imply

$$C \geq \frac{1}{2} - \frac{\pi}{12} - \frac{3}{Q}\,\frac{3\pi/(40Q) - \pi Q/6 + 8Q^3/27}{(3\pi/(2Q))^2 - 4}. \tag{34}$$

Here the numerator is exactly a squared residual norm.

For an entirely rational enclosure, let $c_j = 4^{-j}\binom{2j}{j}$, so $c_0 = 1$ and $c_{j+1} = c_j(2j+1)/(2j+2)$. The binomial series gives

$$Q = \frac{1}{\sqrt{2}} \sum_{j=0}^{\infty} \frac{c_j}{4j+1}.$$

Induction shows $c_j \le (j+1)^{-1/2}$: after squaring the induction step, the required polynomial difference is $3j+2 > 0$. The tail after $j = 1000$ is less than $\frac{1}{4}\int_{1000}^{\infty} x^{-3/2}dx < 1/62$ before the factor $1/\sqrt{2}$. With $S = \sum_{j=0}^{1000} c_j/(4j+1)$, exact rational summation of this recurrence gives

$$\frac{23}{25} < \frac{S}{14143/10000} < Q < \frac{S + 1/62}{14142/10000} < \frac{47}{50}.$$

The square-root bounds follow by squaring. One may certify $157/50 < \pi < 22/7$, for example, from $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and alternating rational arctangent sums. Set $q_- = 23/25$, $q_+ = 47/50$, $\pi_- = 157/50$, $\pi_+ = 22/7$. All quantities in the following lower bound have their indicated signs:

$$R_+ = \frac{3\pi_+}{40q_-} - \frac{\pi_- q_-}{6} + \frac{8q_+^3}{27} > 0, \qquad G_- = \left(\frac{3\pi_-}{2q_+}\right)^2 - 4 > 0.$$

Substitution in (34) gives

$$C \ge \frac{1}{2} - \frac{\pi_+}{12} - \frac{3R_+}{q_- G_-} = \frac{912399186767}{3884615949375}.$$

Subtracting $7/30$ gives the second fraction in (33). Every comparison in this recurrence, including the final fractions, uses exact rational arithmetic. ∎

Lemma 8 (Cubic spectral bound)**.** *The coefficient C_3 in* (27) *satisfies*

$$C_3 \ge \frac{10963971328544500687}{36532685426422500000} > \frac{7}{24}. \tag{35}$$

Proof. Use the notation of (30) at $p = 3$. The beta recurrences give

$$I_4 = \frac{4}{7} I_1, \qquad I_3 = \frac{2Q_3}{5}, \qquad I_2 = \frac{\sqrt{2}}{3}, \qquad Q_3 I_1 = \frac{\pi}{3\sqrt{3}}.$$

With $c_3 = \sqrt{2}Q_3/5$ and $\rho_3 = q^2 - 3c_3 q$, expansion gives

$$\|\rho_3\|_2^2 = I_4 - 6c_3 I_3 + 9c_3^2 I_2 = \frac{4\pi}{21\sqrt{3}Q_3} - \frac{6\sqrt{2}Q_3^2}{25}.$$

The constant term in (32) is $2/3 - \pi/(5\sqrt{3})$. Thus

$$C_3 \ge \frac{2}{3} - \frac{\pi}{5\sqrt{3}} - \frac{5}{2Q_3} \frac{4\pi/(21\sqrt{3}Q_3) - 6\sqrt{2}Q_3^2/25}{(3\pi/(2Q_3))^2 - 3}.$$

The positive central-binomial expansion is now

$$Q_3 = \frac{1}{\sqrt{2}} \sum_{j=0}^{\infty} \frac{c_j}{3j+1}, \qquad c_j = 4^{-j} \binom{2j}{j}.$$

The bound $c_j \le (j+1)^{-1/2}$ proved above bounds the tail after $j = 1000$ by $2/(3\sqrt{1000}) < 2/93$, before the factor $1/\sqrt{2}$. Exact summation of the first 1001 terms with the bounds

$$14142/10000 < \sqrt{2} < 14143/10000$$

gives $49/50 < Q_3 < 101/100$. Set

$$q_- = \frac{49}{50}, \quad q_+ = \frac{101}{100}, \quad s_2 = \frac{14142}{10000}, \quad s_3 = \frac{1732}{1000}, \quad \pi_- = \frac{157}{50}, \quad \pi_+ = \frac{22}{7}.$$

The radical bounds follow by squaring; the rational bounds for π follow from the arctangent certificate used above. The residual upper bound and spectral lower bound are

$$R_+ = \frac{4\pi_+}{21 s_3 q_-} - \frac{6 s_2 q_-^2}{25} > 0, \qquad G_- = \left(\frac{3\pi_-}{2q_+}\right)^2 - 3 > 0.$$

Consequently

$$C_3 \ge \frac{2}{3} - \frac{\pi_+}{5 s_3} - \frac{5R_+}{2 q_- G_-} = \frac{10963971328544500687}{36532685426422500000}.$$

Its difference from $7/24$ is $308604745837938187/36532685426422500000 > 0$. All sums and comparisons above are finite rational operations. ∎

Corollary 3.1 (Quartic first-jet rigidity)**.** *For a jointly C^2 planar quartic common-mean family with $P(s_0, Z) = a|Z|^4$, $a > 0$, every parameter direction satisfies*

$$\partial_{s_j} P(s_0, Z) = 3 b_j(s_0) P(s_0, Z).$$

Proof. A constant amplitude rescaling and (18) reduce to (21) with $p = 4$. Then

$$K_n = \frac{n^2 + 30}{8(6 - n^2)}, \qquad D_n = \frac{7n^2 - 6}{8(6 - n^2)}, \qquad D_1 = \frac{1}{40}, \quad -\frac{19}{8} \le D_n < 0 \quad (n \ge 3 \text{ odd}).$$

Add $2/3$ of the circular difference equation in (28) to the radial one. The resulting weight is

$$W_n = Cn^2 - \frac{1}{4} + \frac{2}{3} D_n. \tag{36}$$

For $n = 1$, $W_1 = C - 7/30 > 0$. For odd $n \ge 3$, $W_n \ge 9C - 11/6 > 4/15$. All sums are absolutely convergent by (23). Since $w_n \ge 0$, the equality $\sum W_n w_n = 0$ forces every Fourier coefficient to vanish. Continuity and homogeneity give $g = 0$. Undoing normalization gives the stated identity, separately for each parameter direction. ∎

3.5 A finite, frequency-independent isolation theorem

Vanishing of the first jet does not by itself exclude a deformation flat at the base parameter. We now prove a finite statement for both $p = 3$ and $p = 4$. Fix one of these two exponents. For $f \in C^2(S^1)$ put

$$W_f(\rho e_\theta) = \rho^p f(\theta), \quad W_f(0) = 0, \qquad V_f(Z) = |Z|^p + W_f(Z), \qquad \|f\|_1^2 = \langle f^2 + (f')^2 \rangle.$$

Use the two fixed base critical manifolds N_R, N_C at this exponent p. For every sufficiently small $\delta = \|f\|_{C^2}$ their normal reductions are defined by the unique small normal solution

$$\Pi_{E_m} \nabla J_f(m + \psi_f(m)) = 0, \quad E_m = (T_m N)^\perp, \quad a_N(f, m) = J_f(m + \psi_f(m)). \tag{37}$$

Here N is either N_R or N_C, and the Hilbert norm is invariant under planar rotations and translations of the time variable.

Theorem 4 (Finite isolation of the planar radial cubic and quartic)**.** *For each $p \in \{3, 4\}$ there is $\delta_0 > 0$ such that $\|f\|_{C^2} < \delta_0$ and the three exact conditions*

$$\langle (1 + f)^{-2/p} \rangle = 1, \tag{38}$$

$$a_{N_R}(f, m) = J_0(N_R) \quad (m \in N_R), \tag{39}$$

$$a_{N_C}(f, m) = J_0(N_C) \quad (m \in N_C) \tag{40}$$

imply $f = 0$. The reductions in these conditions exist for every sufficiently small C^2 perturbation; their exact values are additional conditions.

Proof. All constants denoted C_* below depend only on the chosen exponent, the two base manifolds, their normal inverses, and fixed bounded tubes. They are independent of f and of its Fourier frequencies. Write $C = C_p$, $K_n = K_{p,n}$ and $D_n = D_{p,n}$ for the response coefficients at this exponent.

The mechanical perturbation and its normal inverse. The Cartesian derivatives of W_f in the orthonormal polar frame are

$$\nabla W_f = \rho^{p-1}(p f e_\theta + f' e_\theta^\perp),$$
$$D^2 W_f = \rho^{p-2} \begin{pmatrix} p(p-1)f & (p-1)f' \\ (p-1)f' & pf + f'' \end{pmatrix}.$$

They extend continuously by zero at the origin. Hence $f \mapsto W_f$ is bounded linear from $C^2(S^1)$ into C^2 on every fixed configuration ball. Write

$$J_f = J_0 - \ell_f, \qquad \ell_f(q, T) = T \int_0^1 W_f(q(u))\, du.$$

The explicit Hessian is

$$D^2 \ell_f[(\eta, \tau), (\xi, \sigma)] = T \int_0^1 \eta \cdot D^2 W_f(q) \xi \, du$$

$$+\tau\int_0^1 \nabla W_f(q)\cdot\xi\,du+\sigma\int_0^1 \nabla W_f(q)\cdot\eta\,du.$$

Thus, on a fixed tube with positive periods,

$$\|D^2\ell_f\|\le C_*\delta. \tag{41}$$

The base action has a locally Lipschitz Hessian in both cases. At $p=4$ this follows from its smooth quartic potential integral and a bounded third derivative on the fixed tube. At $p=3$ an additional origin estimate is needed. Write $\Phi_3(z)=|z|^3$. For $z\neq 0$,

$$D^2|z|^3=3|z|I+3\frac{zz^{\mathsf T}}{|z|},$$

with continuous value zero at $z=0$. Set $M(z)=zz^{\mathsf T}/|z|$ away from zero and $M(0)=0$. If $r=|x|\ge s=|y|>0$, then

$$M(x)-M(y)=\frac{xx^{\mathsf T}-yy^{\mathsf T}}{r}+yy^{\mathsf T}\left(\frac{1}{r}-\frac{1}{s}\right).$$

The first term has norm at most $(r+s)|x-y|/r$, and the second at most $(s/r)|r-s|$. The zero case is immediate, and reversing x,y covers the other ordering. Thus $\|M(x)-M(y)\|\le 3|x-y|$ and

$$\|D^2|x|^3-D^2|y|^3\|\le 12|x-y| \qquad (x,y\in\mathbb{R}^2). \tag{42}$$

The integral Taylor formulas consequently give, including segments crossing the origin,

$$\begin{aligned}\left|\Phi_3(x+h)-\Phi_3(x)-D\Phi_3(x)[h]-\tfrac12 D^2\Phi_3(x)[h,h]\right|&\le 2|h|^3,\\ |D\Phi_3(x+h)-D\Phi_3(x)-D^2\Phi_3(x)h|&\le 6|h|^2.\end{aligned} \tag{43}$$

The constants are $12\int_0^1 t(1-t)\,dt=2$ and $12\int_0^1 t\,dt=6$. The H^1-to-C^0 embedding transfers (42) to the loop-loop potential block of the action Hessian. Its period cross terms use the locally Lipschitz base gradient; the kinetic and period terms are smooth and Lipschitz on the fixed bounded tube. Therefore, for either exponent,

$$\|D^2J_0(y+h)-D^2J_0(y)\|\le A_*\|h\|_E. \tag{44}$$

This gives a quadratic gradient remainder and a cubic action remainder without requiring a third derivative of the cubic potential at zero.

If L_m is the base normal Riesz Hessian, then $\sup_m\|L_m^{-1}\|<\infty$ by Lemma 6 and compactness. No differentiability of time translation on the whole Hilbert space is required: the finite-dimensional base manifold is C^2, its tangent projections are C^1, and the Hessian coefficients vary continuously in the sup norm.

Let $\beta_f(m)=d\ell_f(m)|_{E_m}$, identified with its Riesz vector. The normal equation is

$$-\beta_f(m)+L_mz+h_m(z)=0,$$

where, uniformly for $\|z\|, \|w\| \le r$,

$$h_m(0) = 0, \qquad \|h_m(z) - h_m(w)\| \le (A_* r + B_* \delta)\|z - w\|, \qquad \|\beta_f(m)\| \le B_0 \delta.$$

Here A_* is the base Hessian Lipschitz bound in (44), and B_* comes from (41). Choose r and then δ so that, with $K = \sup\|L_m^{-1}\|$,

$$K(A_* r + B_* \delta) \le \tfrac{1}{2}, \qquad K B_0 \delta \le r/2.$$

The map $z \mapsto L_m^{-1}(\beta_f(m) - h_m(z))$ is a contraction of the closed r-ball, uniformly over both manifolds. It gives exactly the normal solution in (37). Moreover,

$$\|\psi_f\| \le C_* \|\beta_f\|, \quad \psi_f = L_m^{-1}\beta_f + r_f, \quad \|r_f\| \le C_* \delta \|\beta_f\|. \tag{45}$$

For the last estimate use $\|h_m(\psi_f)\| \le (A_*/2)\|\psi_f\|^2 + B_* \delta \|\psi_f\|$ and $\|\beta_f\| \le B_0 \delta$. The graph is C^1 in the base point and the C^2 angular parameter by the implicit function theorem.

The tame quadratic action remainder. Taylor expansion in the orbit variable, with f fixed, gives

$$\begin{gathered} a_N(f, m) = J_0(m) - \ell_f(m) - \frac{1}{2}\beta_f(m)[L_m^{-1}\beta_f(m)] + R_N(f, m), \\ |R_N(f, m)| \le C_* \delta \|\beta_f(m)\|^2. \end{gathered} \tag{46}$$

Indeed the base Taylor remainder is $O(\|\psi_f\|^3)$ and the perturbation remainder after its linear term is $O(\delta\|\psi_f\|^2)$. Put $v = L_m^{-1}\beta_f$ and $r_f = \psi_f - v$. The quadratic part cancels exactly as

$$\tfrac{1}{2}\langle L_m \psi_f, \psi_f\rangle - \beta_f[\psi_f] = -\tfrac{1}{2}\beta_f[v] + \tfrac{1}{2}\langle L_m r_f, r_f\rangle.$$

Now use (45), the uniform bound on L_m, and $\|\beta_f\| \le C_* \delta$. This proves the indicated multiplicative remainder using two angular derivatives of f only.

On a radial base orbit in direction θ, let

$$b_f(\theta)^2 = f(\theta)^2 + f(\theta + \pi)^2 + f'(\theta)^2 + f'(\theta + \pi)^2.$$

The first variation of ℓ_f and the displayed gradient formula give

$$\|\beta_f(m_\theta)\| \le C_* b_f(\theta), \qquad \langle b_f^2\rangle = 2\|f\|_1^2, \qquad \sup_\theta b_f(\theta) \le C_* \delta. \tag{47}$$

The period component uses only f and the loop component uses f, f'; restriction to a normal space cannot increase the dual norm. On the circle, uniform angular traversal and Cauchy–Schwarz instead give

$$\|\beta_f(m_C)\| \le C_* \|f\|_1. \tag{48}$$

The phase-averaged radial remainder and the circle remainder in (46) are therefore bounded by

$$C_* \|f\|_{C^2} \|f\|_1^2. \tag{49}$$

This is the required uniform scale; no small remainder in a stronger norm is substituted for it.

Control of the even part in H^1. Decompose

$$e(\theta) = \tfrac{1}{2}(f(\theta) + f(\theta+\pi)), \qquad o(\theta) = \tfrac{1}{2}(f(\theta) - f(\theta+\pi)).$$

They are orthogonal in both L^2 and H^1. Since $\ell_f(m_\theta) = 2T_R e(\theta)/(p+2)$, (39), (46), and (47) imply

$$|e(\theta)| \le C_* b_f(\theta)^2 \le C_* \delta b_f(\theta). \tag{50}$$

A distinct derivative argument avoids differentiating the remainder. Choose a rotation-equivariant normal chart and write $m_\theta + \psi_f(m_\theta) = R_\theta(m_0 + \chi(\theta))$ with $\chi(\theta) \in E_{m_0}$. For fixed f the function

$$F(\theta, \chi) = J_f(R_\theta(m_0 + \chi))$$

is C^2; planar rotation acts smoothly as a bounded operator on H^1. The normal equation gives $D_\chi F = 0$ on the C^1 graph, so its reduced derivative has no χ' term. The remaining derivative is exactly

$$-\ell_{f'(\theta+\cdot)}(m_0 + \chi(\theta)).$$

At $\chi = 0$ this is $-2T_R e'(\theta)/(p+2)$. The homogeneous function $\rho^p f'(\arg Z)$ is C^1, with gradient bounded on the loop range by $C_*\delta$, because $f \in C^2$. Its loop/period integral is therefore Lipschitz with constant $C_*\delta$. Constancy in (39) gives

$$|e'(\theta)| \le C_*\delta \|\chi(\theta)\| \le C_*\delta b_f(\theta). \tag{51}$$

This calculation differentiates the direct finite-dimensional rotation map, not the possibly nondifferentiable translation curve with values in $C^2(S^1)$. Integrating (50) and (51) yields

$$\|e\|_1 \le C_*\delta \|f\|_1, \qquad \|f\|_1 \le 2\|o\|_1, \qquad \|e\|_1 \le 2C_*\delta\|o\|_1 \tag{52}$$

for sufficiently small δ.

Finite forms of the three response equations. In the rest of the proof E denotes an error, allowed to change between lines, satisfying $|E| \le C_*\delta\|o\|_1^2$. The radial averaged normal quadratic form is bounded bilinearly on $H^1(S^1)$:

$$\left|\left\langle \beta_g(m_\theta)[L_{m_\theta}^{-1}\beta_h(m_\theta)]\right\rangle\right| \le C_*\|g\|_1\|h\|_1.$$

This follows from (47), the inverse bound and Cauchy–Schwarz in θ. The circle form has the same bound by (48). Replacing $f = e + o$ by o in either quadratic form therefore costs at most $C_*(\|e\|_1\|o\|_1 + \|e\|_1^2)$, which is an error E by (52).

Taylor's formula for (38), on $|f| < 1/2$, gives

$$0 = -\frac{2}{p}\langle f\rangle + \frac{p+2}{p^2}\langle f^2\rangle + O(\delta\langle f^2\rangle).$$

Orthogonality, $\langle o\rangle = 0$, and (52) imply

$$\langle e\rangle = \frac{p+2}{2p}\langle o^2\rangle + E. \tag{53}$$

For an odd perturbation, ℓ_o vanishes on the entire radial base manifold. Thus $d\ell_o$ annihilates its tangent kernel, and the forced fixed-energy displacement computed in Proposition 3.1 solves the full Hessian equation modulo that kernel. Its pairing is therefore the normal-inverse pairing in (46). The longitudinal and transverse pairings give, with their signs,

$$-\tfrac{1}{2}\beta_o[L_m^{-1}\beta_o] = \frac{T_R}{p+2}\left(\frac{p+1}{p}o(\theta)^2 + Co'(\theta)^2\right).$$

Here $J_f = J_0 - \ell_f$, so the displacement solves $L_m\psi = +d\ell_o$; the quadratic action is minus one half the potential-force pairing times the period. Averaging (46) and using (39) yields

$$2\langle e\rangle = \frac{p+1}{p}\langle o^2\rangle + C\langle (o')^2\rangle + E. \tag{54}$$

Subtract twice (53) to obtain

$$C\langle (o')^2\rangle - \frac{1}{p}\langle o^2\rangle = E.$$

Similarly the circle linear term is $2T_C\langle e\rangle/(p+2)$, and (29) gives

$$-\tfrac{1}{2}\beta_o[L_C^{-1}\beta_o] = \frac{T_C}{p+2}\sum_{n\ \text{odd}} K_n w_n.$$

Therefore (40) and Parseval give

$$\sum_{n\ \text{odd}} D_n w_n = E, \qquad \sum_{n\ \text{odd}} (Cn^2 - \tfrac{1}{p})w_n = E, \qquad \sum_{n\ \text{odd}} (1+n^2)w_n = 2\|o\|_1^2. \tag{55}$$

Every fixed factor absorbed into E preserves its small factor δ and its H^1-squared scale.

Absorption by the strict response gap. Put $\lambda_3 = 1$, $\lambda_4 = 2/3$, and define

$$W_{p,n} = C_p n^2 - \frac{1}{p} + \lambda_p D_{p,n}.$$

For $p = 4$ these are the weights in (36). For $p = 3$ one has

$$D_{3,n} = -1 - \frac{25}{6(n^2-5)}, \qquad D_{3,1} = \frac{1}{24}, \qquad D_{3,n} \geq -\frac{49}{24} \quad (n \geq 3 \text{ odd}).$$

The first-mode weights and the bounds for every odd $n \geq 3$ are

p	$W_{p,1}$	$W_{p,n}$ $(n \geq 3)$
3	$C_3 - 7/24$	$C_3 n^2 - 19/8$ or larger
4	$C_4 - 7/30$	$C_4 n^2 - 11/6$ or larger

Both high-mode bounds are H^1-coercive. For $x = n^2 \geq 9$,

$$\frac{7}{24}x - \frac{19}{8} - \frac{1}{40}(1+x) = \frac{4}{15}(x-9) \geq 0,$$
$$\frac{7}{30}x - \frac{11}{6} - \frac{2}{75}(1+x) = \frac{31}{150}(x-9) \geq 0.$$

The exact lower bounds of Lemmas 8 and 7 therefore give

$$W_{p,n} \geq c_{*,p}(1+n^2),$$
$$c_{*,3} = \frac{308604745837938187}{730653708528450000000}, \qquad c_{*,4} = \frac{11977597159}{15538463797500}. \tag{56}$$

Here $c_{*,3} < 1/40$ and $c_{*,4} < 2/75$, and each is half the corresponding certified first-mode gap. Add λ_p times the first equation in (55) to the second. The absolutely convergent sum and (56) imply

$$2c_{*,p}\|o\|_1^2 \leq C_*\delta\|o\|_1^2.$$

One fixed sufficiently small C^2 neighborhood makes $C_*\delta < 2c_{*,p}$, and forces $o = 0$. Then (52) gives $e = 0$, hence $f = 0$. The neighborhood is independent of Fourier support. This proves the theorem. ∎

3.6 Rigidity on a connected parameter domain

Theorem 5 (Finite radial cubic and quartic family rigidity)**.** *Let $p \in \{3, 4\}$ and let $U \subset \mathbb{R}^m$ be connected and open. Let $P \in C^2(U \times \mathbb{R}^2)$ be positive away from $Z = 0$ and p-homogeneous in Z, with local uniform normal coercivity. Assume that at every $s \in U$ all invariant probabilities on $|W|^2/2 + P(s, Z) = 1$ give the same vector mean $b(s)$ of $P_s(s, Z)$. If one member is $P(s_0, Z) = a_0|Z|^p$, $a_0 > 0$, then*

$$P(s, Z) = a(s)|Z|^p \qquad (s \in U)$$

for a positive C^2 function a. The same conclusion holds on a connected parameter interval, with relative neighborhoods at included endpoints.

Proof. A fixed amplitude rescaling makes the radial member $|Z|^p$; as shown above this preserves the common-mean property. Apply (18) once, with reference point s_0. The resulting V has all actual force means zero and constant angular volume. It is jointly C^2.

Near any parameter s_* at which $V(s_*, Z) = |Z|^p$, apply Lemma 5 to J_s and to each of the two compact base manifolds. At every nearby critical point, (13) vanishes in every parameter direction. The compact envelope argument, along straight parameter segments in a small ball, makes the two reduced actions exactly their respective base values throughout the ball.

Set $f_s(\theta) = V(s, e_\theta) - 1$. Joint C^2 regularity gives continuity into $C^2(S^1)$. Explicitly,

$$\partial_\theta V(s, e_\theta) = D_Z V(s, e_\theta) \cdot e_\theta^\perp,$$

$$\partial_{\theta\theta} V(s, e_\theta) = D_Z^2 V(s, e_\theta)[e_\theta^\perp, e_\theta^\perp] - D_Z V(s, e_\theta) \cdot e_\theta.$$

Uniform continuity of these functions on the compact angular circle shows $\|f_s\|_{C^2} \to 0$ as $s \to s_*$. This is continuity, not parameter differentiability with values in C^2; no derivative $D_s D_Z^2 V$ is used. At a fixed s, the normal reduction made directly in the finite-dimensional parameter and the reduction used for Theorem 4 solve the same normal equation in the same fibers. Their small solutions coincide by normal uniqueness. Thus the exact action values just obtained, together with constant angular volume, are exactly the three conditions of that theorem. It follows that $V(s, Z) = |Z|^p$ on a neighborhood of s_*.

The set of such parameters is nonempty, closed by continuity, and open by this local argument. Connectedness makes it all of U. Undoing the one fixed normalization gives the asserted positive amplitude function. The interval case uses the same local argument and relative open–closed conclusion. ∎

Theorem 5 proves the radial finite-rigidity assertion of Theorem 1 for $p = 3, 4$. The degree-six assertion is proved separately in Theorem 17. The present proof uses two fixed compact periodic families and the quantified remainder (49). It does not infer finite rigidity from the first-jet corollary and does not require uniform inverses over unrelated periodic families.

4 A nonradial quartic neighborhood and connected-family rigidity

We work in the coefficient space $\mathcal{V}_4$ with its ordinary Euclidean inner product in the basis $(x^4, x^3y, x^2y^2, xy^3, y^4)$. Write $J(x, y) = (-y, x)$ and

$$\mathcal{R}P(Z) = \nabla P(Z) \cdot JZ.$$

This is a fixed linear operator on coefficient space. At the base potential (3),

$$\mathcal{R}P_0 = 20x^3y + 40xy^3. \tag{57}$$

The normal Hessian has diagonal entries $12x^2 + 24y^2$ and $24x^2 + 192y^2$, and determinant $288x^4 + 576x^2y^2 + 4608y^4 > 0$ away from zero. Thus P_0 is normally strictly convex. Its only real projective angular critical lines are the coordinate axes; their angular second derivatives are 20 and -40, and their values are 1 and 16.

4.1 The constrained direction left by the straight orbits

The two simple critical lines continue smoothly in coefficient space. Let their continued critical values be $C_x(P)$ and $C_y(P)$. Together with $\mathcal{I}(P)$, set $G = (C_x, C_y, \mathcal{I})$. At P_0, for

Duhamel's inequality bounds the positive part of h'_u by an exponentially decaying initial term plus the convolution of $|k|^2 \in L^1$ with $e^{-a\sigma}$. Its positive part is therefore integrable. As $h_u \geq 0$, its downward variation is also finite, and h_u has a limit. Lemma 12 supplies a cluster point $u \in M$; for this choice the limit is zero. No rate in (105), positivity of E_∞, or finite-length statement is inferred. ∎

9 Supporting first-jet results and sextic resonant admission

This section records a supporting cubic first-jet consequence and proves sextic first-jet rigidity and the resonant estimates used in the finite sextic proof. The cubic and quartic finite classifications were established in Theorem 5; the sextic finite classification is proved in Theorem 17. The hypotheses are joint C^2 regularity, positivity and local uniform normal coercivity, homogeneity in two normal variables, and the common actual-force mean property at every nearby parameter. The parameter dimension is arbitrary. Each assertion is applied separately along a parameter direction. The normalization, compact reduction, and response identities proved in Section 3 apply without polynomiality or evenness.

9.1 The cubic first-jet consequence

Theorem 13 (Planar cubic first-jet rigidity)**.** *If $P(s_0, Z) = a|Z|^3$, with $a > 0$, then*

$$\partial_{s_j} P(s_0, Z) = \frac{5}{2} b_j(s_0) P(s_0, Z)$$

for every parameter component.

Proof. After a fixed amplitude rescaling and (18), write the centered first jet as $g(\rho e_\theta) = \rho^3 \gamma(\theta)$. Lemma 6 applies to the cubic radial family, including its origin crossings: the scalar orbit is C^3, which gives a C^2 phase manifold in H^1, while the potential $|Z|^3$ itself is C^2. The circle is nonresonant because $\sqrt{5} \notin \mathbb{Z}$. Thus Proposition 3.1 gives the two identities (28). In this case

$$D_{3,1} = \frac{1}{24}, \qquad -\frac{49}{24} \leq D_{3,n} < 0 \quad (n \geq 3 \text{ odd}).$$

The second bound follows directly from $D_{3,n} = -1 - 25/[6(n^2-5)]$. Add the two response identities. At $n = 1$ the combined weight is $C_3 - 7/24 > 0$. For odd $n \geq 3$ it is at least $9C_3 - 19/8 > 1/4$. Absolute convergence and nonnegative w_n force all modes to vanish. Hence $g = 0$, and undoing normalization gives the coefficient $5/2$. ∎

9.2 Why three responses alone do not decide the sextic case

Lemma 13 (A sextic coefficient enclosure)**.** *The coefficient C_6 satisfies*

$$\frac{3237314787097591129}{21151420009845104000} < C_6 < \frac{196057}{1200000}, \qquad \frac{32}{243} < C_6 < \frac{32}{195} < \frac{1}{6}. \tag{106}$$

Proof. For $p = 6$, the constant term in (32) is $1/3 - \pi\sqrt{3}/32$. The beta recurrences give

$$\|\rho_6\|_2^2 = \frac{5}{8}I_4 - \frac{48}{7}c_6 I_3 + 18c_6^2 I_2, \qquad I_4 = \frac{\pi\sqrt{3}}{24Q_6}, \quad I_2 = \frac{\pi}{6\sqrt{2}}, \quad c_6 = \frac{\sqrt{2}Q_6}{8}.$$

Since $0 \le q \le 1$, $I_3 \ge I_4$. Replacing I_3 by I_4 in its negative term therefore gives

$$\|\rho_6\|_2^2 \le \frac{5\pi\sqrt{3}}{192Q_6} - \frac{\pi\sqrt{3/2}}{14} + \frac{3\pi Q_6^2}{32\sqrt{2}}.$$

The central-binomial expansion with denominators $6j+1$, using the first 1001 terms and tail $2/(6\sqrt{1000}) < 1/93$, gives $17/20 < Q_6 < 87/100$. Put

$$q_- = \frac{17}{20}, \quad q_+ = \frac{87}{100}, \quad s_{2,-} = \frac{14142}{10000}, \quad s_{2,+} = \frac{14143}{10000}, \quad s_{3,-} = \frac{1732}{1000}, \quad s_{3,+} = \frac{1733}{1000}.$$

Using $\pi_- = 157/50$, $\pi_+ = 22/7$, define

$$R_+ = \frac{5\pi_+ s_{3,+}}{192q_-} - \frac{\pi_- s_{3,-}}{14s_{2,+}} + \frac{3\pi_+ q_+^2}{32s_{2,-}} > 0, \qquad G_- = \left(\frac{3\pi_-}{2q_+}\right)^2 - 6 > 0.$$

Equations (31) and (32) give

$$C_6 > \frac{1}{3} - \frac{\pi_+ s_{3,+}}{32} - \frac{4R_+}{q_- G_-} = \frac{3237314787097591129}{21151420009845104000}.$$

The strict inequality follows from the strict input enclosures. The nonnegative residual correction also gives

$$C_6 \le \frac{1}{3} - \frac{\pi\sqrt{3}}{32} < \frac{1}{3} - \frac{(157/50)(1732/1000)}{32} = \frac{196057}{1200000}.$$

Direct rational comparison proves the remaining inequalities in (106). ∎

Proposition 9.1 (A limitation of the three scalar response tests)**.** *For $p = 6$, the two scalar equations* (28) *admit nonzero nonnegative weights supported on $n = 1, 3, 5$. This is a consistency statement for these equations, not an admitted first jet of a common-mean potential family.*

Proof. Write $C = C_6$ and $w_1 = 1$. The circle coefficients are

$$D_{6,1} = \frac{1}{84}, \qquad D_{6,3} = -\frac{73}{12}, \qquad D_{6,5} = -\frac{217}{204}.$$

Solving the radial and circular difference equations for the other weights gives

$$w_3 = \frac{3(243C - 32)}{14(5451C - 32)}, \qquad w_5 = \frac{17(32 - 195C)}{14(5451C - 32)}.$$

The determinant is $4(5451C - 32)/153 > 0$, and both weights are positive by (106). Let $\gamma = \cos\theta + \sqrt{w_3}\cos 3\theta + \sqrt{w_5}\cos 5\theta$ and define A_{even} by (26). The two scalar difference identities ensure that its angular average also satisfies the volume and circle identities. These assignments do not check force means on other invariant measures. The periodic measures constructed next impose an additional first-order constraint. ∎

9.3 Exact resonant detection of every odd sextic mode above three

For the radial sextic Hamiltonian $H_0 = |W|^2/2 + \rho^6 = 1$, choose positive angular momentum L. Let $A < B$ be the squared inner and outer radial turning points. In the range

$$0 < A \le A_* := \frac{121}{625}, \qquad \frac{L^2}{2} = A - A^4, \qquad (A + B)(A^2 + B^2) = 1, \tag{107}$$

the last equation has a unique root $B(A) \in [9/10, 1]$. Indeed its left side increases strictly in both positive variables, is below one at $B = 9/10$ for the indicated A range, and is above one at $B = 1$ for $A > 0$. It follows that B is continuous and strictly decreasing, with $B(0) = 1$ and $B > A$ throughout the closed range.

Put

$$\begin{gathered} u = A + (B - A)\sin^2\psi, \quad 0 \le \psi \le \frac{\pi}{2}, \\ Q_0 = A^2 + AB + B^2, \qquad Q(u) = u^2 + (A + B)u + Q_0. \end{gathered} \tag{108}$$

The factorization

$$u - u^4 - L^2/2 = (u - A)(B - u)Q(u)$$

comes from (107). The energy and angular momentum equations then give, on the outgoing half radial orbit,

$$\frac{dt}{d\psi} = \frac{1}{\sqrt{2Q(u)}}, \qquad \frac{d\theta}{d\psi} = \frac{\sqrt{A - A^4}}{u\sqrt{Q(u)}}, \qquad \rho^6\frac{dt}{d\psi} = \frac{u^3}{\sqrt{2Q(u)}}. \tag{109}$$

The turning-point singularities have been removed. Let $\Theta(A)$ be the angular advance on this half orbit, starting at $\theta(0) = 0$.

For the limit $A \downarrow 0$ it is useful to write

$$\begin{gathered} d = \frac{A + B}{Q_0}, \quad e = \frac{1}{Q_0}, \quad k(u) = \sqrt{1 + du + eu^2}, \\ h(u) = k(u)^{-1}, \qquad D(u) = \frac{d + eu}{k(u)(1 + k(u))}. \end{gathered}$$

The relation $A - A^4 = ABQ_0$ follows by multiplying the turning-point equation by A. For $u > 0$, $D(u) = (1 - h(u))/u$. Substitution $t = \tan\psi$ evaluates

$$\int_0^{\pi/2} \frac{\sqrt{AB}}{A + (B - A)\sin^2\psi}\, d\psi = \int_0^{\infty} \frac{\sqrt{AB}}{A + Bt^2}\, dt = \frac{\pi}{2}.$$

Thus

$$\delta(A) := \frac{\pi}{2} - \Theta(A) = \sqrt{AB} \int_0^{\pi/2} D(u)\, d\psi. \tag{110}$$

The rationalized expression for D extends continuously and positively at $u = 0$, where its value is $d/2$. On the whole closed parameter/angle rectangle its denominator stays positive and its integral is bounded. It follows that $\delta(A) > 0$ for $A > 0$ and $\Theta(A) \to \pi/2$ as $A \downarrow 0$.

Define the continuous endpoint extension of the ratio

$$\mathcal{R}(A) = \frac{\displaystyle\int_{\pi/3}^{\pi/2} h(u)/u\, d\psi}{\displaystyle\int_0^{\pi/2} D(u)\, d\psi}. \tag{111}$$

The numerator is nonsingular even at $A = 0$, because u has a positive lower bound on this tail interval. For $A > 0$ the ratio is exactly $[\Theta(A) - \theta(\pi/3; A)]/\delta(A)$. Also put

$$M(A) = \int_0^{\pi/2} \frac{u^3}{\sqrt{2Q(u)}}\, d\psi, \qquad M_0(A) = \int_0^{\pi/3} \frac{u^3}{\sqrt{2Q(u)}}\, d\psi.$$

Lemma 14 (Uniform rational orbital bounds)**.** *For every* $0 \le A \le A_*$,

$$\mathcal{R}(A) < \frac{3}{5}, \qquad \frac{1}{2}M(A) - \frac{3}{2}M_0(A) > \frac{1}{200}, \qquad \Theta(A_*) < \frac{2\pi}{5}. \tag{112}$$

The value of $\mathcal{R}$ *at zero is the continuous ratio in* (111)*.*

Proof. We give the complete finite interval certificate, including the direction of each bound. All operations described here are rational operations or integer square roots. Let $S = 10^{12}$ and define outward rounding by

$$\lfloor x \rfloor_S = S^{-1}\lfloor Sx \rfloor, \qquad \lceil x \rceil_S = S^{-1}\lceil Sx \rceil.$$

Use

$$\pi_- = \frac{3141592653589}{10^{12}}, \qquad \pi_+ = \frac{3141592653590}{10^{12}}.$$

For a rational x, alternating sums of $\arctan x$ through index 22 at $x = 1/5$, and through index 8 at $x = 1/239$, together with their next terms, certify $\pi_- < \pi < \pi_+$ from Machin's identity. For $x \ge 0$ let

$$\mathrm{sr}_-(x) = \frac{1}{S}\,\mathrm{isqrt}\big(\lfloor S^2 x \rfloor\big), \qquad \mathrm{sr}_+(x) = \mathrm{sr}_-(x) + S^{-1}.$$

Squaring gives $\mathrm{sr}_-(x)^2 \le x \le \mathrm{sr}_+(x)^2$.

Divide $[0, A_*]$ at $A_j = jA_*/8$, $0 \le j \le 8$. For every positive endpoint, enclose $B(A_j)$ by 50 rational bisections of $[9/10, 1]$ using the sign of $(A_j + B)(A_j^2 + B^2) - 1$, and round the lower and upper brackets outward at scale S. For $A_0 = 0$ use the exact bracket $B = 1$. Since B decreases, on $[A_j, A_{j+1}]$ use the rectangle

$$A \in [a_-, a_+] = [A_j, A_{j+1}], \qquad B \in [b_-, b_+] = [\underline{B}(A_{j+1}), \overline{B}(A_j)].$$

These satisfy $0 \le a_- \le a_+ < b_- \le b_+$.

For each of the three intervals $[0, \pi/2]$, $[0, \pi/3]$ and $[\pi/3, \pi/2]$, use $N = 256$ equal angle cells. At a node $\psi = \pi t$ use the exact values of $\sin^2 \psi$ if $t = 0$ or $t = 1/2$. For every other node let

$$S_m(x) = \sum_{k=0}^{m} \frac{(-1)^k x^{2k+1}}{(2k+1)!}, \quad s_- = \lfloor S_{11}(\pi_- t) \rfloor_S, \quad s_+ = \lceil S_{10}(\pi_+ t) \rceil_S.$$

The alternating sine series and monotonicity of sine on $[0, \pi/2]$ give $0 \le s_- \le \sin\psi \le s_+ \le 1$. Rounding their squares outward gives node bounds for $\sin^2 \psi$. For a whole angle cell use the lower bound at its left node, denoted σ_-, and the upper bound at its right node, denoted σ_+. Monotonicity of $\sin^2$ makes these whole-cell bounds.

On a parameter rectangle and angle cell define

$$\begin{aligned}
u_- &= a_- + (b_- - a_-)\sigma_-, & u_+ &= a_+ + (b_+ - a_+)\sigma_+, \\
q_{0,-} &= a_-^2 + a_- b_- + b_-^2, & q_{0,+} &= a_+^2 + a_+ b_+ + b_+^2, \\
d_- &= (a_- + b_-)/q_{0,+}, & d_+ &= (a_+ + b_+)/q_{0,-}, \\
e_- &= 1/q_{0,+}, & e_+ &= 1/q_{0,-}, \\
k_- &= \mathrm{sr}_-(1 + d_- u_- + e_- u_-^2), & k_+ &= \mathrm{sr}_+(1 + d_+ u_+ + e_+ u_+^2), \\
Q_- &= u_-^2 + (a_- + b_-)u_- + q_{0,-}, & Q_+ &= u_+^2 + (a_+ + b_+)u_+ + q_{0,+}.
\end{aligned}$$

All these enclosures follow from positivity and from the monotonicity of $A(1-\sigma) + B\sigma$ in A, B, σ when $B > A$. The following are whole-cell bounds:

$$\begin{aligned}
D(u) &\ge D_- := \left\lfloor \frac{d_- + e_- u_-}{k_+(1 + k_+)} \right\rfloor_S, \\
h(u)/u &\le H_+ := \left\lceil \frac{1}{u_- k_-} \right\rceil_S && \text{on tail cells,} \\
\frac{u^3}{\sqrt{2Q(u)}} &\ge M_- := \left\lfloor \frac{u_-^3}{\mathrm{sr}_+(2Q_+)} \right\rfloor_S, \\
\frac{u^3}{\sqrt{2Q(u)}} &\le M_+ := \left\lceil \frac{u_+^3}{\mathrm{sr}_-(2Q_-)} \right\rceil_S.
\end{aligned}$$

Every denominator is positive; tail cells have $u_- > 0$ even on the rectangle touching $A = 0$. In particular, no monotonicity of D itself has been assumed. Summing over the

respective full, early, and tail grids gives

$$\mathcal{R}(A) \le R_j^+ := \frac{\pi_+ \sum_{\text{tail}} H_+/(6N)}{\pi_- \sum_{\text{full}} D_-/(2N)},$$
$$\frac{1}{2}M(A) - \frac{3}{2}M_0(A) \ge G_j^- := \frac{\pi_-}{4N}\sum_{\text{full}} M_- - \frac{\pi_+}{2N}\sum_{\text{early}} M_+.$$

These formulas include all integration error by bounding the integrand throughout each cell. Finite rational arithmetic gives the following coarser rational bounds; every table entry is an integer numerator over 10^6:

j	10^6 times an upper bound for R_j^+	10^6 times a lower bound for G_j^-
0	506383	34017
1	509599	30790
2	513624	27412
3	518438	23872
4	524028	20157
5	530387	16258
6	537514	12165
7	545415	7870

Thus every ratio is below $3/5$ and every signed mass is above $1/200$.

For the endpoint angle use $A = A_*$, the same bracket $[b_-, b_+]$ for $B(A_*)$, and 1024 equal cells on $[0, \pi/2]$. In each cell bound its integrand from (109) above by

$$\left\lceil \frac{\operatorname{sr}_+(A_* - A_*^4)}{u_- \operatorname{sr}_-(u_-^2 + (A_* + b_-)u_- + A_*^2 + A_* b_- + b_-^2)} \right\rceil_S, \quad u_- = A_* + (b_- - A_*)\sigma_-.$$

Multiplication of the sum by $\pi_+/(2 \cdot 1024)$ gives a rational upper bound less than

$$1254/1000 < 2\pi_-/5.$$

This proves the third inequality. A second rational evaluation with 16 parameter rectangles, 128 angle cells, and scale 10^{10} gives the same three strict inequalities. The interval construction above covers the whole closed parameter interval. ∎

Lemma 15 (Periodic detection of an odd harmonic)**.** *For every odd integer $n \ge 5$ there is a periodic invariant probability μ_n of $H_0 = 1$ such that*

$$\int \rho^6 e^{in\theta}\, d\mu_n \neq 0.$$

Proof. Let $m = (n-1)/2$ and set $\Theta_n = m\pi/n = \pi/2 - \pi/(2n)$. This lies in $[2\pi/5, \pi/2)$. Lemma 14 puts $\Theta(A_*)$ strictly below Θ_n, while (110) gives $\Theta(A) > \Theta_n$ for sufficiently small positive A. Continuity supplies $A_n \in (0, A_*)$ with $\Theta(A_n) = \Theta_n$. After n full radial cycles the total angular advance is $2n\Theta_n = 2m\pi$. Since $n - 2m = 1$, m and n are coprime.

Position and velocity return, giving a genuine positive-angular-momentum periodic orbit. Let μ_n be its physical-time probability.

At apoapsis $n\Theta_n = m\pi$. For $\psi \in [\pi/3, \pi/2]$, (111) and (112) give

$$0 \le n(\Theta_n - \theta(\psi)) \le n\mathcal{R}(A_n)\delta(A_n) < \frac{3\pi}{10} < \frac{\pi}{3}.$$

Consequently $(-1)^m \cos(n\theta(\psi)) = \cos(n(\Theta_n - \theta(\psi))) \ge 1/2$ on the tail, and is at least -1 elsewhere. The signed outgoing force integral is thus bounded below by $M(A_n)/2 - 3M_0(A_n)/2 > 1/200$. Reflection about apoapsis replaces θ by $2\Theta_n - \theta$, preserving the cosine and reversing the sine. Each successive full radial cycle adds $2\Theta_n$, which preserves the nth harmonic. The full orbit therefore satisfies

$$(-1)^m \int \rho^6 \cos(n\theta)\, d\mu_n > 0, \qquad \int \rho^6 \sin(n\theta)\, d\mu_n = 0.$$

The normalization by the finite positive orbit period preserves the strict sign. This proves the claimed nonzero complex coefficient. Every n uses an interior parameter $A_n > 0$; no limiting zero-angular- momentum orbit is used as a periodic detector. ∎

Theorem 14 (Planar sextic first-jet rigidity)**.** *If $P(s_0, Z) = a|Z|^6$, with $a > 0$, then*

$$\partial_{s_j} P(s_0, Z) = 4b_j(s_0)P(s_0, Z)$$

for every parameter component.

Proof. A fixed amplitude scaling and (18) give $V(s_0, Z) = |Z|^6$ and zero actual force means. Fix a parameter direction and write $g(\rho e_\theta) = \rho^6\gamma(\theta)$. The line means give half-turn oddness. For every odd $n \ge 5$, let μ_n be supplied by Lemma 15. Simultaneously rotate its position and momentum by ϕ. Radial symmetry preserves invariance, so the zero mean hypothesis implies

$$F_n(\phi) := \int \rho^6 \gamma(\theta + \phi)\, d\mu_n = 0 \qquad (0 \le \phi \le 2\pi).$$

Fubini on the two finite measures gives, with $\widehat{\gamma}_n = (2\pi)^{-1} \int \gamma(\phi) e^{-in\phi} d\phi$,

$$\frac{1}{2\pi} \int_0^{2\pi} F_n(\phi) e^{-in\phi}\, d\phi = \widehat{\gamma}_n \int \rho^6 e^{in\theta}\, d\mu_n.$$

The second factor is nonzero, so $\widehat{\gamma}_n = 0$. This projection isolates one rotation frequency even if that orbit has nonzero coefficients at higher multiples of n. It does not split a zero force mean term by term in the original angular series. Reality and half-turn oddness now leave only modes 1 and 3, by Fourier completeness in L^2 and continuity of γ.

Proposition 3.1 applies at $p = 6$ because $\sqrt{8} \notin \mathbb{Z}$. Its circular difference equation is

$$\frac{1}{84} w_1 - \frac{73}{12} w_3 = 0, \qquad \text{hence } w_3 = \frac{w_1}{511}.$$

If $w_1 > 0$, its radial difference equation forces

$$(C_6 - \tfrac{1}{6}) + \frac{9C_6 - \frac{1}{6}}{511} = 0, \qquad C_6 = \frac{32}{195},$$

contrary to Lemma 13. If $w_1 = 0$, the circle equation also gives $w_3 = 0$. Thus $g = 0$. Since the normalization exponent in (18) is 3, undoing it gives exactly the factor 4 in the theorem. ∎

9.4 Scope of the supporting consequences

Under the original jointly convex compact-tail assumptions, suppose the common frozen potential on the effective minimizing set M is a positive multiple of $|Z|^3$ or of $|Z|^6$, and assume common means only on a neighborhood of the compact tail set K. Lemma 12 supplies a cluster point $u_0 \in K \cap M$. Common means hold near u_0 and $b(u_0) = 0$, so the corresponding first-jet theorem gives $P_s(u_0, Z) = 0$ for every Z. For any $u \in M$, constancy of the frozen potential along $u_t = u_0 + t(u - u_0)$ gives $(u-u_0)^{\mathsf{T}} P_{ss}(u_t, Z)(u-u_0) = 0$. Positivity of the original joint Hessian implies $P_{ss}(u_t, Z)(u - u_0) = 0$, as in (11); integrating $\frac{d}{dt} P_s(u_t, Z) = 0$ gives

$$P_s(u, Z) = P_s(u_0, Z) = 0 \qquad (u \in M).$$

Proposition 8.1 therefore applies in both residual-energy regimes. This uses convexity of the original potential and does not require common means outside the specified neighborhood of K, or joint convexity of the amplitude-normalized potential.

The first-jet proof above uses infinitely many distinct detecting periodic families, one for each odd mode above three. By itself it supplies neither a uniform normal inverse for these families nor a uniform nonlinear reduction radius. The finite sextic proof in Theorem 17 therefore uses a different continuation argument: exact minimum actions in one fixed physical annulus, followed by the coherent response estimate (2). It does not infer finite rigidity from vanishing of the first jet.

10 Comparison with critical-line criteria and scope

10.1 What the two finite classifications add

For orientation, two simpler automatic-corrector criteria can be stated and checked directly.

First, let P be a binary polynomial of even degree p and suppose $\mathcal{R}P$ is nonzero with p distinct real projective zero lines. At each such line the straight nonlinear oscillator is periodic. By the virial identity, common means force

$$F_j = P_{s_j} - \frac{p+2}{2} b_j P$$

member; it does not classify sextic families whose normalized shapes never meet that locus. No assertion is made that the collection of tests used here exhausts all invariant measures of a general planar well.

Other isoperiodic and action-rigidity problems impose substantially different data and hypotheses. One-dimensional isoperiodic shears [1] preserve period data for scalar systems. The Suris-potential rigidity result [16] concerns discrete twist-map families with high regularity. The Zoll-near-a-minimum theorem [9] assumes a compact symplectic Morse–Bott minimum and a Zoll sequence. A quartic degenerate normal origin does not satisfy that nondegenerate normal-Hessian hypothesis; the Morse–Bott family used in our periodic-loop reduction is a different object. These works are relevant comparisons, not sources of the finite mechanical classifications proved here.

A Sextic mechanical actions and finite shape constraints

We use normalized angular measure throughout this section: $\langle g\rangle = (2\pi)^{-1}\int_0^{2\pi} g(\theta)\,d\theta$ and $\|g\|_{H^1}^2 = \langle g^2+(g')^2\rangle$. All the statements concern positive, positively homogeneous potentials of degree six. Antipodal symmetry and polynomial dependence are not assumed. Joint convexity is not used in any of the arguments in this section.

A.1 Normalization of the actual parameter force

Let $P(s,Z)$ be jointly C^2 on an open finite-dimensional parameter set times $\mathbb{R}^2$, positive for $Z\neq 0$, and suppose that (CM) holds at every parameter under consideration. Define

$$J_6(s) = \langle P(s,e_\theta)^{-1/3}\rangle, \qquad V(s,Z) = J_6(s)^3 P(s,Z). \tag{113}$$

If $P(s_0,Z)$ is a positive multiple of $|Z|^6$, then $V(s_0,Z)=|Z|^6$. Translate the base parameter so that $s_0=0$.

Lemma 16 (Normalized sextic force)**.** *The normalized family is jointly C^2, satisfies $\langle V(s,e_\theta)^{-1/3}\rangle = 1$, and has*

$$\int V_{s_j}(s,Z)\,d\mu = 0 \tag{114}$$

for every invariant probability on $|W|^2/2+V(s,Z)=1$ and every parameter component. Consequently every nearby critical point of the free-period action (12)*, with $p=6$, has zero parameter derivative.*

Proof. Formula (16) becomes

$$\Omega_P(s,E) = \frac{3\pi^2}{2} J_6(s) E^{4/3}.$$

The microcanonical calculation in Lemma 2 therefore identifies the common unit-energy force mean as $b_j = -\frac{3}{4}\partial_{s_j} \log J_6$. Put $c = J_6^3$. The map $(Z, W) \mapsto (c^{-1/6}Z, W)$ takes the unit shell for P to the unit shell for cP and conjugates the flows by a constant change of time. It transports invariant probabilities bijectively. Homogeneity gives

$$(cP)_{s_j}(c^{-1/6}Z) = P_{s_j}(Z) + (\partial_{s_j} \log c)P(Z).$$

The sextic virial identity is $\int P\, d\mu = 1/4$ on the unit shell. Thus the new mean is $-\frac{3}{4}\partial_{s_j} \log J_6 + \frac{1}{4}\partial_{s_j} \log c = 0$. Local positivity on the angular circle allows two parameter derivatives under its integral, proving all regularity assertions. Finally, (13) identifies the parameter derivative at a periodic critical point with minus its integrated actual force. ∎

The word "actual" in this observation matters: it is the derivative of the same family whose actions are being varied. No assertion about a general centered observable is substituted for (114).

A.2 A canonical section and a uniform base twist

Write $e_\theta^\perp = (-\sin\theta, \cos\theta)$. On the radially maximal, large-radius branch of $Q \cdot W = 0$, the unit-energy section has coordinates

$$Q = r(s, \theta, L)e_\theta, \qquad W = \frac{L}{r(s, \theta, L)}e_\theta^\perp, \qquad \frac{L^2}{2r^2} + r^6 V(s, e_\theta) = 1. \tag{115}$$

The section radius is the outer root. Its energy derivative is $6r^5V(s, e_\theta) - L^2/r^3 > 0$ in the compact range used below. The implicit function theorem makes r jointly C^2. Moreover,

$$W \cdot dQ = L\, d\theta. \tag{116}$$

This equality includes parameter-dependent section radii: their variations at fixed angle are radial and pair to zero with W.

Let $\mathcal{P}_s$ be the next large-radius apoapsis return map and $G_s = \mathcal{P}_s^{-1}$ its inverse. At a radial apoapsis $d_t(Q \cdot W) = |W|^2 - 6V = -6$, including the point where $W = 0$. Thus the section is regular at zero angular momentum. The Cartesian flow of a C^2 potential is C^1, even for trajectories passing through $Q = 0$. Transversality and compactness give a C^1 return map and return time on a fixed collar of the base angular-momentum interval below, for every sufficiently small parameter. Choosing the large-radius branch excludes the intervening small-radius zero of $Q \cdot W$. At the radial base, choose the lift

$$\mathcal{P}_0(\theta, L) = (\theta + \Delta(L), L), \qquad \Delta(0) = \pi, \qquad G_0(\theta, L) = (\theta + d(L), L), \quad d = -\Delta.$$

Lemma 17 (Twist through the fifth resonance)**.** *There is a unique positive angular momentum L_5 in the turning-point range* (107) *for which $\Delta(L_5) = 4\pi/5$. On $0 \le L \le L_5$, and on a fixed slightly larger collar,*

$$d'(L) = -\Delta'(L) > 3/8. \tag{117}$$

After restricting the parameter neighborhood, the inverse physical return maps are positive twist maps on that collar, uniformly in s.

Proof. Use the squared turning radii A, B and the regularization in (107)–(110). To avoid confusing the angular deficit with a perturbation norm, write

$$\eta(A) = \frac{\pi}{2} - \Theta(A) = \sqrt{AB}\int_0^{\pi/2} D(A, \sin^2\psi)\, d\psi, \qquad \Delta = 2\Theta = \pi - 2\eta.$$

Here $0 \le A \le 121/625$, $9/10 \le B \le 1$, and, with $\sigma \in [0,1]$,

$$u = A(1-\sigma) + B\sigma, \qquad Q_0 = A^2 + AB + B^2,$$
$$d_* = (A+B)/Q_0, \qquad e_* = 1/Q_0,$$
$$k = (1 + d_* u + e_* u^2)^{1/2}, \qquad D = \frac{d_* + e_* u}{k(1+k)}.$$

All these expressions, including D, are regular at $A = u = 0$. Implicit differentiation of $(A+B)(A^2+B^2) = 1$ gives

$$B' = -\frac{3A^2 + 2AB + B^2}{A^2 + 2AB + 3B^2}, \qquad Q_0' = 2A + B + (A+2B)B',$$
$$d_*' = \frac{1+B'}{Q_0} - \frac{(A+B)Q_0'}{Q_0^2}, \qquad e_*' = -\frac{Q_0'}{Q_0^2}, \qquad u' = 1 - \sigma + B'\sigma,$$
$$k' = \frac{d_*' u + d_* u' + e_*' u^2 + 2e_* u u'}{2k}.$$

In the last formulas a prime denotes differentiation with respect to A at fixed σ. Put $N = d_* + e_* u$ and $R = k(1+k)$; then

$$N' = d_*' + e_*' u + e_* u', \qquad R' = (1+2k)k', \qquad D_A = N'/R - NR'/R^2.$$

Consequently, for $A > 0$,

$$\sqrt{AB}\,\eta'(A) = \int_0^{\pi/2} \mathcal{W}(A, \sin^2\psi)\, d\psi, \qquad \mathcal{W} = \frac{B + AB'}{2} D + ABD_A. \tag{118}$$

Here is a finite rational verification of the strict bound needed in (118). Divide $[0, 121/625]$ and $[0,1]$ into 32 equal cells each. At each A endpoint bracket B by 60 exact bisections of $[9/10, 1]$, using the increasing polynomial $(A+B)(A^2+B^2) - 1$; use $B(0) = 1$ exactly. Monotonicity of B gives its interval on each A cell. Evaluate the displayed formulas on each of the 32^2 rectangles by interval addition, multiplication, and division, rounding each endpoint outward to multiples of S^{-1}, $S = 10^{15}$. For a positive rational interval $[x_-, x_+]$ use

$$\big[S^{-1}\,\mathrm{isqrt}(\lfloor S^2 x_- \rfloor),\ S^{-1}\big(\mathrm{isqrt}(\lfloor S^2 x_+ \rfloor) + 1\big)\big]$$

for its square-root enclosure. Every divisor in the recurrence has a strictly positive lower endpoint. The smallest resulting lower bound for $\mathcal{W}$ is

$$\frac{170846255452907}{10^{15}} > \frac{1}{10}. \tag{119}$$

These rational recurrences give a whole-rectangle bound with outward enclosures throughout.

Since $L^2 = 2(A - A^4)$, $dA/dL = L/(1 - 4A^3)$ and $L/\sqrt{AB} = \sqrt{2Q_0}$. Equations (118)–(119) imply

$$-\Delta'(L) = \frac{2\sqrt{2Q_0}}{1 - 4A^3} \int_0^{\pi/2} \mathcal{W}(A, \sin^2 \psi)\, d\psi > \frac{\pi\sqrt{2Q_0}}{10} > \frac{3}{8}.$$

The last step uses $Q_0 \geq 81/100$, $\sqrt{2Q_0} > 5/4$ and $\pi > 3$. The regular Cartesian return map supplies the same derivative at $L = 0$, by its continuous limit, and a two-sided collar there. Lemma 14 gives $\Theta(121/625) < 2\pi/5$, whereas $\Theta(0) = \pi/2$. Strict monotonicity gives the unique fifth resonance in the interior, with a positive margin to the outer endpoint. This proves the collar assertion. C^1 dependence of G_s then preserves a positive uniform twist bound for all sufficiently small s. ∎

A.3 Compact periodic families and physical annulus boundaries

Lemma 18 (Three fixed periodic families)**.** *For $V_0 = |Z|^6$, the radial family N_R, the positively oriented circular family N_C, and the fifth-resonance family N_5 are compact, connected, boundaryless Morse–Bott manifolds of the free-period action. Their Hessian kernels are respectively the two directions of time shift and rotation, the one time-shift direction, and the two directions of time shift and rotation. In a normalized common-mean family each admits, on one parameter neighborhood, a C^1 normal continuation consisting entirely of periodic critical points with its original action and period. The radial and fifth-resonance continuations meet the apoapsis section in two invariant graphs*

$$\Gamma_-(s) = \{L = u_-(s, \theta)\}, \qquad \Gamma_+(s) = \{L = u_+(s, \theta)\}, \qquad u_- < u_+, \tag{120}$$

with inverse-return rotation numbers $-1/2$ and $-2/5$.

Proof. The radial and circular kernel computations of Lemma 6 apply with $p = 6$; the circular nonresonance condition is $\sqrt{8} \notin \mathbb{Z}$. In particular the radial transverse monodromy has nonzero shear, rather than an extra periodic Jacobi direction. For N_5, fixing energy and intersecting a linearized periodic orbit with the transverse apoapsis section reduces its periodicity condition to the kernel of

$$D\mathcal{P}_0^5 - I = \begin{pmatrix} 0 & 5\Delta'(L_5) \\ 0 & 0 \end{pmatrix}.$$

Lemma 17 makes this kernel exactly the angle direction. The missing flow direction is time translation. To see explicitly that the free-period coordinate adds no kernel, use (20): the period equation imposes zero energy variation, and section projection removes time shift. The displayed shear forces the angular-momentum variation to vanish; the surviving angle variation is the infinitesimal spatial rotation of the base orbit and has no period variation. Subtraction of that rotation leaves only a time shift, also with zero period variation. The Hessian is self-adjoint Fredholm by (14).

The central energy–angular-momentum level at L_5 is an embedded invariant two-torus filled by periodic orbits. Taking the actual manifold of its parametrized loops, with the finite time–rotation identifications already divided out, gives a compact embedded critical manifold in $H^1 \times \mathbb{R}$; evaluation at time zero is its inverse onto the phase-space torus. Thus no redundant parametrization is being counted as a kernel direction. The analogous radial identification is given explicitly in Lemma 6.

Lemma 16 verifies the critical parameter-derivative hypothesis of Lemma 5 for each of these three fixed manifolds. That lemma supplies their complete normal graphs and constant reduced actions. The sextic virial relation is

$$\oint W \cdot dQ = \frac{3}{2} T_{\mathrm{per}}, \tag{121}$$

so their marked periods are constant as well.

For completeness the normal graphs are genuine phase-space tori. Choose the Hilbert metric invariant under time translations. Uniqueness of the normal equation makes its graph equivariant under those translations. Every graph point solves $\ddot{q} = -T_{\mathrm{per}}^2 \nabla V(s, q)$ in unit-period coordinates. A graph which is C^1 into H^1, hence into C^0, has a right-hand side C^1 into C^0; the differential equation and periodic derivative reconstruction upgrade it to C^1 into C^2. Evaluation of position and physical velocity is consequently a C^1 perturbation of the base torus embedding. It is an embedding after shrinking the neighborhood. Equivalently, equal initial data and the same fixed marked period give the same parametrized loop by uniqueness of the initial-value problem; the corresponding linearized assertion gives immersion. Time-translation equivariance makes its image flow invariant.

Compact transversality now continues each entire base crossing circle with the large-radius section as one C^1 circle; the part of the torus away from that crossing set stays away from the section branch. Its angle projection is C^1 close to the identity, hence it is a graph. The positive base separation L_5 persists uniformly. The two and five section-crossing counts, and the lifted winding numbers, are integer quantities constant under these continuations. They give exactly the rotation numbers asserted in (120). Only three fixed periods have been continued; no parameter neighborhood depending on an unbounded denominator has been intersected. ∎

A.4 Preservation of every rational minimum action

Proposition A.1 (One physical annulus for all denominators)**.** *There is one neighborhood of the radial parameter on which the following holds. The actual inverse apoapsis action has a global generating extension $\bar{h}_s$ for an exact positive C^1 twist map of the cylinder. Its minimal average action satisfies*

$$\beta_s(r) = h_0(2\pi r), \qquad -\frac{1}{2} \le r \le -\frac{2}{5}, \tag{122}$$

where $h_0(d)$ is the actual backward one-cycle action of the radial potential. For every rational p/q strictly inside this interval the physical annulus between (120) *contains an essential invariant circle consisting of p/q-periodic minimizing points. Each such orbit has the radial base marked physical action and period.*

Proof. By (116) the inverse return map is exact symplectic. Its positive twist makes its endpoint angles (θ, Θ) local coordinates on the graph. Choose fixed open increment intervals $I_0 \Subset I_1$, with the closed base range $[-\pi, -4\pi/5]$ strictly inside I_0, such that the physical generator is defined for $\Theta - \theta \in I_1$ at every sufficiently small s. The intervals can be chosen so that every map point between the two continued boundaries has its increment in I_0; compactness and the collar margin in Lemma 17 give this simultaneously. Fix the additive constant of this generator by taking the actual backward abbreviated action:

$$h_s(\theta, \Theta) = \int_{\text{backward arc}} W \cdot dQ, \qquad dh_s = L_{\text{out}}\, d\Theta - L_{\text{in}}\, d\theta. \tag{123}$$

It is C^2 in its angle variables and jointly C^1 in parameter and angles. The return maps converge in C^1, so $h_s - h_0$ converges to zero in the angular C^2 norm. The actual arc integral also fixes the convergence of its constant term. At the base, h_0 depends only on $d = \Theta - \theta$ and

$$h_0'(d) = L, \qquad h_0''(d) = 1/d'(L) > 0.$$

Extend h_0 to a C^2 uniformly convex function h_0^{ext} on $\mathbb{R}$, equal to h_0 on I_1 and quadratic off a larger compact interval. One may extend its positive continuous second derivative to a positive constant and integrate, preserving its value and first derivative on I_1. For a fixed smooth cutoff ζ supported in I_1 and equal to one on I_0, set

$$\bar{h}_s(\theta, \Theta) = h_0^{\text{ext}}(\Theta - \theta) + \zeta(\Theta - \theta)\big(h_s(\theta, \Theta) - h_0(\Theta - \theta)\big). \tag{124}$$

For one sufficiently small parameter neighborhood both $-\partial_\theta \partial_\Theta \bar{h}_s$ and $\partial_\Theta^2 \bar{h}_s$ have positive uniform lower bounds. Quadratic behavior at infinity and the mixed-derivative bound make the momentum equations a global exact positive C^1 twist diffeomorphism $\bar{G}_s$. It agrees with G_s on the whole physical annulus, including its invariant boundary graphs. This construction uses the fixed canonical section and does not straighten the moving boundaries.

We first localize *every* periodic orbit of $\bar{G}_s$ whose rotation lies strictly between $-1/2$ and $-2/5$. An invariant essential graph separates the cylinder into invariant upper and lower components. If g_b is the increasing lift of the map restricted to that graph, positive twist gives $\theta_{j+1} > g_b(\theta_j)$ on an orbit above it. Induction gives $\theta_j \geq g_b^j(\theta_0)$ and hence an orbit rotation at least that of the graph. Below the graph the reversed inequalities give rotation at most its rotation. Apply this first to the upper and then to the lower boundary. An intermediate-rotation periodic orbit can lie in neither outside component, and cannot lie on a boundary of a different rotation. It therefore lies in the physical annulus. In particular the extension introduces no extraneous periodic orbit of an intermediate rotation.

Vary the physical action in (123) at fixed endpoint angles and fixed energy. The usual first variation gives

$$\partial_{s_j} h_s = W_{\text{out}} \cdot \partial_{s_j} Q_{\text{out}} - W_{\text{in}} \cdot \partial_{s_j} Q_{\text{in}} - \int_{\text{backward arc}} V_{s_j}\, dt.$$

The endpoint terms vanish because the section momenta are tangential and the fixed-angle endpoint variations radial. Thus

$$\partial_{s_j} h_s(\theta, \Theta) = -\int_{\text{backward arc}} V_{s_j}\, dt. \tag{125}$$

The integral has oriented negative time. Alternatively, varying the free-time action $\int(|\dot{Q}|^2/2 - V + 1)\, dt$ proves the same formula: the unit-energy constraint cancels the endpoint-time terms. When the arcs concatenate into a periodic orbit, (114) gives

$$\sum_{i=0}^{q-1} \partial_{s_j} h_s(\theta_i, \theta_{i+1}) = 0. \tag{126}$$

The actual-action normalization in (123) is essential here; a freely chosen parameter-dependent constant would change this equation.

Fix a rational $p/q \in (-1/2, -2/5)$, with $q > 0$. On configurations $\theta_q = \theta_0 + 2\pi p$, modulo simultaneous translation by 2π, put

$$\mathcal{A}_s(\theta_0, \ldots, \theta_{q-1}) = \sum_{i=0}^{q-1} \bar{h}_s(\theta_i, \theta_{i+1}), \qquad b_s(p/q) = \frac{1}{q} \min \mathcal{A}_s.$$

Uniform quadratic growth in the increments makes sublevel sets compact on this quotient. A minimum exists, and its unconstrained critical point equations are exactly the momentum matching equations for a q-periodic orbit of $\bar{G}_s$. The preceding localization makes every such minimizer physical. At each of them every parameter derivative of $\mathcal{A}_s$ vanishes by (126).

For this fixed p, q, minimizers on a compact parameter segment belong to a common compact sublevel set. The minimum is locally Lipschitz. Compare a minimizing configuration at s with one at $s + h$ in the two minimum inequalities. Compactness gives

a subsequential limiting minimizer at s; continuity of the parameter derivative makes both bounding difference quotients tend to zero. This applies to both one-sided quotients. Along every parameter segment the minimum is therefore constant. The compact sublevel set may depend on q; the physical extension, its barriers, and the parameter neighborhood do not. It follows that, on the same neighborhood for all denominators,

$$\min \mathcal{A}_s = q\, h_0(2\pi p/q). \tag{127}$$

Indeed at $s = 0$ strict convexity and Jensen's inequality give the minimum at constant increments.

For an exact positive C^1 twist diffeomorphism with a C^2 coercive generator, the continuous convex minimal average action agrees at a rational rotation with the periodic minimum divided by its denominator. At a rational rotation it is differentiable precisely when an essential invariant circle consists entirely of periodic minimizing orbits; see [22, Proposition 2.5], [19], and the C^1 twist setting of [18, Section 4]. Equality (127) on the dense rationals and continuity give (122), including both endpoints. In the interior this function is C^2, so the rational-circle assertion follows from the stated criterion and the physical localization. The endpoint value at $-2/5$ follows by continuity; localization of an arbitrary endpoint minimizer is unnecessary.

Finally, a minimizing p/q configuration has backward action $q\beta_s(p/q)$. Its forward physical action is the negative of this quantity. Equation (121) therefore gives

$$T_{\mathrm{per}} = -\frac{2}{3} q\beta_0(p/q).$$

Both the marked action and period are exactly the radial values. If a representation is an iterate, both quantities include that iterate; no claim about a primitive period is needed. ∎

A.5 The exact brake circle and its boundary integral

Set

$$Q_6 = \int_0^1 \frac{dx}{\sqrt{2(1-x^6)}}, \qquad T = 2Q_6. \tag{128}$$

Thus the radial base period is $2T$ and its apoapsis-to-apoapsis brake leg has positive time T.

Proposition A.2 (Exact mechanical brake boundary)**.** *For every sufficiently small normalized common-mean parameter, the lower invariant graph in* (120) *is exactly $L = 0$. There is an increasing C^1 lift F_s such that*

$$G_s(\theta, 0) = (F_s(\theta), 0), \quad F_s(\theta + 2\pi) = F_s(\theta) + 2\pi, \quad F_s(F_s(\theta)) = \theta - 2\pi. \tag{129}$$

Every inverse brake leg has the same oriented time $-T$. With

$$b_s(\theta) = \partial_L \pi_\theta G_s(\theta, 0) > 0, \qquad b_0 = d'(0), \tag{130}$$

one has the exact identity

$$\mathcal{I}(V(s,\cdot)) := \left\langle \sqrt{\frac{F_s'}{b_s}} \right\rangle = b_0^{-1/2}. \tag{131}$$

Only the original C^2 potential regularity is required.

Proof. The natural mechanical action is invariant under time reversal $q(u) \mapsto q(-u)$. The base radial manifold has that symmetry. Choose its normal-reduction metric invariant under the compact group generated by time translations and reversal. Normal uniqueness in Lemma 5 makes the continued graph reversal invariant. Its phase-space torus is therefore invariant under $(Q, W) \mapsto (Q, -W)$. On the canonical apoapsis section this is $(\theta, L) \mapsto (\theta, -L)$. Since the intersection is a single graph, it satisfies $u_- = -u_-$ and hence $u_- = 0$.

At $L = 0$ the momentum is zero. Thus every rest launch from the unit-energy position boundary is one of these continued brake orbits. Its marked period is $2T$ and it has exactly two large-radius section crossings. Reversal at the other rest endpoint makes the second leg the reverse of the first. Both legs consequently have time T. The winding and crossing counts give the lift relation (129); its derivative is positive by C^1 closeness to the base translation. The twist positivity follows from Lemma 17.

Fix s and suppress it temporarily. For the generator chosen in (123), zero momenta at both endpoints imply

$$h(\theta, F(\theta)) = c, \quad h_\theta(\theta, F(\theta)) = h_\Theta(\theta, F(\theta)) = 0, \qquad c = -\frac{3}{2}T = h_0(-\pi). \tag{132}$$

The value follows also directly from the open-arc virial identity. Its endpoint term $Q \cdot W$ vanishes at both ends, so the abbreviated action is $3/2$ times the oriented leg time.

Define the increasing C^1 circle lift and its inverse by

$$\chi(\theta) = \frac{\theta + F(\theta) + \pi}{2}, \qquad \phi = \chi^{-1}.$$

Then $\chi(F(\theta)) = \chi(\theta) - \pi$. In particular $F(\phi(u)) = \phi(u - \pi)$. Use the global extension in (124) and set $\widetilde{h}(u, v) = \bar{h}(\phi(u), \phi(v))$. We do not assert that this function is C^2: ϕ is only C^1. The uniform positivity of $\bar{h}_{\Theta\Theta}$ and (132) nevertheless give, for all real u, v,

$$\widetilde{h}(u, v) - c \geq c_1 |v - u + \pi|^2, \tag{133}$$

$$\widetilde{h}(u, u - \pi + z) = c + \tfrac{1}{2} A(u) z^2 + o(z^2), \tag{134}$$

$$A(u) = h_{\Theta\Theta}(\phi(u), F(\phi(u)))\phi'(u - \pi)^2 > 0.$$

The remainder is uniform in u. These facts use Taylor's formula for the original C^2 generator and the uniform first-order expansion of the C^1 map ϕ. Its periodic positive derivative is bounded above and below, which also transfers the global quadratic lower bound.

Mechanical reversal of a backward arc, reversing both its spatial path and momentum, gives the local action symmetry

$$h(\theta, \Theta) = h(\Theta, \theta - 2\pi)$$

near the brake graph. Both arcs are uniquely determined in the physical collar and have the same oriented abbreviated action. Thus locally $\widetilde{h}(u, v) = \widetilde{h}(v, u - 2\pi)$. Apply (134) to $(u, u - \pi + z)$ and to $(u - \pi + z, u - 2\pi)$, and let $z \to 0$. Continuity alone proves

$$A(u - \pi) = A(u). \tag{135}$$

No derivative of A or second derivative of ϕ has been taken.

We next extract the boundary coefficient from exact discrete minima. For odd $n \geq 5$ let $d_n = -\pi + \pi/n$ and $p = -(n-1)/2$. The transformation by χ is a bijection of the configuration spaces with winding p, since it commutes with 2π translation. Their boundary condition becomes $u_n = u_0 - (n-1)\pi$. Put

$$x_i = u_i + i\pi, \qquad z_i = x_{i+1} - x_i, \qquad x_n = x_0 + \pi.$$

A configuration with $z_i = \pi/n$ has excess action at most C/n. For any minimizing configuration, (133) therefore gives

$$\sum_{i=0}^{n-1} z_i^2 \leq C/n, \qquad \max_i |z_i| \leq C n^{-1/2}, \qquad \sum_{i=0}^{n-1} |z_i| \leq C. \tag{136}$$

Consequently the uniform remainder in (134) sums to $o(1/n)$ for every minimizer. By (135), $A(u_i) = A(x_i)$, so the excess is, to that error, $\frac{1}{2}\sum_i A(x_i) z_i^2$.

Let $\mathcal{J} = \int_0^\pi \sqrt{A(x)}\, dx$ and let $\mathcal{L}$ be a primitive of $\sqrt{A}$. Its positive continuous, π-periodic derivative and (136) give

$$\sum_i \sqrt{A(x_i)} z_i = \mathcal{L}(x_n) - \mathcal{L}(x_0) + o(1) = \mathcal{J} + o(1).$$

Indeed the error is at most $\omega(\max |z_i|) \sum |z_i|$, where ω is a modulus of continuity of $\sqrt{A}$. This signed telescoping estimate permits negative steps. Cauchy–Schwarz now gives $\sum_i A(x_i) z_i^2 \geq \mathcal{J}^2/n + o(1/n)$. Conversely choose successive points equally spaced in $\mathcal{L}$: $\mathcal{L}(x_{i+1}) - \mathcal{L}(x_i) = \mathcal{J}/n$. Then $\max |z_i| \leq C/n$, and continuity gives the matching upper bound. The exact minimum therefore has the asymptotic formula

$$\min \sum_{i=0}^{n-1} \widetilde{h}(u_i, u_{i+1}) = nc + \frac{\mathcal{J}^2}{2n} + o(1/n). \tag{137}$$

By (127) and its endpoint value this minimum is $nh_0(d_n)$. Since $h_0'(-\pi) = 0$ and $h_0''(-\pi) = 1/b_0$, comparison with its ordinary second-order Taylor expansion gives

$$(\mathcal{J}/\pi)^2 = 1/b_0. \tag{138}$$

This argument uses just a continuous coefficient in a discrete quadratic-energy limit; it needs no smooth minimizing selector.

Finally the generating equations imply $b(\theta) = -1/h_{\theta\Theta}(\theta, F(\theta))$. Differentiate $h_\Theta(\theta, F(\theta)) = 0$ once to obtain

$$h_{\Theta\Theta}(\theta, F(\theta)) = \frac{1}{b(\theta)F'(\theta)}.$$

From $\phi(u - \pi) = F(\phi(u))$ one has $\phi'(u - \pi) = F'(\theta)\phi'(u)$ for $\theta = \phi(u)$. Hence

$$\sqrt{A(u)}\, du = \sqrt{F'(\theta)/b(\theta)}\, d\theta.$$

The full-circle integral on the left is $2\mathcal{J}$ by (135). Equation (138) proves (131). ∎

Lemma 19 (Jacobi realization of the boundary data)**.** *Write $R(\theta) = V(e_\theta)^{-1/6}$ and let $Q_\theta(t)$ be the backward brake trajectory, $-T \le t \le 0$, launched from $(R(\theta)e_\theta, 0)$. For the ordinary Jacobi equation*

$$U'' + D^2V(Q_\theta(t))U = 0$$

let Y_θ, X_θ have data

$$Y_\theta(0) = 0, \quad Y_\theta'(0) = e_\theta^\perp/R(\theta), \qquad X_\theta(0) = R'(\theta)e_\theta + R(\theta)e_\theta^\perp, \quad X_\theta'(0) = 0.$$

Then

$$b(\theta) = \frac{e_{F(\theta)}^\perp \cdot Y_\theta(-T)}{R(F(\theta))}, \qquad F'(\theta) = \frac{e_{F(\theta)}^\perp \cdot X_\theta(-T)}{R(F(\theta))}. \tag{139}$$

In particular, if $q'' = -6q^5$, $q(0) = 1$, $q'(0) = 0$, and $j'' + 6q^4 j = 0$, $j(0) = 0$, $j'(0) = 1$, then $b_0 = j(T) > 0$.

Proof. The section energy equation is even in L, so its initial position has zero L derivative at $L = 0$, and its initial velocity derivative is $e_\theta^\perp/R(\theta)$. These are exactly the data for Y. The endpoint variation due to return-time variation is the endpoint velocity times that variation and vanishes at a brake endpoint. Taking its angular component proves the first formula. The same argument for the initial angle gives X and the second formula. Only the continuous Hessian and ordinary C^1 dependence of a flow are used, including at any origin crossing. At the base, q is even, j is odd, and $F_0(\theta) = \theta - \pi$; the first formula therefore reduces to $b_0 = -j(-T) = j(T)$. ∎

A.6 Uniform one-cycle estimates and finite Fourier constraints

For a fixed nearby member write

$$V_f(\rho e_\theta) = \rho^6(1 + f(\theta)), \qquad \delta = \|f\|_{C^2},$$
$$e(\theta) = \frac{f(\theta) + f(\theta + \pi)}{2}, \quad o(\theta) = \frac{f(\theta) - f(\theta + \pi)}{2}.$$

These are finite shape differences, rather than parameter derivatives. In particular none of the estimates below presupposes that o is an admissible tangent vector to a common-mean family.

Lemma 20 (Uniform action remainder on one radial cycle)**.** *For $d \in [-\pi, -4\pi/5]$ the physical inverse generator satisfies*

$$h_f(\theta, \theta + d) - h_0(d) = \Lambda_d f(\theta) + \mathcal{R}_f(\theta, d), \qquad \langle |\mathcal{R}_f(\cdot, d)| \rangle \le C\|f\|_{H^1}^2, \tag{140}$$

uniformly over this closed interval, where

$$\Lambda_d f(\theta) = -\int_{\text{backward base arc}} |Q|^6 f(\arg Q)\, dt. \tag{141}$$

The weighted integrand is set to zero at $Q = 0$.

Proof. Use the energy-one free-time action on H^1 paths whose endpoints lie on the rays of angles θ and $\theta + d$, with a negative oriented time τ near the backward base return time:

$$\mathcal{H}_f(q, \tau) = \int_0^1 \left(\frac{|\dot{q}|^2}{2\tau} - \tau |q|^6 (1 + f(\arg q)) + \tau \right) du.$$

Endpoint radii vary freely near their positive outer values. The natural endpoint conditions are zero radial momentum at both ends; the τ equation sets the energy equal to one. A smoothly varying rotation along the parameter u trivializes the endpoint-ray path spaces over the compact d interval, with uniformly equivalent H^1 norms.

The full base Hessian on this path space times $\mathbb{R}$ has zero kernel. In fact, an infinitesimal critical arc with fixed initial angle, initial radial momentum zero and energy variation zero is determined by its initial angular-momentum variation. The final apoapsis condition fixes its time variation by transversality. The remaining fixed final angle condition is $d'(L)$ times the angular-momentum variation. Lemma 17 forces that variation, and hence all the others, to vanish. This argument applies also at $L = 0$. The Hessian is self-adjoint Fredholm: up to an overall negative sign for $\tau < 0$, its leading derivative form becomes coercive after an L^2 correction and a period-coordinate correction, and the remaining terms are compact or finite rank. Its inverse is therefore bounded, uniformly on the compact (θ, d) family.

For $W_f(Z) = |Z|^6 f(\arg Z)$, the polar-frame formulas are

$$\nabla W_f = \rho^5 (6 f e_\theta + f' e_\theta^\perp), \qquad D^2 W_f = \rho^4 \begin{pmatrix} 30f & 5f' \\ 5f' & 6f + f'' \end{pmatrix}. \tag{142}$$

They extend continuously by zero at the origin. Thus the perturbing action has Hessian norm at most $C\delta$ on a fixed tube, whereas the base action, with polynomial potential $|Z|^6$, has a locally Lipschitz Hessian. If $\ell_f(q,\tau) = \tau \int W_f(q)$ and $\gamma_f = d\ell_f$ at a base arc, the uniformly invertible critical equation gives a unique displacement ψ with

$$\|\psi\|_{H^1\times\mathbb{R}} \le C\|\gamma_f\|, \qquad h_f = h_0 - \ell_f(\text{base}) + O(\|\gamma_f\|^2).$$

For example, subtract the invertible base Hessian in the critical equation; its remaining derivative is $O(\|\psi\| + \delta)$. Contraction on a fixed small ball proves the displacement estimate. The base cubic action remainder and the perturbation remainder $O(\delta\|\psi\|^2)$ then prove the displayed action estimate. These steps require no third derivative of f.

Write the base arc as $Q_{\theta,d}(u) = \rho_d(u)e_{\theta+\vartheta_d(u)}$ off its zeros. Cauchy–Schwarz and (142) bound $\|\gamma_f\|^2$ by a fixed constant times

$$B_f(\theta,d)^2 = \int_0^1 \left\{\rho_d^{10}\big(f(\theta+\vartheta_d)^2 + f'(\theta+\vartheta_d)^2\big) + \rho_d^{12} f(\theta+\vartheta_d)^2\right\} du.$$

At a zero all weighted integrands are zero. Arc times and radii are uniformly bounded, so $\sup B_f \le C\delta$ and Fubini's theorem gives $\langle B_f(\cdot,d)^2\rangle \le C\|f\|_{H^1}^2$ uniformly in d. This proves (140). Notice that the angular averaging uses only rotated *base* arcs; it never changes variables by a possibly singular perturbed angular trajectory. ∎

Lemma 21 (Finite radial and circular reductions)**.** *For a sufficiently small normalized common-mean shape f, set $a = \|o'\|_2$ and write*

$$o(\theta) = \sum_{n\ge1,\ n\ \text{odd}} (\alpha_n \cos n\theta + \beta_n \sin n\theta), \qquad w_n = \alpha_n^2 + \beta_n^2.$$

Then

$$\|e\|_{H^1} \le C\delta\|f\|_{H^1}, \qquad \|f\|_{H^1} \le Ca, \tag{143}$$

and, with the coefficient C_6 of (27)*,*

$$\sum_{n\ \text{odd}} (C_6 n^2 - 1/6) w_n = E_1, \tag{144}$$

$$\sum_{n\ \text{odd}} D_n w_n = E_2, \qquad D_n = \frac{9n^2-8}{12(8-n^2)}, \tag{145}$$

$$|E_1| + |E_2| \le C\delta a^2.$$

All sums converge absolutely.

Proof. The normal reductions of N_R, N_C exist for every sufficiently small C^2 shape perturbation. By uniqueness, for shapes belonging to the normalized family they are the continuations of Lemma 18; their reduced actions are exactly their base values. We give the finite estimates which turn those equalities into the displayed formulas.

Let L_m be the invertible base normal Hessian, $\ell_f(q, T_{\mathrm{per}}) = T_{\mathrm{per}} \int W_f(q)$, and $\gamma_f = d\ell_f|_{(T_m N)^\perp}$. The perturbation estimate (142), the locally Lipschitz base Hessian and the uniform normal inverse give, exactly as in (45)–(46),

$$\begin{gathered} \|\psi_f\| \le C\|\gamma_f\|, \qquad \psi_f = L_m^{-1}\gamma_f + r_f, \qquad \|r_f\| \le C\delta\|\gamma_f\|, \\ a_N(f,m) = J_0(m) - \ell_f(m) - \tfrac{1}{2}\gamma_f[L_m^{-1}\gamma_f] + R_N(f,m), \\ |R_N(f,m)| \le C\delta\|\gamma_f\|^2. \end{gathered} \tag{146}$$

These estimates apply to $p = 6$ because its base potential is smooth; the kernel hypotheses have been supplied in Lemma 18. More explicitly, writing the normal equation as $-\gamma_f + L_m z + H_m(z) = 0$ gives $\|DH_m(z)\| \le C(\|z\| + \delta)$. Contraction yields the first line. In the action expansion put $v = L_m^{-1}\gamma_f$ and $r_f = \psi_f - v$; the exact cancellation

$$\tfrac{1}{2}\langle L_m\psi_f, \psi_f\rangle - \gamma_f[\psi_f] = -\tfrac{1}{2}\gamma_f[v] + \tfrac{1}{2}\langle L_m r_f, r_f\rangle$$

leaves the cubic base remainder and $O(\delta\|\psi_f\|^2)$ perturbation remainder. This proves the second line with its multiplicative δ.

On a radial orbit in direction θ, put

$$B_f(\theta)^2 = f(\theta)^2 + f(\theta+\pi)^2 + f'(\theta)^2 + f'(\theta+\pi)^2.$$

Then $\|\gamma_f(m_\theta)\| \le CB_f(\theta)$, $\langle B_f^2\rangle = 2\|f\|_{H^1}^2$ and $\sup B_f \le C\delta$. On the circular family $\|\gamma_f\| \le C\|f\|_{H^1}$ by uniform angular traversal. Thus the angularly averaged radial remainder and the circular remainder are bounded by $C\delta\|f\|_{H^1}^2$.

On the radial base manifold $\ell_f(m_\theta) = T_R e(\theta)/4$, where $T_R = 2T$. The constant reduced action and (146) give $|e(\theta)| \le CB_f(\theta)^2 \le C\delta B_f(\theta)$. For its derivative use a rotation-equivariant chart $m_\theta + \psi_f(m_\theta) = R_\theta(m_0 + \xi(\theta))$. Differentiate the exact reduced action, rather than its Taylor remainder. The normal equation cancels the term containing ξ'; the remaining term is $-\ell_{f'(\theta+\cdot)}(m_0 + \xi(\theta))$. At $\xi = 0$ this is $-T_R e'(\theta)/4$. The function $|Z|^6 f'(\arg Z)$ is C^1, with C^1 norm at most $C\delta$ on the relevant ball, by $f \in C^2$. Hence $|e'(\theta)| \le C\delta\|\xi(\theta)\| \le C\delta B_f(\theta)$. Integration proves the first assertion in (143). Since o has mean zero, Poincaré's inequality gives $\|o\|_{H^1} \le \sqrt{2}\|o'\|_2$. The orthogonality of e, o in H^1 then absorbs the small even term and proves $\|f\|_{H^1} \le Ca$.

The angular average of the radial normal quadratic form, and the circular normal quadratic form, in (146) are bilinearly bounded on angular H^1, by the preceding forcing estimates, the uniform inverse and Cauchy–Schwarz. Replacing $f = e + o$ by o in either such form therefore costs at most $C\delta a^2$. The volume normalization expands to

$$\langle e\rangle = \frac{2}{3}\langle o^2\rangle + O(\delta a^2). \tag{147}$$

For arbitrary odd input o, its potential integral vanishes on the whole radial base manifold. Its first variation therefore annihilates the tangent kernel. The longitudinal and transverse

forced Jacobi solutions used to prove Proposition 3.1 are consequently solutions of the normal Hessian equation modulo that kernel, regardless of whether o arises from a family derivative. Their direct quadratic pairing, with $p = 6$, is

$$-\tfrac{1}{2}\gamma_o[L_m^{-1}\gamma_o] = \frac{T_R}{8}\left(\frac{7}{6}o(\theta)^2 + C_6 o'(\theta)^2\right).$$

Combining this with the radial linear term $T_R e/4$ yields

$$2\langle e\rangle = \frac{7}{6}\langle o^2\rangle + C_6\langle (o')^2\rangle + O(\delta a^2). \tag{148}$$

The circular forced solutions in (29), now with $p = 6$, similarly compute the direct normal quadratic form as

$$-\tfrac{1}{2}\gamma_o[L_C^{-1}\gamma_o] = \frac{T_C}{8}\sum_{n\ \text{odd}} K_{6,n} w_n, \qquad K_{6,n} = \frac{56 + n^2}{12(8 - n^2)}.$$

Its linear term is $T_C\langle e\rangle/4$, so $2\langle e\rangle = \sum K_{6,n} w_n + O(\delta a^2)$. Subtracting twice (147) from these two identities, and using $\langle o^2\rangle = \frac{1}{2}\sum w_n$ and $a^2 = \frac{1}{2}\sum n^2 w_n$, proves (144)–(145). The radial coefficients grow quadratically, the circular coefficients are bounded, and $o \in H^1$, proving absolute convergence. ∎

Lemma 22 (Finite constraints from the original common means)**.** *For every sufficiently small normalized common-mean shape as above, let $a = \|o'\|_2$ and $\widehat{f}_n = \langle f(\theta)e^{-in\theta}\rangle$. There are fixed constants $C, c > 0$, independent of its angular Fourier support, such that*

$$\|f\|_{H^1} \le Ca, \qquad |\langle f\rangle| \le Ca^2, \tag{149}$$

$$|\widehat{f}_n| \le Ca^2 \qquad (n \ge 5 \ \textit{odd}), \tag{150}$$

$$w_n = 4|\widehat{f}_n|^2 \qquad (n \ge 1 \ \textit{odd}),$$

$$\sum_{n\ge 5,\ n\ \text{odd}} n^2 w_n \ge ca^2. \tag{151}$$

In addition $\|e\|_{H^1} \le C\delta a$, and the exact brake conditions, fixed leg time, and boundary identity of Proposition A.2 hold for this same shape.

Proof. The norm assertion is (143). Taylor expansion of $\langle (1 + f)^{-1/3}\rangle = 1$ gives

$$\langle f\rangle = \frac{2}{3}\langle f^2\rangle + O(\delta\langle f^2\rangle) = O(a^2). \tag{152}$$

Fix odd $n \ge 5$, put $m = (n - 1)/2$ and $d_n = -2\pi m/n$, and consider the constant-increment competitor

$$S_n(\theta) = \frac{1}{n}\sum_{j=0}^{n-1} h_f(\theta + jd_n, \theta + (j + 1)d_n) - h_0(d_n).$$

It is admissible for the global minimum problem, and the cutoff is one at these increments. Equation (127), including the fifth-resonance endpoint, gives $S_n(\theta) \geq 0$ for every θ. If $T_r(d) > 0$ denotes the base forward time for one radial cycle, the open-arc virial identity and rotation averaging give

$$\langle \Lambda_d f \rangle = \frac{T_r(d)}{4} \langle f \rangle.$$

The uniform one-cycle remainder and (152) therefore show $0 \leq \langle S_n \rangle \leq Ca^2$, with a constant independent of n. Every Fourier coefficient of a nonnegative function has absolute value at most its mean. Cyclic averaging by jd_n leaves the nth Fourier coefficient unchanged, because $nd_n \in 2\pi\mathbb{Z}$. Hence

$$|\widehat{\Lambda_{d_n} f}_n| \leq |\widehat{S}_n| + |\widehat{\mathcal{R}_f(\cdot, d_n)}_n| \leq Ca^2. \tag{153}$$

The orbital bounds in Lemma 14 supply a uniform lower bound for the corresponding multiplier, rather than just its nonvanishing. At the unique nth resonance in Lemma 15, the signed outgoing integral of $\rho^6 \cos(n\theta)$ exceeds $1/200$. Reflection about apoapsis doubles this real contribution and cancels the sine contribution over a full radial cycle. Starting at a different apoapsis or using backward time only changes its phase, sign, or complex conjugate. Thus Fubini in (141) gives

$$\widehat{\Lambda_{d_n} f}_n = \kappa_n \widehat{f}_n, \qquad |\kappa_n| > 1/100.$$

These are unnormalized one-cycle action integrals, so no probability period factor is inserted. Together with (153) this proves (150). The identity for w_n follows from the real Fourier convention and the fact that f and o have the same odd coefficients.

It remains to force a definite amount of high derivative energy. For the circular coefficients, $D_1 = 1/84$ and $D_n \leq -3/4$ for odd $n \geq 3$. Thus (145) implies

$$\sum_{n \geq 3,\ n \text{ odd}} w_n \leq w_1/63 + C\delta a^2.$$

The radial equation then bounds $\sum n^2 w_n$ by a constant times $\sum w_n + C\delta a^2$. Since $2a^2 = \sum n^2 w_n$, absorption gives

$$a^2 \leq C w_1. \tag{154}$$

Now use the strict enclosure of Lemma 13: $32/243 < C_6 < 32/195 < 1/6$. Set

$$\lambda = \frac{12}{73}(9C_6 - 1/6) > 0, \qquad K = \frac{520}{511}(32/195 - C_6) > 0.$$

Add λ times (145) to (144). The mode-three coefficient is zero, the mode-one coefficient is $-K$, and for every odd $n \geq 5$ the remaining coefficient is at most $C_6 n^2$, since $D_n < 0$. Consequently

$$K w_1 \leq C_6 \sum_{n \geq 5,\ n \text{ odd}} n^2 w_n + C\delta a^2.$$

Use (154) and shrink the fixed C^2 neighborhood once more to absorb the last term. This proves (151). All normal-form errors were bounded by $C\delta a^2$ before this absorption, so the resulting constants are independent of frequency. The last brake assertions were proved for all nearby normalized parameters in Proposition A.2. ∎

Corollary A.1 (A finite cubic loss)**.** *The shapes in Lemma 22 also satisfy $a \le C\delta^3$.*

Proof. By (150), $w_n \le Ca^4$ for odd $n \ge 5$. Parseval and the C^2 bound give $\sum n^4 w_n \le 2\delta^2$. For an integer $M \ge 5$ split (151) at M:

$$ca^2 \le Ca^4 M^3 + 2\delta^2 M^{-2}.$$

If $a = 0$ there is nothing to prove. Otherwise $a \le C\delta$, and for small δ one can choose an integer M comparable to $(\delta/a^2)^{2/5}$, which is at least five. The right side is then at most $Ca^{8/5}\delta^{6/5}$. Division and raising to the power $5/2$ give the assertion. ∎

The finite constraints just obtained follow from the original all-parameter common-mean hypothesis through actual mechanical actions. Their derivation neither differentiates an invariant probability nor requires an inverse bound for a family of orbits with growing periods. They will be combined below with the mechanical boundary identity (131).

B The scalar response at a radial sextic well

This section fixes the normalization of the radial response and proves the scalar estimates used below. The Fourier variable is real throughout. In particular, the division by the half-leg mass in this section does not assert the existence of a bounded inverse mass operator on arbitrary data.

B.1 Radial coordinates and the half-leg mass

The reference potential is $|x|^6$, at energy one. Let

$$q'' = -6q^5, \qquad q(0) = 1, \qquad q'(0) = 0, \qquad Q = \int_0^1 \frac{\mathrm{d}r}{\sqrt{2(1-r^6)}}, \qquad T = 2Q. \tag{155}$$

Thus $q(Q) = 0$, $q(T) = -1$, and

$$\tfrac{1}{2}q'^2 + q^6 = 1, \qquad q'(Q) = -\sqrt{2}, \qquad q(T-t) = -q(t). \tag{156}$$

Write $L = \partial_t^2 + 6q^4$, and define

$$Lj = 0, \quad j(0) = 0, \quad j'(0) = 1, \qquad Ly = -q^5, \quad y(0) = y'(0) = 0 \quad (0 \le t \le Q). \tag{157}$$

The constant and the endpoint Green factor are

$$b = j(T) = \sqrt{2}j'(Q), \qquad K(t) = bq(t) + j(t) = j(T-t). \tag{158}$$

Indeed $qj' - q'j = 1$; hence $j(Q) = 1/\sqrt{2}$ and $j'(T) = -1$. Reflection in Q and uniqueness give the last identity in (158), and its derivative at Q gives the middle one. The rational enclosure proved in Lemma 24 below is

$$1.05618 < b < 1.05619, \qquad b > \frac{21}{20}, \qquad \sqrt{b} < \frac{11}{10}. \tag{159}$$

In particular b is positive. Positivity of j before Q follows from $(j/q)' = q^{-2}$. Since $j'' = -6q^4 j$ and $j'(Q) > 0$, we also have $0 < j' \le 1$ and $0 < j(t) \le t$ on $(0, Q]$.

On the positive half-leg introduce

$$u = \frac{j}{q}, \qquad z = -\frac{y}{q}, \qquad I(t) = \int_0^t q(s)^6 \, \mathrm{d}s = \frac{2t - q(t)q'(t)}{8}, \qquad \omega(u) = q(t(u))^8. \tag{160}$$

Both u and z increase from zero to infinity. The exact identities are

$$\frac{\mathrm{d}u}{\mathrm{d}t} = q^{-2}, \qquad \frac{\mathrm{d}z}{\mathrm{d}t} = Iq^{-2}, \qquad z_u = I, \qquad z_{uu} = \omega, \qquad I(Q) = \frac{Q}{4}. \tag{161}$$

The identity for z follows by integrating $(qy' - q'y)' = -q^6$. Near zero,

$$q = 1 - 3t^2 + \tfrac{15}{2}t^4 + O(t^6), \quad j = t - t^3 + O(t^5), \quad y = -\tfrac{1}{2}t^2 + \tfrac{3}{2}t^4 + O(t^6), \quad z = \tfrac{1}{2}t^2 + O(t^6). \tag{162}$$

At Q, q has a simple zero and $j(Q) > 0$, so $u \asymp q^{-1}$, $z \asymp q^{-1}$, and $\omega = O(u^{-8})$.

For $k = 0, 1, 2, 3$ define the raw half-leg moments by any of the equivalent absolutely convergent integrals

$$\begin{aligned} a_k(\lambda) &= \int_0^\infty u^k \omega(u) e^{-i\lambda z(u)} \, \mathrm{d}u \\ &= \int_0^Q j(t)^k q(t)^{6-k} e^{-i\lambda z(t)} \, \mathrm{d}t = \int_0^\infty m_k(x) e^{-i\lambda x} \, \mathrm{d}x, \qquad m_k(z(u)) = \frac{u^k \omega(u)}{I(u)}. \end{aligned} \tag{163}$$

Here and below $I(u)$ denotes $I(t(u))$. In particular $m_0(z(t)) = q(t)^8 / I(t)$. All absolute moments in u of degree at most six exist.

Lemma 23 (Nonvanishing half-leg mass)**.** *For every $\lambda > 0$, $\operatorname{Im} a_0(\lambda) < 0$. Also $a_0(0) > 0$ and $a_0(-\lambda) = \overline{a_0(\lambda)}$. Consequently a_0 has no real zeros, and there are constants $0 < c < C < \infty$ such that*

$$c(1 + |\lambda|)^{-1/2} \le |a_0(\lambda)| \le C(1 + |\lambda|)^{-1/2}. \tag{164}$$

Moreover $a_0 \in C^6(\mathbb{R})$ and the ratios

$$n_k(\lambda) = \frac{a_k(\lambda)}{a_0(\lambda)}, \qquad 0 \le k \le 3, \tag{165}$$

are bounded on the real axis.

Proof. The density m_0 is positive and strictly decreasing: q decreases strictly and I increases strictly. It is integrable, is comparable to $x^{-1/2}$ at zero, and is $O(x^{-8})$ at infinity. For $\lambda > 0$, absolute integrability permits grouping successive sine half-periods:

$$\int_0^\infty m_0(x)\sin(\lambda x)\,\mathrm{d}x = \frac{1}{\lambda}\sum_{\ell=0}^{\infty}\int_0^\pi \sin v\left[m_0\left(\frac{v+2\ell\pi}{\lambda}\right) - m_0\left(\frac{v+(2\ell+1)\pi}{\lambda}\right)\right]\mathrm{d}v > 0. \tag{166}$$

Every summand is strictly positive. This proves the sign and nonvanishing statements.

At the rest endpoint the substitution $v = \sqrt{2z(t)}$ turns the mass integral into a quadratic-phase integral with smooth amplitude equal to one at zero. Subtract a smooth cutoff equal to one there. For the remaining amplitude, division by v gives an integrable derivative; at the other endpoint the transformed amplitude and its derivatives have power decay. One integration by parts, and the Gaussian integral with a positive damping factor followed by its limit, give

$$a_0(\lambda) = e^{-i\pi/4}\sqrt{\frac{\pi}{2}}\lambda^{-1/2} + O(\lambda^{-1}) \qquad (\lambda \longrightarrow +\infty). \tag{167}$$

A fully quantitative version is given in (192) below. Nonvanishing and continuity on the remaining compact interval prove (164). Differentiating the time integral up to six times is justified by $q^6|y/q|^r \in L^1(0,Q)$, $0 \le r \le 6$.

For $k = 1,2,3$, the function $F_k = u^k\omega/I$ has finite total variation. This follows at zero from (162) and at infinity from the analytic simple-zero expansions of q, j, I; on intermediate compact intervals it is smooth. Its value at infinity is zero and its value at zero is finite. Integration by parts using $z_u = I$ therefore gives $a_k(\lambda) = O(|\lambda|^{-1})$. Division by (164), and continuity on compact intervals, prove (165). ∎

B.2 The centered profile and its exact moment identities

There is no choice of an argument branch in the phase normalization

$$\zeta_\lambda = \frac{\overline{a_0(\lambda)}}{|a_0(\lambda)|}, \qquad g_\lambda(u) = \zeta_\lambda\omega(u)e^{-i\lambda z(u)}. \tag{168}$$

Write

$$A = \int_0^\infty g_\lambda\,\mathrm{d}u = |a_0(\lambda)| > 0, \quad C = \int_0^\infty u g_\lambda\,\mathrm{d}u, \quad D = \int_0^\infty u^2 g_\lambda\,\mathrm{d}u, \quad E = \int_0^\infty u^3 g_\lambda\,\mathrm{d}u. \tag{169}$$

Define on $[0,T]$ the continuous, phase-centered profile

$$B_\lambda(t) = \zeta_\lambda q(t)^4 e^{-i\lambda z(t)} \quad (0 \le t < Q), \qquad B_\lambda(Q) = 0, \qquad B_\lambda(T-t) = -\overline{B_\lambda(t)}. \tag{170}$$

Its quotient by q extends continuously by zero at Q, and

$$g_\lambda(u)\,\mathrm{d}u = q(t)^2 B_\lambda(t)\,\mathrm{d}t \quad (0 < t < Q), \qquad \int_0^T q^2 B_\lambda\,\mathrm{d}t = 0. \tag{171}$$

Let P, V be complex-valued and W real-valued solutions with zero initial position and velocity of

$$LP = -qB_\lambda, \qquad LV = jB_\lambda, \qquad LW = -\operatorname{Re}(\overline{B_\lambda}V) - \frac{j}{q}\operatorname{Re}(\overline{B_\lambda}P). \tag{172}$$

The last source is continuous at Q. The quotient-normalized scalar response is exactly

$$\mathcal{Q}(\lambda) = \frac{1}{\sqrt{b}}\left\{\frac{W(T)}{2b} - \frac{|P(T)|^2}{8} + \frac{3|V(T)|^2}{8b^2} - \frac{\operatorname{Re}(P(T)\overline{V(T)})}{4b}\right\}. \tag{173}$$

These choices fix both its sign and its multiplicative constant. At $\lambda = 0$ one has $B_0 = \operatorname{sgn}(q)|q|^4$. Complex conjugation shows that $\mathcal{Q}$ is even.

Proposition B.1 (Half-leg and Volterra identities)**.** *Put*

$$\begin{aligned} J_3 &= \int_0^\infty\int_0^\infty |u-v|^3 \operatorname{Re}(g_\lambda(u)\overline{g_\lambda(v)})\,\mathrm{d}u\,\mathrm{d}v, \\ H(u) &= \int_u^\infty (v-u)g_\lambda(v)\,\mathrm{d}v, \qquad \mathscr{E} = \int_0^\infty |H(u)|^2\,\mathrm{d}u. \end{aligned} \tag{174}$$

All these integrals converge absolutely. The endpoint identities are

$$P(T) = -bA - 2\operatorname{Re}C, \qquad V(T) = 2i\operatorname{Im}(bC + D), \tag{175}$$

$$\begin{aligned} W(T) = {} & A\operatorname{Re}(b^2C + 3bD + E) + b\{2\operatorname{Re}(C^2) - |C|^2\} \\ & + 3\operatorname{Re}(CD) + \tfrac{1}{2}J_3. \end{aligned} \tag{176}$$

Consequently the two equivalent scalar identities are

$$\sqrt{b}\,\mathcal{Q} = -\frac{b^2A^2}{8} + \frac{3}{2}A\operatorname{Re}D + \frac{A\operatorname{Re}E}{2b} + \frac{3\operatorname{Re}(C\overline{D})}{2b} + \frac{3(\operatorname{Im}D)^2}{2b^2} + \frac{J_3}{4b}, \tag{177}$$

$$\sqrt{b}\,\mathcal{Q} = -\frac{b^2A^2}{8} + \frac{3}{2}A\operatorname{Re}D + \frac{A\operatorname{Re}E}{b} + \frac{3(\operatorname{Im}D)^2}{2b^2} + \frac{3}{b}\mathscr{E}. \tag{178}$$

Proof. The decay $|g_\lambda(u)| = O(u^{-8})$ gives finite third absolute moments, $H(u) = O(u^{-6})$, and the required Fubini bounds. Variation of constants, with $qj' - q'j = 1$, gives

$$P(t) = -j(t)\int_0^t q^2B_\lambda\,\mathrm{d}s + q(t)\int_0^t qjB_\lambda\,\mathrm{d}s, \qquad V(t) = j(t)\int_0^t qjB_\lambda\,\mathrm{d}s - q(t)\int_0^t j^2B_\lambda\,\mathrm{d}s.$$

Reflection and (171) yield (175). In particular $P(T)$ is real, $V(T)$ is purely imaginary, and their cross term in (173) is zero.

For the second response use $u = j/q$ on each open half of $[0,T]$ and $\mathrm{d}\mu = q^2B_\lambda\,\mathrm{d}t$. The preceding formulas give

$$V(t) + u(t)P(t) = -q(t)\int_0^t (u(t) - u(s))^2\,\mathrm{d}\mu(s).$$

Both minus signs in the equation for W are thus accounted for in

$$W(T) = \operatorname{Re} \int_0^T (b + u(t)) \overline{\mathrm{d}\mu(t)} \int_0^t (u(t) - u(s))^2 \, \mathrm{d}\mu(s). \tag{179}$$

This is an ordinary absolutely convergent double integral. Reflection obeys $u(T-t) = -b - u(t)$ and, when each half is given positive time measure, $\mathrm{d}\mu(T-t) = -\overline{\mathrm{d}\mu(t)}$. After splitting (179) into two same-half triangles and a cross-half rectangle, the same-half coefficients on $u > v > 0$ add to $(b+u-v)(u-v)^2$. Their integral is

$$bA \operatorname{Re} D - b|C|^2 + \tfrac{1}{2} J_3.$$

The symmetrized cross-half coefficient is $\frac{1}{2}(u+v)(b+u+v)^2$, multiplying $\operatorname{Re}(g_\lambda(u) g_\lambda(v))$. Its integral is

$$\operatorname{Re}(b^2 AC + 2bAD + 2bC^2 + AE + 3CD).$$

Adding proves (176).

Substitution in (173) first cancels $bA \operatorname{Re} C/2$. The terms involving C alone reduce to $-3(\operatorname{Im} C)^2/2$ from the first two summands and $+3(\operatorname{Im} C)^2/2$ from $|V(T)|^2$, and cancel as well. The remaining mixed term is $3 \operatorname{Re}(C\overline{D})/(2b)$. This proves (177).

Finally, for $v, w \geq 0$ set $G(v,w) = \int_0^{\min(v,w)} (v-u)(w-u) \, \mathrm{d}u$. Direct polynomial expansion on $v \geq w$, and then symmetry, gives

$$|v-w|^3 = v^3 + w^3 - 3v^2 w - 3vw^2 + 12 G(v,w).$$

Multiply by $\operatorname{Re}(g_\lambda(v)\overline{g_\lambda(w)})$ and integrate. Fubini gives

$$J_3 = 2A \operatorname{Re} E - 6 \operatorname{Re}(C\overline{D}) + 12\mathscr{E}. \tag{180}$$

Its substitution in (177) proves (178) without boundary integrations at infinity. ∎

B.3 A rational certificate for the complete finite phase interval

The finite computation uses rational interval arithmetic and explicit analytic remainder estimates. The enclosure rules and coverage argument below prove a bound on the complete phase interval. The corresponding variation bounds are established in Lemma 25.

Lemma 24 (Continuous finite-phase certificate)**.** *The radial data satisfy* (159)*, and*

$$\sqrt{b}\, \mathcal{Q}(\lambda) < -\frac{1}{1000}, \qquad \mathcal{Q}(\lambda) < -\frac{1}{1100} \qquad (|\lambda| \leq 128). \tag{181}$$

Proof. All arithmetic enclosures have endpoints in $2^{-120}\mathbb{Z}$. Addition, multiplication, and division are rounded outwards; division requires a strictly positive denominator interval. Products take the minimum and maximum of the four endpoint products. Squaring an interval containing zero uses lower endpoint zero. Integer square roots provide outward square-root enclosures. Rational comparisons, rather than printed decimals, determine every assertion.

Validated radial integration. The system is exactly (155)–(157), with state (q, q', j, j', y, y'). The step-center enclosures satisfy $|q| < 11/10$ and that the other five components have modulus less than two. In a complex polydisc of radius $1/10$ for q and radius one for the other components, $|q| \le 6/5$ and the other moduli are at most three. The position right sides are bounded by three, the acceleration right sides by sixty, and the weighted row-sum Lipschitz bound is less than fifty. For example, the following rational bounds hold, with $x = 6/5$ and $M = 3$,

$$6x^4M + x^4M + x^3M^2 < 60, \qquad \frac{24x^3M + 4x^3M + 3x^2M^2}{10} + 6x^4 + x^4 + 2x^3M < 50.$$

They dominate the needed rows of this subsystem. On the complex time disk of radius $1/100$, the integral map preserves the polydisc and is a contraction. Every component is therefore analytic and bounded by three on that disk.

Position Taylor coefficients are generated recursively through degree 20. If $[f]_r$ is the coefficient of order r, the recurrence is

$$[q]_{r+2} = \frac{-6[q^5]_r}{(r+1)(r+2)}, \quad [j]_{r+2} = \frac{-6[q^4 j]_r}{(r+1)(r+2)}, \quad [y]_{r+2} = \frac{-6[q^4 y]_r - [q^5]_r}{(r+1)(r+2)}.$$

Products are coefficient convolutions using outward intervals. The velocity polynomials have degree 19. With step $h = 1/2000$ the Cauchy tail, for positions and velocities, is bounded by

$$3\frac{(1/20)^{20}}{1 - 1/20} < 10^{-24}. \tag{182}$$

The same remainder is valid on the whole time cell. Interval Horner evaluation on $[0, h]$ encloses its complete trajectory, retaining all accumulated initial and rounding errors. Evaluation on $\{h\}$ gives the next initial enclosure.

The interval recurrence gives

$$q(0.8586576711) > 0, \qquad q(0.8586576712) < 0.$$

Energy conservation and monotonicity identify the enclosed zero as Q. Evaluation of the last validated Taylor polynomial on this entire root interval and $b = \sqrt{2}j'(Q)$ give $1.05618 < b < 1.05619$. The remaining two assertions in (159) follow by rational comparisons and outward square root.

Improper crossing tails. The moment integrations stop at the exact time $t_c = 4/5$. Let r be the validated upper bound for $q(t_c)$; its enclosure satisfies $r < 1/10$. On $[t_c, Q)$, $|q'| \ge 1$, $0 < q \le r$, and $0 \le j \le t < Q < 1$. For the raw kth moment the omitted absolute integral is at most

$$\varepsilon_k = \frac{r^{7-k}}{7-k}, \qquad 0 \le k \le 3. \tag{183}$$

Both real and imaginary enclosures are widened by this amount. For $u, v \geq 0$, $|u - v|^3 \leq u^3 + v^3$. The total absolute mass is $Q/4 < 1/4$, and the total third absolute moment is $\int_0^Q q^3 j^3 \,\mathrm{d}t < 1$. The union of the two omitted regions in J_3 is consequently bounded by

$$\varepsilon_J = \frac{\varepsilon_3}{2} + 2\varepsilon_0. \tag{184}$$

These bounds are uniform in phase, even where z diverges.

Whole-cell, whole-phase integration. For each of the 256 closed phase intervals

$$\Lambda_\ell = [\ell/2, (\ell+1)/2], \qquad 0 \leq \ell \leq 255,$$

use the entire phase interval and the entire radial time-cell enclosure to enclose $\phi = \lambda y/q = -\lambda z$. A rational enclosure for π follows from $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, using alternating sums through orders 40 and 12 respectively. Subtracting an integer multiple of 2π gives a reduced interval whose exact rational midpoint has modulus less than four. At that midpoint use the sine polynomial of degree 41 and cosine polynomial of degree 40. The errors obey

$$\frac{4^{43}}{43!} < 10^{-24}, \qquad \frac{4^{42}}{42!} < 10^{-24}.$$

Widen each trigonometric interval by the radius of the reduced phase interval, using the global Lipschitz constant one, and intersect with $[-1, 1]$. The result encloses $e^{i\phi}$ for every time and every phase in the product of the two cells.

Put $d_k = q^{6-k} j^k e^{i\phi}$ and $M_k(t) = \int_0^t d_k(s)\,\mathrm{d}s$. On a cell with left endpoint t_i, its whole-cell enclosure is $M_k(t_i) + [0, h] d_k([t_i, t_i + h])$. The full moment increment is enclosed by $h\, d_k([t_i, t_i + h])$. To compute J_3 simultaneously use the exact chronological derivative

$$\frac{\mathrm{d}J_3(t)}{\mathrm{d}t} = 2\,\mathrm{Re} \sum_{k=0}^{3} (-1)^k \binom{3}{k} d_{3-k}(t) \overline{M_k(t)}. \tag{185}$$

Here $J_3(t)$ is the double integral restricted to $[0, t]^2$ in physical time, before centering; multiplication by the common unit phase does not change it. Integrating its whole-cell interval range, with the preceding enclosures for $M_k(t)$, encloses the increment. Thus quadrature error is included directly; no sampled-phase or sampled-time error estimate is being assumed. There are 1600 time cells before t_c.

After adding (183) and (184), compute $A = |M_0(Q)|$, $\zeta = \overline{M_0(Q)}/A$ and $(C, D, E) = \zeta(M_1(Q), M_2(Q), M_3(Q))$. Every phase box verifies a positive lower bound for A before division. Substitute these enclosures in the exact formula (177). The largest upper endpoint among all 256 boxes is the rational number

$$-\frac{1822618750900789488191762325916601}{13292279957849158729038070602803445 76} < -\frac{1}{1000}. \tag{186}$$

The boxes share endpoints and cover all of $[0, 128]$; hence the assertion holds continuously on that interval. Evenness covers negative phases, and $\sqrt{b} < 11/10$ gives the second assertion of (181). ∎

B.4 Explicit estimates beyond the finite interval

Derivatives in this subsection, unless supplied with a subscript t, are in the variable u. Put $s = \sqrt{2(1-q^6)} = -q'$ in physical time. Then

$$q_u = -sq^2, \qquad \omega' = -8sq^9, \qquad \omega'' = 48q^{10}(3-4q^6). \tag{187}$$

The Wronskian gives

$$h_0 = usq = sj = 1 - qj', \qquad 0 \le h_0 \le 1. \tag{188}$$

This function is increasing, since $h_0' = sq + 2uq^2(4q^6-1)$. If $q^6 \ge 1/4$ this is positive; otherwise $u \le 1/(sq)$ gives $h_0' \ge 6q^7/s > 0$. These inequalities extend to zero by continuity.

The logarithmic slope $e = u\omega'/\omega = -8h_0$ decreases. For $F_1 = u\omega/I$,

$$u(\log F_1)' = 1 + e - F_1, \qquad F_1 = \frac{1}{I}\int_0^u \omega(v)(1+e(v))\,\mathrm{d}v \ge 1 + e(u).$$

Thus F_1 decreases from one to zero, and its total variation is one. Also ω/I is decreasing and at most $1/u$. We will use the stronger estimate

$$I \ge uq^4 = u\sqrt{\omega}, \qquad \frac{\omega}{I^2} \le \frac{1}{u^2}. \tag{189}$$

Indeed $(I - uq^4)' = q^4(q^4 - 1 + 4h_0) \ge 0$, because $h_0 \ge 1 - q$ and $1 - q^4 \le 4(1-q)$. The difference is zero at the origin.

Lemma 25 (Quantitative radial variations)**.** *Let $v = \sqrt{2z(u)}$ and $m_0(z(u)) = \omega/I$. Then*

$$\mathrm{TV}_{v>0}\left(m_0(v^2/2) - \frac{1}{v}\right) < \frac{11}{2}, \quad \mathrm{TV}_{u>0}\,\frac{u^2\omega}{I} < \frac{6}{25}, \quad \mathrm{TV}_{u>0}\,\frac{u^3\omega}{I} < \frac{8}{125}. \tag{190}$$

All three functions have zero limits at both endpoints.

Proof. Total variation is invariant under the increasing changes between t, u, and v. We describe a rational certificate in the bounded t coordinate, including its endpoint estimates. Denote the first function by r and the last two by F_2, F_3.

The analytic initial interval. On the complex disk $|t| \le 1/10$, the second-order integral equations preserve $|q-1| < 1/10$, $|j| < 1/5$, $|y| < 1/10$ and are contractions: a row-sum Lipschitz bound less than 50 is multiplied by $(1/10)^2/2$. For clarity, the rational image and improvement estimates, with $R = 1/10$ and $X = 11/10$, are

$$3R^2X^5 < \tfrac{1}{20}, \quad R + 3R^2X^4/5 < \tfrac{1}{5}, \quad \tfrac{1}{2}R^2(6X^4/10 + X^5) < \tfrac{1}{10}, \quad \frac{R^2X^5}{2(1-3R^2X^4)} < \tfrac{1}{100}.$$

In particular

$$|q(t)-1| \le \tfrac{1}{20}(|t|/R)^2, \qquad |y(t)| \le \tfrac{1}{100}(|t|/R)^2.$$

The analytic even functions $\overline{I} = I/t$ and $Y = -2y/t^2$ satisfy

$$|\overline{I}-1| < \tfrac{1}{5}, \qquad |Y-1| < \tfrac{1}{10}, \qquad |q-1| < \tfrac{1}{20}.$$

For the first two inequalities one may use respectively $(21/20)^5/10 < 1/5$ and $(1/4 + 3/50)(21/20)^4/6 < 1/10$ in their integral formulas. Therefore

$$S(\xi) = \frac{q^8}{\overline{I}} - \sqrt{q/Y}, \qquad \xi = t^2,$$

is analytic for $|\xi| \le 1/100$, with $S(0) = 0$ and $|S| < 4$. The square-root branch has value one at zero; the displayed disks exclude zero. If $\mathcal{R}(\xi) = S(\xi)/\xi = \sum_{n\ge 0} R_n \xi^n$, Cauchy's bound gives $|R_n| \le 400\,100^n$. Now $r(t) = t\mathcal{R}(t^2)$. The radial coefficient recurrences in the preceding proof, followed by formal reciprocal and square-root recurrences, give $R_0 = -18$ and the exact rational coefficients through R_{12}. For a reciprocal of $A(\xi) = \sum A_n \xi^n$ use $B_0 = A_0^{-1}$ and $B_n = -A_0^{-1} \sum_{k=1}^{n} A_k B_{n-k}$; for a square root with $A_0 = 1$ use $B_0 = 1$ and $B_n = (A_n - \sum_{k=1}^{n-1} B_k B_{n-k})/2$. On $0 \le t \le 1/20$, the omitted derivative series is at most

$$400 \sum_{n\ge 13} (1+2n)4^{-n} < \tfrac{1}{100}.$$

Outward Horner evaluation of $\sum_{n=0}^{12}(1+2n)R_n t^{2n}$, with this enlargement, is contained in $[-18001/1000, -16724/1000]$. Hence r decreases on this whole interval. For $k = 2, 3$, the elementary bounds $q \ge 99/100$, $j' \ge 99/100$, $j \le t$, $I \ge (9/10)t$, and $q'/q \ge -7t$ give

$$\frac{(F_k)'_t}{F_k} \ge \frac{99k/100 - 10/9}{t} - 7(8-k)t > 0 \qquad (0 < t \le 1/20).$$

These bounds also follow directly from the integral equations and $0 < q \le 1$, $0 < j \le t$ on this short interval.

The middle interval. Use the whole-cell radial enclosures of the preceding proof, of width $1/2000$, on $[1/20, 4/5]$. The exact formulas to enclose are

$$r = q^8/I - 1/v, \qquad r'_t = 8q^7 q'/I - q^{14}/I^2 + I/(q^2 v^3),$$
$$F_k = j^k q^{8-k}/I, \qquad (F_k)'_t = kj^{k-1}j' q^{8-k}/I + (8-k)j^k q^{7-k} q'/I - j^k q^{14-k}/I^2.$$

All denominators have positive lower bounds. The intervals in which a derivative sign is not needed are

$$[51/200, 263/1000] \quad \text{for } r, \qquad [189/1000, 97/500] \quad \text{for } F_2,$$
$$[293/1000, 299/1000] \quad \text{for } F_3.$$

Before and after its stated window, the sign of r_t' is respectively negative and positive, and the sign of $(F_k)_t'$ is respectively positive and negative. These assertions are checked on every entire cell. Within each window sum the cell length times an upper enclosure for the absolute derivative. No number or uniqueness of critical points inside a window is asserted or required.

The terminal interval. At $t = 4/5$ the radial certificate verifies $q < 1/10$, $I > 1/5$, and $j > 1/2$. These inequalities persist to Q; also $-3/2 < q' \le -1$, $0 < j' \le 1$, and $|y| < 3$. The negative logarithmic q term in $(F_k)_t'$ dominates the positive j term, since $2k - 10(8-k) < 0$ for $k = 2, 3$. For r_t' the two negative terms have total magnitude at most $60q^7 + 25q^{14}$. Since $v^2 = -2y/q \le 6/q$, the positive term is at least $1/(75\sqrt{q}) \ge 1/75$, whereas $60/10^7 + 25/10^{14} < 1/75$. Thus $r_t' > 0$. The endpoint expansions give zero limits for all three functions at Q, as well as at zero.

For completeness, combining the monotone pieces with the absolute variations inside the windows gives the following exact upper bounds:

$$\begin{aligned}
\mathrm{TV}(r) &\le \frac{706586486994460008928430489279125025 9}{132922799578491587290380706028034457 6} < \frac{11}{2},\\
\mathrm{TV}(F_2) &\le \frac{153486672487317054631906581269494847}{664613997892457936451903530140172288} < \frac{6}{25},\\
\mathrm{TV}(F_3) &\le \frac{841792707781437918431642120049432 83}{132922799578491587290380706028034457 6} < \frac{8}{125}.
\end{aligned}$$

For r this bound is the sum of the negatives of lower enclosures of its two window-end values and the enclosed window variation. For F_k it is the sum of upper enclosures of their two window-end values and the window variation. This specifies the complete variation enclosure and proves (190). ∎

Lemma 26 (Large-phase moments and energy)**.** *For $\lambda \ge 128$ the centered moments and energy satisfy*

$$\frac{3}{4\sqrt{\lambda}} \le A \le \frac{2\sqrt{2}}{\sqrt{\lambda}}, \qquad |C| \le \frac{2}{\lambda}, \qquad |D| \le \frac{6}{25\lambda}, \qquad |E| \le \frac{8}{125\lambda}, \qquad \mathscr{E} < 20\lambda^{-5/2}. \tag{191}$$

All estimates except the lower bound for A hold for every $\lambda > 0$.

Proof. Change variables to $v = \sqrt{2z}$ and write $r(v) = m_0(v^2/2) - 1/v$. Subtracting the Fresnel integral and integrating by parts gives

$$\left| a_0(\lambda) - e^{-i\pi/4}\sqrt{\frac{\pi}{2}}\lambda^{-1/2} \right| \le \frac{\mathrm{TV}(r)}{\lambda} < \frac{11}{2\lambda}. \tag{192}$$

Indeed the difference amplitude is $vr(v)$; division by the derivative of the quadratic phase leaves $r(v)$, whose endpoint values vanish. One first integrates on compact subintervals

and then takes the improper limits. Since $\pi > 25/8$ and $\sqrt{\lambda} \geq \sqrt{128} > 11$, this proves $A \geq 3/(4\sqrt{\lambda})$.

For the upper bound split the u integral at $h > 0$. The initial piece has modulus at most h, since $\omega \leq 1$. On the rest, one integration by parts with decreasing amplitude $\omega/I \leq 1/u$ gives at most $2/(\lambda h)$. The choice $h = \sqrt{2/\lambda}$ gives $A \leq 2\sqrt{2}/\sqrt{\lambda}$. Integration by parts using F_1, its endpoint value one, and its variation one gives $|C| \leq 2/\lambda$. The vanishing endpoint values and variations of F_2, F_3 give the bounds for D, E in the same way. Centering changes no absolute values.

To estimate the energy, set

$$a_*(u) = \frac{\omega}{I^2}, \qquad p_*(u) = -\frac{(\omega/I)'}{I} = -\frac{\omega'}{I^2} + \frac{\omega^2}{I^3}.$$

Both are positive decreasing functions. This is immediate for a_*. For p_*,

$$I^4 p_*' = -I^2\omega'' + 4I\omega\omega' - 3\omega^3. \tag{193}$$

If $\omega'' \geq 0$ all terms are nonpositive. Otherwise $q^6 > 3/4$ and $q > 19/20$. From $I \leq u \leq t/q^2$ and $t^2 \leq (1-q)/(3q^5)$, which follow respectively from $\omega \leq 1$, $j \leq t$, and $1 - q(t) = 6\int_0^t (t-s)q(s)^5\,\mathrm{d}s \geq 3t^2 q(t)^5$, one obtains

$$I^2|\omega''| \leq 16q(1-q)(4q^6 - 3) \leq \tfrac{4}{5}, \qquad 3\omega^3 \geq 3(19/20)^{24} > \tfrac{4}{5}.$$

Equation (193) is strictly negative in this case as well. Also $p_* \leq -a_*'$, so $\int_u^\infty p_*\,\mathrm{d}v \leq a_*(u)$.

Two integrations by parts in $H(u)$, for $u > 0$, produce the second amplitude $G_u(v) = a_*(v) - (v-u)p_*(v)$. Its initial value is $a_*(u)$, its final value is zero, and

$$\mathrm{TV}_{v\geq u} G_u \leq a_*(u) + \int_u^\infty p_*\,\mathrm{d}v + \int_u^\infty (v-u)(-p_*')\,\mathrm{d}v \leq 3a_*(u).$$

Together with (189) this proves

$$|H(u)| \leq \frac{4a_*(u)}{\lambda^2} \leq \frac{4}{\lambda^2 u^2}. \tag{194}$$

For small u use instead the exact identity $H(u) = C - uA + \int_0^u (u-v)g_\lambda(v)\,\mathrm{d}v$ to obtain

$$|H(u)| \leq \frac{2}{\lambda} + \frac{2\sqrt{2}}{\sqrt{\lambda}}u + \frac{u^2}{2}. \tag{195}$$

Integrate the square of (195) up to $\lambda^{-1/2}$ and that of (194) thereafter. The result is

$$\mathscr{E} \leq \lambda^{-5/2}\left(4 + 4\sqrt{2} + \frac{10}{3} + \frac{\sqrt{2}}{2} + \frac{1}{20} + \frac{16}{3}\right) < 20\lambda^{-5/2}.$$

The last strict comparison follows already from $\sqrt{2} < 3/2$. ∎

Theorem 15 (Uniform negative scalar response)**.** *For every real λ,*

$$\mathcal{Q}(\lambda) \le -\frac{1}{1100(1+|\lambda|)}. \tag{196}$$

More precisely, $\mathcal{Q}(\lambda) < -1/1100$ for $|\lambda| \le 128$ and $\mathcal{Q}(\lambda) < -1/(200|\lambda|)$ for $|\lambda| \ge 128$. There are therefore constants $c_* > 0$ and $C_* < \infty$ such that

$$\mathcal{Q}(\lambda) \le -c_*|a_0(\lambda)|^2, \qquad |a_k(\lambda)| \le C_*|a_0(\lambda)| \quad (0 \le k \le 3). \tag{197}$$

Proof. By (178) and Lemma 26, the possible positive terms, divided by A^2, are at most

$$\frac{12}{25\sqrt{\lambda}} + \frac{32}{375b\sqrt{\lambda}} + \frac{96}{625b^2\lambda} + \frac{320}{3b\lambda\sqrt{\lambda}}.$$

For $\lambda \ge 128$, replace λ, $\sqrt{\lambda}$, and b by the lower bounds 128, 11, and 21/20. The resulting rational number is exactly $274/2205 < 1/8$. Hence

$$\sqrt{b}\,\mathcal{Q}(\lambda) < -\frac{(b^2-1)A^2}{8} \le -\frac{369}{51200\lambda}, \qquad \mathcal{Q}(\lambda) < -\frac{369}{56320\lambda} < -\frac{1}{200\lambda}.$$

The interval in Lemma 24 and this tail estimate overlap at 128. Evenness covers the negative half-axis and gives (196). The upper half of (164) and the bounded ratios in Lemma 23 imply (197). ∎

B.5 The second moment ratio as a two-sided clock kernel

The boundedness of n_2 can be strengthened in a form that permits a change of clock at an actual brake involution. This requires a kernel for the ratio itself, rather than an inverse kernel for a_0. We use the Fourier convention

$$\widehat{k}(\lambda) = \int_{\mathbb{R}} k(t)e^{-i\lambda t}\,\mathrm{d}t, \qquad k(t) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{k}(\lambda)e^{i\lambda t}\,\mathrm{d}\lambda$$

when the latter integral converges absolutely.

Lemma 27 (Second-ratio symbol and kernel)**.** *Let $n = a_2/a_0$. Then $n \in C^4(\mathbb{R})$, and on both real tails*

$$n(\lambda) = \frac{1}{i\lambda} - \frac{24}{(i\lambda)^2} + R(\lambda), \qquad |\partial_\lambda^j R(\lambda)| \le C_j|\lambda|^{-3-j}, \quad 0 \le j \le 4. \tag{198}$$

There is a real, integrable, two-sided kernel k with $n = \widehat{k}$. More precisely,

$$k(t) = \mathbf{1}_{\{t>0\}}e^{-t}(1-23t) + \rho(t), \qquad \rho \in C^1(\mathbb{R}), \qquad |\rho(t)| + |\rho'(t)| \le C(1+|t|)^{-4}. \tag{199}$$

Thus k is C^1 on each open half-line, its jump at zero is one, and its regular derivative k'_{reg} satisfies

$$|k(t)| + |k'_{\mathrm{reg}}(t)| \le C(1+|t|)^{-4} \quad (t \ne 0), \qquad \int_{\mathbb{R}} (1+|t|)|k(t)|\,\mathrm{d}t + \int_{\mathbb{R}} |t|\,|k'_{\mathrm{reg}}(t)|\,\mathrm{d}t < \infty. \tag{200}$$

In particular k has finite weighted variation, with its distributional derivative given by $Dk = \delta_0 + k'_{\mathrm{reg}}(t)\,\mathrm{d}t$. For any self-adjoint operator A_β with unitary group $U_t = e^{itA_\beta}$, the following strong integrals define bounded operators, and their truncated integrals converge in operator norm:

$$N_2 = n(A_\beta) = \int_{\mathbb{R}} k(t)U_{-t}\,\mathrm{d}t, \qquad N_2^* = \int_{\mathbb{R}} k(t)U_t\,\mathrm{d}t. \tag{201}$$

Moreover, for $L \geq 1$, each integral restricted to $|t| > L$ has operator norm at most CL^{-3}.

Proof. We first justify the differentiated endpoint expansions needed for the symbol assertion. At the rest point the polynomial ODE makes q analytic and even, j analytic and odd, and I analytic and odd. Equation (161) makes z analytic in t^2, with nonzero derivative $1/2$ in that variable. Analytic inversion and (162) therefore give convergent fractional-power expansions, including their differentiated remainders,

$$m_0(x) = (2x)^{-1/2}(1 - 36x + O(x^2)), \qquad m_2(x) = \sqrt{2x}(1 - 28x + O(x^2)). \tag{202}$$

In particular these are more than zeroth-order estimates.

At the crossing set $\tau = Q - t$. Analyticity of the ODE and energy give

$$q = \sqrt{2}\tau + O(\tau^7), \qquad I = I_Q + O(\tau^7), \qquad \frac{\mathrm{d}z}{\mathrm{d}t} = \frac{I_Q}{2\tau^2} + O(\tau^4), \qquad I_Q = Q/4.$$

The remainders here are analytic. Integration gives $z = I_Q/(2\tau) + c + O(\tau^5)$, with no logarithmic term. Thus $1/z$ is an analytic function of τ with nonzero derivative at zero and has an analytic inverse. Consequently q and j are analytic in $1/z$ near zero, $u = j/q$ is meromorphic there, and

$$\frac{\mathrm{d}^r m_0}{\mathrm{d}x^r} = O(x^{-8-r}), \qquad \frac{\mathrm{d}^r m_2}{\mathrm{d}x^r} = O(x^{-6-r}) \quad (x \longrightarrow \infty) \tag{203}$$

for every fixed $r \geq 0$. In particular $x^j m_0, x^j m_2 \in L^1(0,\infty)$ for $0 \leq j \leq 4$, so $a_0, a_2 \in C^4(\mathbb{R})$, and real-axis nonvanishing of a_0 gives $n \in C^4(\mathbb{R})$.

Here are details of the use of the endpoint expansions in Fourier asymptotics. Insert a smooth cutoff equal to one near zero. For each of the four differentiated transforms and the undifferentiated one, expand $x^j m_\ell(x)$, $\ell = 0, 2$, to any desired finite order using (202). A term $cx^{\alpha+r-1}$ in m_ℓ contributes

$$c(-i)^j \Gamma(\alpha + r + j)(i\lambda)^{-\alpha-r-j}$$

to $\partial_\lambda^j a_\ell$, with $\alpha = 1/2$ for $\ell = 0$ and $\alpha = 3/2$ for $\ell = 2$. This formula follows by exponential damping of the gamma integral. The cutoff changes it by a remainder decreasing to arbitrary order. For a local remainder whose derivatives are bounded by the corresponding powers of x, split at $x = |\lambda|^{-1}$: its initial integral has the asserted order, and repeated integration by parts on dyadic intervals beyond that point gives the same bound by a

convergent geometric sum. The cutoff away from zero has integrable derivatives of every order by (203); repeated integration by parts there gives arbitrarily rapid decay. This proves the expansions with remainders differentiated through order four. For negative λ use the conjugate branch of the fractional power, consistently with $a_\ell(-\lambda) = \overline{a_\ell(\lambda)}$.

In particular, with $c_0 = \Gamma(1/2)/\sqrt{2}$,

$$a_0(\lambda) = c_0(i\lambda)^{-1/2}\left(1 - \frac{18}{i\lambda} + O(|\lambda|^{-2})\right), \qquad a_2(\lambda) = c_0(i\lambda)^{-3/2}\left(1 - \frac{42}{i\lambda} + O(|\lambda|^{-2})\right).$$

The same statements hold after up to four differentiations, with the corresponding decrease in orders. Quotient differentiation, justified by (164), proves (198). In particular the leading ratio coefficient is exactly $2\Gamma(3/2)/\Gamma(1/2) = 1$, and the next coefficient is $-42 + 18 = -24$. No assertion about complex zeros of a_0 is used.

Subtract the rational reference symbol

$$n_{\mathrm{ref}}(\lambda) = \frac{1}{1 + i\lambda} - \frac{23}{(1 + i\lambda)^2}, \qquad r(\lambda) = n(\lambda) - n_{\mathrm{ref}}(\lambda). \tag{204}$$

It matches the two displayed terms in (198), so $r \in C^4(\mathbb{R})$ and $r^{(j)} = O(|\lambda|^{-3-j})$ for $0 \le j \le 4$. Both r and λr are integrable, as are their fourth derivatives. All lower derivative boundary terms vanish at both infinities. Define

$$\rho(t) = \frac{1}{2\pi}\int_{\mathbb{R}} r(\lambda) e^{i\lambda t}\,\mathrm{d}\lambda.$$

The integrability of λr gives $\rho \in C^1$. Four integrations by parts applied to r and to $i\lambda r$ give $|\rho(t)| + |\rho'(t)| \le C|t|^{-4}$ for $|t| \ge 1$; their L^1 norms give the bounded estimates for smaller t. For the derivative estimate, $(\lambda r)^{(4)} = \lambda r^{(4)} + 4r^{(3)} \in L^1$. Thus no fifth derivative or fifth moment is needed. Conjugate symmetry of r makes ρ real.

The inverse transform of n_{ref} is $\mathbf{1}_{\{t>0\}} e^{-t}(1 - 23t)$. Fourier inversion therefore gives (199) and the regularity, jump, and variation assertions. Finally, spectral calculus and $k \in L^1$ prove (201); the tail estimate follows from $\|U_t\| = 1$ and $\int_{|t|>L} |k(t)|\,\mathrm{d}t \le CL^{-3}$. ∎

The precise feature of this lemma used in a clock change is the ordinary derivative bound on each of the two open half-lines, together with the unitary tail estimate. An increasing clock fixing zero preserves each half-line, so it does not transport the jump across its location. The kernel is real but is neither required to be positive nor required to be supported on one side. In particular (201) acts boundedly on arbitrary inputs to N_2, including response vectors of the form $a_0(A_\beta)h$.

C Actual transport at the sextic brake boundary

Throughout this section the normalized potential is

$$V_f(Z) = |Z|^6(1 + f(\arg Z)), \qquad \delta = \|f\|_{C^2(\mathbb{S}^1)}, \quad \beta = f', \quad h = f'',$$
$$a = \|o'\|_2, \qquad o(\theta) = \tfrac{1}{2}[f(\theta) - f(\theta + \pi)].$$

Angular norms use $\mathrm{d}\theta/(2\pi)$. We work in the neighborhood in which Section A supplies an exact brake orbit from every energy-one rest point and a common leg time T. Let $Q_f(t,\theta)$ be that orbit, $\rho = |Q_f|$, $\alpha = \arg Q_f$, and $\ell = Q_f \times \partial_t Q_f$. The endpoint map on the circle is denoted F_b. Its forward and inverse lifts differ by 2π, but the circle map is the same involution. Let q be the radial solution with $q(0) = 1$, $q'(0) = 0$, and let $Q = T/2$ be its first zero. Thus $q(T) = -1$ and $q'(T) = 0$. The radial transverse operator and its second fundamental solution are

$$\mathscr{L} = \partial_t^2 + 6q^4, \qquad \mathscr{L} j = 0, \quad j(0) = 0, \quad j'(0) = 1.$$

We write $b = j(T) > 0$ and $K(t) = j(T-t) = bq(t) + j(t)$. In particular

$$G_E(t,s) = j(t)q(s) - q(t)j(s), \qquad G_E(t,t) = 0, \qquad \partial_t G_E(t,t) = 1. \tag{205}$$

The longitudinal radial position Green function is defined in the same way for $\partial_t^2 + 30q^4$; G_0 denotes their diagonal matrix. Only position Green functions, rather than their velocity derivatives, are used in the two-source estimates below.

C.1 Uniform geometry, including critical launch sets

Lemma 28 (Relative brake geometry)**.** *For all sufficiently small δ, put $\beta_0 = \beta(\theta)$ and $\beta_- = \beta(\theta + \pi)$. Then*

$$|\beta_0 + \beta_-| \le C\delta|\beta_0|, \qquad \beta(F_b(\theta)) = -\beta_0 + O(\delta|\beta_0|), \qquad F_b' = 1 + O(\delta). \tag{206}$$

The two antipodal slopes vanish together. If $\beta_0 = 0$, the whole leg is the exact straight asymmetric oscillator. If $\beta_0 \neq 0$, its angular lift is strictly monotone and

$$\operatorname{sgn} \ell = -\operatorname{sgn} \beta_0, \qquad c|\beta_0| \min(t, T-t, 1) \le |\ell(t,\theta)| \le C|\beta_0| \min(t, T-t, 1). \tag{207}$$

Its total angular change is $-\operatorname{sgn}(\beta_0)\pi + O(|\beta_0|)$ and it has no extra winding. Its minimum radius is comparable to $|\beta_0|$.

Let $x = x(t,\theta)$ be the signed line oscillator with potential $(1+f(\theta))x_+^6 + (1+f(\theta+\pi))x_-^6$, launched at its positive energy-one rest point. Uniformly on the whole leg,

$$\|Q_f - x e_\theta\|_{C_t^1} \le C|\beta_0|, \qquad \rho \asymp \sqrt{x^2 + \beta_0^2}, \qquad \|x - q\|_{C_t^1} \le C\delta. \tag{208}$$

On either inward head, at distance from the origin satisfying $q \ge D\delta$ in its inward time coordinate, with fixed sufficiently large D,

$$\rho \asymp q, \quad |\rho - q| \le C\delta, \quad q|\rho_t - q_t| \le C\delta, \quad |\beta(\alpha) - \beta(\theta)| \le C\delta|\beta(\theta)|/\rho. \tag{209}$$

The last inequality uses the inward launch label on each head. On any fixed interior time interval, the whole actual leg obeys

$$|\beta(\alpha)| \le C|\beta_0|(1 + \delta/\rho). \tag{210}$$

All constants are uniform as a nonzero β_0 tends to zero.

Proof. Set $c_\pm = 1 + f(\theta)$ and $1 + f(\theta + \pi)$ respectively, $R_\pm = c_\pm^{-1/6}$, and $B = \max(|\beta_0|, |\beta_-|)$. The reference line crosses zero at R_+Q and turns at $\tau_x = (R_+ + R_-)Q$. On that line the actual transverse force is $-\kappa|x|^5 E_\theta$, where $\kappa = \beta_0$ on $x > 0$ and $\kappa = -\beta_-$ on $x < 0$. Cartesian force comparison, with its uniform Lipschitz constant, gives $\|Q_f - xe_\theta\|_{C^1} \le CB$ through the crossing. Let

$$\psi'' + 6c_{\operatorname{sgn} x}|x|^4\psi = -\kappa|x|^5, \qquad \psi(0) = \psi'(0) = 0.$$

Taylor expansion is needed only for the radial polynomial force. Its quadratic remainder is $O(B^2)$; the perturbing Hessian is bounded by $C\delta$, so its force increment is $O(\delta B)$. A bounded linear propagator therefore gives

$$Q_f = xe_\theta + \psi E_\theta + O_{C^1}(\delta B). \tag{211}$$

No angular third derivative is involved.

For $W = x\psi' - x'\psi$, the exact scalar equations imply $W' = -\beta_0|x|^6$ on the positive half and $W' = -\beta_-|x|^6$ on the negative half. With $I_0 = \int_0^Q q^6 \, \mathrm{d}t > 0$, this yields

$$W(\tau_x) = -I_0(R_+^7\beta_0 + R_-^7\beta_-).$$

The seventh power includes physical time scaling. The actual next apoapsis differs from τ_x by $O(B)$, because its event derivative is uniformly separated from zero. Since $|\ell'| \le C\delta$, (211) and the exact final rest condition give $0 = -I_0(R_+^7\beta_0 + R_-^7\beta_-) + O(\delta B)$. As $R_\pm^7 = 1 + O(\delta)$, this first implies $|\beta_0 + \beta_-| \le C\delta B$ and then $B \le 2|\beta_0|$. If $\beta_0 = 0$, both transverse line forces vanish and uniqueness makes the orbit the line oscillator. Otherwise endpoint position comparison also gives $\operatorname{dist}(F_b(\theta), \theta + \pi) \le C|\beta_0|$. Using $\|\beta'\|_\infty \le \delta$ proves the second relation in (206). Ordinary C^1 flow and transverse-event dependence give its final relation.

The matching just proved makes $\kappa = \beta_0 + O(\delta|\beta_0|)$ on both halves. If $y_c'' + 6c_{\operatorname{sgn} x}|x|^4 y_c = -|x|^5$ has zero initial data, then

$$Q_f = xe_\theta + \beta_0 y_c E_\theta + O_{C^1}(\delta|\beta_0|).$$

The Wronskian $xy_c' - x'y_c$ is uniformly close to $-\int_0^t \operatorname{sgn}(q)|q|^6 \, \mathrm{d}s$, which is strictly negative on every compact subinterval of $(0, T)$. This proves (207) away from the ends. Near the initial rest point, direct force comparison gives $|\alpha(t) - \theta| \le C|\beta_0|t^2$ and hence $\beta(\alpha) = \beta_0 + O(\delta|\beta_0|t^2)$. The exact torque equation

$$\alpha_t = \ell/\rho^2, \qquad \ell_t = -\rho^6\beta(\alpha) \tag{212}$$

proves the initial bound $|\ell| \asymp |\beta_0|t$ with the stated sign. Apply this argument to the reversed rest launch at $F_b(\theta)$ and use (206) for the final endpoint layer.

On the interior, bounded speed and $|\ell| \le \rho|\partial_t Q_f|$ give $\rho \ge c|\beta_0|$. At the reference crossing, the line comparison gives $\rho \le C|\beta_0|$. Together with $|Q_f - xe_\theta| \le C|\beta_0|$, this

proves the radius comparison in (208). The projection onto e_θ is strictly decreasing: near the first rest point its acceleration is negative, near the last it is positive and its final velocity is zero, and on the intervening compact interval its velocity is close to the strictly negative radial one. Thus the projection crosses zero once. Monotonicity of the angular lift and the endpoint angle estimate rule out extra winding and give the stated total change.

On a head with $q \geq D\delta$, Cartesian C^1 proximity gives $\rho \asymp q$ and $|\rho - q| \leq C\delta$. Differentiating the norm as $\rho_t = Q_f \cdot \partial_t Q_f/\rho$ shows $|\rho_t - q_t| \leq C\delta/q$ there. The line comparison bounds the angular distance from the corresponding launch ray by $C|\beta_0|/\rho$, which proves (209). For (210), use the nearer positive or negative reference ray when $\rho \geq C|\beta_0|$, together with (206). In the remaining region, the bounded angular distance and the Lipschitz bound for β give $|\beta(\alpha)| \leq C(|\beta_0| + \delta)$, which is the same estimate because $\rho \leq C|\beta_0|$. ∎

Lemma 29 (Positive integrated kernels)**.** *For $p \geq 0$ and an angular function g, put*

$$\mathcal{T}_p g(\theta) = \int_0^T \rho^p |g(\alpha)| \, \mathrm{d}t, \qquad \mathcal{T}_{p,r} g(\theta) = \int_{\rho \leq r} \rho^p |g(\alpha)| \, \mathrm{d}t.$$

If $\mathcal{M}$ is the uncentered circular maximal operator, then almost everywhere

$$\mathcal{T}_p g \leq C_p[\mathcal{M}g(\theta) + \mathcal{M}g(F_b(\theta))], \tag{213}$$

$$\mathcal{T}_{p,r} g \leq C_p r^{p+1}[\mathcal{M}g(\theta) + \mathcal{M}g(F_b(\theta))]. \tag{214}$$

In particular their L^2 operator bounds are C_p and $C_p r^{p+1}$. The radius estimate is uniform in $0 < r < r_*$, including arbitrary positive-measure critical launch sets.

Proof. For $B = |\beta(\theta)| > 0$, let d_0, d_1 be lifted angular distances from the initial and final endpoints. Apply the line comparison both forward and backward to obtain $\rho|\sin d_i| \leq CB$. The total angular change is less than $4\pi/3$, so the smaller distance is less than $2\pi/3$ and

$$\rho \leq C \min\{1, B/\min(d_0, d_1)\}.$$

On the fixed interior time interval, monotone change of variable and (207) bound the weighted time density by

$$C_p B^{-1} \min\{1, (B/\min(d_0, d_1))^{p+2}\}.$$

Near the initial endpoint, $d_0 \asymp Bt^2$ and the density is at most $C_p(Bd_0)^{-1/2}$ for $0 < d_0 < CB$; the analogous statement holds at the final endpoint. Split the two endpoint distances. Each resulting one-sided kernel is bounded by

$$C_p \left[B^{-1/2} s^{-1/2} \mathbf{1}_{0<s<C_0 B} + B^{-1} \min\{1, (B/s)^{p+2}\}\right].$$

Its dyadic annuli are controlled by the endpoint maximal function with summable weights $2^{-j/2}$ and $2^{-j(p+1)}$. Reduction of the lifted interval to the circle has uniformly bounded multiplicity. This proves (213).

Choose r_* below the radius on the fixed endpoint time intervals. On $\rho \le r$ only the interior estimate is needed, and its density is

$$C_p B^{-1} \min\{r, B/d\}^{p+2} = C_p r^{p+1} \kappa^{-1} \min\{1, (\kappa/d)^{p+2}\}, \qquad \kappa = B/r.$$

The same annulus sum is uniform for every $\kappa > 0$; if κ exceeds the circumference, use the whole-circle average. No assumption $B \le r$ is needed. If $B = 0$, the straight orbit samples only θ and $\theta + \pi = F_b(\theta)$, and its speed near the crossing is bounded above and below. Its contribution is Cr^{p+1} times those two values. This also handles non-discrete critical sets. Finally $F_b' = 1 + O\delta$ makes composition with F_b bounded on L^2, and the maximal inequality proves the norm statements. One may first use continuous inputs and then extend by these positive bounds; this specifies the operators on L^2 equivalence classes. ∎

The time integral in these estimates is taken before the angular norm. No instantaneous pullback estimate or bound for a raw same-time product is required.

C.2 A response scale adapted to the actual slope

Let ϕ_s be the complete flow of the C^1 vector field β on the circle. Its half-density representation is

$$U_s g(\theta) = \sqrt{\phi_s'(\theta)}\, g(\phi_s(\theta)).$$

We use the complexified L^2 space for spectral calculus; all mechanical coefficients and final quadratic forms are real.

Lemma 30 (Generator and graph core)**.** *The strongly continuous unitary group U_s has self-adjoint generator*

$$A_\beta = -i(\beta \partial_\theta + \beta'/2), \qquad U_s = e^{isA_\beta}.$$

Smooth periodic functions are a graph core. With

$$\Lambda = (1 + A_\beta^2)^{1/2}, \qquad B_\beta = (1 + iA_\beta)^{-1/2}, \qquad S_g = \|B_\beta g\|_2,$$

one has $S_g = \|\Lambda^{-1/2} g\|_2$. The actual brake involution acts unitarily by $Jg = \sqrt{F_b'}\, g \circ F_b$, and $J = J^* = J^{-1}$.

Proof. Change of variable gives unitarity and the C^1 flow gives strong continuity. Differentiation on smooth functions identifies the displayed expression. For completeness,

let $D = iA_\beta$ and let J_ϵ be periodic convolution with a smooth approximate identity φ_ϵ. Distributionally,

$$[D, J_\epsilon]u(x) = \int \left[(\beta(x) - \beta(y))\varphi_\epsilon'(x-y) + \tfrac{1}{2}(\beta'(x) + \beta'(y))\varphi_\epsilon(x-y)\right] u(y)\,\mathrm{d}y.$$

These commutators are uniformly bounded by $C\|\beta'\|_\infty$ on L^2. On smooth inputs their leading terms cancel and they converge strongly to zero; density gives the same conclusion for every L^2 input. If u and its distributional Du belong to L^2, the same formula gives $DJ_\epsilon u = J_\epsilon Du + [D, J_\epsilon]u \to Du$. Closedness of the transport generator, already identified on smooth functions, puts u in its domain. Conversely, testing the strong group derivative against smooth functions identifies its distributional derivative with Du. Thus the two domains coincide, and the preceding convolution argument proves the graph-core assertion. The equality of response norms is the spectral identity $|(1+i\lambda)^{-1/2}|^2 = (1+\lambda^2)^{-1/2}$. The assertions about J follow from $F_b^2 = \mathrm{id}$ and the chain rule. None uses a second derivative of β or of F_b. ∎

C.3 Signed origin operators

Fix a smooth nondecreasing η equal to zero on $[0,1]$ and one on $[2,\infty)$, and put

$$\lambda_r(t) = 1 - \eta(|q(t)|/r), \qquad r \geq D\delta.$$

For sufficiently large fixed D, its support has $\rho \leq Cr$ and is separated from both rest endpoints. On the derivative support $\rho \asymp r$ and $|\lambda_r'| \leq C/r$.

Lemma 31 (Signed layer operator)**.** *Suppose an amplitude $c(t,\theta)$ is measurable, locally absolutely continuous in time off critical crossings, and obeys*

$$|c| \leq M\rho^{-k}, \qquad |c_t| \leq M\rho^{-k-1}, \qquad \nu = p - k \geq 2.$$

Its weighted expression is interpreted at critical crossings by its integrable extension, with the value at the crossing instant irrelevant. Then

$$T_c g = \int_0^T \lambda_r c\rho^p g(\alpha)\,\mathrm{d}t$$

satisfies

$$\|T_c\| \leq CMr^{\nu+1}, \qquad \|T_c A_\beta\| \leq CMr^{\nu+2}, \qquad \|T_c g\|_2 \leq CMr^{\nu+1}S_g. \tag{215}$$

The middle operator is the bounded extension from the smooth graph core. If M is replaced by a label majorant $d(\theta) \in L^2$, the final estimate holds in angular L^1 with M replaced by $\|d\|_2$.

Proof. The first bound is Lemma 29 with exponent ν. On a noncritical launch, (207) and (210) imply

$$\left|\frac{\rho^2\beta(\alpha)}{\ell}\right| \le C(\rho^2+\delta\rho)$$

on the interior layer. For smooth g, integrate only the $\beta g'$ part of $A_\beta g$ in time. With $d=\lambda_r c\rho^{p+2}\beta(\alpha)/\ell$, the exact identity is

$$T_c A_\beta g = i\int_0^T \left[d_t - \tfrac{1}{2}\lambda_r c\rho^p\beta'(\alpha)\right] g(\alpha)\,\mathrm{d}t. \tag{216}$$

Both fixed endpoint terms vanish. The coefficient is

$$(\lambda_r c)_t\rho^{p+2}\frac{\beta(\alpha)}{\ell} + (p+2)\lambda_r c\rho^{p+1}\rho_t\frac{\beta(\alpha)}{\ell} + \tfrac{1}{2}\lambda_r c\rho^p\beta'(\alpha) - \lambda_r c\rho^{p+2}\beta(\alpha)\frac{\ell_t}{\ell^2}.$$

The first three terms are bounded by $CM(\rho^{\nu+1}+\delta\rho^\nu)$; the cutoff derivative obeys the same estimate since $\rho\asymp r$ on its support. By the exact torque equation the last term is at most $CM\rho^{\nu+8}(1+\delta/\rho)^2$, and is smaller than the same majorant. Positive kernels give $\|T_cA_\beta\|\le CM(r^{\nu+2}+\delta r^{\nu+1})$, which proves the second bound.

There is no division by ℓ on the critical launch set. There the orbit samples only θ and $\theta+\pi$, both zeros of β. A C^1 function has derivative zero almost everywhere on its zero set; antipodal translation preserves the exceptional null set. Thus $A_\beta g$ vanishes at both sampled labels for almost every critical launch, proving the generator estimate there directly.

The bounded extensions and the graph core give $T_c^*L^2\subset\mathrm{Dom}(A_\beta)$ and $A_\beta T_c^* = (T_cA_\beta)^*$. Spectral Cauchy–Schwarz gives

$$\|\Lambda^{1/2}T_c^*y\|_2^2 \le \|T_c^*y\|_2^2 + \|T_c^*y\|_2\|A_\beta T_c^*y\|_2 \le CM^2r^{2\nu+2}\|y\|_2^2.$$

Duality proves the last estimate in (215). For a label majorant d, divide the amplitude by d where positive and define it as zero elsewhere. The normalized amplitude has the uniform bounds above; angular Cauchy–Schwarz after this operator bound gives the L^1 assertion. The label multiplier is not differentiated. ∎

Lemma 32 (One-generator Green pairing)**.** *Let two amplitudes satisfy the preceding bounds with data (p,k,M) and (p',k',N), and put $\nu=p-k\ge 2$, $\mu=p'-k'\ge 2$. Let G be C^1 on the closed chronological triangle, with bounded first derivatives and $G(t,t)=0$. Then*

$$\left|\left\langle \int_{s<t}\lambda_r(t)c(t)\rho(t)^pG(t,s)\lambda_r(s)d(s)\rho(s)^{p'}\overline{g(\alpha(t))}\,h_1(\alpha(s))\,\mathrm{d}s\,\mathrm{d}t\right\rangle\right| \le CMNr^{\nu+\mu+2}S_gS_{h_1}. \tag{217}$$

The corresponding angular output has the same L^1 bound.

Proof. Before inserting a generator, the absolute form is bounded by the angular integral of two complete positive time integrals. Angular Cauchy–Schwarz and Lemma 29 give $CMNr^{\nu+\mu+2}\|g\|_2\|h_1\|_2$. Insert A_β in the first input only. The time integration in (216) now also differentiates G; its moving lower endpoint contributes zero because $G(s,s)=0$. The derivative of G has bounded size and gains at least one radial power. The resulting bound is $CMNr^{\nu+\mu+3}\|g\|_2\|h_1\|_2$. Inserting the generator in the second input only gives the same bound, with its moving upper endpoint again zero. The critical-set treatment is the one in the preceding lemma.

Represent the form as $\langle g, \mathcal{K}h_1\rangle$. These separate bounds give bounded $\mathcal{K}$, $A_\beta\mathcal{K}$ and $\mathcal{K}A_\beta$, with the adjoint-domain interpretation used above. Consequently both $\Lambda\mathcal{K}$ and $\mathcal{K}\Lambda$ have norm at most $CMNr^{\nu+\mu+2}$ for $r\le 1$. Let P_n be the spectral projection of Λ onto $[1,n]$ and interpolate

$$\Lambda^z P_n\mathcal{K}P_n\Lambda^{1-z}, \qquad 0\le \Re z\le 1.$$

On the two boundary lines the estimates are those of $\mathcal{K}\Lambda$ and $\Lambda\mathcal{K}$, because Λ^{it} is unitary. The three-lines theorem, followed by the limit on finite-spectral vectors, proves $\|\Lambda^{1/2}\mathcal{K}\Lambda^{1/2}\|\le CMNr^{\nu+\mu+2}$, which is (217). For the L^1 output insert an arbitrary angular test of modulus at most one in the outer amplitude; it preserves all time bounds, and duality applies. In particular no derivative of that test is taken.

The proof does not bound $A_\beta\mathcal{K}A_\beta$. Applying both integration-by-parts operations in one calculation would generally expose a diagonal term after the first Green zero had been used. The two separate boundary estimates and interpolation avoid that term. ∎

The exact polar Hessian formula is

$$D^2(|Z|^6 f)=\rho^4\left[30f\,e_\alpha\otimes e_\alpha+5f'(e_\alpha\otimes E_\alpha+E_\alpha\otimes e_\alpha)+(6f+f'')E_\alpha\otimes E_\alpha\right]. \tag{218}$$

Its frame coefficients are bounded and their time derivatives are $O(1/\rho)$, since $|\alpha_t|=|\ell|/\rho^2\le C/\rho$ on the layer. The actual regularizer

$$A_{\mathrm{reg}}=\frac{x}{x^2+\beta(\theta)^2} \tag{219}$$

obeys $|A_{\mathrm{reg}}|\le C/\rho$ and $|\partial_t A_{\mathrm{reg}}|\le C/\rho^2$ by (208). Hence two-layer position blocks with one regularizer have bound $Cr^9(S_h+a)^2$, and those without it have bound $Cr^{10}(S_h+a)^2$. Here $S_f+S_{f'}\le C\|f\|_{H^1}\le Ca$ by Lemma 22. These bounds concern signed complete layer integrals or zero-diagonal Green pairings. They do not assert a retarded velocity estimate or a supremum over terminal time.

C.4 Exact inward clocks and their densities

We retain the geometry and notation of Lemma 28. In particular, $q(0)=1$, $q'(0)=0$, $q''+6q^5=0$, $q(Q)=0$, $T=2Q$, and j is the solution of

$$j''+6q^4 j=0, \qquad j(0)=0, \quad j'(0)=1.$$

Write

$$b = j(T), \qquad K(t) = j(T-t) = bq(t) + j(t), \qquad u(t) = \frac{j(t)}{q(t)},$$
$$I(t) = \int_0^t q(s)^6\, ds, \qquad z_0(t) = \int_0^t \frac{I(s)}{q(s)^2}\, ds \quad (0 \le t < Q). \tag{220}$$

The Wronskian gives $u' = q^{-2}$. The map z_0 increases from zero to infinity; its inverse is denoted by t_0. We use the following radial clock densities:

$$m_0(z) = \frac{q(t_0(z))^8}{I(t_0(z))}, \qquad m_k(z) = u(t_0(z))^k m_0(z) \quad (0 \le k \le 3), \qquad m_4^{\mathrm{src}}(z) = \frac{m_0(z)}{q(t_0(z))^2}. \tag{221}$$

Thus $m_k(z)\, dz = q(t)^6 u(t)^k\, dt$, whereas $m_4^{\mathrm{src}}(z)\, dz = q(t)^4\, dt$. The subscript on m_4^{src} denotes the power of the source and is not a moment index. The group U_s and its self-adjoint generator A_β are those of Lemma 30; our convention is $U_s = e^{isA_\beta}$. Throughout this subsection,

$$B_\beta = (1 + iA_\beta)^{-1/2} = \int_0^\infty \frac{e^{-z}}{\sqrt{\pi z}} U_{-z}\, dz, \qquad S_g = \|B_\beta g\|_2. \tag{222}$$

All group integrals are strong integrals after application to a vector. Inner products and norms use angular Lebesgue measure with the same normalization as the preceding invariant identities.

Lemma 33 (Exact clock and normalized endpoint bounds)**.** *There are fixed $D_0 > 1$ and $c > 0$ with the following properties. Put $\delta = \|f\|_{C^2}$, $\beta = f'$, and $R_\theta = (1 + f(\theta))^{-1/6}$. For $\delta > 0$, the actual inward arc on which $\rho \ge D_0\delta$ has an increasing clock Z_θ satisfying*

$$\alpha(t, \theta) = \phi_{-Z_\theta(t)}(\theta).$$

The clock is defined, uniformly in the label, for $0 \le Z_\theta \le 2L_\delta$, where $L_\delta = c/\delta$. At a critical label we choose

$$Z_\theta(t) = R_\theta^6 z_0(t/R_\theta). \tag{223}$$

Let t_θ be the inverse clock, and set

$$q_\theta(z) = q(t_\theta(z)), \quad u_\theta(z) = u(t_\theta(z)), \quad \mathcal{J}_z(\theta) = \phi'_{-z}(\theta).$$

There is a positive density K_0 such that, exactly,

$$q(t)^2 \rho(t, \theta)^4 g(\alpha(t, \theta))\, dt = K_0(\theta, z)(U_{-z}g)(\theta)\, dz. \tag{224}$$

Define

$$k_4 = K_0/q_\theta^2, \quad K_k = u_\theta^k K_0, \quad r_\theta = \frac{u_\theta}{\sqrt{2z}}, \quad d_\theta = \sqrt{2z} K_0, \quad e_\theta = d_\theta/q_\theta. \tag{225}$$

The coefficients r_θ, d_θ extend absolutely continuously to zero, with

$$r_\theta(0) = R_\theta^{-2}, \qquad d_\theta(0) = R_\theta^2, \qquad |r_\theta| + |d_\theta| + |r'_\theta| + |d'_\theta| \le C \quad (0 \le z \le 1). \tag{226}$$

On $1 \le z \le 2L_\delta$, uniformly in the label,

$$
\begin{aligned}
&q_\theta \asymp \rho \asymp (1+z)^{-1}, \quad t'_\theta \le C(1+z)^{-2}, \quad |q'_\theta| \le C(1+z)^{-2},\\
&|u_\theta| \le C(1+z), \quad |u'_\theta| \le C,\\
&|r_\theta| \le C(1+z)^{1/2}, \quad |r'_\theta| \le C(1+z)^{-1/2}, \quad |K_0| \le C(1+z)^{-8}, \quad |K'_0| \le C(1+z)^{-9},\\
&|d_\theta| \le C(1+z)^{-15/2}, \quad |d'_\theta| \le C(1+z)^{-17/2},\\
&|e_\theta| \le C(1+z)^{-13/2}, \quad |e'_\theta| \le C(1+z)^{-15/2},\\
&|k_4| \le C(1+z)^{-6}, \qquad |k'_4| \le C(1+z)^{-7}.
\end{aligned} \tag{227}
$$

All first derivatives here and below are clock derivatives. The coefficient families are jointly measurable and have the displayed label-independent majorants; no label derivative of these coefficients is asserted.

Proof. At a noncritical label put $B = |\beta(\theta)|$, $b_\theta(t) = |\beta(\alpha(t,\theta))|$, and $\ell_+(t) = |\ell(t,\theta)|$. The slope-carrier estimate in Lemma 28 ensures that $\beta(\alpha)$ has the sign of $\beta(\theta)$ on the retained arc. The torque equation and the angular equation therefore give

$$
\ell_+(t) = \int_0^t \rho(s)^6 b_\theta(s)\, ds, \qquad Z'_\theta(t) = \frac{\ell_+(t)}{\rho(t)^2 b_\theta(t)}, \qquad t'_\theta(z) = \frac{\rho(t_\theta(z))^2 b_\theta(t_\theta(z))}{\ell_+(t_\theta(z))}. \tag{228}
$$

Integration of $d\alpha/\beta(\alpha)$ proves the claimed exact reparameterization. In particular this construction does not compare two nearby angular evaluation maps in operator norm.

The geometry gives $|b_\theta/B - 1| \le C\delta/\rho$ and $|\rho - q| \le C\delta$. Since $I(t) \asymp t$ on $[0, Q]$,

$$
|\ell_+(t)/B - I(t)| \le C\delta t \le C\delta I(t).
$$

Consequently, on a retained arc with D_0 sufficiently large,

$$
\frac{Z'_\theta}{z'_0} = 1 + O(\delta/q), \qquad |Z_\theta - z_0(t)| \le C\delta z_0(t)/q(t), \qquad |t_\theta(z) - t_0(z)| \le C\delta. \tag{229}
$$

The integral estimate uses the decrease of q. Near zero its inverse version is the sharper $|t_\theta(z) - t_0(z)| \le C\delta\sqrt{z}$. Away from zero it follows by multiplying the clock error by q^2/I and using $z_0 q/I \le C$. Since $q \asymp (1+z_0)^{-1}$, choosing c small fixes a common clock interval $[0, 2c/\delta]$ strictly within the radius-$D_0\delta$ heads.

The exact density in (224) is

$$
K_0(\theta, z) = q(t)^2 \rho(t)^6 \frac{b_\theta(t)}{\ell_+(t)} \mathcal{J}_z(\theta)^{-1/2}, \qquad t = t_\theta(z). \tag{230}
$$

The factor $\mathcal{J}_z^{-1/2}$ is indispensable. At a critical label the actual positive arc is $\rho(t) = R_\theta q(t/R_\theta)$, $\alpha = \theta$. Formula (223) gives instead

$$
K_0(\theta, z) = R_\theta^{-1} q(t)^2 \frac{q(t/R_\theta)^6}{I(t/R_\theta)} \mathcal{J}_z(\theta)^{-1/2}. \tag{231}
$$

This is a legitimate change of variable even if it is not the limit of the chosen clocks at neighboring noncritical labels. The flow fixes the label, but its half-density need not equal one there.

We give endpoint derivative estimates directly from the equations. On a fixed short initial interval put

$$A(t) = \rho(t)/R_\theta, \quad \widehat{B}(t) = b_\theta(t)/B, \quad \widehat{L}(t) = \ell_+(t)/(BR_\theta^6 t).$$

The initial rest condition, radial equation $\rho'' = \ell_+^2/\rho^3 - 6\rho^5(1+f(\alpha))$, and the chain rule give

$$A = 1+O(t^2),\ A' = O(t), \qquad \widehat{B} = 1+O(\delta t^2),\ \widehat{B}' = O(\delta t), \qquad \widehat{L} = t^{-1}\int_0^t A(s)^6\widehat{B}(s)\,ds.$$

The last integrand equals one at zero and has derivative $O(t)$. Subtracting its average from its endpoint value proves $\widehat{L} = 1+O(t^2)$ and $\widehat{L}' = O(t)$, without differentiating that integrand twice. Thus

$$Z' = R_\theta^4 t E_Z(t), \quad Z = \tfrac{1}{2}R_\theta^4 t^2\overline{E}_Z(t), \quad E_Z = \widehat{L}/(A^2\widehat{B}), \quad \overline{E}_Z = 2t^{-2}\int_0^t sE_Z(s)\,ds,$$

where both normalized factors are $1+O(t^2)$ with derivatives $O(t)$. Furthermore, exactly,

$$tK_0 = q(t)^2A(t)^6\widehat{B}(t)\widehat{L}(t)^{-1}\mathcal{J}_{Z(t)}^{-1/2} = 1+O(t^2), \qquad (tK_0)' = O(t).$$

Here $\partial_t\mathcal{J}_{Z(t)}^{-1/2} = \frac{1}{2}\beta'(\alpha)Z'\mathcal{J}_{Z(t)}^{-1/2}$ uses only f''. The radial equation also gives $u(t)/t = 1+O(t^2)$ with derivative $O(t)$. Consequently

$$r_\theta = R_\theta^{-2}(u(t)/t)\overline{E}_Z^{-1/2}, \qquad d_\theta = R_\theta^2\overline{E}_Z^{1/2}(tK_0)$$

have time derivatives $O(t)$. Division by $Z' \asymp t$ proves (226). For a critical label the same calculation uses $E_Z = (I(\sigma)/\sigma)/q(\sigma)^2$ and $tK_0 = q(t)^2q(\sigma)^6/[I(\sigma)/\sigma]\,\mathcal{J}_Z^{-1/2}$, where $\sigma = t/R_\theta$, and never divides by B.

For completeness, the comparison estimates for the densities, needed below as well, are

$$|K_k - m_k| \le C\delta(1+z)m_k, \qquad |(K_k-m_k)'| \le C\delta m_k \quad (z\ge z_*), \qquad 0\le k\le 3, \quad (232)$$

where $z_* > 0$ is fixed. The first follows from (229), the product formula (230), $e^{-\delta z} \le \mathcal{J}_z \le e^{\delta z}$, and $|zm_k'| \le Cm_k$. To check the second, the exact logarithmic derivative is

$$\partial_z \log K_k = \left(k\frac{j_t}{j} + (2-k)\frac{q_t}{q} + 6\frac{\rho_t}{\rho}\right)t_\theta' - \tfrac{1}{2}\beta'(\alpha) - \frac{\rho^8 b_\theta^2}{\ell_+^2}. \qquad (233)$$

The radial expression is obtained by setting $\rho = q$, $b_\theta/\ell_+ = 1/I$ and $t_\theta' = q^2/I$. On $z\ge z_*$, I and j are bounded below. Use $|\rho - q| \le C\delta$, $q|\rho_t - q_t| \le C\delta$, $b_\theta/\ell_+ = I^{-1}(1+O(\delta/q))$,

and $|t_\theta - t_0| \le C\delta$. Each logarithmic-derivative difference is $O(\delta)$: the possible q^{-1} factor is paired with $t'_\theta = O(q^2)$. Since $|m'_k| \le Cm_k/z$, this proves (232). Formula (231) gives the same bounds at critical labels.

Finally the radial energy equation gives, with $\tau = Q - t$, $q \asymp \tau$, $I \asymp 1$, $z_0 \asymp \tau^{-1}$, and $m_k = O(z^{k-8})$, $m'_k = O(z^{k-9})$ at infinity. On the retained range $\delta z \le 2c$, (232) implies $K_0 = O(z^{-8})$ and $K'_0 = O(z^{-9})$. The inverse-clock formula, $u_z = t'_\theta/q^2$, and the product rule now give every estimate in (227). ∎

C.5 Causal factorization with boundary measures

All the distributions in the next two lemmas are extended by zero to negative clocks. In particular a derivative is a causal distributional derivative, and its initial atom is retained. Put

$$p_0(z) = \frac{\mathbf{1}_{z>0}}{\sqrt{\pi z}}, \qquad p(z) = e^{-z}p_0(z),$$
$$I_A = p_0 *, \quad B = p *, \quad D_A = \partial_z I_A, \quad D_B = e^{-z} D_A e^z.$$

These definitions concern scalar convolution; B is not the sampled Hessian.

Lemma 34 (One- and two-clock factorization)**.** *The following statements hold, also for jointly measurable families indexed by the launch label.*

(i) If a causal kernel is $k = ap + r$, where a is independent of the clock, r is locally absolutely continuous, $r, r' \in L^1(0,\infty)$, and $r(0)$ is finite, then

$$k = p * \nu, \qquad \nu = a\delta_0 + p * (r' + r) + r(0)p. \tag{234}$$

In particular its operator $\int k(\theta, z)U_{-z}\, dz$ factors on the right through B_β. If $|a|$, $|r(0)|$, $|r| + |r'|$ have common majorants with total size C_*, the remaining operator has norm at most C_*. If the common majorants are instead multiplied by a label function $d \in L^2$, the corresponding scalar functional has norm at most $C_*\|d\|_2$ on the response space, and the resulting output has L^1 norm at most $C_*\|d\|_2 S_g$.

(ii) Suppose $F_\theta = (p_0 \otimes p_0) * \mu_{A,\theta}$, where $|\mu_{A,\theta}| \le \Lambda_A$ and $\Lambda_A([0,R]^2) \le C(1+R)^m$. There is a common positive measure Λ_B with square mass at most $C(1+R)^{m+1}$ and $F_\theta = (p \otimes p) * \mu_{B,\theta}$, $|\mu_{B,\theta}| \le \Lambda_B$. Let the separated multipliers $a_\theta(z), b_\theta(w)$ be locally absolutely continuous with common bounds A_0, A_1, B_0, B_1 for their values and first derivatives. Define

$$W_A(s) = A_0(s) + \int_0^\infty e^{-v} A_1(s+v)\, dv, \qquad W_B(t) = B_0(t) + \int_0^\infty e^{-v} B_1(t+v)\, dv.$$

If $L = \int W_A(s) W_B(t)\, d\Lambda_B(s,t) < \infty$, then

$$a_\theta(z) b_\theta(w) F_\theta(z,w) = (p \otimes p) * \nu_\theta, \qquad |\nu_\theta| \le \Lambda, \qquad \Lambda([0,\infty)^2) \le L. \tag{235}$$

Its angular bilinear form is bounded by LS_gS_h. The same bound holds with $JU_{-z}g$ in one slot, or with a bounded label multiplier outside the slots, with the corresponding multiplier norm.

(iii) If $\Lambda_B([0,R]^2) \le C(1+R)^M$ and, for $p_*, q_* > M$,

$$\begin{aligned} A_0(s) &\le C\epsilon(1+s)^{-p_*}, & A_1(s) &\le C\epsilon\{s^{-1/2}\mathbf{1}_{0<s<1} + (1+s)^{-p_*-1}\},\\ B_0(t) &\le C(1+t)^{-q_*}, & B_1(t) &\le C\{t^{-1/2}\mathbf{1}_{0<t<1} + (1+t)^{-q_*-1}\}, \end{aligned} \tag{236}$$

then $L \le C\epsilon$. Removing a smooth clock tail at scale $R \ge 2$ in either coefficient has cost $C\epsilon R^{-\gamma}S_gS_h$, where $\gamma = \min(p_* - M, q_* - M) > 0$. These constants are unchanged by an additional smooth cutoff at any larger scale.

Proof. Since $p * p = e^{-z}\mathbf{1}_{z>0}$, causal integration of $r' + r$, including its initial value, proves (234). If $d\nu_\theta = c(\theta, s)\, d\Lambda(s)$, with $|c| \le 1$, the associated operator is exactly

$$\int c(\theta, s)(U_{-s}B_\beta g)(\theta)\, d\Lambda(s).$$

The coefficient is outside the unitary group. Minkowski proves the operator bound. Angular Cauchy–Schwarz proves the functional bound when $|c| \le d(\theta)$, and insertion of an arbitrary bounded angular dual test proves the stated L^1 bound.

Here is a direct multiplier calculation which avoids constants growing exponentially with the clock length. For a causal measure μ, changing the order in $I_A(aI_A\mu)$ gives

$$I_A(aI_A\mu)(x) = \int_{[0,x]} \frac{1}{\pi} \int_0^1 \frac{a(s + (x-s)t)}{\sqrt{t(1-t)}}\, dt\, d\mu(s).$$

The diagonal trace of the inner integral is $a(s)$. Differentiation therefore gives $D_A(aI_A\mu) = a\mu + K_a\mu$, where the second term has kernel

$$K_a(x, s) = \mathbf{1}_{x>s} \frac{1}{\pi} \int_0^1 a'(s + (x-s)t) \frac{\sqrt{t}}{\sqrt{1-t}}\, dt.$$

In particular every initial atom of μ survives in $a\mu$. Conjugation by e^x proves the exact Bessel identity

$$D_B(aB\mu) = a\mu + \int_{s<x} e^{-(x-s)} K_a(x, s)\, dx\, d\mu(s). \tag{237}$$

For fixed s, Tonelli and $v = (x-s)t$ show that the variation of the second term is at most

$$\int_0^\infty J_0(v) A_1(s+v)\, dv, \quad J_0(v) = \frac{1}{\pi} \int_0^1 \frac{e^{-v/t}}{\sqrt{t(1-t)}}\, dt \le e^{-v}.$$

Together with the diagonal term this is $W_A(s)$ and, by replacing all factors by their majorants, constructs a common positive output measure.

The conversion from Abel to Bessel kernels is also polynomially controlled. Direct Laplace transformation, or integration of the difference of two exponentials, gives

$$D_B - D_A = c_0 *, \qquad c_0(t) = \frac{1 - e^{-t}}{2\sqrt{\pi} t^{3/2}}, \quad c_0 \geq 0, \quad \|c_0\|_1 = 1.$$

For $k = c_0 * p_0$ one has

$$p * (\delta_0 + k) = p_0, \qquad k \geq 0, \qquad k(t) \leq C(1+t)^{-1/2}, \qquad \int_0^R k \leq C\sqrt{1+R}. \tag{238}$$

For $t \leq 1$ the bound is the beta integral with $c_0(v) \leq Cv^{-1/2}$. For $t \geq 1$, split its convolution at $v = t/2$ and use $\|c_0\|_1 = 1$ on the first part and $c_0(v) \leq Cv^{-3/2}$ on the second. Thus

$$\mu_B = (\delta_0 + k)_z * (\delta_0 + k)_w * \mu_A$$

has the asserted common majorant and square growth. All these are locally finite causal convolutions. Apply (237) in each coordinate to obtain (235). Local truncation followed by the finite weighted majorant justifies convergence in total variation even if μ_B has infinite total mass.

Writing $d\nu_\theta = c(\theta, s, t) d\Lambda(s, t)$, substitution into the form leaves

$$\int d\Lambda(s,t) \left\langle c(\cdot, s, t) U_{-s} B_\beta g, U_{-t} B_\beta h \right\rangle.$$

Cauchy–Schwarz and unitarity prove (ii). An external J acts on the entire first field, giving $JU_{-s}B_\beta g$; it is never commuted past the multiplier or past another operator.

For (iii), the effective weights have the displayed polynomial decay. On dyadic rectangles of sizes $2^j \geq 2^k \geq 1$, their weighted mass is at most $C\epsilon 2^{(M-p_*)j} 2^{-q_* k}$. Summing first in k and then in j, and treating the other order symmetrically, proves the bound and the tail rate. Unit-size rectangles include all axis atoms. If a smooth removed tail ψ_R vanishes below R and equals one above $2R$, its effective weight is bounded by

$$C\epsilon(1+s)^{-p_*}\{\mathbf{1}_{s \geq R/2} + e^{-R/4}\}.$$

Indeed for $s < R/2$ the derivative integral starts at $v \geq R/2$; for $s \geq R/2$ use $|\psi_R'| \leq C/R$. The exponentially small term is smaller than the stated polynomial rate. The same product-rule estimate handles an additional, more remote cutoff. Approximation and the common positive majorants justify all operations for the asserted absolutely continuous coefficients. ∎

Lemma 35 (The chronological corner measures)**.** *For $z, w > 0$ set*

$$C_{\mathrm{can}}(z,w) = (1 - \sqrt{z/w})\mathbf{1}_{z<w}, \qquad L(z,w) = (\sqrt{z/w} - 1)\mathbf{1}_{w<z},$$
$$R_r(z,w) = (r(z) - r(w))\mathbf{1}_{w<z}, \qquad R_r^{[2]}(z,w) = (r(z) - r(w))^2 \mathbf{1}_{w<z},$$

$$H_0(z,w) = (w^{-1/2} - z^{-1/2})\mathbf{1}_{w<z}, \qquad S_r(z,w) = z^{-1/2}(r(z) - r(w))\mathbf{1}_{w<z}.$$

Suppose r is locally absolutely continuous with $|r'(s)| \le C(1+s)^{-1/2}$. Each displayed kernel has a locally finite causal double Abel derivative, dominated by a common positive measure when r varies under this bound. The following exponents are valid in Lemma 34:

base kernel	*Abel square-mass bound for $R \ge 1$*	*Bessel exponent M*
C_{can}, L	CR	2
R_r, S_r	$CR^{3/2}$	5/2
$R_r^{[2]}$	CR^2	3
H_0	$C\sqrt{R}$	3/2
1	CR	2
$w^{-1/2}$ *or* $z^{-1/2}$	$C\sqrt{R}$	3/2
$(zw)^{-1/2}$	C	1

The same bounds hold for transposed kernels. In particular the exponent 3/2 for H_0 is not being assigned to S_r.

Proof. We prove the boundary formulas as well as the variation bounds. For $a(z,w) = D_{A,z}C_{\mathrm{can}}(z,w)$, beta integration gives

$$a(z,w) = \begin{cases} (\pi z)^{-1/2} - \sqrt{\pi}/(2\sqrt{w}), & z < w, \\ -\dfrac{1}{2\sqrt{\pi}} \displaystyle\int_0^w (1 - \sqrt{s/w})(z-s)^{-3/2}\, ds, & w < z. \end{cases}$$

The two values agree at $w = z$, since $\int_0^1 (1-\sqrt{s})(1-s)^{-3/2}\, ds = \pi - 2$. For fixed z, $a(z,\cdot)$ starts at zero, decreases to $(1-\pi/2)/\sqrt{\pi z}$ at $w = z$, and increases to $1/\sqrt{\pi z}$ as $w \to \infty$. Its derivative on $w < z$ is

$$-\frac{1}{4\sqrt{\pi} w^{3/2}} \int_0^w \sqrt{s}(z-s)^{-3/2}\, ds,$$

and on $w > z$ it is $\sqrt{\pi}/(4w^{3/2})$. Both are locally integrable, including at $w = z$. Thus its total variation is $(\pi - 1)/\sqrt{\pi z}$. Convolving that derivative with p_0 proves

$$\int_{[0,R]^2} |D_{A,w} D_{A,z} C_{\mathrm{can}}| \le \frac{4(\pi-1)}{\pi} R.$$

Causal inversion follows from the half-integral semigroup and the zero initial trace. The identity

$$L = \sqrt{z}\, w^{-1/2} - 1 + C_{\mathrm{can}}, \qquad D_{A,z} D_{A,w} L = \tfrac{\pi}{2}\, dz \otimes \delta_0(dw) - \frac{dz\, dw}{\pi\sqrt{zw}} + D_{A,z} D_{A,w} C_{\mathrm{can}} \quad (239)$$

proves the L claim. This edge measure cannot be omitted: for $w < z$, $I_{A,w} L = \sqrt{\pi z} - 2\sqrt{w/\pi}$ has the nonzero initial trace $\sqrt{\pi z}$.

For R_r the ordinary causal mixed derivative is

$$\lambda_r = r'(z)\, dz \otimes \delta_0(dw) - \mathrm{diag}_*(r'(s)\, ds). \quad (240)$$

Double ordinary integration recovers $[r(z)-r(0)]-[r(\min(z,w))-r(0)]$, proving this identity including its sign. Its square variation is $O(\sqrt{1+R})$. Therefore $D_{A,z}D_{A,w}R_r = (p_0\otimes p_0)*\lambda_r$ has square variation $O((1+R)^{3/2})$. Replacing r' by its positive bound constructs a common majorant. Similarly,

$$\partial_z\partial_w R_r^{[2]} = 2(r(z)-r(0))r'(z)\,dz\otimes\delta_0(dw) - 2r'(z)r'(w)\mathbf{1}_{w<z}\,dz\,dw. \tag{241}$$

There is no diagonal atom because the first derivative has zero trace there. This measure has square variation $O(1+R)$, hence its double Abel half integral has growth $O((1+R)^2)$.

For the nonzero-initial corner, first integrate in w. On $w<z$, $I_{A,w}H_0 = \sqrt{\pi} - 2\sqrt{w}/(\sqrt{\pi}\sqrt{z})$, so that

$$D_{A,w}H_0 = \sqrt{\pi}\,\delta_0(dw) + f_0(z,w)\,dw, \qquad f_0(z,w) = -\frac{1-\sqrt{(1-z/w)_+}}{\sqrt{\pi zw}}.$$

The expression for $w\ge z$ follows by differentiating

$$\frac{2}{\sqrt{\pi}}\arcsin\sqrt{z/w} - \frac{2(\sqrt{w}-\sqrt{w-z})}{\sqrt{\pi}\sqrt{z}}.$$

The formulas match continuously at $w=z$ and create no diagonal atom. A second direct integration gives

$$I_{A,z}f_0 = -w^{-1/2} + \frac{\mathfrak{h}(z/w)}{\pi\sqrt{w}}, \qquad \mathfrak{h}(v) = \int_0^{\min(v,1)} \frac{\sqrt{1-t}}{\sqrt{t}\sqrt{v-t}}\,dt.$$

For $v\le 1$ substitution $t=vs$ writes this as $\int_0^1 \sqrt{1-vs}/\sqrt{s(1-s)}\,ds$. Thus $\mathfrak{h}$ decreases from π to 2 on $[0,1]$, and its original formula shows a decrease from 2 to zero thereafter. Differentiation under the nonnegative integrals and Tonelli show $\int_0^\infty |\mathfrak{h}'(v)|\,dv = \pi$; the function is continuous at one, so this accounts for its full variation. Its initial trace cancels that of $-w^{-1/2}$. Consequently

$$D_{A,z}D_{A,w}H_0 = z^{-1/2}\,dz\otimes\delta_0(dw) + \frac{\mathfrak{h}'(z/w)}{\pi w^{3/2}}\,dz\,dw. \tag{242}$$

The interior term is nonpositive and its absolute integral in z is $w^{-1/2}$. Its square variation is at most $2\sqrt{R}$, and the edge contributes exactly $2\sqrt{R}$.

For the divided corner S_r, ordinary integration by parts gives

$$\begin{aligned}\partial_z\partial_w S_r = {} & \left(\frac{r'(z)}{\sqrt{z}} - \frac{r(z)-r(0)}{2z^{3/2}}\right)dz\otimes\delta_0(dw)\\ & - \operatorname{diag}_*\left(\frac{r'(z)}{\sqrt{z}}\,dz\right) + \frac{r'(w)}{2z^{3/2}}\mathbf{1}_{w<z}\,dz\,dw.\end{aligned} \tag{243}$$

Indeed its first w derivative is $(r(z)-r(0))z^{-1/2}\delta_0 - r'(w)z^{-1/2}\mathbf{1}_{w<z}$. The edge coefficient tends to zero at $z=0$, so there is no corner atom. The estimate $|r(z)-r(0)|\le$

$C\min(z,\sqrt{z})$ bounds the variation in a square by $C\sqrt{R}$ for $R \le 1$ and $C(1+\log R)$ for $R \ge 1$. For the bulk term this follows by integrating $z^{-3/2}\int_0^z |r'(w)|\,dw$; the edge and diagonal have the same bounds. Applying p_0 in both coordinates gives square growth $CR^{3/2}$ near zero and $CR(1+\log R) \le CR^{3/2}$ at infinity. This proves the stated safe exponent using only one derivative of r.

Finally the Abel measures for 1, $w^{-1/2}$ and $(zw)^{-1/2}$ are, respectively,

$$\frac{dz\,dw}{\pi\sqrt{zw}}, \qquad z^{-1/2}\,dz \otimes \delta_0(dw), \qquad \pi\delta_{(0,0)}.$$

Causal inversion and (238) finish the proof of every table entry. The explicit formulas also give common positive majorants, rather than merely separate total-variation bounds for individual labels. ∎

C.6 One-source estimates and relative clock moments

Fix a smooth function χ equal to one on $[0,1]$ and zero on $[2,\infty)$, with $0 \le \chi \le 1$, and write $\chi_\delta(z) = \chi(z/L_\delta)$. A retained source is extended by zero after this clock support. Its other half is parametrized by the inward clock at $F_b(\theta)$. A physical-time cutoff which is constant near each rest endpoint and whose first clock derivative is $O((1+z)^{-1})$ on its transition band is also allowed below.

Lemma 36 (One-source head calculus)**.** *Let $Y(t,\theta)$ have a uniform C^2 time norm at most M on the leg, with no assumption on $Y(0,\theta)$ or $Y(T,\theta)$. For $C_g = \rho^4 g(\alpha)$ and either retained head cutoff H, one has*

$$\left\| \int_0^T HC_g Y\,dt \right\|_2 \le CMS_g. \tag{244}$$

The same bound holds componentwise and after multiplication by fixed bounded C^2 radial time factors. If the C^2 time majorant is a label function $d \in L^2$, the left side with the L^1 norm is bounded by $C\|d\|_2 S_g$.

Let $w(t) = K(t)j(t)$ and $A = A_{\mathrm{reg}}$. If $|Y| + |Y_t| \le d(\theta)$ with $d \in L^2$, then

$$\left| \left\langle \int_0^T HwA\rho^4 a(t,\theta) g(\alpha) Y\,dt \right\rangle \right| \le C\epsilon \|d\|_2 S_g, \tag{245}$$

provided the source coefficient, in either inward clock, satisfies

$$|a_\pm(\theta,z)| \le C\epsilon(1+z)^m, \qquad |\partial_z a_\pm(\theta,z)| \le C\epsilon(1+z)^{m-1}, \quad 0 \le m \le 2, \tag{246}$$

and is bounded with bounded first derivative near zero. There is no requirement that the two source coefficients be physical reflections of one another.

Proof. Near rest write $k_4 = a_0(\theta, z)/\sqrt{2z}$. Lemma 33 gives bounded a_0, a_0' and also bounded $t_\theta(z)/\sqrt{2z}$ and its first derivative. Taylor's formula with integral remainder, which needs only Y_{tt}, gives

$$Y(t) = Y(0) + tY_t(0) + R_Y(t), \qquad |R_Y(t)| \le Mt^2/2, \quad |(R_Y)_t(t)| \le Mt.$$

The clock kernel $k_4 Y(t_\theta)$ therefore has an explicit $z^{-1/2}$ leading term, a finite constant term, and an $O(M\sqrt{z})$ remainder with derivative $O(Mz^{-1/2})$. Subtract the appropriate multiple of p and apply (234), retaining the finite initial value of the remainder. At large clocks, $k_4 = O(z^{-6})$, $k_4' = O(z^{-7})$, and $t_\theta' = O(z^{-2})$; all remaining majorants are integrable uniformly through the cutoff. The same Taylor calculation at T proves the negative-half statement. Writing the reflected bare composition exactly as

$$g(F_b(\theta)) = F_b'(\theta)^{-1/2}(Jg)(\theta) \tag{247}$$

keeps its bounded label factor outside J. The one-clock lemma now proves (244); its functional variant with arbitrary bounded angular dual tests proves the L^1 assertion.

For (245) there is a stronger endpoint cancellation:

$$w(t_\theta(z))k_4(\theta, z) = K(t_\theta(z))r_\theta(z)d_\theta(z)/q_\theta(z). \tag{248}$$

It has finite initial value, and its first derivative is bounded by $C(1 + z^{-1/2})$ near zero. The same identity holds on the negative head because $w(T - t) = w(t)$. Multiplication by AY preserves a finite initial value and an integrable derivative: A is bounded at rest, $A_z = O(1 + z^{-1/2})$, and $Y_z = O(dz^{-1/2})$. At infinity $A = O(z)$, $A_z = O(1)$, so $wAk_4 = O(z^{-5})$ and $(wAk_4)' = O(z^{-6})$. Even multiplication by (246) leaves both kernel and derivative integrable. Formula (234) with its finite boundary term, and angular Cauchy–Schwarz with d, prove the claim. ∎

Lemma 37 (Relative comparison of the four moments)**.** *Define*

$$\mathsf{A}_{k,f} g = \int_0^\infty \chi_\delta(z) K_k(\theta, z) U_{-z} g \, dz, \qquad \mathsf{M}_{k,\chi} g = \int_0^\infty \chi_\delta(z) m_k(z) U_{-z} g \, dz.$$

For $0 \le k \le 3$*,*

$$\|\mathsf{A}_{k,f} g - \mathsf{M}_{k,\chi} g\|_2 \le C\delta S_g, \qquad \|\mathsf{A}_{k,f} g\|_2 + \|\mathsf{M}_{k,\chi} g\|_2 \le C S_g. \tag{249}$$

The complete radial moments $\mathsf{M}_k = \int_0^\infty m_k U_{-z} \, dz$ *have the same bound, and for* $R \ge 2$*,*

$$\left\| \int_0^\infty (1 - \chi(z/R)) m_k(z) U_{-z} g \, dz \right\|_2 \le C R^{k-7} S_g. \tag{250}$$

Proof. We spell out the rest-endpoint cancellation required for a relative bound. On a short initial interval let $x(t) = R_\theta q(t/R_\theta)$. The equations used in Lemma 33 also give

$$\rho - x = O(B^2 t^4), \quad \rho_t - x_t = O(B^2 t^3), \quad \widehat{B} - 1 = O(\delta t^2), \quad \widehat{B}_t = O(\delta t).$$

For the first two estimates, $f(\alpha)-f(\theta)=O(B^2t^2)$ and $\ell_+^2/\rho^3=O(B^2t^2)$ are integrated twice in the bounded radial difference equation, from identical initial data. The last two estimates use only $\beta'(\alpha)\alpha_t/B$. Subtracting the corresponding radial normalized equations consequently gives $O(\delta t^2)$ differences with $O(\delta t)$ derivatives. The same statement at critical labels follows directly from (231). Thus the four density differences and the derivatives of their remainders at zero are

k	K_k-m_k	derivative of the displayed remainder
0	$(R_\theta^2-1)(2z)^{-1/2}+O(\delta\sqrt{z})$	$O(\delta z^{-1/2})$
1	$O(\delta z)$	$O(\delta)$
2	$(R_\theta^{-2}-1)\sqrt{2z}+O(\delta z^{3/2})$	$O(\delta\sqrt{z})$
3	$(R_\theta^{-4}-1)2z+O(\delta z^2)$	$O(\delta z)$

(251)

The leading general coefficient is $(2z)^{(k-1)/2}R_\theta^{-2(k-1)}$; for $k=1$ it is exactly one. This derivation controls derivatives of normalized quantities by their equations, rather than by differentiating an unspecified Taylor error.

For $k=0$ subtract $\sqrt{\pi/2}(R_\theta^2-1)p$. For the other three differences no singular term is required. Equations (232) and (251) show that each remaining kernel vanishes at zero and has a common L^1 majorant of size $C\delta$ for its value and derivative. The cutoff derivative is bounded by C/L_δ and preserves this estimate. Apply (234) to prove the first bound. For the model kernels subtract the universal $z^{-1/2}$ term for $k=0$; for $k=1$ retain the finite boundary value $m_1(0)=1$; for $k=2,3$ the boundary value is zero. The radial endpoint and infinity estimates prove the second bound. Finally the tail kernel has zero initial value and $\|k_{\mathrm{tail}}\|_1+\|k'_{\mathrm{tail}}\|_1\le CR^{k-7}$. The same factorization proves (250). ∎

C.7 Two-source calculus on the growing heads

For a real source C let P_C, v_C solve

$$(\partial_t^2+6q^4)P_C=-qC,\qquad (\partial_t^2+6q^4)v_C=jC$$

with zero initial data. On retained sources define

$$\begin{aligned}\mathcal{Q}_A(C)=b^{-1/2}\Big\{&-\frac{1}{2b}\left\langle\int_0^T KC(v_C+jAP_C)\,dt\right\rangle\\&-\frac{\|P_C(T)\|_2^2}{8}+\frac{3\|v_C(T)\|_2^2}{8b^2}-\frac{\langle P_C(T),v_C(T)\rangle}{4b}\Big\}.\end{aligned}\tag{252}$$

Its symmetric polarization is denoted by $\mathcal{Q}_A(C,D)$. The definition with $A=1/q$ is used only where a retained source is present, or for the unitary model defined below. In particular it does not define an uncut actual quotient form at the origin.

Proposition C.1 (Head source and Green bounds)**.** *Let C,D be two-sided retained sources of the form*

$$C=\chi_\delta(Z)\rho^4a_\pm(\theta,Z)g(\alpha),\qquad D=\chi_\delta(Z)\rho^4b_\pm(\theta,Z)h(\alpha),$$

where the coefficients satisfy (246) *with parameters* (ϵ, m) *and* (ϵ', n)*, respectively,* $0 \le m, n \le 2$*. Then*

$$|\mathcal{Q}_A(C,D)| + |\mathcal{Q}_{1/q}(C,D)| \le C\epsilon\epsilon' S_g S_h,$$
$$|\mathcal{Q}_A(C,D) - \mathcal{Q}_{1/q}(C,D)| \le C\delta\epsilon\epsilon' S_g S_h,$$
$$\left|\left\langle \int_0^T wACP_D\, dt \right\rangle\right| \le C\epsilon\epsilon' S_g S_h. \tag{253}$$

Each ordered block obeys the corresponding bound before cancellation between the two halves.

For two time amplitudes with uniform C^2 *norms* M, N*, the field*

$$F(\theta) = \int_{s<t} W(t,\theta)C(t,\theta)G(t,s)D(s,\theta)Y(s,\theta)\, ds\, dt, \quad G(t,s) = j(t)q(s) - q(t)j(s), \tag{254}$$

has $\|F\|_1 \le CMN\epsilon\epsilon' S_g S_h$*. Nonzero values of* W, Y *at either rest endpoint are allowed. The same bound with* $M = 1$ *holds for* $W = wA$*, using its clock bounds* $|W| \le C(1+z)$*,* $|W_z| \le C$ *at large clocks and its integrable* $O(1 + z^{-1/2})$ *initial derivative. In particular a uniform* C^2 *time bound* $N = O(\delta)$ *on* Y *gives a factor* δ *in this last Green estimate.*

Proof. We exhibit all kernel types and their separated coefficients. The proof will also give uniform tail estimates used later. Label subscripts are suppressed on a same-half block; all coefficients in a clock variable include its cutoff and are extended continuously by zero beyond $2L_\delta$. Extend r_θ constantly beyond that point. This preserves the global bound $|r'_\theta(z)| \le C(1+z)^{-1/2}$ needed for the corner measures; no bound on the constant terminal value is required. Causality then reconstructs each kernel exactly on the supported square. On a cross block the outer negative clock z is at $p = F_b(\theta)$ and the inner positive clock w is at θ. The label factor $F_b'(\theta)^{-1/2}$ in (247) is always absorbed into the outer coefficient. On a negative same-half block the exact change of label $\theta = F_b(\eta)$ has Jacobian $F_b'(\eta)$, also kept in a coefficient. None of these label-only factors is differentiated.

First control the actual regularizer. With $\kappa = A - 1/q$, direct algebra gives

$$\kappa = -\frac{x(x-q) + \beta(\theta)^2}{q(x^2 + \beta(\theta)^2)}, \qquad \kappa_t = \frac{q_t}{q^2} - \frac{x_t}{x^2} + \frac{x_t\beta(\theta)^2(3x^2 + \beta(\theta)^2)}{x^2(x^2+\beta(\theta)^2)^2}. \tag{255}$$

On a retained head $|x| \asymp |q|$, $|q| \ge D_0\delta$, and $|x-q| + |x_t - q_t| \le C\delta$. Hence $|\kappa| \le C\delta/|q|^2$ and $|\kappa_t| \le C\delta/|q|^3$. For either the positive coefficient or the negative coefficient $\kappa_-(\theta, z) = \kappa(T - t_\theta(z), F_b(\theta))$, it follows that

$$|\kappa_\pm| \le C\delta(1+z)^2, \quad |(\kappa_\pm)_z| \le C\delta(1+z) \quad (z \ge 1). \tag{256}$$

Near zero they are $O(\delta)$ and have derivative $O(\delta(1 + z^{-1/2}))$. The positive derivative is in fact $O(\delta)$, by the initial rest condition, but that improvement is unnecessary. Since $q(b+u) = K$,

$$q(b+u) = O(1), \quad [q(b+u)]_z = O(z^{-2}),$$
$$q(b+u)^2 = O(z), \quad [q(b+u)^2]_z = O(1) \qquad (z \ge 1).$$

The same estimates with κ replaced by A lose one growth power and remove the factor δ, because $A = 1/q + \kappa$ and δz is bounded on the head.

The regularizer difference is exactly

$$\mathcal{Q}_A(C) - \mathcal{Q}_{1/q}(C) = -\frac{1}{2b\sqrt{b}} \left\langle \int_0^T Kj\kappa C P_C \, dt \right\rangle. \tag{257}$$

For its positive-positive triangle use the exact formula

$$P_C(t) = -q(t) \int_0^t [u(t) - u(s)] q(s)^2 C(s) \, ds.$$

Apart from the fixed prefactor, its kernel is $K\kappa u(z)(u(z) - u(w))K_0(z)K_0(w)\mathbf{1}_{w<z}$. The identity $u = \sqrt{2z}r$, $K_0 = d/\sqrt{2z}$ splits this into

$$[K\kappa r^2 d](z)d(w)L(z,w) + [K\kappa r d](z)d(w)R_r(z,w). \tag{258}$$

In particular the leading corner at rest is $\kappa(0)L/(2\sqrt{b})$, where $\kappa(0) = R_\theta/(R_\theta^2 + \beta(\theta)^2) - 1$. It is generally nonzero. Its initial edge is precisely (239), not an interior derivative with that edge removed.

For the negative-negative triangle put $t = T - \tau$, $s = T - \sigma$. Then $\tau < \sigma$, hence $z < w$, and

$$q(T-\tau) = -q(\tau), \quad j(T-\tau) = q(\tau)(b + u(\tau)), \quad K(T-\tau) = q(\tau)u(\tau). \tag{259}$$

After the exact label change described above, the kernel is

$$[-F_b' q(b+u)\kappa_- r d](z)d(w)\{r(z)C_{\mathrm{can}}(z,w) + (r(w) - r(z))\mathbf{1}_{z<w}\}. \tag{260}$$

The minus sign follows from $u(T-\tau) = -(b + u(\tau))$; the order is the transpose of the positive triangle. This records both the orientation and the reflected measure.

The cross rectangle has the two separated terms

$$\begin{aligned}[q_p(b+u_p)^2\kappa(T-t_p,\theta)r_p d_p](z)[d_\theta/\sqrt{2}](w)w^{-1/2} \\ + [q_p(b+u_p)\kappa(T-t_p,\theta)r_p d_p](z)[r_\theta d_\theta](w).\end{aligned} \tag{261}$$

Indeed before insertion of $K_0(z)K_0(w)$ the radial factor is $q_p u_p(b+u_p)\kappa(T-t_p,\theta)(b+u_p+u_\theta)$. The transported factors are $JU_{-z}g$ and $U_{-w}h$, with the outside factor $F_b'^{-1/2}$ already specified. The following table lists value decay powers of the separated outer and inner coefficients, before source growth; their derivatives have one additional inverse power at infinity and at most an integrable inverse square root at zero:

block/base	p_*	q_*	M
positive/negative canonical	9/2	15/2	2
positive/negative R_r	5	15/2	5/2
cross $w^{-1/2}$	4	15/2	3/2
cross 1	5	7	2

(262)

Every outer coefficient contains δ. For example the slowest coefficient in (261) has powers $1+2+1/2-15/2=-4$; the canonical coefficient has $0+2+1-15/2=-9/2$. The product rule, the clock estimates, and (256) prove all derivative entries. Subtracting $m,n\le 2$ from the corresponding decay powers still leaves them strictly larger than M. Lemmas 34 and 35 now prove the second line of (253). Replacing κ by A increases each outer decay power in the table by one and removes the small factor; this proves its last line.

For the quotient form the two Volterra responses must first be combined. Their exact identity is

$$v_C(t)+\frac{j(t)}{q(t)}P_C(t)=-\frac{1}{q(t)}\int_0^t G(t,s)^2C(s)\,ds. \tag{263}$$

Thus its ordered kernel, apart from its fixed prefactor, is

$$K(t)G(t,s)^2C(t)C(s)/q(t).$$

On the positive triangle its clock form is

$$(b+u(z))(u(z)-u(w))^2K_0(z)K_0(w)\mathbf{1}_{w<z}.$$

The exact useful decomposition is

$$(u(z)-u(w))^2K_0(z)K_0(w)\mathbf{1}_{w<z}=d(z)d(w)\{r(z)^2L(z,w)-r(w)^2C_{\mathrm{can}}(w,z)+R_r^{[2]}(z,w)\}. \tag{264}$$

On the negative triangle the outer radial factor is $-u(z)$, the order is $z<w$, and transposition gives the same three types. The exact cross rectangle is $-u_p(z)(b+u_p(z)+u_\theta(w))^2K_0(p,z)K_0(\theta,w)$, namely

$$\begin{aligned}&-[r_pd_p(b+u_p)^2/\sqrt{2}](z)d_\theta(w)w^{-1/2}\\&\quad-2[r_pd_p(b+u_p)](z)[r_\theta d_\theta](w)-\sqrt{2}[r_pd_p](z)[\sqrt{w}\,r_\theta(w)^2d_\theta(w)].\end{aligned} \tag{265}$$

The associated table is

block/base	p_*	q_*	M
triangle L	$11/2$	$15/2$	2
triangle C_{can}	$13/2$	$13/2$	2
triangle $R_r^{[2]}$	$13/2$	$15/2$	3
cross $w^{-1/2}$	5	$15/2$	$3/2$
cross 1, second term	6	7	2
cross 1, third term	7	6	2

(266)

For a transposed block interchange the variables as necessary. The displayed exact products and (227) prove these entries and their derivative bounds. After either source-growth loss the exponents remain strictly above M. This proves the quotient ordered estimate. Its endpoints are linear combinations of moment densities u^kK_0, $k\le 2$, on the two halves. With source growth their infinity powers are z^{k+m-8}, at worst z^{-4}, and their derivatives

are integrable. At zero subtract the coefficient $a_{\pm}(0)R_{\theta}^{2}/\sqrt{2z}$ for $k=0$; retain the finite initial value for $k=1$; use the vanishing $\sqrt{z}$ term for $k=2$. Formula (234) bounds the endpoint fields by $C\epsilon S_{g}$. This proves the endpoint products and hence the first line of (253).

It remains to verify the nonzero-initial Green assertion. Write $e=d/q$. On the positive triangle, after clock conversion,

$$G(t_{z},t_{w})k_{4}(z)k_{4}(w)\mathbf{1}_{w<z}=\frac{e(z)e(w)}{\sqrt{2}}\{r(z)H_{0}(z,w)+S_{r}(z,w)\}. \tag{267}$$

This follows simply by writing its left side as $e(z)e(w)[r(z)/\sqrt{w}-r(w)/\sqrt{z}]/\sqrt{2}$. The negative triangle has the bracket $r(w)H_{0}(w,z)+S_{r}(w,z)$ and the exact reflected label factor. For the cross rectangle the corresponding expression is

$$\frac{e_{p}(z)e_{\theta}(w)}{2}\left\{\frac{b}{\sqrt{zw}}+\frac{\sqrt{2}r_{p}(z)}{\sqrt{w}}+\frac{\sqrt{2}r_{\theta}(w)}{\sqrt{z}}\right\}. \tag{268}$$

Multiplying these identities by $W(z)Y(w)$ retains separability. For bounded physical-time W,Y, the complete decay table is

block/base	p_{*}	q_{*}	M
positive H_{0}	6	13/2	3/2
positive S_{r}	13/2	13/2	5/2
negative H_{0}	13/2	6	3/2
negative S_{r}	13/2	13/2	5/2
cross $(zw)^{-1/2}$	13/2	13/2	1
cross $w^{-1/2}$	6	13/2	3/2
cross $z^{-1/2}$	13/2	6	3/2

(269)

For $W=wA$, subtract one from every outer exponent in this table. They still exceed their respective M, even after losses of two powers in each source. The time amplitudes contribute derivatives $O(Mz^{-1/2})$, $O(Nz^{-1/2})$ near zero and $O(Mz^{-2})$, $O(Nz^{-2})$ at infinity. This verifies the multiplier hypotheses. For $W=wA$ use its stated clock bounds instead of a uniform physical-time derivative. The edge in (242) and all three terms in (243) are thus explicitly included.

The estimates so far bound the signed angular pairing. Insert an arbitrary bounded measurable dual test $\sigma(\theta)$, $|\sigma|\le 1$, in the outer coefficient throughout. It changes neither its time regularity nor any majorant. Uniform duality proves the L^{1} output bound in (254). At no point has a coefficient been passed through U or J. ∎

Lemma 38 (Negative-head endpoint factors)**.** *On a negative retained head write $t=T-\tau$ and use the inward clock at $p=F_{b}(\theta)$. After the half-density conversion the four source factors KjC_{g}, KqC_{g}, $Kj^{2}AC_{g}$, $KjqAC_{g}$ have the following exact densities, with*

the reflected outside factor from (247) *understood:*

time factor multiplying $\rho^4 g(\alpha)$	*clock density*
Kj	$u(b+u)K_0$
Kq	$-uK_0$
Kj^2A	$[A(T-\tau,\theta)q(\tau)]u(b+u)^2K_0$
$KjqA$	$-[A(T-\tau,\theta)q(\tau)]u(b+u)K_0$

(270)

Each defines an operator of norm at most CS_g *on input* g*. The assertion remains valid after source coefficients* (246)*, with bound* $C\epsilon S_g$*, or multiplication by bounded* C^1 *time amplitudes, with their norm as an additional factor. The positive-head factors* qj, j^2, q^2, jq *have densities* uK_0, u^2K_0, K_0, uK_0*. Their corresponding bound, including the source coefficients* (246)*, holds after multiplication by bounded* C^2 *time amplitudes, with their* C^2 *norm as an additional factor, by the endpoint treatment of Lemma 36.*

Proof. Substitute (259) into each factor and multiply by $k_4 = K_0/q^2$. This proves the table, including both minus signs. The retained coefficient $A(T-\tau,\theta)q(\tau)$ is bounded, with derivative $O(z^{-1})$ at infinity and $O(1+z^{-1/2})$ at zero; this follows either from (255) or from $A = 1/q+\kappa$ on the retained physical half. The slowest density in the table is $O(z^{-5})$, with derivative $O(z^{-6})$. Source growth of order two still leaves integrable z^{-3} decay and derivative. At zero the factor $uK_0 = rd$ cancels the inverse square root exactly. Every negative density therefore has a finite initial value and an integrable first derivative, also after a bounded C^1 time amplitude, whose clock derivative is $O(z^{-1/2})$. Formula (234), with its $k(0)p$ term, proves the negative claim. The positive table follows directly from the definition of K_0 and uses the explicit initial singular term when K_0 itself occurs. ∎

Lemma 39 (Relative same-half chronology)**.** *Let the positive source have density* $\chi_\delta K_0 U_{-z} h$ *after multiplication by* q^2dt*, and reflect this source by the exact unitary* J *on the other half. In the quotient form, replacing this positive density and its physical* u *clock by* $\chi_\delta m_0 U_{-z} h$ *and* $u(t_0(z))$ *changes the sum of its two same-half ordered blocks by at most* $C\delta S_h^2$*.*

Proof. For a scalar signed measure a on the nonnegative u axis, write $A = \int a$, $C_1 = \int ua$, $D_2 = \int u^2 a$, $E_3 = \int u^3 a$, and $H_a(s) = \int (u-s)_+ a(u)$. Direct integration gives

$$|u-v|^3 = u^3 + v^3 - 3uv(u+v) + 12\int_0^{\min(u,v)} (u-s)(v-s)\,ds.$$

Consequently

$$\int_{u>v} (b+u-v)(u-v)^2 a(u)a(v) = b(AD_2 - C_1^2) + AE_3 - 3C_1D_2 + 6\int_0^\infty H_a(s)^2\,ds. \quad (271)$$

There is no diagonal contribution since its polynomial vanishes there. Applying this identity labelwise and then integrating gives its Hilbert-valued version. The exact J disappears from the negative same-half pairing by unitarity; radial reflection leaves precisely the left side of (271). The moment products are already controlled by Lemma 37.

For the remaining energy, set $u_0(z) = u(t_0(z))$ and define

$$H_{f,s}h = \int \chi_\delta(z)(u_\theta(z) - s)_+ K_0(\theta, z) U_{-z} h \, dz,$$
$$H_{0,s}h = \int \chi_\delta(z)(u_0(z) - s)_+ m_0(z) U_{-z} h \, dz.$$

We prove

$$\|(H_{f,s} - H_{0,s})h\|_2 \le C\delta(1+s)^{-5} S_h, \qquad \|H_{0,s}h\|_2 \le C(1+s)^{-6} S_h. \tag{272}$$

Here are the weak-derivative bounds including the moving threshold. Put $E_0 = K_0 - m_0$, $E_1 = u_\theta K_0 - u_0 m_0$. At small clocks, (251) and the inverse clock estimates give

$$|E_0| \le C\delta z^{-1/2}, \quad |E_0'| \le C\delta z^{-3/2}, \quad |E_1| \le C\delta z, \quad |E_1'| \le C\delta,$$
$$|u_\theta - u_0| \le C\delta\sqrt{z}, \qquad |u_\theta' - u_0'| \le C\delta z^{-1/2}.$$

Where both positive parts are active, the kernel difference is $E_1 - sE_0$. For small $s > 0$ activity requires $z \ge cs^2$; integration of $C\delta(1+sz^{-3/2})$ over that region is $O(\delta)$. The radial threshold z_s has size s^2, and the two thresholds can differ only in $|z - z_s| \le C\delta s^2$. In that strip the kernel difference is $O(\delta)$ and its weak derivative is $O(s^{-2})$, hence its value and derivative have a common integral bound $C\delta$. Both positive-part kernels vanish at their thresholds, so these first derivatives have no threshold atom. At $s = 0$ the difference is E_1 with zero initial value.

For $z \ge 1$ the estimates are

$$|E_0| \le C\delta z^{-7}, \quad |E_0'| \le C\delta z^{-8}, \quad |E_1| \le C\delta z^{-6}, \quad |E_1'| \le C\delta z^{-7},$$
$$|u_\theta - u_0| \le C\delta z^2, \qquad u_0 \asymp z, \qquad u_0' \asymp 1.$$

For $s \ge 1$, the common active region begins at $z \ge cs$; there the L^1 bounds for $E_1 - sE_0$ and its derivative are $C\delta s^{-5}$ and $C\delta s^{-6}$. The one-active region is contained in $|z - z_s| \le C\delta s^2$, $z_s \asymp s$; its kernel is $O(\delta s^{-6})$ and its derivative $O(s^{-8})$. Since δs is uniformly small on a nonempty supported strip, these imply the same integral bounds. The cutoff derivative adds only its controlled $O(\delta)$ factor. Thus the difference kernel has zero initial value and a common $W^{1,1}$ bound $C\delta(1+s)^{-5}$. The model kernel has

$$\|\chi_\delta(u_0 - s)_+ m_0\|_1 + \|[\chi_\delta(u_0 - s)_+ m_0]'\|_1 + |[(u_0 - s)_+ m_0](0)| \le C(1+s)^{-6}.$$

Its initial value is one at $s = 0$ and zero for $s > 0$; its variation remains bounded as a small positive threshold approaches zero. Applying (234), with that boundary value, proves (272). Square integration in s gives an $O(\delta S_h)$ difference of the tail fields and an $O(S_h)$ bound for each. Their energies therefore differ by $O(\delta S_h^2)$. This and the moment estimates prove the lemma. All identities hold first for bounded inputs; the common majorants extend them by density to arbitrary L^2 inputs. ∎

C.8 The retained principal form and the unitary tails

Define the full radial-clock model by the Hilbert-valued source

$$\mathscr{C}_0(t) = q(t)^4 U_{-z_0(t)} h \quad (0 < t < Q), \qquad \mathscr{C}_0(T-t) = q(t)^4 J U_{-z_0(t)} h. \tag{273}$$

At Q it has the continuous L^2 extension zero. Let P_0, v_0 be its radial zero-data responses, as in (252), and set $\mathcal{Q}_{\mathrm{mod}}(h) = \mathcal{Q}_{1/q}(\mathscr{C}_0)$. This last expression is well defined: $\|\mathscr{C}_0(t)\|_2$ vanishes as $|q|^4$ and the radial responses are bounded, so the single quotient is integrable. The cut model $\mathcal{Q}_{\mathrm{mod},R}$ inserts $\chi(z/R)$ in each inward source slot, keeping J exactly.

Proposition C.2 (Retained-head comparison)**.** *Let $h = f''$ and suppose the normalization supplies $\|f\|_{H^1} \le Ca$. Write*

$$\mathscr{B}(t,\theta) = E_\theta \cdot D^2\big(|Q|^6 f(\arg Q)\big)(Q_f(t,\theta))E_\theta.$$

Let $\mathscr{B}_{\mathrm{clk}}$ be its cut by χ_δ on each actual inward head, with the original launch frame on the second half. Then

$$|\mathcal{Q}_A(\mathscr{B}_{\mathrm{clk}}) - \mathcal{Q}_{\mathrm{mod}}(h)| \le C(a^2 + aS_h + \delta S_h^2). \tag{274}$$

Only retained actual heads occur on the left before the unitary model is restored. More precisely, uniformly in $R \ge 2$,

$$|\mathcal{Q}_{\mathrm{mod}}(h) - \mathcal{Q}_{\mathrm{mod},R}(h)| \le CR^{-7/2} S_h^2. \tag{275}$$

Let η be smooth, zero on $[0,1]$ and one on $[2,\infty)$. For a fixed sufficiently large D put $r = D\delta$ and $H_r(t) = \eta(|q(t)|/r)$. Then

$$|\mathcal{Q}_A(\mathscr{B}_{\mathrm{clk}}) - \mathcal{Q}_A(H_r\mathscr{B})| \le C_D \delta^{7/2}(S_h + a)^2. \tag{276}$$

This compares two legal retained-head forms; it makes no assertion here about adding the missing physical-origin source.

Proof. We first compare the source and frame. Put $s = \sin(\alpha - \theta)$, $c_1 = \cos(\alpha - \theta)$. The exact radial, mixed and tangential entries of the perturbing Hessian in the moving polar frame are $30\rho^4 f$, $5\rho^4 f'$, and $\rho^4(6f + h)$. Therefore

$$\mathscr{B} = \rho^4\{(1 - s^2)h(\alpha) + (6 + 24s^2)f(\alpha) + 10sc_1 f'(\alpha)\}. \tag{277}$$

On the positive clock $\alpha_z = -\beta(\alpha)$; on the negative clock this holds with initial label $F_b(\theta)$, while the frame angle remains the original θ. Lemma 28 gives $|\sin(F_b(\theta) - \theta)| \le C|\beta(\theta)|$ and $|\beta(F_b(\theta))| \le C|\beta(\theta)|$. The flow equation and $\|\beta'\|_\infty \le \delta$ give, on both heads,

$$\begin{aligned} |s| &\le C\delta(1+z), \qquad & |s_z| + |(c_1)_z| &\le C\delta, \\ |s^2| &\le C\delta^2(1+z)^2, & |(s^2)_z| &\le C\delta^2(1+z), \\ |sc_1| &\le C\delta(1+z), & |(sc_1)_z| &\le C\delta. \end{aligned} \tag{278}$$

At critical labels this is direct. Hence the difference from the cut principal scalar source $C_{h,\mathrm{clk}} = \chi_\delta \rho^4 h(\alpha)$ has response size at most $C(a + \delta^2 S_h)$ in the source classes of Proposition C.1. Polarization yields

$$|\mathcal{Q}_A(\mathscr{B}_{\mathrm{clk}}) - \mathcal{Q}_A(C_{h,\mathrm{clk}})| \le C\{S_h(a + \delta^2 S_h) + (a + \delta^2 S_h)^2\} \le C(aS_h + a^2 + \delta^2 S_h^2). \quad (279)$$

The original launch frame has been retained throughout.

The regularizer comparison in (253) costs $C\delta S_h^2$. The scalar source has exact bare brake reflection. Replacing that bare reflection by J changes only a bounded label-only multiplier, of size $O(\delta)$, on the negative source. The same polarized head bound, now for $\mathcal{Q}_{1/q}$, proves another $C\delta S_h^2$ cost. Thus it remains to compare a source with its actual positive half and its exact J reflection to the cut model.

Here are all the endpoint and cross-half identities used in this comparison. If the four positive-half moments are A_0, C_1, D_2, E_3, then

$$P(T) = -bA_0 - (1 - J)C_1, \qquad v(T) = (1 + J)(bC_1 + D_2). \quad (280)$$

These follow by inserting $G(T, t) = K(t)$ in the two zero-data responses and using (259). The cross-half ordered contribution, inside the overall factor $b^{-1/2}$ and including its coefficient $1/(2b)$, is

$$-\frac{1}{2b}\{b^2\langle JC_1, A_0\rangle + 2b\langle JD_2, A_0\rangle + 2b\langle JC_1, C_1\rangle + \langle JE_3, A_0\rangle + 3\langle JD_2, C_1\rangle\}. \quad (281)$$

Indeed its unsymmetrized kernel is $-u(z)(b + u(z) + u(w))^2$ against the two $q^2 C\, dt$ measures. Using $J^* = J$ symmetrizes it to $-\frac{1}{2}(u + v)(b + u + v)^2$; expansion gives exactly the five displayed terms. Lemma 37 bounds the change of each moment by $C\delta S_h$, while every moment is $O(S_h)$. Thus all endpoints and the cross rectangle change by $C\delta S_h^2$. The two remaining triangles are exactly Lemma 39. This proves comparison with the cut model at scale L_δ.

To restore the full model, use the radial versions of r, d on the whole half-line. Their bounds and the causal corner measures used in (266) are global. Extend no actual flow past its retained head. Split the removed two-slot cutoff as

$$1 - \chi(z/R)\chi(w/R) = (1 - \chi(z/R)) + \chi(z/R)(1 - \chi(w/R)).$$

For the three triangle rows of (266), the margins $\min(p_* - M, q_* - M)$ are $7/2, 9/2, 7/2$; for its three cross rows they are $7/2, 4, 4$. Lemma 34(iii) therefore gives $CR^{-7/2}S_h^2$ for the ordered tails. Moment tails through order three cost $CR^{-4}S_h^2$ in their products by (250); endpoint moments alone cost $CR^{-5}S_h^2$. This proves (275). Taking $R = L_\delta$ and combining the preceding estimates proves (274). Every restored integral is unitary; no infinite bare-flow integral has been formed.

Finally choose D so large that $H_r > 0$ implies a legal actual head with $z \le L_\delta$. The two cutoffs then differ only in a tail $z \ge c_D/\delta$ inside $z \le 2L_\delta$. Their difference

$d_{\mathrm{cut}} = \chi(z/L_\delta) - \eta(q_\theta(z)/r)$ has $|d_{\mathrm{cut}}| \le C$, $|(d_{\mathrm{cut}})_z| \le C/(1+z)$: on the transition of η, $q_\theta \asymp r$, $z \asymp r^{-1}$, and $|q'_\theta|/r \le C/z$. For two source-growth orders $m, n \in \{0,1,2\}$ in (277), the quotient tail bound is

$$C\delta^{m+n} R^{-\min(p_*-M-m, q_*-M-n)}, \qquad R \asymp \delta^{-1}.$$

It is bounded by $C\delta^{\min(p_*-M, q_*-M)}$, hence by $C\delta^{7/2}$. The regularizer table (262) has minimal margin $5/2$ and an additional factor δ, giving the same result. Endpoint tails are at most $C\delta^5$. Polarization, with at least one slot carrying d_{cut}, proves (276). ∎

C.9 Bounded ratio reversal under the actual involution

Lemma 40 (Actual reversal of the bounded second-moment ratio)**.**
Let $N_2 = (a_2/a_0)(A_\beta)$, with the bounded ratio and signed two-sided kernel supplied by Lemma 27. For the actual brake involution,

$$\|JN_2J - N_2^*\|_{L^2\to L^2} \le C\delta. \tag{282}$$

More generally the same conclusion holds for any real two-sided kernel k which is C^1 on the two open half-lines, has a finite jump at zero, and satisfies $|k(t)| + |k'_{\mathrm{reg}}(t)| \le C(1+|t|)^{-4}$ there.

Proof. The actual conjugated flow $\psi_t = F_b \circ \phi_t \circ F_b$ is C^1 as a map, with half-density group $W_t = JU_tJ$. Its velocity is the continuous function

$$\gamma(\theta) = \frac{\beta(F_b(\theta))}{F_b'(\theta)} = -\beta(\theta) + O(\delta|\beta(\theta)|).$$

This formula follows by differentiating only in time and using $F_b'(F_b(\theta))F_b'(\theta) = 1$. We do not differentiate F_b' or assert that γ is C^1. For a noncritical label, both trajectories stay in the same noncritical interval, and there is an exact increasing clock

$$\psi_t(\theta) = \phi_{-\sigma_\theta(t)}(\theta), \qquad \sigma_\theta(0) = 0, \qquad \sigma'_\theta(t) = -\frac{\gamma(\psi_t(\theta))}{\beta(\psi_t(\theta))} = 1 + O(\delta). \tag{283}$$

This holds for positive and negative t. The impossibility of reaching a zero of the Lipschitz slope field in finite flow time, or the explicit conjugation, justifies the full noncritical trajectory. At a critical label both position flows fix that label, and we choose $\sigma_\theta(t) = t$. The resulting convention is measurable and suffices for the change of variable.

The map derivatives, computed without differentiating the new clock in the label, are

$$\psi_t'(\theta) = F_b'(\phi_t(F_b(\theta)))\phi_t'(F_b(\theta))F_b'(\theta).$$

Since $F_b' = 1 + O(\delta)$ and $e^{-\delta|t|} \le \phi_t' \le e^{\delta|t|}$, for $s = \sigma_\theta(t)$ one has

$$\left|\sqrt{\psi_t'(\theta)} - \sqrt{\phi'_{-s}(\theta)}\right| \le C\delta(1+|s|)e^{C\delta|s|}. \tag{284}$$

Indeed $|t| \asymp |s|$, and each of the two slope-flow derivatives is within $C\delta|t|e^{\delta|t|}$ or $C\delta|s|e^{\delta|s|}$ of one. The critical convention satisfies the same estimate using the displayed map derivatives.

It suffices to estimate $JN_2^*J - N_2$. For $\delta > 0$ truncate both unitary integrals to $|t| \le L = \delta^{-1}$. Each omitted integral has norm at most CL^{-3}. For a bounded input the first retained integral is, pointwise,

$$\int_{-L}^{L} k(t)\sqrt{\psi_t'(\theta)}h(\psi_t(\theta))\,dt.$$

Change to $s = \sigma_\theta(t)$ on each open half-line. Its kernel at $h(\phi_{-s}(\theta))$ is

$$\frac{k(t_\theta(s))\sqrt{\psi'_{t_\theta(s)}(\theta)}}{\sigma_\theta'(t_\theta(s))}.$$

The inverse clock has derivative $1 + O(\delta)$ and $|t_\theta(s) - s| \le C\delta|s|$. It preserves the sign of s. The half-line derivative bound on k therefore gives

$$\left|\frac{k(t_\theta(s))}{\sigma_\theta'(t_\theta(s))} - k(s)\right| \le C\delta(1+|s|)^{-3}.$$

In particular the jump at zero is not moved across zero, so it introduces no extra interior term. Multiplication by (284) has an integrable majorant of size $C\delta(1+|s|)^{-3}$ on the truncated window. The two terminal strips have length $O(\delta L)$ and kernel size $O(L^{-4})$, hence total mass $O(\delta L^{-3})$.

For $|s| \le CL$, the bare pullback norm is at most $e^{\delta|s|/2}$, a uniform constant. Minkowski with the universal integrable majorants proves an operator error $C\delta$ on this finite window. The two discarded tails are estimated only as unitary integrals. Since $L^{-3} = \delta^3$, the required bound follows. The case $\delta = 0$ is exact. Taking adjoints proves (282) in its stated orientation.

In particular this estimate acts on the already smoothed vector $A_0 = \mathsf{M}_0 h$. If $e = (1+J)A_0$ and $D_2 = \mathsf{M}_2 h = N_2 A_0$, then the exact ordered identity is

$$JD_2 + N_2^* A_0 = N_2^* e + (JN_2J - N_2^*)JA_0. \tag{285}$$

Thus its reversal error is $C\delta\|A_0\|_2$, not an unweighted error applied to h. Neither an inverse of M_0 on arbitrary data nor a reversal identity for the unbounded generator has been used. ∎

C.10 Whole-leg source calculus and the buffered comparison

For an angular input g, write $C_g(t,\theta) = \rho^4 g(\alpha)$. In particular $C_h = \rho^4 h(\alpha)$ retains the full actual source. The following consequences join the retained-head calculus to the physical origin; the resulting constants are independent of Fourier support.

Lemma 41 (Whole-leg source estimates)**.** *For a time amplitude with $\|Y(\cdot,\theta)\|_{C^2} \le M$ uniformly,*

$$\left\| \int_0^T C_g Y \,\mathrm{d}t \right\|_2 \le C M S_g. \tag{286}$$

If its C^2 majorant is instead $d(\theta) \in L^2$, the same integral has angular L^1 norm at most $C\|d\|_2 S_g$. If W, Y have uniform C^2 bounds M, N, respectively, then

$$\left\| \int_{s<t} W(t) C_g(t) G_E(t,s) C_{g_1}(s) Y(s) \,\mathrm{d}s \,\mathrm{d}t \right\|_1 \le C M N S_g S_{g_1}. \tag{287}$$

These statements hold componentwise for fixed radial matrix kernels. For the full fixed-frame Hessian $B = E_\theta \cdot D^2(|Z|^6 f)(Q_f)E_\theta$, its one-source endpoint integrals have norm $C(S_h + a)$. Moreover, with $w = Kj$,

$$\left| \left\langle \int_0^T w A_{\mathrm{reg}} B Z_a \,\mathrm{d}t \right\rangle \right| \le C\|d\|_2 (S_h + a), \tag{288}$$

$$\left| \left\langle \int_{s<t} w(t) A_{\mathrm{reg}}(t) B(t) G_E(t,s) B(s) Y(s) \,\mathrm{d}s \,\mathrm{d}t \right\rangle \right| \le C M (S_h + a)^2, \tag{289}$$

provided $\|Z_a(\cdot,\theta)\|_{C^1} \le d(\theta)$ in the first estimate and $\|Y\|_{C^2} \le M$ uniformly in the second.

Proof. Choose $r = D\delta$ and a buffered partition

$$H_r = \eta(|q|/r), \quad Z_r = 1 - \eta(2|q|/r), \quad W_r = \eta(2|q|/r) - \eta(|q|/r). \tag{290}$$

Then $H_r + W_r + Z_r = 1$. The deep support $|q| < r$ of Z_r is disjoint from the exterior support $|q| > r$ of H_r. The transition W_r lies in the legal retained-head clock range, and $W_r + Z_r$ lies in a fixed physical-time layer of radius Cr. Derivatives of its cutoff amplitudes are $O(1/r)$ only where $\rho \asymp r$.

For (286), the heads use the one-source estimate of Lemma 36, including the initial inverse-square-root coefficient and its finite causal value. The layer uses Lemma 31 with $p = 4, k = 0$, at cost $Cr^5 M S_g$. For a label majorant divide by d before applying the uniform estimate and then use angular Cauchy–Schwarz. This yields L^1, not an L^2 multiplication assertion.

For (287), the two retained heads use the nonzero-initial causal kernels of Lemma 35. The pairs of coefficient decay powers and their safe Bessel measure exponents are

$$(6, 13/2; 3/2),\ (13/2, 13/2; 5/2),\ (13/2, 6; 3/2),\ (13/2, 13/2; 5/2),$$

$$(13/2, 13/2; 1),\ (6, 13/2; 3/2),\ (13/2, 6; 3/2).$$

Both coefficient powers exceed the indicated measure exponent. A two-layer term is bounded by $Cr^{10} M N S_g S_{g_1}$ by Lemma 32. In a deep-layer/exterior pair chronology is fixed, since the positive exterior precedes the deep layer and the negative exterior follows it. The identity $G_E(t,s) = j(t)q(s) - q(t)j(s)$ separates it into two products of complete

one-source integrals, one controlled by the head estimate and one by the r^5 signed-layer estimate. The transition remains in the head calculation. Inserting an arbitrary bounded angular dual test in W proves the stated L^1 norm.

For B, use (218); its inputs are h, f, f', whose response norms sum to $S_h + Ca$. Their frame coefficients meet the retained-head bounds and the signed-layer amplitude bounds. In (288), $w = Kj$ cancels the initial inverse-square-root density at both rest ends. Thus a C^1 time amplitude suffices: the kernel has a finite initial value, an integrable inverse-square-root first derivative, and far-clock decay z^{-5} before the two allowed polar growth powers. The label-majorized one-source estimate gives $C\|d\|_2(S_h + a)$ on the heads. On the origin use Lemma 31 with $p = 4, k = 1$ and its L^1 version.

For (289), the actual outer multiplier wA_{reg} has far-clock size $O(1 + z)$ and first derivative $O(1)$, not a uniform physical-time C^1 bound. The corresponding head powers are

$$(5, 13/2; 3/2),\ (11/2, 13/2; 5/2),\ (11/2, 6; 3/2),\ (11/2, 13/2; 5/2),$$
$$(11/2, 13/2; 1),\ (5, 13/2; 3/2),\ (11/2, 6; 3/2).$$

They still exceed the measure exponents after the at most two polar growth powers in either slot; the coefficients carrying that growth also carry the corresponding powers of δ. The amplitude Y supplies its factor M and the required first clock derivative, including a nonzero initial value. A two-layer term has bound $CMr^9(S_h + a)^2$. For the two disjoint mixed rectangles use

$$w(t)A_{\mathrm{reg}}(t)G_E(t, s)Y(s) = [wA_{\mathrm{reg}}j](t)[qY](s) - [wA_{\mathrm{reg}}q](t)[jY](s).$$

If the outer factor is in the layer, its norm is $Cr^4(S_h+a)$ and the positive-head inner norm is $CM(S_h+a)$. If the outer factor is in the negative head, it is one of the densities explicitly checked in Lemma 38; the inner layer has norm $CMr^5(S_h + a)$. These are products of complete time integrals, so angular Cauchy–Schwarz is legitimate. This proves the last assertion. ∎

Proposition C.3 (Full principal comparison)**.** *Let P_B, v_B solve $\mathscr{L}P_B = -qB$ and $\mathscr{L}v_B = jB$ with zero data, and define*

$$\mathcal{Q}_{\mathrm{reg}}(f) = b^{-1/2}\Big\{ -\frac{1}{2b}\Big\langle \int_0^T KB(v_B + jA_{\mathrm{reg}}P_B)\,\mathrm{d}t \Big\rangle - \frac{1}{8}\|P_B(T)\|_2^2 + \frac{3}{8b^2}\|v_B(T)\|_2^2 - \frac{1}{4b}\langle P_B(T), v_B(T)\rangle \Big\}. \tag{291}$$

For the coherent model $\mathcal{Q}_{\mathrm{mod}}$ defined by the radial clock and the actual involution J, one has

$$|\mathcal{Q}_{\mathrm{reg}}(f) - \mathcal{Q}_{\mathrm{mod}}(h)| \le C(a^2 + aS_h + \delta S_h^2). \tag{292}$$

Proof. First cut the two actual source slots with H_r from (290). Terms with both slots in $W_r + Z_r$ are bounded by $Cr^9(S_h + a)^2$ when a regularizer occurs and by $Cr^{10}(S_h + a)^2$ otherwise. Endpoint products use the single-layer r^5 bound. For a transition/exterior pair both slots remain in the head clock range. The weighted causal tail estimates give $Cr^{7/2}(S_h + a)^2$. This power includes polar growth: a coefficient δ^{m+n} multiplying tail powers $p - m, q - n$ contributes

$$\delta^{m+n} r^{\min(p-M-m,q-M-n)} = C_D r^{\min(p-M+n,q-M+m)},$$

where M is the base measure exponent. The quotient margins are at least $7/2$; regularizer margins are at least $5/2$ with an extra $\delta \asymp r$. Endpoint margins are at least five.

For the disjoint deep/exterior rectangles, the complete chronological kernel has the exact four-product decomposition

$$\begin{aligned} &K(t)B(t)G_E(t,s)[j(s) - j(t)A_{\mathrm{reg}}(t)q(s)]B(s) \\ &= [KjB](t)[qjB](s) - [KqB](t)[j^2B](s) \\ &\quad - [Kj^2A_{\mathrm{reg}}B](t)[q^2B](s) + [KjqA_{\mathrm{reg}}B](t)[jqB](s). \end{aligned}$$

On the positive exterior head the inner densities are respectively uK_0, u^2K_0, K_0, uK_0, with the actual density K_0, not its radial replacement. On the negative exterior put $t = T - \tau$ and use inward q, j, u; the outer densities are

$$u(b+u)K_0, \quad -uK_0, \quad [A_{\mathrm{reg}}(T-\tau)q]u(b+u)^2K_0, \quad -[A_{\mathrm{reg}}(T-\tau)q]u(b+u)K_0.$$

They have the finite initial values and integrable derivative bounds proved in Lemma 38; their norms are $C(S_h + a)$. The outer layer factors have norm at most $Cr^4(S_h + a)$, and inner layer factors at most $Cr^5(S_h + a)$. Thus these rectangles cost at most $Cr^4(S_h + a)^2$. This establishes a total source-cut cost $Cr^{7/2}(S_h + a)^2$.

The physical-time to clock-cut change is another legal head tail with the same bound. On the retained heads the source/frame comparison, actual regularizer comparison, relative moments and ordered same-half calculus of Proposition C.2 cost $C(a^2 + aS_h + \delta S_h^2)$. Bare reflected composition is changed to J with its exact bounded label coefficient, whose deviation is $O\delta$; no derivative of F_b' is taken. Finally restore only the unitary model tails. The causal tail estimates give $C\delta^{7/2}S_h^2$. Since $r = D\delta$ and $\delta \le 1$, these costs prove (292). The actual regularizer was never replaced by $1/q$ inside the physical origin layer. ∎

D The full sextic boundary response and finite classification

We retain the notation of Sections A–C. In particular $h = f''$, $\beta = f'$, $a = \|o'\|_2$, $S_h = \|B_\beta h\|_2$, and every actual brake leg has the same time T. Write $\mathcal{F}(Z) = |Z|^6 f(\arg Z)$

and

$$D = D^2V_f(Q_f) - D^2V_0(qe_\theta), \qquad C_f = \rho^4 h(\alpha), \qquad \Pi = E_\theta \otimes E_\theta.$$

All matrices below are expressed in the fixed launch frame (e_θ, E_θ). Subscripts R, E denote their radial and transverse components, not derivatives.

D.1 The full lower matrix and the weak first mean

Lemma 42 (A pointwise lower-order majorant)**.** *Let*

$$L^{\mathrm{mat}} = D - C_f\Pi, \qquad d_0(\theta) = |f(\theta)| + |f(\theta+\pi)| + |\beta(\theta)|.$$

Then $\sup_t |L^{\mathrm{mat}}(t,\theta)| \le Cd_0(\theta)$, $\|d_0\|_2 \le Ca$, *and* $\|d_0\|_\infty \le C\delta$. *The same pointwise bound holds for* $c = [D^2V_0(Q_f) - D^2V_0(qe_\theta)]_{EE}$ *and* $B - C_f$, *where* $B = [D^2\mathcal{F}(Q_f)]_{EE}$. *If a field* Z *is uniformly bounded, the zero-data radial response to* $-L^{\mathrm{mat}}Z$ *has a* C^2*-time majorant* $Cd_0\|Z\|_\infty$.

Proof. For $\rho > 0$, choose the closer of $\theta, \theta+\pi$, denoted θ_σ. Lemma 28 gives

$$\mathrm{dist}(\alpha, \theta_\sigma) \le C\min(1, |\beta(\theta)|/\rho), \qquad |\beta(\theta_\sigma)| \le C|\beta(\theta)|.$$

The Lipschitz bounds for f and β imply

$$\rho^4|f(\alpha)| \le C\{|f(\theta)| + |f(\theta+\pi)| + \delta|\beta(\theta)|\rho^3\} \le Cd_0, \qquad \rho^4|\beta(\alpha)| \le C|\beta(\theta)|.$$

In (218), the remaining difference from $C_f\Pi$ is $\rho^4 h(\alpha)(E_\alpha \otimes E_\alpha - \Pi)$. Its norm is at most $C\delta|\beta(\theta)|\rho^3$. The radial polynomial Hessian is Lipschitz on the bounded orbit region and $|Q_f - qe_\theta| \le Cd_0$ by the line comparison. This proves all matrix bounds, including $B - C_f = L^{\mathrm{mat}}_{EE} - c$. They extend continuously through a critical crossing. The response bound follows first in C^1 from the bounded radial propagator and then in C^2 from its continuous-source equation. No derivative of L^{mat} is asserted. The norm bounds use Lemma 22. ∎

Let $X = \partial_\theta Q_f$ be the actual launch-angle Jacobi field, expressed in the fixed launch frame, and let $R = Q_f - xe_\theta$. Define

$$P_* = X_E - x = E_\theta \cdot \partial_\theta R, \qquad Z_* = X_R - x_\theta = e_\theta \cdot \partial_\theta R.$$

The field Q_f is C^1 in the label, and X is continuous there. Only time derivatives of X will be used.

Lemma 43 (Weak actual-Hessian identity)**.** *For every bounded time-only weight* w,

$$\left\langle \int_0^T wB\,\mathrm{d}t \right\rangle = -\left\langle \int_0^T wA_{\mathrm{reg}}[BP_* + (D^2\mathcal{F}(Q_f))_{ER}Z_*]\,\mathrm{d}t \right\rangle + O(\|w\|_\infty a^2). \tag{293}$$

Moreover $\|Z_*\|_{L^2_\theta C^1_t} \le Ca$ *and* $\|\int |A_{\mathrm{reg}}(D^2\mathcal{F})_{ER}|\,\mathrm{d}t\|_2 \le Ca$. *The mean radial-polynomial term obeys*

$$\left|\left\langle \int_0^T w(t)c(t,\theta)\,\mathrm{d}t \right\rangle\right| \le C\|w\|_\infty a^2. \tag{294}$$

Proof. The line equation differentiated once in θ gives $\|x_\theta\|_{C^1} \le C|\beta(\theta)|$; its coefficient is $30c_{\operatorname{sgn} x}|x|^4$ and its forcing is $-6f'_{\operatorname{sgn} x}x^5$, continuous across zero. For $\epsilon > 0$ set $A_\epsilon = x/(x^2+\beta^2+\epsilon^2)$. Direct differentiation gives

$$\partial_\theta A_\epsilon = \frac{x_\theta(\beta^2+\epsilon^2-x^2) - 2x\beta\beta'}{(x^2+\beta^2+\epsilon^2)^2}, \qquad |\partial_\theta A_\epsilon|\rho^5 \le C|\beta|(\rho^3+\delta\rho^2).$$

These weighted estimates also hold on the critical set. Put $H = D^2\mathcal{F}(Q_f)$ and $P = \nabla\mathcal{F}(Q_f)$. The ordinary chain rule, requiring only $\mathcal{F} \in C^2$, is $E_\theta \cdot HX = \partial_\theta(E_\theta \cdot P) + e_\theta \cdot P$. Using $X = xE_\theta + x_\theta e_\theta + \partial_\theta R$ and integrating in the label yields the exact identity

$$\langle B\rangle = \big\langle A_\epsilon e_\theta \cdot P - (\partial_\theta A_\epsilon)E_\theta \cdot P - A_\epsilon x_\theta H_{ER} - A_\epsilon E_\theta \cdot H\partial_\theta R + (1 - A_\epsilon x)B\big\rangle.$$

The interval $[0,T]$ is fixed, so multiplication by $w(t)$ introduces no moving-endpoint label term.

Put $k = |f| + |f'|$. Since $|P| \le C\rho^5 k(\alpha)$, the derivative term is bounded after positive time integration by $C\langle|\beta|[\mathcal{T}_3 k + \delta\mathcal{T}_2 k]\rangle \le Ca^2$. The polar formula and line geometry give

$$|H_{ER}| \le C\rho^4|\beta(\alpha)| + C\delta|\beta(\theta)|\rho^3.$$

Thus the x_θ term is at most $C\langle|\beta|\mathcal{T}_3|\beta|\rangle + C\delta\langle\beta^2\rangle$, and the residual term is at most $C\delta(a^2+\epsilon^2)$. The same estimate proves the stated positive integral bound for $A_{\mathrm{reg}}H_{ER}$. The radial row of the Jacobi equation has source $-D_{RR}X_R - D_{RE}X_E$ and initial radial value R'_θ; here $R_\theta = (1+f(\theta))^{-1/6}$. Lemma 42 and bounded propagation give $\|X_R\|_{L^2C^1} \le Ca$, and hence the claimed bound on Z_*.

It remains to estimate the radial-force term. Write $B_0 = |\beta(\theta)|$ and first work where $|x| \ge L_0B_0$, for a fixed large L_0. The actual angle is within $CB_0/|x|$ of θ_σ. The elementary short-arc integral bound gives

$$|f(\alpha) - f(\theta_\sigma)| \le C(B_0/|x|)\mathcal{M}|f'|(\theta_\sigma).$$

The polar gradient formula and $|\rho - |x|| \le CB_0$ consequently yield

$$|e_\theta \cdot [\nabla\mathcal{F}(Q_f) - \nabla\mathcal{F}(xe_\theta)]| \le CB_0\rho^4\big[|f(\theta_\sigma)| + \mathcal{M}|f'|(\theta_\sigma) + |f'(\alpha)|\big]. \tag{295}$$

Multiply by $|A_{\mathrm{reg}}| \le C/\rho$ and use the positive kernels and angular Cauchy–Schwarz to obtain Ca^2. On $|x| \le L_0B_0$, the radius is at most CB_0 and $|A_{\mathrm{reg}}| \le 1/(2B_0)$. The pointwise small-radius kernel gives an actual force integral at most $CB_0^5[\mathcal{M}k(\theta) + \mathcal{M}k(F_b(\theta))]$. The line contribution is at most $CB_0^5(|f(\theta)| + |f(\theta+\pi)|)$, since its crossing speed is bounded below. Their means are $C\delta^4a^2$. At $B_0 = 0$ the two paths agree and the inner set is the single crossing instant.

On the reference line the remaining expression is exactly

$$A_{\mathrm{reg}}e_\theta \cdot \nabla\mathcal{F}(xe_\theta) = 6|x|^4 f(\theta_\sigma) - \frac{6\beta^2|x|^4 f(\theta_\sigma)}{x^2+\beta^2}.$$

The second term has mean integral $O(\delta a^2)$. Replacing x by q in the first costs Ca^2, since the signed piecewise function $|x|^4 f(\theta_{\operatorname{sgn} x})$ is Lipschitz with constant $C(|f(\theta)| + |f(\theta+\pi)|)$, and $\|x-q\|_{C^1}$ has the same majorant. Its remaining angular mean is $6\langle f\rangle \int w|q|^4\,\mathrm{d}t = O(\|w\|_\infty a^2)$. This proves the radial-force estimate. Dominated convergence, using $|A_\epsilon H| \le C\delta\rho^3$ and the preceding positive bounds, now gives (293) as $\epsilon \downarrow 0$.

For (294), set $r = e_\theta \cdot (Q_f - xe_\theta)$. Expansion only of the radial polynomial force gives

$$r'' + 30x^4 r = -N_0 - e_\theta \cdot [\nabla\mathcal{F}(Q_f) - \nabla\mathcal{F}(xe_\theta)], \qquad |N_0| \le C|Q_f - xe_\theta|^2.$$

Estimate (295) without the regularizer, using the positive $p = 4$ kernel; the inner region uses $p = 5$ and gives $C\delta^5 a^2$. Bounded propagation yields $\|r\|_{L^1_\theta C^1_t} \le Ca^2$. The line oscillator is C^2 in its two scalar coefficients, so $x - q = u_+(t)f(\theta) + u_-(t)f(\theta+\pi) + O_{C^1}(|f(\theta)|^2 + |f(\theta+\pi)|^2)$. Its mean is $O_{C^1}(a^2)$. Finally

$$c = 24q^3 e_\theta \cdot (Q_f - qe_\theta) + O(|Q_f - qe_\theta|^2),$$

whose mean is $O(a^2)$ by the just proved longitudinal estimate and Lemma 42. This proves (294). ∎

Lemma 44 (Full first-insertion mean)**.** *Let $\mathscr{L}P = -qB$ with zero data and $w = Kj$. Then*

$$\left|\left\langle \int wB\,\mathrm{d}t \right\rangle + \left\langle \int wA_{\mathrm{reg}} BP\,\mathrm{d}t \right\rangle\right| \le C(a^2 + aS_h + \delta S_h^2). \tag{296}$$

Proof. Set $Z = X_E - q$ and $D_* = P - P_*$. Ordinary Jacobi comparison gives $\|Z\|_{C^2_t} \le C\delta$ uniformly: its initial displacement is $R_\theta - 1$, its initial derivative is zero, and its equation has a continuous source of size $C\delta$. Exact subtraction gives

$$\mathscr{L}D_* = BZ + cX_E + D_{RE}X_R + \mathscr{L}x, \qquad D_*(0) = D_*'(0) = 0.$$

Write $D_* = \Phi + Z_a$, with $\mathscr{L}\Phi = BZ$ and $\mathscr{L}Z_a = cX_E + D_{RE}X_R + \mathscr{L}x$, both with zero data. The latter has a C^1-time label majorant d_a satisfying $\|d_a\|_2 \le Ca$, by Lemma 42 and $|\mathscr{L}x| \le C(|f(\theta)| + |f(\theta+\pi)|)$. The low-profile bound (288) gives

$$|\langle \int wA_{\mathrm{reg}} BZ_a\,\mathrm{d}t\rangle| \le Ca(S_h + a).$$

The other term is exactly

$$\left\langle \int_{s<t} w(t)A_{\mathrm{reg}}(t)B(t)G_E(t,s)B(s)Z(s)\,\mathrm{d}s\,\mathrm{d}t \right\rangle.$$

Apply (289) with the small actual amplitude Z, of C^2-time norm $C\delta$, to bound it by $C\delta(S_h + a)^2$. Lemma 43 bounds the remaining $A_{\mathrm{reg}}H_{ER}Z_*$ product by Ca^2. Substitution into (293) proves (296) with its stated plus sign. All source terms have remained on the actual full leg, and the small factor came from the actual field $X_E - q$. ∎

D.2 Endpoint fields and the exact coefficient expansion

Put $R_\theta = (1+f(\theta))^{-1/6}$ and let z_f be the Jacobi field for the initial angular momentum of the inverse return, written on the positive-time leg. Its initial data are $z_f(0) = 0$, $z_f'(0) = -E_\theta/R_\theta$. Let

$$z_0 = -jE_\theta, \qquad \mathcal{K}_D Z(t) = -\int_0^t G_0(t,s)D(s)Z(s)\,\mathrm{d}s, \qquad E = [(1+f)^{1/6} - 1]z_0.$$

Then $z_f = z_0 + E + \mathcal{K}_D z_f$ exactly. Set

$$\Gamma(\theta) = \frac{E_{F_b(\theta)}}{R_{F_b(\theta)}} \quad \text{in the launch frame,} \qquad \Gamma_0 = -E_\theta.$$

The actual endpoint derivatives are

$$b_f = \Gamma \cdot z_f(T), \qquad F_b' = \Gamma \cdot X(T). \tag{297}$$

The return-time correction to position is zero because the return velocity vanishes. These formulas therefore retain the actual event, without differentiating a return-map Jacobi field in the potential. Geometry gives

$$\|\Gamma - \Gamma_0\|_2 \le Ca, \qquad \|\Gamma - \Gamma_0\|_\infty \le C\delta. \tag{298}$$

Lemma 45 (Complete endpoint remainder)**.** *Let $E_1 = (f/6)z_0$, $E_2 = -5f^2 z_0/72$. The exact expansion*

$$z_f = z_0 + E + \mathcal{K}_D z_0 + \mathcal{K}_D E + \mathcal{K}_D^2 z_0 + \mathcal{K}_D^2 E + \mathcal{K}_D^3 z_f \tag{299}$$

has, after keeping $z_0 + E_1 + \mathcal{K}_D z_0 + E_2 + \mathcal{K}_D E_1 + \mathcal{K}_D^2 z_0$, an angular L^1 endpoint remainder at most $C\delta(S_h + a)^2$. Also

$$\|b_f - b\|_2 + \|F_b' - 1\|_2 \le C(S_h + a). \tag{300}$$

Proof. Write $\mathcal{K}_D = \mathcal{K}_C + \mathcal{K}_L$ using Lemma 42. A low response has C^2-time label majorant Cd_0 whenever its input is uniformly bounded. All actual Jacobi fields have uniform C^2-time bounds by their continuous coefficient equations.

To estimate the complete third insertion, integrate its outer actual coefficient first. The matrix

$$W(T,s) = \int_s^T G_0(T,t)D(t)G_0(t,s)\,\mathrm{d}t$$

satisfies $W_{ss} = G_0(T,s)D(s) - W(T,s)A_0(s)$, where $A_0 = \mathrm{diag}(30q^4, 6q^4)$; the multiplication by A_0 is on the right. Thus $\|W\|_{C^2} \le C\delta$, with terminal orders $\delta(d^3, d^2, d)$ for W, W_s, W_{ss}, $d = T - s$. This equation differentiates no entry of D. In $\mathcal{K}_D(\mathcal{K}_C + \mathcal{K}_L)^2 z_f$ there are four orderings. The two-principal term is bounded by $C\delta S_h^2$ using (287), with outer amplitude W and inner actual $z_{f,E}$. For $\mathcal{K}_D\mathcal{K}_C\mathcal{K}_L z_f$, the remaining one-source

amplitude has C^2-time majorant $C\delta d_0$, so the L^1 version of (286) gives $C\delta a S_h$. For the reversed mixed ordering integrate the low middle factor:

$$V(T,r)=\int_r^T W(T,s)L^{\mathrm{mat}}(s)G_0(s,r)\,\mathrm{d}s.$$

Its C^2-time majorant is $C\delta d_0$, with terminal orders $C\delta d_0(d^5,d^4,d^3)$. The same one-source bound gives $C\delta a S_h$. The two-low term is bounded pointwise by $C\delta d_0^2$ and hence in angular L^1 by $C\delta a^2$. Consequently $\|\mathcal{K}_D^3 z_f(T)\|_1\le C\delta(S_h+a)^2$.

With the fixed radial outer endpoint kernel, the same four orderings give

$$\|\mathcal{K}_D^2 z_0(T)\|_1\le C(S_h+a)^2.$$

The scalar coefficient in E is constant in physical time, so it commutes with this Volterra operator and $\|\mathcal{K}_D^2E(T)\|_1\le C\delta(S_h+a)^2$. The remaining terms of (299) are $E-E_1-E_2=O(f^3)z_0$ and $\mathcal{K}_D(E-E_1)=O(f^2)\mathcal{K}_D z_0$. Their endpoint norms are bounded respectively by $C\delta a^2$ and $C\|f^2\|_2(S_h+a)\le C\delta a(S_h+a)$. This proves the remainder assertion.

Finally the exact equation $z_f=z_0+E+\mathcal{K}_D z_f$ uses one full principal source with a uniformly C^2 actual amplitude. Equations (286) and (298) give the first part of (300). The analogous launch-angle equation $X=qE_\theta+E_X+\mathcal{K}_D X$ has a radial homogeneous initial correction whose coefficients are $R_\theta-1$ and R'_θ, both of L^2 size Ca. The same estimates and (297) give its second part. ∎

Define the two scalar insertion groups, with the actual endpoint factor left unexpanded, by

$$L_f=\Gamma\cdot[z_0+E_1+\mathcal{K}_D z_0](T)-b, \tag{301}$$

$$B_{2,f}=\Gamma\cdot[E_2+\mathcal{K}_D E_1+\mathcal{K}_D^2 z_0](T). \tag{302}$$

They have sup norms $C\delta, C\delta^2$, respectively, and $\|B_{2,f}\|_1\le C(S_h+a)^2$.

Lemma 46 (Means and endpoint products)**.** *Let $\mathscr{L}P=-qB$ and $\mathscr{L}v=jB$ have zero data, and put $p=F_b'-1$. Then*

$$\left|\langle L_f\rangle-\left\langle\int KjA_{\mathrm{reg}}BP\,\mathrm{d}t\right\rangle\right|\le C(a^2+aS_h+\delta S_h^2), \tag{303}$$

$$\left|\langle B_{2,f}\rangle-\left\langle\int KBv\,\mathrm{d}t\right\rangle\right|\le C(a^2+aS_h+\delta S_h^2), \tag{304}$$

$$\|p+P(T)\|_2+\|L_f+v(T)\|_2\le C(a+\delta S_h). \tag{305}$$

Replacing the three quadratic endpoint products p^2, L_f^2, pL_f by $P(T)^2, v(T)^2, P(T)v(T)$ therefore costs at most $C(a^2+aS_h+\delta S_h^2)$.

Proof. Write $z_1=\mathcal{K}_D z_0$. Its radial component has L^2C^1 norm Ca, and its transverse component is $v+v_c$, where $\mathscr{L}v_c=jc$. Equation (294) gives $|\langle v_c(T)\rangle|\le Ca^2$, while

$\|z_1(T)\|_2 \le C(S_h + a)$ by the whole-leg one-source estimate. There is also a stronger mean cancellation for the initial endpoint group. Write $F_b(\theta) = \theta + \pi + \Delta(\theta)$, with $|\Delta| \le C|\beta|$. Directly,

$$\Gamma \cdot (z_0 + E_1)(T) = b(1 + f(F_b(\theta)))^{1/6} \cos \Delta(\theta)[1 + f(\theta)/6].$$

Slope matching and the C^2 Taylor formula give $f(F_b(\theta)) - f(\theta+\pi) = O(\beta^2)$. The two linear f means are $O(a^2)$, and the quadratic remainder is bounded by $C(f(\theta)^2 + f(\theta+\pi)^2 + \beta^2)$. Thus the mean of this expression minus b is $O(a^2)$, and its L^2 norm is Ca. The identity

$$L_f + v(T) = [\Gamma \cdot (z_0 + E_1)(T) - b] - v_c(T) + (\Gamma - \Gamma_0) \cdot z_1(T)$$

and (296) prove (303), including the positive regularizer sign. They also prove the second norm in (305), using the sup bound on $\Gamma - \Gamma_0$ for that norm.

For the first norm, use $\|X_E - q\|_{C^2} \le C\delta$ and write

$$(\mathcal{K}_C X)_E = P_C - \int_0^t G_E(t,s) C_f(s)[X_E(s) - q(s)] \, \mathrm{d}s, \qquad \mathscr{L} P_C = -q C_f.$$

At T the second term has L^2 size $C\delta S_h$ by (286); the full low response has size Ca. Consequently $X_E(T) = -R_\theta + P_C(T) + O_{L^2}(a + \delta S_h)$. Using (297) and (298) gives $p = -P_C(T) + O_{L^2}(a + \delta S_h)$. The difference $P - P_C$ has C^2-time majorant Cd_0 because $B - C_f$ is a full low coefficient. This proves the first norm.

For (304), expand $\mathcal{K}_D^2 z_0$ into its four principal/low orderings. A principal/low term has angular L^1 size CaS_h by the label-majorized whole-leg one-source bound. For the low/principal ordering, integrating the outer low coefficient gives a C^2-time majorant Cd_0 and terminal orders $Cd_0(d^3, d^2, d)$; the same bound applies. The two-low term has size Ca^2. The exact pure-principal sign is

$$\Gamma_0 \cdot \mathcal{K}_C^2 z_0(T) = \int K C_f v_C \, \mathrm{d}t, \qquad \mathscr{L} v_C = j C_f.$$

The E_2 term has L^1 size Ca^2; $\mathcal{K}_D E_1 = (f/6)\mathcal{K}_D z_0$ has size $Ca(S_h + a)$. Replacing Γ by Γ_0 in the second insertion costs $C\delta(S_h + a)^2$. To replace C_f by B, put $\ell_0 = B - C_f$ and $\mathscr{L} v_{\ell_0} = j\ell_0$. Exactly,

$$\int KBv - \int KC_f v_C = \int KC_f v_{\ell_0} + \int K\ell_0 v_C + \int K\ell_0 v_{\ell_0}.$$

The first amplitude has C^2 majorant Cd_0. In the second term integrate the outer low factor first, obtaining the same majorant; the third is pointwise bounded by Cd_0^2. Their angular L^1 norms are therefore CaS_h, CaS_h, Ca^2. This proves (304) on the whole leg.

All four endpoint fields in (305) have L^2 size $C(S_h + a)$. Difference of squares and angular Cauchy–Schwarz give a total product error $C(a+\delta S_h)(S_h+a) \le C(a^2+aS_h+\delta S_h^2)$. The cross product has a plus sign because both approximations are negative. ∎

Proposition D.1 (Complete invariant reduction)**.** *The exact normalized boundary invariant satisfies*

$$|I(f) - I(0) - \mathcal{Q}_{\mathrm{reg}}(f)| \le C(a^2 + aS_h + \delta S_h^2). \tag{306}$$

Consequently exact common means give

$$|\mathcal{Q}_{\mathrm{mod}}(h)| \le C_I(a^2 + aS_h + \delta S_h^2). \tag{307}$$

Proof. By the boundary invariant proved in Section A, $I(f) = \langle\sqrt{F_b'/b_f}\rangle$ and $I(0) = b^{-1/2}$. Put $p = F_b' - 1$ and $l = b_f - b$. Their sup norms are $O\delta$, and (300) controls their L^2 norms. The elementary Taylor formula is

$$\sqrt{\frac{1+p}{b+l}} = b^{-1/2}\left[1 + \frac{p}{2} - \frac{l}{2b} - \frac{p^2}{8} + \frac{3l^2}{8b^2} - \frac{pl}{4b}\right] + O\big(\delta(p^2+l^2)\big).$$

Since F_b has degree one, $\langle p\rangle = 0$ exactly. Lemma 45 gives $l = L_f + B_{2,f} + R$ with $\|R\|_1 \le C\delta(S_h + a)^2$. In the quadratic terms replacing l by L_f costs $C\delta\|B_{2,f}\|_1 + C\delta\|R\|_1$; this uses an actual sup bound times an angular L^1 bound. We obtain

$$I(f) - I(0) = b^{-1/2}\left\{ - \frac{\langle L_f + B_{2,f}\rangle}{2b} - \frac{\langle p^2\rangle}{8} + \frac{3\langle L_f^2\rangle}{8b^2} - \frac{\langle pL_f\rangle}{4b}\right\} + O\big(\delta(S_h+a)^2\big).$$

Insert Lemma 46. The surviving terms are precisely (291), including both negative Volterra contributions and the negative cross-endpoint coefficient. This proves (306). Combine it with $I(f) = I(0)$ and Proposition C.3 to obtain (307). ∎

D.3 Centering and the coherent model sign

Lemma 47 (Full fixed-period centering)**.** *For M_0 from the radial half-mass kernel and the actual involution J,*

$$\|(I + J)M_0 h\|_2 \le C(a + \delta S_h). \tag{308}$$

Proof. The exact launch Jacobi field has $X'(0) = X'(T) = 0$, because both rest conditions hold at a time independent of the launch label. The transverse Wronskian therefore gives

$$(qX_E' - q'X_E)' = -q[C_f X_E + (L^{\mathrm{mat}}X)_E],$$
$$\int_0^T q^2 C_f \,\mathrm{d}t = -\int_0^T qC_f(X_E - q)\,\mathrm{d}t - \int_0^T q(L^{\mathrm{mat}}X)_E\,\mathrm{d}t.$$

The first amplitude has uniform C^2-time size $C\delta$, so (286) bounds it in L^2 by $C\delta S_h$. The second integral is bounded by Cd_0 pointwise. Thus the whole mass $A_{\mathrm{tot}} = \int q^2 C_f\,\mathrm{d}t$ has L^2 norm $C(a + \delta S_h)$.

Use the symmetric fixed-time cutoff $H_r = \eta(|q|/r)$, $r = D\delta$, and let $A_{\mathrm{head}}h = \int_0^Q H_r q^2 C_f\,\mathrm{d}t$. Exact scalar reversal gives

$$(I + R_b)A_{\mathrm{head}}h = A_{\mathrm{tot}} - \int_0^T (1 - H_r)q^2 C_f\,\mathrm{d}t, \qquad R_b g = g \circ F_b.$$

The omitted complete origin integral, without splitting it into two half-integrals, has norm $Cr^5 S_h$ by Lemma 31. The relative moment comparison, its fixed-time/clock transition band and the response-relative unitary tail give $\|(A_{\mathrm{head}}-M_0)h\|_2 \le C\delta S_h$ and $\|A_{\mathrm{head}}h\|_2 \le CS_h$. Finally $\|J-R_b\| \le C\delta$ by $F_b' = 1+O\delta$. These estimates prove (308). This proof uses the uncut Wronskian and never restores an infinite bare-flow tail. ∎

Lemma 48 (The centered coherent sign)**.** *For sufficiently small δ,*

$$\mathcal{Q}_{\mathrm{mod}}(h) \le -\frac{1}{2200}S_h^2 + C_M(aS_h + \delta S_h^2). \tag{309}$$

Proof. Write $A_0 = M_0 h$, $C_1 = M_1 h$, $D_2 = M_2 h$, $E_3 = M_3 h$. The model (273) gives exactly

$$P_0(T) = -bA_0 - C_1 + JC_1, \qquad v_0(T) = (I+J)(bC_1 + D_2).$$

Its cross-half Volterra contribution inside the overall factor $b^{-1/2}$ is

$$-\frac{1}{2b}\left[b^2\langle JC_1, A_0\rangle + 2b\langle JD_2, A_0\rangle + 2b\langle JC_1, C_1\rangle + \langle JE_3, A_0\rangle + 3\langle JD_2, C_1\rangle\right].$$

Expand the three endpoint products and use $J = J^*$ and $J^2 = I$. All JC_1 terms cancel. The complete remaining J-dependent part is

$$-\frac{3}{4}\langle JA_0, D_2\rangle - \frac{1}{2b}\langle JA_0, E_3\rangle + \frac{3}{4b^2}\langle JD_2, D_2\rangle. \tag{310}$$

The J-free endpoint polynomial is

$$-\frac{b^2}{8}\|A_0\|^2 + \frac{1}{2}\|C_1\|^2 + \frac{1}{4}\langle A_0, D_2\rangle + \frac{3}{2b}\langle C_1, D_2\rangle + \frac{3}{4b^2}\|D_2\|^2;$$

the same-half Volterra terms are also J-free.

The scalar mass and ratio bounds give $\|A_0\| \asymp S_h$ and $\|C_1\|+\|D_2\|+\|E_3\| \le CS_h$. Put $e = (I+J)A_0$ and $N_2 = (a_2/a_0)(A_\beta)$, so $D_2 = N_2 A_0$. By Lemma 40, $\|JN_2J - N_2^*\| \le C\delta$. The ordered identity

$$JD_2 + N_2^* A_0 = N_2^* e + (JN_2J - N_2^*)JA_0$$

and Lemma 47 show that the ideal replacements $JA_0 \mapsto -A_0$, $JD_2 \mapsto -N_2^* A_0$ change the form by at most $C(aS_h + \delta S_h^2)$. No inverse of M_0 is applied to an arbitrary vector, and the ratio reversal acts on the already smoothed A_0.

These are exactly the phase-centered replacements in the scalar profile. If V is the unit phase of M_0, then $N_2^* M_0 = M_2^* V^2$; the ideal expression is therefore $\int \mathcal{Q}(\lambda)\,\mathrm{d}\mu_h(\lambda)$, with the same $b^{-1/2}$ normalization and scalar phase as in Theorem 15. Its sign is $\mathcal{Q}(\lambda) \le -[1100(1+|\lambda|)]^{-1}$. Since $1+|\lambda| \le \sqrt{2}\sqrt{1+\lambda^2}$ and $S_h^2 = \int (1+\lambda^2)^{-1/2}\,\mathrm{d}\mu_h$, that spectral integral is at most $-S_h^2/2200$. This proves (309). ∎

Theorem 16 (Closed sextic response estimate)**.** *Every normalized member sufficiently close in angular C^2 to the radial member of an original C^2 common-mean family satisfies*

$$\|(1+iA_{f'})^{-1/2}f''\|_2 \le C_*\|o'\|_2. \tag{311}$$

The constants are uniform in one parameter neighborhood and independent of angular Fourier support.

Proof. Combine (307) and (309). Let $c_0 = 1/2200$ and $C = \max(1, C_I, C_M)$. Then

$$c_0 S_h^2 \le Ca^2 + 2CaS_h + 2C\delta S_h^2.$$

Restrict $\delta \le c_0/(8C)$ as well as all fixed geometric thresholds. The last term is at most $c_0 S_h^2/4$, and $2CaS_h \le c_0 S_h^2/4 + 4C^2a^2/c_0$. Consequently

$$S_h^2 \le (2C/c_0 + 8C^2/c_0^2)a^2.$$

This argument does not divide by a. Every physical-origin term has already been bounded signedly; there is no unweighted Sobolev norm left to absorb. ∎

D.4 Finite isolation and the connected family

Lemma 49 (The response scale controls a half derivative)**.** *For $\beta \in C^1$ in a bounded set,*

$$\|g\|_{H^{-1/2}} \le C\|(1+iA_\beta)^{-1/2}g\|_2.$$

Thus (311) *implies $\|f - \langle f\rangle\|_{H^{3/2}} \le Ca$.*

Proof. The map $u \mapsto A_\beta u$ is bounded from H^1 to L^2, uniformly in the stated C^1 set. Equivalently the identity map embeds H^1 into the graph space of A_β. Interpolation of this Hilbert-scale embedding with the identity on L^2 gives $\|\Lambda^{1/2}u\|_2 \le C\|u\|_{H^{1/2}}$. Duality yields $\|g\|_{H^{-1/2}} \le C\|\Lambda^{-1/2}g\|_2$. The equality of response norms is Lemma 30. Apply the result to $g = f''$ and use the nonzero Fourier modes to obtain the final assertion. ∎

Theorem 17 (Finite radial sextic family rigidity)**.** *Let $S \subset \mathbb{R}^m$ be connected and open, with m finite, and let $P \in C^2(S \times \mathbb{R}^2)$ be positive away from $Z = 0$ and positively homogeneous of degree six in Z. Suppose that for every $s \in S$ all invariant probability measures on $\{|W|^2/2 + P(s,Z) = 1\}$ give the same vector mean to $P_s(s,Z)$. If $P(s_*,Z) = \kappa_*|Z|^6$ for one $s_* \in S$ and $\kappa_* > 0$, then*

$$P(s,Z) = \kappa(s)|Z|^6 \quad (s \in S), \qquad \kappa \in C^2(S), \quad \kappa > 0.$$

Joint convexity, polynomial form, central symmetry and angular C^3 regularity are not required for this frozen classification.

Proof. Normalize as in Section A, so the radial member is $|Z|^6$ and the angular volume is fixed. On a small parameter neighborhood write $V(s,\rho e_\theta)=\rho^6(1+f(s,\theta))$. All action and brake hypotheses used above follow from the original common-mean assumption by that section. Theorem 16 and Lemma 49 give $\sum_{n\ge1,\ n\ \mathrm{odd}} n^3 w_n \le C_2 a^2$, where $w_n = 4|\widehat{f}(n)|^2$ for positive odd n. Lemma 22 gives

$$ca^2 \le \sum_{n\ge5,\ n\ \mathrm{odd}} n^2 w_n, \qquad w_n \le C_1 a^4 \quad (n\ge5,\ n\ \mathrm{odd}).$$

For every fixed integer $M\ge5$,

$$ca^2 \le C_1' a^4 M^3 + C_2 a^2/M.$$

Choose M once with $C_2/M \le c/2$. If $a>0$, this forces $a^2 \ge c/(2C_1'M^3) > 0$, whereas $a \le C\delta$. For sufficiently small δ this is impossible. If $a=0$, $\|f\|_{H^1}\le Ca$ gives $f=0$ directly. This proves local finite isolation, including arbitrarily high angular frequencies.

The normalized radial locus is nonempty and relatively closed by angular C^2 continuity. Apply the same local argument at any radial member to see that it is relatively open. Connectedness makes it all of S. Undoing the fixed volume normalization gives the stated positive amplitude; $\kappa(s)=P(s,e_0)$ is C^2. Only the frozen action, mechanical and transport arguments were used. In particular no convexity of the joint parameter–position Hessian or of the normalized family was used. Joint convexity of the original potential remains a separate hypothesis of the actual trajectory selection theorem. ∎

E Ordinary means and the original vertical Peano class

This appendix supplies the endpoint entry used in the physical second variation and in the energy completion. It precedes both arguments logically and uses only the original angular C^1 norm. Write $I=(0,\pi/2)$ and $\delta=\pi/2-\theta$.

Theorem 18 (Original ordinary-null entry)**.** *Let $g(\rho e_\theta)=\rho^4 a(\theta)$ be EO, with $a\in C^1(\mathbb{S}^1)$, and suppose its ordinary torus means vanish for all sufficiently small positive c. Put $A=a(\pi/2)$. Then*

$$a(\pi/2-\delta) - A\sin^3(\pi/2-\delta) = o(\delta^2). \tag{312}$$

If the interval of ordinary nullity is fixed, the left side has modulus at most $C\|a\|_{C^1}\delta^2$ on a fixed collar. This is a value statement; no derivative quotient or second angular derivative is asserted.

E.1 The complete scalar-measure reduction

Subtract Ag_{ref}, where $g_{\mathrm{ref}}(x,y)=y^3\sqrt{x^2+y^2}$, and call the remaining force g_0. Its clock-weighted parabolic observable $F_0(u,v)=(u^2+v^2)g_0(uv,(u^2-v^2)/2)$ is C^1, homo-

$$\partial_{\theta\theta} V(s, e_\theta) = D_Z^2 V(s, e_\theta)[e_\theta^\perp, e_\theta^\perp] - D_Z V(s, e_\theta) \cdot e_\theta.$$

Uniform continuity of these functions on the compact angular circle shows $\|f_s\|_{C^2} \to 0$ as $s \to s_*$. This is continuity, not parameter differentiability with values in C^2; no derivative $D_s D_Z^2 V$ is used. At a fixed s, the normal reduction made directly in the finite-dimensional parameter and the reduction used for Theorem 4 solve the same normal equation in the same fibers. Their small solutions coincide by normal uniqueness. Thus the exact action values just obtained, together with constant angular volume, are exactly the three conditions of that theorem. It follows that $V(s, Z) = |Z|^p$ on a neighborhood of s_*.

The set of such parameters is nonempty, closed by continuity, and open by this local argument. Connectedness makes it all of U. Undoing the one fixed normalization gives the asserted positive amplitude function. The interval case uses the same local argument and relative open–closed conclusion. ∎

Theorem 5 proves the radial finite-rigidity assertion of Theorem 1 for $p = 3, 4$. The degree-six assertion is proved separately in Theorem 17. The present proof uses two fixed compact periodic families and the quantified remainder (49). It does not infer finite rigidity from the first-jet corollary and does not require uniform inverses over unrelated periodic families.

4 A nonradial quartic neighborhood and connected-family rigidity

We work in the coefficient space $\mathcal{V}_4$ with its ordinary Euclidean inner product in the basis $(x^4, x^3y, x^2y^2, xy^3, y^4)$. Write $J(x, y) = (-y, x)$ and

$$\mathcal{R}P(Z) = \nabla P(Z) \cdot JZ.$$

This is a fixed linear operator on coefficient space. At the base potential (3),

$$\mathcal{R}P_0 = 20x^3y + 40xy^3. \tag{57}$$

The normal Hessian has diagonal entries $12x^2 + 24y^2$ and $24x^2 + 192y^2$, and determinant $288x^4 + 576x^2y^2 + 4608y^4 > 0$ away from zero. Thus P_0 is normally strictly convex. Its only real projective angular critical lines are the coordinate axes; their angular second derivatives are 20 and -40, and their values are 1 and 16.

4.1 The constrained direction left by the straight orbits

The two simple critical lines continue smoothly in coefficient space. Let their continued critical values be $C_x(P)$ and $C_y(P)$. Together with $\mathcal{I}(P)$, set $G = (C_x, C_y, \mathcal{I})$. At P_0, for

$h = ax^4 + bx^3y + cx^2y^2 + dxy^3 + ey^4$,

$$DC_x[h] = a, \quad DC_y[h] = e, \quad D\mathcal{I}[h] = -\tfrac{1}{2}\int_0^{2\pi} h(e_\theta)P_0(e_\theta)^{-3/2}\,\mathrm{d}\theta.$$

The coefficient of c in the last expression is nonzero; reflection symmetry kills the b, d coefficients. Hence

$$\operatorname{rank} DG(P_0) = 3, \qquad \ker DG(P_0) = \operatorname{span}\{x^3y, xy^3\}. \tag{58}$$

Modulo the rotation (57), the only remaining direction is represented by $g = x^3y$.

Consider jets $P_r = P_0 + rg + (r^2/2)B + o(r^2)$ that preserve G through second order. Write $B = ax^4 + bx^3y + cx^2y^2 + dxy^3 + ey^4$. At a nondegenerate angular critical point, the second derivative of its value is $B - (\partial_\theta g)^2/\partial_\theta^2 P_0$. Therefore the axial constraints give $a = 1/20$, $e = 0$. Differentiating the volume twice gives

$$\int BP_0^{-3/2}\,\mathrm{d}\theta = \tfrac{3}{2}\int g^2P_0^{-5/2}\,\mathrm{d}\theta, \qquad c = \frac{(3/2)J_* - I_4/20}{I_{22}}, \tag{59}$$

where

$$I_4 = 4\int_0^\infty \frac{t^4\,\mathrm{d}t}{(t^4+12t^2+16)^{3/2}},$$
$$I_{22} = 4\int_0^\infty \frac{t^2\,\mathrm{d}t}{(t^4+12t^2+16)^{3/2}},$$
$$J_* = 4\int_0^\infty \frac{t^6\,\mathrm{d}t}{(t^4+12t^2+16)^{5/2}}.$$

The coefficients b, d remain free. Their contributions to the full curved-orbit average below vanish by reflection; this cancellation is taken before reducing the full integral to a quarter orbit.

4.2 A separated curved brake orbit

The parabolic coordinates

$$x = uv, \quad y = (u^2 - v^2)/2, \quad p_u = vW_x + uW_y, \quad p_v = uW_x - vW_y, \quad S = u^2 + v^2 \tag{60}$$

are canonical away from the origin. On physical energy $H = 1$, multiplication by S gives

$$K = S(H-1) = \tfrac{1}{2}(p_u^2 + p_v^2) + U(u) + U(v), \quad U(t) = t^{10} - t^2, \qquad \mathrm{d}t = S\,\mathrm{d}\tau. \tag{61}$$

This separation of the quartic base is already known; see [21, Example 8.5 in the preprint]. Choose

$$v = v_0 = 5^{-1/8}, \quad p_v = 0, \quad \tfrac{1}{2}p_u^2 + U(u) = E_* := \tfrac{4}{5}v_0^2.$$

Since $U'(v_0) = 0$ and $U''(v_0) = 16$, this is a periodic orbit. Let Q be its fictitious quarter period, from the positive turning point of u to its zero. Its physical trajectory lies on

$$y = x^2/(2v_0^2) - v_0^2/2,$$

and stays away from the origin because $|(x, y)| = S/2 \geq v_0^2/2$.

Let $z > 1$ be the unique positive root above one of

$$z^5 - 5z - 4 = (z + 1)^2(z^3 - 2z^2 + 3z - 4) = 0. \tag{62}$$

With $u = v_0 Y$, $Y^2 = z\cos^2\psi$, $0 \leq \psi \leq \pi/2$, and $w = \cos^2\psi$, energy conservation gives the regular endpoint formula

$$\frac{\mathrm{d}\tau}{\mathrm{d}\psi} = \left[2\left\{\frac{z^4}{5}(1 + w + w^2 + w^3 + w^4) - 1\right\}\right]^{-1/2}. \tag{63}$$

The transverse variational equation is $\eta'' + 16\eta = 0$. The full return multipliers are $\exp(\pm 16iQ)$. The exact bounds in Appendix P.1 give $2 < 8Q/\pi < 3$, so neither multiplier is one. The fixed-energy orbit is nondegenerate. Positive time change does not change its return map. The implicit-function theorem for a transverse Poincaré section therefore yields a smooth coefficient-dependent orbit and its smooth action, denoted $\mathcal{A}(P)$, near P_0.

4.3 Second variation in the genuine mechanical direction

For $g = x^3 y$, the perturbation of K is

$$F(u, v) = Sg(x, y) = \tfrac{1}{2}(v^3 u^7 - v^7 u^3).$$

Along $v = v_0$,

$$F_u = \tfrac{7}{2}v_0^3 u^6 - \tfrac{3}{2}v_0^7 u^2, \qquad F_v = \tfrac{3}{2}v_0^2 u^7 - \tfrac{7}{2}v_0^6 u^3. \tag{64}$$

The actual fixed-energy first variations solve

$$\xi_u'' + U''(u)\xi_u = -F_u, \qquad \xi_v'' + 16\xi_v = -F_v, \tag{65}$$

with quarter boundary conditions

$$\xi_u'(0) = \xi_u'(Q) = 0, \qquad \xi_v'(0) = 0, \quad \xi_v(Q) = 0.$$

Here these are the derivatives of a continued periodic orbit, not arbitrarily chosen solutions of a variational equation. For the scalar longitudinal oscillator, $F(\cdot, v_0)$ is odd. Reflection $u \mapsto -u$, $r \mapsto -r$ makes its period an even function of r, so the first period derivative is zero. The fixed-energy first variation has zero velocity variation at the zero crossing, which gives the second Neumann condition. Transversely, nonresonance makes the periodic solution unique, and the odd half-period symmetry of F_v gives the mixed Neumann–Dirichlet conditions. Its equilibrium energy variation is zero. The longitudinal and transverse variations thus combine into the fixed-energy derivative of the full periodic orbit with zero first period derivative.

By Lemma 1, the second action derivative for the jets under consideration is

$$\mathcal{A}'' = -\int_0^{4Q} (SB + F_u\xi_u + F_v\xi_v)\,\mathrm{d}\tau. \tag{66}$$

The factor S and its variation are essential: $\int g\,\mathrm{d}t = \int F\,\mathrm{d}\tau$. Replacing F_u, F_v by derivatives of g would omit the clock variation.

The longitudinal contribution has an exact scalar elimination. Put

$$a(u) = \frac{v_0^3 u^2}{20(u^4 + v_0^4)}.$$

Then $F(u, v_0) = U'(u)a(u)$. Define an auxiliary potential by $\widetilde{U}_r(y - ra(y)) = U(y)$. Its first jet is F and its second jet is $B_{\mathrm{aux}} = U''a^2 + 2U'aa'$. At fixed energy its abbreviated action is

$$2\int_{-u_{\max}}^{u_{\max}} \sqrt{2(E_* - U(y))}(1 - ra'(y))\,\mathrm{d}y,$$

which is constant because a' is odd. The second action variation therefore gives

$$\int_0^Q F_u\xi_u\,\mathrm{d}\tau = -\int_0^Q B_{\mathrm{aux}}\,\mathrm{d}\tau, \qquad \frac{B_{\mathrm{aux}}}{v_0^2} = \frac{R^2(5R^4 + 8R^2 - 5)}{200(R^2+1)^2}, \quad R = Y^2. \tag{67}$$

The reflection-even part of the compensator is $B_{\mathrm{even}} = x^4/20 + cx^2y^2$, and

$$\frac{SB_{\mathrm{even}}}{v_0^2} = \frac{1}{5}(R+1)\left(\frac{R^2}{20} + \frac{c}{4}R(R-1)^2\right). \tag{68}$$

The free reflection-odd terms have zero full integral. The remaining terms have four equal quarter integrals.

For the transverse term write $f = F_v$ and let $L = -\,\mathrm{d}^2/\,\mathrm{d}\tau^2 - 16$ with Neumann–Dirichlet conditions on $[0, Q]$. Then $\xi_v = L^{-1}f$. Its eigenvalues are

$$\lambda_k = ((k + 1/2)\pi/Q)^2 - 16 \quad (k \geq 0).$$

The certified interval for Q makes exactly λ_0 negative. Dropping all positive spectral terms gives the rigorous lower bound

$$\int_0^Q f\xi_v\,\mathrm{d}\tau \geq -\frac{(2/Q)\left(\int_0^Q f\cos(\pi\tau/(2Q))\,\mathrm{d}\tau\right)^2}{16 - (\pi/(2Q))^2}, \qquad \frac{f}{v_0} = \frac{R\sqrt{R}(3R^2 - 7)}{10}. \tag{69}$$

No truncation error from the positive spectrum is needed for this one-sided estimate.

Lemma 9 (Strict constrained curvature)**.** *For $g = x^3y$ and every B satisfying the second-order constraints $G'' = 0$,*

$$\frac{1}{v_0^2}\int_0^Q (SB_{\mathrm{even}} - B_{\mathrm{aux}} + fL^{-1}f)\,\mathrm{d}\tau > \frac{1}{2500}. \tag{70}$$

Consequently $\mathcal{A}'' < -v_0^2/625$.

Proof. Equations (59) and (63)–(69) reduce the assertion to one-dimensional elementary integrals. The rational interval computation in Appendix P.1 gives a lower bound $719741/10^9 > 1/2500$ for the left side. Every endpoint and operation used in that comparison is bounded outward by rational arithmetic. Multiplying by the four quarters and using (66) proves the action inequality. Reflection removes the unconstrained b, d terms, so the bound applies to every admissible B. ∎

4.4 A rank-and-curvature argument through a rank jump

Shrink the coefficient neighborhood of P_0 so that normal strict convexity persists, the two simple and distinct critical values continue, no other real angular critical lines occur, the curved orbit continues, DG has rank three, and $\mathcal{R}P \neq 0$. Define the four smooth rotational invariants

$$\mathcal{F}(P) = (G(P), \mathcal{A}(P)).$$

Rotations carry the continued orbit to the unique continued orbit up to phase. Consequently

$$D\mathcal{F}(P)[\mathcal{R}P] = 0. \tag{71}$$

At P_0, reflection makes $D\mathcal{A}$ vanish on x^3y and xy^3. Thus $D\mathcal{F}(P_0)$ has rank three, rather than four.

Choose a smooth unit vector $v(P)$ in $\ker DG(P)$ perpendicular to $\mathcal{R}P$, by projecting x^3y and normalizing. Let

$$T(P) = DG(P)^*\big(DG(P)DG(P)^*\big)^{-1}$$

be a right inverse. Define

$$\tau(P) = D\mathcal{A}(P)[v(P)], \qquad q(P) = D^2\mathcal{A}(P)[v,v] - D\mathcal{A}(P)T(P)D^2G(P)[v,v]. \tag{72}$$

Differentiating (71) in any direction h gives

$$D^2\mathcal{F}(P)[h, \mathcal{R}P] + D\mathcal{F}(P)[\mathcal{R}h] = 0. \tag{73}$$

Every left-null covector for $D\mathcal{F}$ therefore annihilates Hessian terms with a rotation factor.

At P_0, the projection of $g = x^3y$ perpendicular to (57) is $(4/5)x^3y - (2/5)xy^3$, of squared coefficient norm $4/5$. Since $D\mathcal{A}$ factors through DG there, Lemma 9 and (73) show

$$q(P_0) < -v_0^2/500 < 0.$$

Continuity permits a fixed smaller neighborhood $\mathcal{U}$ on which q never vanishes.

Lemma 10 (Local rotation constraint)**.** *Every C^2 coefficient curve $P(t)$ in $\mathcal{U}$ along which all four values $\mathcal{F}(P(t))$ are constant satisfies $P'(t) \in \mathbb{R}\mathcal{R}P(t)$.*

Proof. At a fixed time write $g = P' = a\mathcal{R}P + bv$, using the rank-three constraint $DG[g] = 0$. If $\tau(P) \neq 0$, then $D\mathcal{A}[g] = b\tau = 0$ immediately gives $b = 0$.

If $\tau(P) = 0$, then $D\mathcal{A}$ vanishes on $\ker DG$. The covector $\lambda = (-D\mathcal{A}T, 1)$ annihilates $D\mathcal{F}$. Twice differentiating the constant-value condition and applying this covector yields

$$0 = \lambda D^2\mathcal{F}[g, g] = b^2 q(P),$$

where (73) removes the mixed and pure rotation terms. Since $q(P) \neq 0$, again $b = 0$. The argument does not differentiate λ, assume constant rank of $D\mathcal{F}$, divide by τ, or bound P''. ∎

If V is a normalized common-mean family with values in $\mathcal{U}$, all four invariants are constant on every selected parameter curve. The volume is constant by normalization. For a continued straight critical-line orbit the mean force is $(1/3)\partial_s \log C_x$, or $(1/3)\partial_s \log C_y$, so its zero mean fixes the corresponding critical value. The curved action is constant by Lemma 1. Therefore Lemma 10, applied to coordinate curves, gives

$$V_{s_j} = c_j(s)\mathcal{R}V. \tag{74}$$

The coefficient map is C^2: its coefficients can be recovered from finitely many fixed normal evaluations, so fourth normal differentiability of the original family is not being assumed. The quotient defining c_j is C^1.

4.5 One entry member determines the entire connected family

Proof of Theorem 2. The local conclusion (74) is equivariant under fixed rotations and thus holds on every rotated neighborhood $\mathcal{U}_\phi = \{W(R_\phi \cdot) : W \in \mathcal{U}\}$. Along any C^1 selected path staying in such a neighborhood,

$$\frac{\mathrm{d}}{\mathrm{d}t} V(s(t)) = \gamma(t)\mathcal{R}V(s(t)), \qquad \gamma(t) = \sum_j c_j(s(t))s_j'(t).$$

Since $\mathcal{R}$ is a fixed linear operator on coefficient space, uniqueness for this linear ordinary differential equation gives

$$V(s(t), Z) = V(s(0), R_{\Theta(t)}Z), \qquad \Theta(t) = \int_0^t \gamma(u)\,\mathrm{d}u.$$

This is a parameter-space identity, not a replacement of the actual mechanical trajectory.

Let $\mathcal{O}_Q = \{Q(R_\theta \cdot) : \theta \in \mathbb{R}\}$ and $S_Q = \{s \in S : V(s) \in \mathcal{O}_Q\}$. The rotation orbit is compact and therefore closed, so S_Q is relatively closed. It contains s_*. If $s \in S_Q$, choose ϕ such that $V(s) = Q(R_\phi \cdot) \in \mathcal{U}_\phi$. A small parameter ball around s maps into $\mathcal{U}_\phi$; integrating along its straight segments puts the entire ball in S_Q. Hence S_Q is open as well. Connectedness gives $S_Q = S$.

The two critical values of Q are distinct, so a rotation preserving Q must preserve each labeled critical line. Its angle is 0 or π modulo 2π. Conversely every binary quartic is invariant under $-I$. Thus the stabilizer is exactly $\{I, -I\}$. The orbit is a smooth circle parametrized by θ modulo π, with nonzero derivative $\mathcal{R}Q$. Local inversion gives C^2 angles; on a simply connected S, the circle-valued map lifts through $\mathbb{R} \to \mathbb{R}/(\pi\mathbb{Z})$. Undoing (1) gives $\alpha = (\mathcal{I}_0/\mathcal{I}(P))^2$. On a nonsimply-connected region even a global $SO(2)$-valued lift may fail, since an odd multiple of π preserves the potential but changes the rotation matrix. This does not affect the orbit-valued statement. ∎

The neighborhood contains exactly two real projective critical lines, throughout. It is therefore outside the maximal-four-critical-line hypothesis for binary quartics. No nearby potential is assumed to retain parabolic separation, coordinate-reflection symmetry, or integrability.

5 The complete invariant-average kernel at the non-radial quartic

The coefficient-neighborhood theorem of Section 4 and a classification of all centered forces at one well are different assertions. We now study the latter question at

$$Q_0(x, y) = P_0(x, y) = x^4 + 12x^2y^2 + 16y^4, \qquad H = \tfrac{1}{2}|W|^2 + Q_0.$$

For $r \in \{0, 1, 2\}$ let $\mathcal{K}^r$ consist of the forces $g(\rho e_\theta) = \rho^4 a(\theta)$, $a \in C^r(\mathbb{S}^1)$, whose mean is zero for every invariant probability on $H = 1$. No central symmetry is imposed. The superscript records angular regularity, not a homogeneity degree. The continuous class is sufficient for the exact criterion below. For a jointly C^2 parameter family the first force is in C^1, and its second force is only continuous; neither will be given an extra spatial derivative.

5.1 Separation and the physical clock

Put

$$\begin{gathered} x = uv, \qquad y = (u^2 - v^2)/2, \qquad S = u^2 + v^2, \\ p_u = vW_x + uW_y, \qquad p_v = uW_x - vW_y. \end{gathered} \tag{75}$$

Away from the origin this is a canonical two-fold covering, with deck transformation $(u, v, p_u, p_v) \mapsto (-u, -v, -p_u, -p_v)$. Direct substitution gives

$$H = \frac{p_u^2 + p_v^2}{2S} + \frac{u^{10} + v^{10}}{S}.$$

At physical energy $E > 0$, the regularized Hamiltonian and clock are

$$S(H - E) = \tfrac{1}{2}p_u^2 + U_E(u) + \tfrac{1}{2}p_v^2 + U_E(v), \qquad U_E(q) = q^{10} - Eq^2, \qquad \mathrm{d}t = S\,\mathrm{d}\tau. \tag{76}$$

P Rational action certificates

The strict numerical inequalities used in the proofs are rational interval comparisons. The interval operations below use exact rational endpoints and outward bounds for every transcendental quantity. The coefficient bounds in Lemmas 7 and 13, the cubic coefficient bound, and the sextic detecting integrals are specified analytically in the relevant proofs. The full-annulus twist certificate and the finite-phase and tail certificates for the sextic scalar response are specified in the preceding appendices. This appendix gives the curved-brake certificate.

P.1 Curved-brake interval certificate

Use $N = 4096$ equal angular cells and outward rounding to multiples of 10^{-12} after interval arithmetic operations. Square roots of positive rational numbers are enclosed by integer square roots: if $x = A/B > 0$ and $S = 10^{12}$, take

$$k = \left\lfloor \sqrt{\lfloor AS^2/B \rfloor} \right\rfloor, \qquad k/S \le \sqrt{x} \le (k+1)/S.$$

These enclosures satisfy the squared endpoint inequalities. The starting bounds are

$$3141592653589/10^{12} < \pi < 3141592653590/10^{12},$$

certified by $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and consecutive alternating partial sums (indices through 22 and 8, respectively). On $[0, \pi/2]$, the sine Taylor polynomials through degrees 23 and 21 bound sine below and above, and the cosine polynomials through degrees 22 and 20 do the same. Monotonicity transports endpoint intervals into enclosures of $\sin^2\psi$ over each cell. The final cell endpoints are set to exactly zero or one where appropriate.

The coefficient c in (59) is computed first. With $t = 2\tan\theta$, $s = \sin^2\theta$, $z_0 = s(1-s)$ and $D_0 = 1 + z_0$, the three integrals become

$$I_4 = \int_0^{\pi/2} \frac{1 - 2z_0}{D_0^{3/2}}\,\mathrm{d}\theta, \quad I_{22} = \frac{1}{2}\int_0^{\pi/2} \frac{z_0}{D_0^{3/2}}\,\mathrm{d}\theta, \quad J_* = \frac{1}{4}\int_0^{\pi/2} \frac{z_0(1-2z_0)}{D_0^{5/2}}\,\mathrm{d}\theta.$$

On each cell, the exact range of $s(1-s)$ includes $1/4$ if the cell crosses $1/2$. Summing interval ranges times interval cell lengths encloses the integrals, and the lower bound for I_{22} is positive.

The cubic factor in (62) is strictly increasing, since its derivative is $3z^2 - 4z + 3 > 0$. Its values at $8/5$ and $17/10$ have opposite signs. Fifty exact bisections, followed by outward rounding, enclose its root. For each orbit cell take $w = \cos^2\psi$ and $R = zw$. The denominator in (63) is increasing in both positive variables z, w and has a certified positive lower endpoint. Its reciprocal square-root interval gives the cell duration. Sum these durations to obtain $[Q_-, Q_+]$.

Evaluate the algebraic expression (68) minus (67) on the interval R of each cell, multiply by its duration, and add. To bound the transverse projection, maintain cumulative time bounds $[t_-, t_+]$. Throughout the next cell, the phase $\pi\tau/(2Q)$ lies in

$$\left[\frac{\pi_- t_-}{2Q_+}, \frac{\pi_+(t_+ + \Delta t_+)}{2Q_-}\right].$$

Intersect its cosine range with $[0, 1]$, use the alternating Taylor bounds, and multiply by the force interval from (69) and the duration. This encloses the full projection without sampling its integrand at isolated points.

The following coarser rational bounds are consequences of those operations. All entries in the two numeric columns are integers divided by 10^9.

Quantity	Lower	Upper
c	-264300268	-262819123
Q	909971128	910253649
$v_0^{-2}\int_0^Q (SB_{\mathrm{even}} - B_{\mathrm{aux}})\,\mathrm{d}\tau$	2127830	2184651
$v_0^{-1}\int_0^Q f\cos(\pi\tau/(2Q))\,\mathrm{d}\tau$	-91331979	-90969906

The finer rational endpoints verify $2 < 8Q/\pi < 3$ and

$$16 - (\pi/(2Q))^2 > 13, \qquad (3\pi/(2Q))^2 - 16 > 0.$$

Writing P_{abs} for the maximum absolute projection endpoint divided by v_0, the interval evaluation gives

$$T_- = -\frac{2P_{\mathrm{abs}}^2}{Q_-\,(16 - (\pi_+/(2Q_-))^2)} > -\frac{1408089}{10^9}.$$

The interval sum is at least $2127830/10^9$, and hence the total corrected curvature is greater than

$$\frac{2127830 - 1408089}{10^9} = \frac{719741}{10^9} > \frac{1}{2500}.$$

These estimates prove Lemma 9. An independently organized check with 8192 cells and outward scale 10^{11} gives a lower bound exceeding 0.0007368543; that decimal is a summary, not an input to the proof.

References

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[2] H. Abo, A. Seigal and B. Sturmfels, Eigenconfigurations of tensors, in *Algebraic and Geometric Methods in Discrete Mathematics*, Contemporary Mathematics **685**, American Mathematical Society, 2017, 1–25. doi:10.1090/conm/685/13717. Author preprint.

quartic trace vanishes, or prove that the trace is injective. The exact reconstruction is one-way on known null inputs; its values on arbitrary data need not be ordinary-null. A numerical observation-tail bound and a separating Gram inequality are still required for the finite comparison of Proposition N.1. The OE and OO linear kernels, general mixed-parity realizability and a nonpolynomial finite-neighborhood classification remain separate questions.

8 Controlled correctors and actual point selection

We first prove a dynamical implication in arbitrary finite normal and selected dimensions. Let $U \subset \mathbb{R}^m$ be open and convex and $P \in C^2(U \times \mathbb{R}^n)$ be jointly convex, p-homogeneous in Z, $p > 2$, and coercive near a compact selected tail $K \Subset U$. Set

$$a = 2/p, \qquad \nu = 1/2 + 1/p, \qquad \sigma = \log t, \qquad \varepsilon = e^{-a\sigma}.$$

The physical equations have damping $(1 + a)/t$. Substitution of $q = \varepsilon Z$, $W = \varepsilon Z_\sigma$ and $k = s_\sigma$ gives exactly

$$Z' = \varepsilon^{-1} W, \qquad W' = -\varepsilon^{-1} P_Z, \qquad s' = k, \qquad k' = -ak - P_s. \tag{94}$$

Here and below primes in this section denote σ derivatives. In particular,

$$H = \tfrac{1}{2}|W|^2 + P(s, Z), \quad E = H + \tfrac{1}{2}|k|^2, \qquad H' = k \cdot P_s, \quad E' = -a|k|^2. \tag{95}$$

The formulas are exact, not an averaged model.

Lemma 11 (Compact-tail and relative-energy estimates)**.** *On such a tail, all rescaled states are bounded, $k \in L^2$, $k \to 0$, and $H \to E_\infty \geq 0$. If the normal state is not identically zero, then $H > 0$ at every finite time and, eventually, $|k| \leq CH$. In the case $E_\infty = 0$,*

$$E = H + O(H^2), \qquad |H'| \leq CH^2, \qquad H(\sigma) \geq \frac{1}{H(S)^{-1} + C(\sigma - S)}.$$

Proof. The nonnegative decreasing energy bounds k, W, Z by coercivity. Its dissipation makes $k \in L^2$. The selected force and hence k' are bounded on the compact state region, so uniform continuity and L^2 integrability give $k \to 0$. Equation (95) then gives $H \to E_\infty$. Local uniqueness for the C^1 vector field makes the zero normal state invariant and prevents an excited solution from reaching it at a finite time.

Homogeneity and compactness give $|P_s| \leq C|Z|^p \leq CH$. Thus $|(logH)'| \leq C|k| \to 0$. For each $0 < d < a$, at sufficiently late S and $S \leq u \leq \sigma$,

$$H(u) \leq H(\sigma) e^{d(\sigma - u)}, \qquad H(\sigma) \geq H(S) e^{-d(\sigma - S)}.$$

Variation of constants in the selected equation gives

$$|k(\sigma)| \le e^{-a(\sigma-S)}|k(S)| + C\int_S^\sigma e^{-a(\sigma-u)}H(u)\,\mathrm{d}u \le e^{-a(\sigma-S)}|k(S)| + \frac{C}{a-d}H(\sigma).$$

The first term divided by $H(\sigma)$ tends to zero. This proves $k = O(H)$ without a positive-energy assumption. The remaining assertions follow from (95) and integration of $|(H^{-1})'| \le C$. ∎

Theorem 12 (A controlled force corrector implies selection)**.** *On positive-energy normal states over a neighborhood of K, suppose there are a field b and a C^1 vector-valued function A satisfying*

$$\mathcal{L}_s A = P_s - H_s b(s), \qquad |A| + |A_s| \le C H_s^\nu. \tag{96}$$

The selected derivative A_s is taken at fixed (Z, W). Then every excited actual compact tail has

$$E_\infty > 0, \quad I(s,H) = D(s)H^\nu \to I_\infty > 0, \quad I - I_\infty = O(\varepsilon), \quad s \to s_\infty \in \operatorname{argmin}_U U_{I_\infty}.$$

No additional exactness or convexity assumption on b is needed.

Proof. Integrating (96) against the microcanonical measure proves $b = -\nabla \log D/\nu$ by Lemma 2. Integrating it against any invariant probability also proves the common-mean property. Lemma 3 defines the positive concave D on all of U and the convex effective energy U_I.

On an excited tail put

$$B = A/H, \qquad I = D(s)H^\nu, \qquad \mathcal{J} = \log I - \nu\varepsilon k \cdot B.$$

Since $\mathcal{L}_s H_s = 0$, we have $\mathcal{L}_s B = P_s/H - b$. The full chain rule, with the fixed-state parameter derivative, gives

$$B' = \varepsilon^{-1}\mathcal{L}_s B + B_s k, \qquad B_s = A_s/H - A \otimes P_s/H^2, \qquad |B| + |B_s| \le C H^{\nu-1}.$$

The integrating factor cancels the mean work: $(\log I)' = \nu k \cdot \mathcal{L}_s B$. Using $\varepsilon' = -a\varepsilon$ and $k' = -ak - P_s$ yields the exact identity

$$\mathcal{J}' = \nu\varepsilon\big[(2ak + P_s)\cdot B - k \cdot (B_s k)\big]. \tag{97}$$

Lemma 11 now gives

$$|\varepsilon k \cdot B| \le C\varepsilon H^\nu, \qquad |\mathcal{J}'| \le C\varepsilon(H^\nu + H^{\nu+1}).$$

The energy is bounded, and $\int_\sigma^\infty \varepsilon\,\mathrm{d}u = \varepsilon(\sigma)/a$. Thus $\mathcal{J}$ has a finite real limit with an $O(\varepsilon)$ tail error. The correction tends to zero at the same rate. Consequently $\log I$ has a finite real limit, $I_\infty > 0$, and $I - I_\infty = O(\varepsilon)$. Since D is bounded above and below on K,

this also bounds H away from zero and proves $E_\infty > 0$. In particular, positivity has not been assumed in order to justify division by the varying energy.

Set $U_\infty = U_{I_\infty}$ and $v = k + \varepsilon A$. Direct differentiation gives

$$s' = v - \varepsilon A, \qquad v' = -av - Hb + \varepsilon A_s k.$$

The action estimate implies $H = U_\infty(s) + O(\varepsilon)$, and $\nabla U_\infty = U_\infty b$. Hence

$$s' = v + r_s, \qquad v' = -av - \nabla U_\infty(s) + r_v, \qquad |r_s| + |r_v| \le C\varepsilon, \quad v \to 0, \quad U_\infty(s) \to E_\infty. \tag{98}$$

Every selected cluster point u satisfies $\nabla U_\infty(u) = 0$: integrate the second equation over a fixed-length interval starting at a subsequence with $s \to u$. The first equation and $v \to 0$ keep s uniformly close to u on that interval; all other integrated terms tend to zero. By convexity, u is a global minimizer on U.

For any such u, put $y = s + v/a$ and $Q_u = |y-u|^2/2 + U_\infty(s)/a^2$. An exact cancellation gives

$$Q_u' = -\frac{(s-u)\cdot \nabla U_\infty(s)}{a} + (y-u)\cdot(r_s + r_v/a) + \frac{\nabla U_\infty(s)\cdot r_s}{a^2}. \tag{99}$$

The first term is nonpositive and the remaining terms are integrable. Since Q_u is bounded below, it has a finite limit. As $U_\infty(s)$ converges and $v \to 0$, the distance $|s-u|$ has a limit. Choosing u to be a cluster point makes that limit zero. This finite-dimensional, integrable-error convex-distance argument is consistent with the classical damped convex-gradient mechanism [3]; no finite-length or isolated-minimum hypothesis has been used. ∎

Proof of Theorem 3. In the radial case $P = \kappa(s)|Z|^p$, $p \in \{3, 4, 6\}$, (7) gives $b = 2\nabla \log \kappa/(p+2)$. The centered derivative is $P_{s_j} = (p+2)b_j P/2$. In the nonradial quartic case, Theorem 2 gives globally

$$P_{s_j} - 3b_j P = c_j \mathcal{R}P, \qquad c_j = \frac{\langle P_{s_j} - 3b_j P, \mathcal{R}P\rangle_{\text{coeff}}}{\|\mathcal{R}P\|^2_{\text{coeff}}}. \tag{100}$$

The denominator is nonzero. On compact parameter subsets the amplitude stays positive and all coefficient derivatives are bounded, so c_j and its first selected derivatives are bounded. This formula does not require a global angle.

The two mechanical identities

$$\mathcal{L}_s(Z\cdot W) = 2H_s - (p+2)P, \qquad \mathcal{L}_s(W\cdot JZ) = -\mathcal{R}P$$

show that

$$A_j = -c_j W\cdot JZ - \frac{b_j}{2} Z\cdot W \tag{101}$$

satisfies (96), with $c_j = 0$ in the radial case. Coercivity gives $|Z||W| \le CH^\nu$. Since the selected derivative is at fixed (Z, W), its bound follows from the coefficient derivatives directly. The corrector extends as zero at the zero normal state and is C^1. Theorem 12

proves all excited-tail claims. On the invariant zero-normal branch, $k' = -ak$, whence $s_\infty = s(\sigma_0) + k(\sigma_0)/a$. Finally $q = \varepsilon Z \to 0$ by boundedness of Z. ∎

For example, in any fixed selected coordinate r, the same corrector controls the signed radial work of the actual tail:

$$-\int_\alpha^\beta r(P_r - Hb_r)\,\mathrm{d}\sigma = -[\varepsilon r A_r]_\alpha^\beta + \int_\alpha^\beta \varepsilon\big[(k_r - ar)A_r + r(A_r)_s \cdot k\big]\,\mathrm{d}\sigma. \tag{102}$$

The right side is $O(e^{-a\alpha})$, uniformly in $\beta \geq \alpha$. It is a signed work estimate; it does not assert absolute integrability of the oscillating force.

8.1 A corrector-free supporting implication

The following implication uses pointwise stationarity only on the effective minimizing set. We give its proof separately from the finite-classification corrector argument in order to keep the residual-energy regimes and hypotheses distinct.

Lemma 12 (Attraction under common means)**.** *Under the compact-tail and original convexity assumptions of this section, suppose* (CM) *holds near K. Every selected cluster point of an excited actual tail belongs to $M = \operatorname{argmax}_U D = \operatorname{argmin}_U U_1$, whether $E_\infty > 0$ or $E_\infty = 0$.*

Proof. For $h = E_\infty > 0$, take $\sigma_j \to \infty$ with $s(\sigma_j) \to u$ and consider the unit-window occupations of $X = (Z, W)$. Compactness gives a weak limit supported on $H_u = h$. For every C^1 test ϕ,

$$\int_{\sigma_j}^{\sigma_j+1} \mathcal{L}_{s(\sigma)}\phi(X(\sigma))\,\mathrm{d}\sigma = [\varepsilon\phi]_{\sigma_j}^{\sigma_j+1} + a\int_{\sigma_j}^{\sigma_j+1} \varepsilon\phi\,\mathrm{d}\sigma \longrightarrow 0.$$

Since $k \to 0$, the parameter freezes uniformly over that interval. The limit is invariant for H_u. Integration of $k' = -ak - P_s$ shows its force mean is zero. Thus $0 = hb(u)$; Lemma 3 makes u a global minimizer.

For $E_\infty = 0$ the preceding unnormalized argument cannot be divided by h. Instead put

$$v = k/E, \quad x = (z, w) = (E^{-1/p}Z, E^{-1/2}W), \quad \gamma = 1/2 - 1/p, \quad \delta = \varepsilon E^{-\gamma}.$$

Lemma 11 gives bounded s, v, x, $E \geq c/(1+\sigma)$, and hence $\delta \to 0$. With $J_s(x) = |w|^2/2 + P(s, z)$, $f_s = P_s(s, z)$, and $\mathcal{D}_x = (z/p, w/2)$, the exact equations are

$$\begin{aligned} J_s(x) &= 1 - E|v|^2/2, & s' &= Ev, \\ v' &= -av - f_s(x) + aE|v|^2 v, & x' &= \delta^{-1}X_s(x) + aE|v|^2\mathcal{D}_x, \\ E' &= -aE^2|v|^2, & \delta' &= -a\delta(1 - \gamma E|v|^2). \end{aligned} \tag{103}$$

For fixed late S define the probability

$$\mu_\sigma = a\int_S^\sigma e^{-a(\sigma-u)}\delta_{x(u)}\,\mathrm{d}u + e^{-a(\sigma-S)}\delta_{x(S)}.$$

Variation of constants in the v equation, splitting at $\sigma - R$, gives

$$v(\sigma) + \frac{1}{a}\int f_{s(\sigma)}\,\mathrm{d}\mu_\sigma \longrightarrow 0. \tag{104}$$

Indeed the remote mass is $O(e^{-aR})$, while on the recent window $\sup|s(u) - s(\sigma)| \le CRE(\sigma - R) \to 0$ and the cubic remainder tends to zero. First take $\sigma \to \infty$ for fixed R, then $R \to \infty$.

The same splitting makes any weak limit along $s(\sigma_j) \to s_*$ supported on $J_{s_*} = 1$. For a C^1 test,

$$\mathcal{L}_{s(u)}\phi(x(u)) = \delta(u)\frac{\mathrm{d}}{\mathrm{d}u}\phi(x(u)) - a\delta(u)E(u)|v(u)|^2\mathcal{D}\phi(x(u)).$$

Integration by parts against the exponential weight gives a vanishing integral: the boundary terms contain δ, and the differentiated weight uses $a\delta + \delta' = a\gamma E|v|^2\delta$. Thus the limit is frozen invariant. The generator criterion used here is immediate by applying the zero-generator integral to ϕ composed with the frozen flow. The common-mean hypothesis and compactness applied to (104) give

$$v + b(s)/a \longrightarrow 0. \tag{105}$$

Introduce energy time by $\mathrm{d}\tau/\,\mathrm{d}\sigma = E$. Its range is unbounded by the reciprocal lower bound. The selected paths are uniformly Lipschitz in τ because $\mathrm{d}s/\,\mathrm{d}\tau = v$. For any translates $\tau_j \to \infty$, local uniform subsequential limits are complete paths $S : \mathbb{R} \to K$. Equation (105) implies

$$\dot{S} = -b(S)/a = -\frac{\nabla U_1(S)}{aU_1(S)}.$$

Along a complete such path, $\mathrm{d}U_1(S)/\,\mathrm{d}\tau = -|\nabla U_1(S)|^2/(aU_1(S))$. Compactness and positivity of U_1 give a finite backward dissipation integral. Its integrand is uniformly continuous, so the gradient tends to zero as $\tau \to -\infty$. A backward cluster point is a global minimizer of the convex U_1. Monotonicity then forces $U_1(S)$ to be that minimum for every finite τ; the complete path is constant. Every selected cluster point of the original tail is the value at zero of such a translated limit, and therefore belongs to M. ∎

Proposition 8.1 (Pointwise-stationary minima)**.** In Lemma 12, assume additionally that $P_s(u, Z) = 0$ for every $u \in M$ and every Z. Then the actual selected tail converges to one point of M in either residual-energy regime and in arbitrary finite selected dimension.

Proof. For each $u \in M$, convexity of $P(\cdot, Z)$ gives $(s-u)\cdot P_s(s, Z) \ge P(s, Z) - P(u, Z) \ge 0$. Set $h_u = |s - u|^2/2$. The exact selected equation yields

$$h_u'' + ah_u' = |k|^2 - (s - u)\cdot P_s(s, Z) \le |k|^2.$$

Duhamel's inequality bounds the positive part of h'_u by an exponentially decaying initial term plus the convolution of $|k|^2 \in L^1$ with $e^{-a\sigma}$. Its positive part is therefore integrable. As $h_u \geq 0$, its downward variation is also finite, and h_u has a limit. Lemma 12 supplies a cluster point $u \in M$; for this choice the limit is zero. No rate in (105), positivity of E_∞, or finite-length statement is inferred. ∎

9 Supporting first-jet results and sextic resonant admission

This section records a supporting cubic first-jet consequence and proves sextic first-jet rigidity and the resonant estimates used in the finite sextic proof. The cubic and quartic finite classifications were established in Theorem 5; the sextic finite classification is proved in Theorem 17. The hypotheses are joint C^2 regularity, positivity and local uniform normal coercivity, homogeneity in two normal variables, and the common actual-force mean property at every nearby parameter. The parameter dimension is arbitrary. Each assertion is applied separately along a parameter direction. The normalization, compact reduction, and response identities proved in Section 3 apply without polynomiality or evenness.

9.1 The cubic first-jet consequence

Theorem 13 (Planar cubic first-jet rigidity)**.** *If $P(s_0, Z) = a|Z|^3$, with $a > 0$, then*

$$\partial_{s_j} P(s_0, Z) = \frac{5}{2} b_j(s_0) P(s_0, Z)$$

for every parameter component.

Proof. After a fixed amplitude rescaling and (18), write the centered first jet as $g(\rho e_\theta) = \rho^3 \gamma(\theta)$. Lemma 6 applies to the cubic radial family, including its origin crossings: the scalar orbit is C^3, which gives a C^2 phase manifold in H^1, while the potential $|Z|^3$ itself is C^2. The circle is nonresonant because $\sqrt{5} \notin \mathbb{Z}$. Thus Proposition 3.1 gives the two identities (28). In this case

$$D_{3,1} = \frac{1}{24}, \qquad -\frac{49}{24} \leq D_{3,n} < 0 \quad (n \geq 3 \text{ odd}).$$

The second bound follows directly from $D_{3,n} = -1 - 25/[6(n^2-5)]$. Add the two response identities. At $n = 1$ the combined weight is $C_3 - 7/24 > 0$. For odd $n \geq 3$ it is at least $9C_3 - 19/8 > 1/4$. Absolute convergence and nonnegative w_n force all modes to vanish. Hence $g = 0$, and undoing normalization gives the coefficient $5/2$. ∎

If $w_1 > 0$, its radial difference equation forces

$$(C_6 - \tfrac{1}{6}) + \frac{9C_6 - \frac{1}{6}}{511} = 0, \qquad C_6 = \frac{32}{195},$$

contrary to Lemma 13. If $w_1 = 0$, the circle equation also gives $w_3 = 0$. Thus $g = 0$. Since the normalization exponent in (18) is 3, undoing it gives exactly the factor 4 in the theorem. ∎

9.4 Scope of the supporting consequences

Under the original jointly convex compact-tail assumptions, suppose the common frozen potential on the effective minimizing set M is a positive multiple of $|Z|^3$ or of $|Z|^6$, and assume common means only on a neighborhood of the compact tail set K. Lemma 12 supplies a cluster point $u_0 \in K \cap M$. Common means hold near u_0 and $b(u_0) = 0$, so the corresponding first-jet theorem gives $P_s(u_0, Z) = 0$ for every Z. For any $u \in M$, constancy of the frozen potential along $u_t = u_0 + t(u - u_0)$ gives $(u-u_0)^{\mathsf{T}} P_{ss}(u_t, Z)(u-u_0) = 0$. Positivity of the original joint Hessian implies $P_{ss}(u_t, Z)(u - u_0) = 0$, as in (11); integrating $\frac{d}{dt} P_s(u_t, Z) = 0$ gives

$$P_s(u, Z) = P_s(u_0, Z) = 0 \qquad (u \in M).$$

Proposition 8.1 therefore applies in both residual-energy regimes. This uses convexity of the original potential and does not require common means outside the specified neighborhood of K, or joint convexity of the amplitude-normalized potential.

The first-jet proof above uses infinitely many distinct detecting periodic families, one for each odd mode above three. By itself it supplies neither a uniform normal inverse for these families nor a uniform nonlinear reduction radius. The finite sextic proof in Theorem 17 therefore uses a different continuation argument: exact minimum actions in one fixed physical annulus, followed by the coherent response estimate (2). It does not infer finite rigidity from vanishing of the first jet.

10 Comparison with critical-line criteria and scope

10.1 What the two finite classifications add

For orientation, two simpler automatic-corrector criteria can be stated and checked directly.

First, let P be a binary polynomial of even degree p and suppose $\mathcal{R}P$ is nonzero with p distinct real projective zero lines. At each such line the straight nonlinear oscillator is periodic. By the virial identity, common means force

$$F_j = P_{s_j} - \frac{p+2}{2} b_j P$$

to vanish on every one of those lines. A homogeneous polynomial vanishing on a real line is divisible by its defining linear factor. The p distinct factors are coprime and exhaust the degree of $\mathcal{R}P$, so $F_j = c_j \mathcal{R}P$. The coefficient quotient gives C^1 functions c_j on a region where this hypothesis holds. The corrector is again

$$A_j = -c_j W \cdot JZ - (b_j/2) Z \cdot W.$$

This is a maximal-critical-line criterion, counting unoriented projective lines, including a line at infinity in an affine chart. It excludes radial wells, for which $\mathcal{R}P = 0$, and the base (3), which has only two such lines instead of four.

Second, in any normal dimension n, near a positive diagonal quartic $P_*(Z) = \sum_i \alpha_i z_i^4$, the critical lines have representatives

$$z_i = \begin{cases} \varepsilon_i/\sqrt{\alpha_i}, & i \in A, \\ 0, & i \notin A, \end{cases} \qquad \varnothing \neq A \subset \{1, \dots, n\}, \quad \varepsilon_i \in \{\pm 1\},$$

modulo simultaneous sign reversal. There are $(3^n - 1)/2$ lines, all simple. The constrained Hessian has eigenvalues $8P_*(e)$ on the active tangent space and $-4P_*(e)$ on the inactive directions, so these branches persist. Evaluation of quartics on this finite set has kernel

$$\operatorname{span}\{\alpha_i z_i^3 z_j - \alpha_j z_i z_j^3 : i < j\} = \{(P_*)_Z \cdot BZ : B^{\mathsf{T}} = -B\}.$$

To verify the kernel, single-coordinate supports kill pure fourth powers; the two relative signs on two-coordinate supports kill the square-square term and impose the displayed relation on the two cubic-linear terms. After subtracting those pair terms, the three independent sign characters on three-coordinate supports kill all square-linear-linear terms. Four-coordinate supports then kill the remaining four-linear terms. A nonzero rank minor and independence of the rotational derivatives persist in a coefficient neighborhood. Since every rotational derivative always vanishes on a critical line, the kernel there remains exactly the rotational space. Common means consequently imply

$$P_{s_j} - 3b_j P = P_Z \cdot B_j Z, \qquad A_j = -W \cdot B_j Z - (b_j/2) Z \cdot W.$$

The C^1 coefficient fields follow from the Gram-matrix inverse of the independent rotational columns. This criterion is polynomial, works in arbitrary normal dimension, and requires the family to lie in its coefficient region. Its elementary critical-line description is related to the eigenvector configurations of homogeneous tensors [2, 8] and to Darboux-point methods [20]; no integrability criterion from those works is invoked here.

Both criteria already imply pathwise normalized-shape rigidity: if $\widetilde{P} = D^p P$, then $\widetilde{P}_{s_j} = \widetilde{P}_Z \cdot B_j Z$. Along a parameter path, solve $O' = -BO$, with $B = \sum_j B_j s_j'$. Skew symmetry keeps O orthogonal and $\mathrm{d}\widetilde{P}(s(t), O(t)Z)/\,\mathrm{d}t = 0$. Thus the fact that an established rotation identity integrates to an orthogonal orbit is not unique to Theorem 2. The

additions here are its deficient two-critical-line neighborhood, the curved-action second-order obstruction through a rank jump, and the one-entry continuation over an entire connected region. The radial theorem instead works in the infinite-dimensional C^2 angular class at a point with a continuum of critical directions. These are different mechanisms from finite evaluation rank.

10.2 Why the four nonradial tests need the polynomial restriction

The following limited boundary statement explains the role of coefficient dimension. It is not a counterexample to (CM).

Proposition 10.1 (Four invariant values admit nonpolynomial compensation)**.** There is a C^2 positive quartic-homogeneous family $V(r, Z)$ for small $|r|$, with $V(0) = P_0$ and

$$V_r(0, \rho e_\theta) = \rho^4 \sin(6\theta),$$

for which the two continued critical values, $\mathcal{I}(V)$, and the continued curved-orbit action are all exactly constant. A positive scalar multiplier makes the original family jointly convex while preserving equality of the force means for the two straight orbital measures, the curved orbital measure, and the microcanonical measure. The construction makes no assertion about the remaining invariant probabilities.

Proof. For a smooth π-periodic angular function β, let $B(\rho e_\theta) = \rho^4 \beta(\theta)$. The four derivatives at P_0 are

$$\beta(0), \quad \beta(\pi/2), \quad -\tfrac{1}{2} \int \beta P_0^{-3/2} \, \mathrm{d}\theta, \quad -\int_{\text{brake}} \rho(t)^4 \beta(\theta(t)) \, \mathrm{d}t.$$

They annihilate $\sin 6\theta$ by its axial zeros and reflection parity. This tangent is not a rotation derivative, since (57) has angular frequencies two and four only.

These four functionals are independent on smooth angular functions. The first two are distinct point masses; the volume weight is smooth and positive. On the parabolic brake orbit, $\mathrm{d}\theta/\,\mathrm{d}u = 2v_0/(u^2 + v_0^2) > 0$. At the nonaxial turning angle θ_*, the turning point is nondegenerate and the pushforward of $\rho^4 \, \mathrm{d}t$ has a positive one-sided density proportional to $|\theta - \theta_*|^{-1/2}$. It cannot be proportional to the smooth volume weight. Choose two smooth bumps controlling the axial evaluations and two bumps away from the axes with different ratios of brake-weighted to volume-weighted integrals. A bump sufficiently close to the turning angle and a bump on a regular interval supply those different ratios. Their four-by-four derivative matrix is invertible.

For their homogeneous extensions B_i, apply the finite-dimensional implicit-function theorem to the four invariants of $P_0 + r\rho^4 \sin 6\theta + \sum_i a_i B_i$. It gives smooth $a(r)$ with $a(0) = a'(0) = 0$ and all four values exactly constant. The orbit stays away from the

origin, so the required return-map smoothness concerns only a smooth finite-dimensional perturbation on that region. The homogeneous potentials are C^3, hence C^2, at the origin; positivity and strict normal convexity persist for small r.

Finally set $P(r, Z) = e^{\lambda r^2} V(r, Z)$. At $r = 0$, its selected Schur complement is

$$V_{rr}(0, Z) + 2\lambda P_0(Z) - \nabla g(Z)^{\mathsf{T}} (P_0)_{ZZ}^{-1} \nabla g(Z), \qquad g = \rho^4 \sin 6\theta.$$

All terms other than $2\lambda P_0$ are bounded in modulus by $C|Z|^4$, and $P_0 \geq c|Z|^4$. Large λ makes this positive for $Z \neq 0$, uniformly on the unit circle. Continuity and homogeneity preserve positivity for small r; at $Z = 0$ the full continuous Hessian is zero. Thus P is jointly convex. Homogeneous transport of the four measures and the virial identity give their common unit-energy force mean $2\lambda r/3$. ∎

The constrained polynomial direction in Lemma 9 has too few second-jet coefficients to cancel the curved-action defect. The smooth angular class supplies extra compensating directions. By contrast, the cubic and quartic radial proof uses an entire compact family of periodic actions and an all-angular coercive response. At degree six, exact minimum actions and the actual brake boundary invariant supply additional constraints, and the response estimate excludes frequency escape. Proposition 10.1 neither supplies all invariant means nor an actual nonselecting trajectory.

10.3 Remaining classification questions

The fixed-base analysis now distinguishes several remaining questions. The doubly even linear component is eliminated, and the polynomial intersection is exactly rotational. The EO linear kernel is finite-dimensional, but neither its zero-vertical dimension nor the existence of a nonzero-vertical full-null fiber is decided. The quartic value trace is not known to vanish or to be injective. The OE injectivity and OO rotation-only linear assertions remain open. Pure-parity two-jet restrictions and the same-family odd-y rank bound do not give a general mixed-parity theorem. Nor does rotationality of one initial derivative exclude higher-order flat departures of a nonlinear family.

The exact reconstruction and mechanical contraction provide a route to a certified finite comparison, with the precise sufficient condition in Proposition N.1. They do not yet supply the numerical horizontal weighted constant, all retained-interval approximation constants or a verified separating margin required by that condition. These are quantitative obligations for the explicit operators, not a license to truncate the unknown angular input without a tail bound.

The present finite family results do not classify connected common-mean families whose shapes never meet a radial member at degree three, four or six, or the certified polynomial neighborhood at degree four. Outside these classes, equality of invariant force means still leaves open the required quantitative and parameter-dependent control of a force primitive. The radial sextic theorem closes finite deformations through a radial

member; it does not classify sextic families whose normalized shapes never meet that locus. No assertion is made that the collection of tests used here exhausts all invariant measures of a general planar well.

Other isoperiodic and action-rigidity problems impose substantially different data and hypotheses. One-dimensional isoperiodic shears [1] preserve period data for scalar systems. The Suris-potential rigidity result [16] concerns discrete twist-map families with high regularity. The Zoll-near-a-minimum theorem [9] assumes a compact symplectic Morse–Bott minimum and a Zoll sequence. A quartic degenerate normal origin does not satisfy that nondegenerate normal-Hessian hypothesis; the Morse–Bott family used in our periodic-loop reduction is a different object. These works are relevant comparisons, not sources of the finite mechanical classifications proved here.

A Sextic mechanical actions and finite shape constraints

We use normalized angular measure throughout this section: $\langle g\rangle = (2\pi)^{-1}\int_0^{2\pi} g(\theta)\,d\theta$ and $\|g\|_{H^1}^2 = \langle g^2+(g')^2\rangle$. All the statements concern positive, positively homogeneous potentials of degree six. Antipodal symmetry and polynomial dependence are not assumed. Joint convexity is not used in any of the arguments in this section.

A.1 Normalization of the actual parameter force

Let $P(s,Z)$ be jointly C^2 on an open finite-dimensional parameter set times $\mathbb{R}^2$, positive for $Z\neq 0$, and suppose that (CM) holds at every parameter under consideration. Define

$$J_6(s) = \langle P(s,e_\theta)^{-1/3}\rangle, \qquad V(s,Z) = J_6(s)^3 P(s,Z). \tag{113}$$

If $P(s_0,Z)$ is a positive multiple of $|Z|^6$, then $V(s_0,Z)=|Z|^6$. Translate the base parameter so that $s_0=0$.

Lemma 16 (Normalized sextic force)**.** *The normalized family is jointly C^2, satisfies $\langle V(s,e_\theta)^{-1/3}\rangle = 1$, and has*

$$\int V_{s_j}(s,Z)\,d\mu = 0 \tag{114}$$

for every invariant probability on $|W|^2/2+V(s,Z)=1$ and every parameter component. Consequently every nearby critical point of the free-period action (12)*, with $p=6$, has zero parameter derivative.*

Proof. Formula (16) becomes

$$\Omega_P(s,E) = \frac{3\pi^2}{2} J_6(s) E^{4/3}.$$

Evaluate the algebraic expression (68) minus (67) on the interval R of each cell, multiply by its duration, and add. To bound the transverse projection, maintain cumulative time bounds $[t_-, t_+]$. Throughout the next cell, the phase $\pi\tau/(2Q)$ lies in

$$\left[\frac{\pi_- t_-}{2Q_+}, \frac{\pi_+(t_+ + \Delta t_+)}{2Q_-}\right].$$

Intersect its cosine range with $[0, 1]$, use the alternating Taylor bounds, and multiply by the force interval from (69) and the duration. This encloses the full projection without sampling its integrand at isolated points.

The following coarser rational bounds are consequences of those operations. All entries in the two numeric columns are integers divided by 10^9.

Quantity	Lower	Upper
c	-264300268	-262819123
Q	909971128	910253649
$v_0^{-2}\int_0^Q (SB_{\mathrm{even}} - B_{\mathrm{aux}})\,\mathrm{d}\tau$	2127830	2184651
$v_0^{-1}\int_0^Q f\cos(\pi\tau/(2Q))\,\mathrm{d}\tau$	-91331979	-90969906

The finer rational endpoints verify $2 < 8Q/\pi < 3$ and

$$16 - (\pi/(2Q))^2 > 13, \qquad (3\pi/(2Q))^2 - 16 > 0.$$

Writing P_{abs} for the maximum absolute projection endpoint divided by v_0, the interval evaluation gives

$$T_- = -\frac{2P_{\mathrm{abs}}^2}{Q_-\,(16 - (\pi_+/(2Q_-))^2)} > -\frac{1408089}{10^9}.$$

The interval sum is at least $2127830/10^9$, and hence the total corrected curvature is greater than

$$\frac{2127830 - 1408089}{10^9} = \frac{719741}{10^9} > \frac{1}{2500}.$$

These estimates prove Lemma 9. An independently organized check with 8192 cells and outward scale 10^{11} gives a lower bound exceeding 0.0007368543; that decimal is a summary, not an input to the proof.

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